Principles of fluid flows

Jinyuan Liu[1]

October 29, 2026

[1]email address: wallturb@gmail.com

Preface: This note, intended for being used as a quick reference, provides a collection of a wide range of equations in fluid mechanics. We try to provide enough details so that all derivations can be reproduced. We also try to use symbols and notations as consistently as possible, and it is unavoidable that this note is biased to the author's own taste and experience. The logical order of the materials is from general to special, which is arguably counter-pedagogical. Thus, this note does not pretend to be an introductory one (on how this World works). This note is still under construction.

Contents

1 Conservation laws **11**

1.1 Reynolds transport theorem and generic conservation laws . . . 11
1.1.1 Reynolds transport theorem in a moving and deforming control volume . 12
1.1.2 Material derivative . . . 13
1.2 Mass conservation . . . 13
1.2.1 Incompressibility, continuity . . . 13
1.3 Momentum conservation . . . 14
1.4 Energy conservation . . . 15
1.4.1 Mechanical energy equation . . . 15
1.4.2 Total energy equation . . . 16
1.4.3 Internal energy equation . . . 17
1.4.4 Heat equation . . . 17
1.4.5 Entropy equation . . . 17
1.5 Bernoulli equation . . . 18
1.6 Constitutive relations . . . 19
1.7 Pressure Poisson . . . 19

2 Vortex dynamics **21**

2.1 Kelvin's theorem . . . 21
2.1.1 Helmholtz's theorems . . . 22
2.2 Vorticity transport equation . . . 22
2.2.1 Enstrophy transport equation . . . 23
2.3 Vorticity–streamfunction formulation . . . 24
2.3.1 In two dimensions . . . 25
2.3.2 In three dimensions (omega y-Laplacian v) . . . 26
2.4 Potential flow . . . 27
2.4.1 2D potential flow in cylindrical coordinates . . . 28
2.5 Stokes flows . . . 29
2.6 Beltrami flows . . . 29
2.6.1 Examples . . . 30
2.7 Point vortex models . . . 31
2.8 Lamb–Oseen similarity solution, Burgers vortex . . . 31
2.9 Heat diffusion: a similarity solution . . . 33

3 Velocity gradient tensor, its decomposition and dynamics **35**
3.1 Pseudo-vector and associated antisymmetric rotation tensor . . . 36
3.2 Strain rate tensor . . . 37
3.3 Velocity field decomposition examples . . . 38
3.4 Dynamics of the velocity gradient tensor . . . 39
3.4.1 Dynamics of $\mathbf{A}$ and its powers . . . 39
3.4.2 Dynamics of $\mathbf{S}$, $\mathbf{\Omega}$, and the eigenvalues of $\mathbf{S^2 + \Omega^2}$ . . . 40
3.4.3 Dynamics of the invariant space . . . 41
3.5 Lagrangian representations . . . 42

4 Laminar wall flows **43**
4.1 Plane Poiseuille–Couette . . . 43
4.2 Laminar Poiseuille channel . . . 44
4.3 Hagen–Poiseuille (circular pipe) flow . . . 46
4.4 The boundary layer theory . . . 47
4.5 Boundary layer equations . . . 47
4.5.1 von Kármán momentum integral . . . 48
4.5.2 The Blasius similarity solution . . . 51
4.5.3 Boundary layer with external pressure gradient, separation . . . 53
4.5.4 Falkner–Skan Equations . . . 53
4.5.5 von Kármán swirling disk flow . . . 54
4.5.6 Laminar planar mixing layer . . . 54
4.5.7 Laminar planar jet and wake . . . 55
4.5.8 Lift and drag . . . 55

5 Turbulent flows **57**
5.1 Mean flow and fluctuations . . . 57
5.1.1 Reynolds average . . . 57
5.1.2 Continuity and momentum . . . 57
5.1.3 Transport equation of the fluctuating velocity . . . 59
5.1.4 Mean-flow and turbulent kinetic energy . . . 59
5.1.5 MKE equation . . . 59
5.1.6 TKE equation . . . 60
5.1.7 Reynolds stress transport equation . . . 62
5.1.8 TKE equations in cylindrical coordinates . . . 64
5.1.9 Dissipation rate transport equation . . . 70
5.1.10 Scalar flux, its mean and variance transport equations . . . 71
5.1.11 Poisson equation for mean and fluctuation pressure . . . 73
5.1.12 Turbulent vorticity and enstrophy . . . 74
5.1.13 Spectral TKE equation . . . 75
5.1.14 Error evolution equation . . . 78
5.2 LES equations . . . 78
5.2.1 Filtering . . . 78
5.2.2 Filtered equations of motion . . . 79
5.2.3 LES energy equations . . . 81
5.2.4 Smagorinsky and Dynamic models . . . 83

5.2.5 Triple decomposition TKE equation . . . 83
5.3 Theory of homogeneous isotropic turbulence . . . 87
5.3.1 Correlation functions . . . 87
5.3.2 Structure function . . . 89
5.3.3 The Kármán–Horwarth equation and the Kolmorogov equation . . . 90
5.3.4 The generalized Kolmogorov equation, Kármán–Horwarth–Monin–Hill . . . 92
5.3.5 Energy spectrum and its evolution equation . . . 92
5.4 Scales of turbulent motions . . . 94
5.4.1 The integral length scale . . . 94
5.4.2 The Kolmogorov scale . . . 94
5.4.3 The Taylor microscale . . . 95
5.4.4 Other useful scales . . . 97
5.5 Turbulent free shear flows . . . 97
5.5.1 Momentum integral . . . 97
5.5.2 Similarity solutions . . . 97
5.5.3 Round jet . . . 98
5.5.4 Plane jet . . . 100
5.5.5 Round wake . . . 101
5.5.6 Plane wake . . . 103
5.5.7 Plane mixing layer . . . 106
5.6 Turbulent wall flows . . . 107
5.6.1 Turbulent channel flow . . . 107

6 Geophysical fluid dynamics 109
6.1 Basics . . . 109
6.1.1 Non-inertial frames, centrifugal and Coriolis forces . . . 109
6.1.2 Absolute velocity . . . 109
6.1.3 Inertial oscillations: buoyancy and Coriolis frequencies . . . 109
6.2 Boussinesq approximation . . . 109
6.3 Balanced flows . . . 109
6.3.1 Hydrostatic and geostrophic balances . . . 109
6.3.2 Thermal wind relations . . . 110
6.3.3 Cyclostrophic wind relations . . . 111
6.3.4 Example: Taylor-Proudman theory . . . 111
6.3.5 Surface and bottom Ekman layers . . . 112
6.4 QGPV theory . . . 114
6.5 Governing equations of unbalanced motions . . . 116
6.5.1 Incompressibility . . . 117
6.5.2 Scalar transport equation . . . 118
6.6 GFD vorticity equations . . . 119
6.6.1 Absolute vorticity equation . . . 119
6.6.2 Potential vorticity equation; Ertal's theorem . . . 120
6.6.3 Relation of PV to Kelvin's theorem in a rotating frame . . . 121
6.7 Turbulence equations for an active scalar . . . 122
6.7.1 Mean flow equations . . . 122
6.7.2 Fluctuation equations . . . 122

6.7.3 MKE, MPE, TKE, TPE, and buoyancy flux equations . . . 123
6.8 Miscellaneous . . . 125
6.8.1 Derivation of Coriolis force . . . 125
6.8.2 Boussinesq approximation . . . 125

7 Waves in GFD **127**
7.1 Internal waves: governing equations and dispersion relation . . . 127
7.1.1 General solutions . . . 129
7.1.2 Hydrostatic approximation . . . 129
7.1.3 Near-inertial waves . . . 129
7.1.4 Vertical modes . . . 130
7.2 Inertial and buoyancy oscillations . . . 131
7.3 Shallow water waves . . . 131
7.3.1 Travelling wave solutions . . . 134
7.4 Deep water waves . . . 135

8 Hydrodynamic stability **137**
8.1 Linearized Navier–Stokes . . . 137
8.1.1 The role of pressure . . . 138
8.1.2 The role of linear mechanisms, the Reynolds–Orr equation . . . 139
8.2 Parallel shear flows . . . 140
8.2.1 Orr–Sommerfeld equations . . . 142
8.2.2 Rayleigh's inflection point criterion . . . 143
8.2.3 Squire's transformation and theorem . . . 145
8.2.4 Taylor–Goldstein equation . . . 146
8.2.5 Howard's semicircle theorem . . . 147
8.2.6 Miles–Howard sufficient condition . . . 148
8.2.7 The Orr–Sommerfeld–Squires system . . . 150
8.2.8 Transient growth . . . 153
8.2.9 Resolvent analysis . . . 155
8.2.10 The Orr and lift-up mechanisms . . . 158
8.2.11 The adjoint OSS operator . . . 158
8.2.12 Axisymmetric parallel shear flows . . . 158
8.3 Centrifugal and rotational instabilities . . . 164
8.3.1 Rayleigh's criterion . . . 164
8.3.2 Axisymmetric circular flows: normal mode analysis . . . 165
8.3.3 Inertial/Coriolis instability . . . 166
8.4 Some classical instabilities . . . 167
8.4.1 Kelvin–Helmholtz and Rayleigh–Taylor instabilities . . . 167
8.4.2 Taylor–Couette instability . . . 170
8.4.3 Rayleigh–Bénard instability . . . 170
8.4.4 The Lorenz system . . . 170
8.5 Adjoints . . . 171
8.5.1 Adjoint matrices, operators, and equations . . . 171
8.5.2 Non-self-adjointness and non-normality . . . 173
8.5.3 Adjoint of the linearised N-S equations (Op's) . . . 173

9 Compressible flows, gas dynamics **177**
9.1 Compressibility . . . 177
9.2 Thermodynamics . . . 177
9.2.1 Useful relations . . . 177
9.2.2 First and second laws of thermodynamics . . . 177
9.3 Conservation laws . . . 178
9.3.1 Speed of sound . . . 180
9.4 Adiabatic isentropic flows . . . 180
9.4.1 Area effects . . . 182
9.5 Shock waves, discontinuities, and jump conditions . . . 182
9.5.1 Jump conditions . . . 183
9.5.2 Rankine–Hugoniot relations / Mach number relations for perfect gas . . . 183
9.6 Euler equation . . . 184
9.6.1 Characteristics . . . 184
9.6.2 Riemann invariants . . . 185
9.6.3 Small perturbation linearized equations; acoustic equation . . . 186
9.7 Kinetic theory, microscopic view of fluid and flow properties . . . 187
9.8 Compressible turbulence . . . 187
9.8.1 Helmholtz velocity decomposition . . . 187
9.8.2 The Farve average . . . 187

10 Topics in Computational Fluid Dynamics **189**
10.1 The projection method for incompressible flows, primitive variables . . . 189
10.1.1 The Leray Projection . . . 189
10.1.2 The projection method . . . 190
10.2 Vorticity–Streamfunction and vorticity–Laplacian formulations, state variables . . . 192
10.3 Spectral collocation method . . . 192
10.3.1 Initial perturbation generation . . . 192

A Vectors, tensors, and their calculus **195**
A.1 Levi-Civita symbol . . . 195
A.1.1 Determinant representation . . . 195
A.1.2 Epsilon identity . . . 195
A.1.3 Contracted epsilon identity . . . 196
A.2 Vector identities . . . 196
A.3 Tensor eigenvalues, invariants, and its application in fluids . . . 197
A.3.1 Discriminant of a cubic equation . . . 198
A.3.2 Application to the velocity gradient tensor . . . 199
A.3.3 Application to the strain rate tensor . . . 200
A.3.4 Application to the Reynolds stress tensor, the invariant map, and the Lumley triangle, . . . 201

B Matrix Analysis **203**
B.1 Unitary matrix . . . 203
B.2 Singular value decomposition and eigenvalue decomposition . . . 203
B.3 Conformal mapping . . . 203

C Fourier analysis **205**
C.1 Fourier series . . . 205
C.2 Fourier transform . . . 207
C.2.1 Orthogonality, completeness, and linear independence . . . 207
C.2.2 Derivatives . . . 208
C.2.3 The inversion theorem; duality . . . 208
C.2.4 Convolution theorem . . . 208
C.2.5 Correlation functions and spectra; Parseval's theorem . . . 209
C.2.6 Symmetries of real FT . . . 210
C.2.7 Fourier transform in multiple dimensions . . . 211
C.2.8 Examples . . . 212
C.3 Discrete Fourier transform . . . 214
C.3.1 Discrete orthogonality . . . 216
C.3.2 Conjugate symmetry of the real DFT . . . 216
C.3.3 Frequency ordering and the Nyquist frequency . . . 216
C.3.4 Aliasing and de-aliasing . . . 217
C.3.5 MATLAB `fft()`, and `FFTW` . . . 219
C.3.6 Examples . . . 219
C.4 The Fast Fourier Transform . . . 220
C.4.1 The matrix form of the DFT . . . 220
C.4.2 Vandermonde matrix . . . 221
C.4.3 An example . . . 222
C.4.4 The idea of the FFT . . . 223
C.4.5 Miscellaneous . . . 224

D Coordinate systems and their transformations **225**
D.1 Cylindrical coordinates . . . 225
D.1.1 Operators in cylindrical coordinates . . . 226
D.1.2 Navier–Stokes in cylindrical coordinates . . . 229
D.1.3 Scalar equation in cylindrical coordinates . . . 230
D.1.4 Vorticity equation in cylindrical coordinates . . . 230
D.1.5 Vorticity–streamfunction in 3D axisymmetric flow . . . 231
D.2 Spherical coordinate . . . 232
D.2.1 From cylindrical to spherical . . . 233
D.3 General curvilinear coordinates . . . 233
D.3.1 Length, area, and volume . . . 233
D.3.2 Jacobian . . . 235
D.3.3 Three major calculus theorems . . . 236
D.3.4 Differential operators in curvilinear coordinate systems . . . 237
D.3.5 Derivatives of unit vectors . . . 239
D.3.6 Examples . . . 241
D.3.7 Galilean transformation and invariance . . . 242

E Special functions **243**
E.1 Polynomials and Taylor's expansions . . . 243
E.2 Trigonometric functions . . . 245
E.2.1 Trigonometric relations . . . 245
E.3 Hyperbolic functions . . . 245
E.4 Chebyshev functions . . . 246
E.4.1 Chebyshev spectral methods . . . 246
E.5 Bessel functions . . . 246

F Fundamentals of Numerical Mathematics **249**
F.1 Numerical Approximation . . . 249
F.2 Numerical Interpolation . . . 249
F.3 Numerical Integration . . . 249
F.4 Matrix Analysis . . . 249
F.5 Filtering . . . 249

Chapter 1

Conservation laws

1.1 Reynolds transport theorem and generic conservation laws

Consider a control volume V (CV) enclosed by a surface $A = \partial V$ and a property ψ (per unit mass ρV) carried by the fluid. We define the total property $\psi(\boldsymbol{x}, t)$ contained in the CV as

$$\Psi_{sys}(\boldsymbol{x}, t) = \int_V \rho(\boldsymbol{x}, t)\psi(\boldsymbol{x}, t)dV, \tag{1.1}$$

such that it is an extensive property and $\psi = d\Psi/dm = (1/\rho)d\Psi/dV$, with $\rho(\boldsymbol{x}, t)$ being a constant for an infinitesimal element. The RTT concerns the change of Ψ_{sys} following material, $d\Psi_{sys}/dt \triangleq D\Psi_{sys}/Dt$, where D is used to denote the material derivative, which is a total derivative.

The rate of change of the total property is

$$\frac{D}{Dt}\left(\int_V \rho(\boldsymbol{x}, t)\psi(\boldsymbol{x}, t)dV\right) = \frac{d}{dt}\left(\int_V \rho(\boldsymbol{x}, t)\psi(\boldsymbol{x}, t)dV\right) + \int_A \rho\psi(\boldsymbol{u}\cdot\boldsymbol{n})dA, \tag{1.2}$$

which says the change along the material is equal to the local change plus the fluxes through the surfaces. This is the Reynolds transport theorem (RTT). Here $\boldsymbol{n}$ is the outer normal of the surface element dA on the control surface (CS).

Using the Gauss's theorem (divergence theorem),

$$\int_A \rho\psi(\boldsymbol{u}\cdot\boldsymbol{n})dA = \int_V \nabla\cdot(\rho\psi\boldsymbol{u})dV. \tag{1.3}$$

In the absence of external force, source, or sink,

$$\frac{D\Psi_{sys}}{Dt} = \frac{d}{dt}\left(\int_V \rho\psi dV\right) + \int_A \rho\psi(\boldsymbol{u}\cdot\boldsymbol{n})dA = \int_V \left[\frac{\partial(\rho\psi)}{\partial t} + \nabla\cdot(\rho\psi\boldsymbol{u})\right]dV = 0. \tag{1.4}$$

Since the control volume is arbitrary and can be shrunk to infinitesimal, we have the differential form

$$\frac{\partial(\rho\psi)}{\partial t} + \nabla\cdot(\rho\psi\boldsymbol{u}) = 0. \tag{1.5}$$

Here, the scalar can be taken to be mass ($\psi = 1$), momentum ($\psi := \boldsymbol{u}$), or any other scalar ($\psi = X_i$, where X_i is the species concentration).

The fixed CV case (1.2) is a special case of a more general, moving CV RTT,

$$\frac{D}{Dt}\left(\int_{V(t)}\psi(\boldsymbol{x},t)\rho(\boldsymbol{x},t)dV\right)=\frac{d}{dt}\left(\int_{V(t)}\psi(\boldsymbol{x},t)\rho(\boldsymbol{x},t)dV\right)+\int_{A(t)}\rho\psi[(\boldsymbol{u}-\boldsymbol{u}_A]\cdot\boldsymbol{n}]dA \quad (1.6)$$

where $\boldsymbol{u}_A$ is the velocity of the CS, derived below.

1.1.1 Reynolds transport theorem in a moving and deforming control volume

In this section, we will show that the RTT in (1.2) is a special case of the more general expression. We start from the 1-D Leibniz rule for integral over the variable interval $x\in[a(t),b(t)]$:

$$\frac{d}{dt}\int_{a(t)}^{b(t)}f(x,t)dx=\int_{a(t)}^{b(t)}\frac{\partial f}{\partial x}dx+f(b(t),t)\frac{db}{dt}-f(a(t),t)\frac{da}{dt} \quad (1.7)$$

$$=\int_{a(t)}^{b(t)}\frac{\partial f}{\partial x}dx+u_bf(b)-u_af(a) \quad (1.8)$$

$$=\int_{a(t)}^{b(t)}\frac{\partial f}{\partial x}dx+\int_{a(t)}^{b(t)}f(\boldsymbol{u}_A\cdot\boldsymbol{n})dx, \quad (1.9)$$

where the boundary velocity is $\boldsymbol{u}_A=u_A\boldsymbol{e}_x$, with $u_a=da/dt, u_b=db/dt$ and the outer normals are $n_a=-1, n_b=1$. It is generalizable to multi-dimensions for an arbitrarily moving CV, as

$$\frac{d}{dt}\int_{V(t)}fdV=\int_{V(t)}\frac{\partial f}{\partial t}dV+\int_{A(t)=\partial V(t)}f(\boldsymbol{u}_A\cdot\boldsymbol{n})dA, \quad (1.10)$$

In the derivation, there are two (time-dependent) volumes being considered, the material volume $sys(t)$ which moves at each point at the fluid velocity $\boldsymbol{u}(\boldsymbol{x},t)$, and the moving control volume $CV(t)$ which moves at its own velocity $\boldsymbol{u}_A$. They coincide at the present time t but not otherwise. Applying the generalized Leibniz rule to each, we have

$$\frac{D\Psi_{sys}}{Dt}=\frac{d}{dt}\int_{sys(t)}\rho\psi dV=\int_{sys(t)}\frac{\partial(\rho\psi)}{\partial t}dV+\int_{\partial sys(t)}\rho\psi(\boldsymbol{u}\cdot\boldsymbol{n})dA \quad (1.11)$$

and

$$\frac{d}{dt}\int_{CV(t)}\rho\psi dV=\int_{CV(t)}\frac{\partial(\rho\psi)}{\partial t}dV+\int_{\partial CV(t)}\rho\psi(\boldsymbol{u}_A\cdot\boldsymbol{n})dA. \quad (1.12)$$

Differentiate these two and apply $sys(t)=CV(t)$ at time t to obtain the moving CV RTT

$$\frac{D\Psi_{sys}}{Dt}=\frac{d}{dt}\int_{CV(t)}\rho\psi dV+\int_{\partial CV(t)}\rho\psi[(\boldsymbol{u}-\boldsymbol{u}_A)\cdot\boldsymbol{n}]dA, \quad (1.13)$$

where $\boldsymbol{u}-\boldsymbol{u}_A$ is the fluid velocity relative to the velocity of the moving CV. When the CV is fixed, $\boldsymbol{u}_A=0$, we revert to the standard RTT (1.2). It is about selecting the CV. One can select a CV that follows the fluid system (such as following a dropping water surface) such that there is no relative velocity between the CV and the material system, but the term $(d/dt)\int_{CV(t)}\rho\psi dV$ needs to be evaluated.

1.1.2 Material derivative

We recall

$$\frac{D\Psi_{sys}}{Dt} = \int_V \left[\frac{\partial(\rho\psi)}{\partial t} + \nabla\cdot(\rho\psi\boldsymbol{u})\right] dV = 0. \tag{1.14}$$

Using mass conservation (1.19) we have

$$\frac{D\Psi_{sys}}{Dt} = \int_V \rho\left(\frac{\partial}{\partial t} + \boldsymbol{u}\cdot\nabla\right)\psi dV \triangleq \int_V \rho\frac{D\psi}{Dt}dV \tag{1.15}$$

where we define

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \boldsymbol{u}\cdot\nabla \tag{1.16}$$

as the material derivative, with $\partial/\partial t$ being the local rate-of-change and $\boldsymbol{u}\cdot\nabla$ being the convective derivative, similar to the mechanisms in RTT. The material derivative is a bridge between the Eulerian and Lagrangian description of fluids. We note that Newton's second law is a Lagrangian description.

The above can also be shown from a differential point of view. Consider the total time derivative of a scalar ψ:

$$\frac{d}{dt}\psi(\boldsymbol{x}(t),t) = \frac{\partial\psi}{\partial t} + \frac{\partial\psi}{\partial\boldsymbol{x}}\cdot\frac{d\boldsymbol{x}}{dt} = \frac{\partial\psi}{\partial t} + \boldsymbol{u}\cdot\nabla\psi. \tag{1.17}$$

1.2 Mass conservation

Taking $\psi = 1$,

$$\frac{\partial}{\partial t}\left(\int_V \rho dV\right) + \int_{A=\partial V}\rho(\boldsymbol{u}\cdot\boldsymbol{n})dA = 0. \tag{1.18}$$

In differential form:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}) = 0. \tag{1.19}$$

More over, using the definition of material derivative, we have

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\boldsymbol{u}) = \frac{\partial\rho}{\partial t} + \boldsymbol{u}\cdot\nabla\rho + \rho(\nabla\cdot\boldsymbol{u}) \tag{1.20}$$

$$= \frac{D\rho}{Dt} + \rho(\nabla\cdot\boldsymbol{u}) \tag{1.21}$$

$$= 0. \tag{1.22}$$

1.2.1 Incompressibility, continuity

It can be seen from (1.21) that solenoidality of the velocity field ($\nabla\cdot\boldsymbol{u} = 0$) is a sufficient and necessary condition for imcompressibility ($D\rho/Dt = 0$, density doesn't change along the material). The physical meaning is as the following.

Consider a cubic finite volume $dV = dxdydz$, the volume change is

$$\delta(dV) = [(u + \partial_x u dx) - u]dydzdt + [(v + \partial_y v dy) - v]dxdzdt + [(w + \partial_z w dz) - w]dxdydt \tag{1.23}$$

$$= (\partial_x u + \partial_y v + \partial_x w)dxdydzdt. \tag{1.24}$$

We have the divergence of a fluid parcel

$$\nabla \cdot \boldsymbol{u} = \partial_x u + \partial_y v + \partial_x w = \frac{1}{dV}\frac{\delta(dV)}{\delta t}, \tag{1.25}$$

which is the volume change rate per unit volumn. The specific volume (not to be confused with viscosity) is defined as $\nu = 1/\rho$. We can also have

$$\nabla \cdot \boldsymbol{u} = \frac{1}{\nu}\frac{D\nu}{Dt}, \tag{1.26}$$

which is stating that the divergence of the velocity filed is the normalized rate of change of specific volume.

One might have already been satisfied with such statements. Actually, there are more to it. The compressibility (change of density) can be due to pressure or thermal effects, which will be discussed more in detail in Chapter 6.

Also ref. Batchelor (1967).

1.3 Momentum conservation

Let $\psi = \boldsymbol{u}$ to be the quantity being transported. The conservation of momentum is:

$$\frac{D}{Dt}\left(\int_V \rho \boldsymbol{u} dV\right) = \frac{\partial}{\partial t}\left(\int_V \rho \boldsymbol{u} dV\right) + \int_{A=\partial V} \rho \boldsymbol{u}(\boldsymbol{u}\cdot\boldsymbol{n}) dA \tag{1.27}$$

where the material derivative (rate of change of momentum) equals to the (vector) force applied on the CV, including surface forces and volumetric forces, per Newton's second law. Hence,

$$\frac{\partial}{\partial t}\int_V \rho \boldsymbol{u} dV + \int_A \rho \boldsymbol{u}(\boldsymbol{u}\cdot\boldsymbol{n}) dA = \int_A \boldsymbol{\sigma}\cdot\boldsymbol{n} dA + \int_V \rho \boldsymbol{f} dV, \tag{1.28}$$

using Gauss's theorem,

$$\frac{\partial}{\partial t}\int_V \rho \boldsymbol{u} dV + \int_V \nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u}) dV = \int_V (\nabla\cdot\boldsymbol{\sigma}) dV + \int_V \rho\boldsymbol{f} dV, \tag{1.29}$$

where $\boldsymbol{\sigma} = \lim_{\Delta A\to 0}\Delta\boldsymbol{F}/\Delta\boldsymbol{A}$ is the stress tensor (force per unit area) and $\boldsymbol{f}$ ($\boldsymbol{f} = \lim_{\Delta V\to 0}\Delta\boldsymbol{F}/\Delta m$) is the body forcing (per unit mass). The body force, for example, can be the gravity $\boldsymbol{f} = \boldsymbol{g}$ or the Coriolis $\boldsymbol{f} = \boldsymbol{u}\times\boldsymbol{f}_c$.

We then have the integration from of the Euler equation:

$$\int_V \left[\partial_t(\rho\boldsymbol{u}) + \nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u})\right] dV = \int_v [\nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{f}] dV \tag{1.30}$$

which is valid for a finite volume V. The differential form is

$$\partial_t(\rho\boldsymbol{u}) + \nabla\cdot(\rho\boldsymbol{u}\boldsymbol{u}) = \nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{f}. \tag{1.31}$$

Using the continuity equation, the above equation converts to

$$\rho\ddot{\boldsymbol{x}} = \nabla\cdot\boldsymbol{\sigma} + \rho\boldsymbol{f}, \tag{1.32}$$

which is a general form of momentum balance/$F = ma$ in continuum mechanics, where $-\rho\ddot{\boldsymbol{x}}$ is D'Alembert's force (with $\boldsymbol{x}$ being the displacement), $\boldsymbol{\sigma}$ is the stress tensor, and $\boldsymbol{f}$ is the body force (per unit volume).

Now we consider the sources of stresses,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_p + \boldsymbol{\tau}. \tag{1.33}$$

- Pressure. It is isotropic and the direction is $-\boldsymbol{n}$ so $\boldsymbol{\sigma}_p = -p\boldsymbol{I}$.
- Viscous stress. For a Newtonian fluid (see section 1.6),

$$\boldsymbol{\tau} = 2\mu\boldsymbol{S} + \lambda(\nabla \cdot \boldsymbol{u})\boldsymbol{\delta} \tag{1.34}$$

$$= 2\mu\boldsymbol{S} - \frac{2}{3}\mu(\nabla \cdot \boldsymbol{u})\boldsymbol{\delta} \tag{1.35}$$

  where δ_{ij} is the Kronecker delta. According to Stokes assumption, the bulk viscosity $\lambda = -2/3\mu$.

- For incompressible flows, $\boldsymbol{\tau} = 2\mu\boldsymbol{S}$ and

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + 2\mu\boldsymbol{S}. \tag{1.36}$$

The momentum conservation equations or the Navier–Stokes equations are

$$\partial_t(\rho\boldsymbol{u}) + \nabla \cdot (\rho\boldsymbol{u}\boldsymbol{u}) = -\nabla p + \mu\nabla^2\boldsymbol{u} + \rho\boldsymbol{f}. \tag{1.37}$$

We note that $\partial_j(2\mu S_{ij}) = \partial_j(\mu\partial_j u_i)$.

1.4 Energy conservation

1.4.1 Mechanical energy equation

Multiply the momentum equation

$$\frac{\partial\rho u_i}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) = \frac{\partial\sigma_{ij}}{\partial x_j} + \rho f_i \tag{1.38}$$

with u_i and contract:

$$\frac{\partial\rho u_i^2/2}{\partial t} + \frac{\partial}{\partial x_j}\left(\frac{1}{2}\rho u_i^2 u_j\right) = u_i\frac{\partial\sigma_{ij}}{\partial x_j} + \rho u_i f_i. \tag{1.39}$$

Here we use the product rule

$$u_i\frac{\partial\sigma_{ij}}{\partial x_j} = \frac{\partial u_i\sigma_{ij}}{\partial x_j} - \sigma_{ij}\frac{\partial u_i}{\partial x_j}, \tag{1.40}$$

and

$$\sigma_{ij}\frac{\partial u_i}{\partial x_j} = \left(-p\delta_{ij} + 2\mu S_{ij} - \frac{2}{3}\mu\frac{\partial u_k}{\partial x_k}\delta_{ij}\right)\frac{\partial u_i}{\partial x_j} \tag{1.41}$$

$$= -p\frac{\partial u_k}{\partial x_k} + 2\mu S_{ij}S_{ij} - \frac{2}{3}\mu\left(\frac{\partial u_k}{\partial x_k}\right)^2 \tag{1.42}$$

After rearrangements, we have the mechanical energy ($u_k^2/2$; per unit mass) equation:

$$\rho\frac{D(\rho u_k^2/2)}{Dt} = \frac{\partial(\rho u_k^2/2)}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i u_k^2/2) = \rho u_j f_j + \frac{\partial}{\partial x_j}(u_i\sigma_{ij}) + p\frac{\partial u_k}{\partial x_k} - \phi, \tag{1.43}$$

where

$$\phi = 2\mu S_{ij}S_{ij} - \frac{2}{3}\mu\left(\frac{\partial u_k}{\partial x_k}\right)^2 = 2\mu\left(S_{ij} - \frac{1}{3}\frac{\partial u_j}{\partial x_j}\delta_{ij}\right)^2, \tag{1.44}$$

the viscous dissipation, is always positive, transfering mechanical energy to internal energy. Note $p(\nabla\cdot\boldsymbol{u})$ is the volume expansion work for the mechanical energy equation (mechanical energy increases when $\nabla\cdot\boldsymbol{u} > 0$), and $-p(\nabla\cdot\boldsymbol{u})$ the compression heating (internal energy increases when $\nabla\cdot\boldsymbol{u} < 0$) for the internal energy.

1.4.2 Total energy equation

The total energy ($E = e + u_k^2/2$; per unit mass) conservation follows the thermodynamic first law (in the Lagrangian framework), that the change of total energy is equal to the work done to the system (control volume) plus heat added,

$$\frac{D}{Dt}\int_V \rho(e + u_i^2/2)\,dV = \int_V \rho u_i f_i\,dV + \int_A u_i\sigma_{ij}\,dA_j - \int_A q_i\,dA_i, \tag{1.45}$$

where the force acting on the surface of the CV is $\boldsymbol{\sigma}\cdot d\boldsymbol{A}$ and heat flux into the CV is $-\boldsymbol{q}\cdot\boldsymbol{n}$, $\boldsymbol{q} = -k\nabla T$ is the heat flux according to Fourier's law, and k is the thermal conductivity. Using the Reynolds transport theorem and the continuity equation,

$$\frac{D}{Dt}\int_V \rho(e + u_i^2/2)\,dV = \int_V \rho\frac{D}{Dt}(e + u_i^2/2)\,dV. \tag{1.46}$$

Finally, we have the total energy equation:

$$\rho\frac{D(e + u_k^2/2)}{Dt} = \frac{\partial\rho(e + u_k^2/2)}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_j(e + u_k^2/2)\right) = \rho u_j f_j + \frac{\partial}{\partial x_j}(u_i\sigma_{ij}) - \frac{\partial q_j}{\partial x_j} \tag{1.47}$$

Here σ_{ij} is the stress tensor,

$$\sigma_{ij} = -p\delta_{ij} + \tau_{ij} \tag{1.48}$$

and the stress work can be broken into the pressure work and the viscous stress work as

$$\frac{\partial}{\partial x_j}(u_i\sigma_{ij}) = -\frac{\partial(u_j p)}{\partial x_j} + \frac{\partial(u_i\tau_{ij})}{\partial x_j}. \tag{1.49}$$

We note that the potential energy is included in $\rho u_i f_j = -\rho u_i\partial_i\Phi = -\rho D\Phi/Dt$ ($\partial_t\Phi = 0$), where $\Phi = gz$ is the gravitational potential. Otherwise, the total energy is defined as $e + u_k^2/2 + gz$.

1.4.3 Internal energy equation

Subtract the mechanical energy equation from the total energy equation, we obtain the internal energy ($e = c_v T$; per unit mass) equation:

$$\rho \frac{De}{Dt} = \frac{\partial(\rho e)}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i e) = -\frac{\partial q_j}{\partial x_j} - p\frac{\partial u_j}{\partial x_j} + \phi, \tag{1.50}$$

with the meanings of each term already explained above.

1.4.4 Heat equation

Heat equation (under Boussinesq): when incompressible and the heating due to the viscous dissipation is neglected, the inernal energy equation turns to

$$\rho c_v \frac{DT}{Dt} = k\nabla^2 T, \tag{1.51}$$

or

$$\frac{DT}{Dt} = \alpha \nabla^2 T, \tag{1.52}$$

where $\alpha = k/(\rho c_v)$ is called thermal diffusivity.

1.4.5 Entropy equation

Following the Tds relation

$$Tds = de + pd\nu = de - \frac{p}{\rho^2} d\rho, \tag{1.53}$$

we have

$$\rho T \frac{DS}{Dt} = \rho \frac{De}{Dt} - \frac{p}{\rho}\frac{D\rho}{Dt} \tag{1.54}$$

$$= -\frac{\partial q_j}{\partial x_j} - p\frac{\partial u_j}{\partial x_j} + \phi + \frac{p}{\rho}\rho\frac{\partial u_j}{\partial x_j} \tag{1.55}$$

$$= -\frac{\partial q_j}{\partial x_j} + \phi \tag{1.56}$$

Hence,

$$\rho \frac{DS}{Dt} = -\frac{1}{T}\frac{\partial q_j}{\partial x_j} + \frac{\phi}{T} \tag{1.57}$$

$$= -\frac{\partial}{\partial x_j}\left(\frac{q_i}{T}\right) + \frac{k}{T^2}\left(\frac{\partial T}{\partial x_j}\right)^2 + \frac{\phi}{T} \tag{1.58}$$

where in the RHS the first term represents reversible entropy transfer since it can take both signs, and the last two terms are always positive (requiring $k, \nu > 0$) and represent the irreversible entropy production.

1.5 Bernoulli equation

Assumptions:

- Inviscid.
- Barotropic.
- Potential force.
- Steady.

Consider the inviscid Euler equation:

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \boldsymbol{F} \tag{1.59}$$

with a conservative force

$$\boldsymbol{F} = -\nabla \Phi. \tag{1.60}$$

Using Eq. (A.20) we have

$$(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \nabla \frac{\boldsymbol{u}^2}{2} - \boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \nabla \frac{\boldsymbol{u}^2}{2} - \boldsymbol{u} \times \boldsymbol{\omega} \tag{1.61}$$

We defined

$$\boldsymbol{L} = \boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \boldsymbol{u} \times \boldsymbol{\omega} \tag{1.62}$$

which is called the Lamb vector. The Euler equation becomes

$$\nabla \frac{\boldsymbol{u}^2}{2} - \boldsymbol{u} \times \boldsymbol{\omega} = -\nabla \left(\frac{p}{\rho}\right) - \nabla \Phi \tag{1.63}$$

and hence

$$\nabla \left(\frac{\boldsymbol{u}^2}{2} + \frac{p}{\rho} + \Phi\right) = \boldsymbol{L} \tag{1.64}$$

which is called the Lamb-Gromyko equation. Define

$$H = \frac{\boldsymbol{u}^2}{2} + \frac{p}{\rho} + \Phi \tag{1.65}$$

we have

$$\nabla H = \boldsymbol{L}. \tag{1.66}$$

If the flow is irrotational, i.e., $\boldsymbol{L} = 0$ and $H = \text{const.}$, we recover a special version (isentropic) of Bernuolli's theorem. It is a combination of momentum and energy conservation. When we take $\Phi = gz$ to be the gravitational potential, $\boldsymbol{F} = -\nabla\Phi = (0, 0, -g)$, the conservation of H implies

$$\frac{\boldsymbol{u}^2}{2} + \frac{p}{\rho} + gz = \text{const.}, \tag{1.67}$$

which is useful for pipe flows when neglecting friction loss.

The Bernoulli equation can be interpreted as energy conservation as well. The total energy equation (1.47), after some rearrangements, turns to

$$\frac{\partial \rho(e+u_k^2/2)}{\partial t}+\frac{\partial}{\partial x_j}\left(\rho u_j\left(e+u_k^2/2+\frac{p}{\rho}\right)\right)=\rho f_j u_j+\frac{\partial}{\partial x_j}(u_i\tau_{ij})-\frac{\partial q_j}{\partial x_j}. \tag{1.68}$$

In steady flows, $\partial_t = 0$. The external force is gravity and $u_i f_i = -u_i \partial\Phi/\partial x_i = -u_i \partial(gz)/\partial x_i$. We have

$$\rho u_j \frac{\partial}{\partial x_j}\left(e+u_k^2/2+\frac{p}{\rho}+gz\right)=0 \tag{1.69}$$

in the absence of viscosity and external heating, implying

$$e+\frac{|\boldsymbol{u}|^2}{2}+\frac{p}{\rho}+gz=h+\frac{|\boldsymbol{u}|^2}{2}+gz=\text{const.} \tag{1.70}$$

along the streamlines (its gradient is perpendicular to $\nabla \boldsymbol{u}$), where we used $h = e + p/\rho$. This enthalpy/energy version of Bernoulli will be more useful in compressible flows.

1.6 Constitutive relations

A note on angular momentum conservation.

A difference between solid and fluid mechanics is that, in solid mechanics, stress is proportional to strain, while in fluid mechanics, stress is proportional to strain rate.

Some comments about stress–strain relation in the solids.

1.7 Pressure Poisson

Take the divergence of the follow equation for incompressible flows ($\nabla \cdot \boldsymbol{u} = 0$):

$$\frac{\partial \boldsymbol{u}}{\partial t}+\boldsymbol{u}\cdot\nabla\boldsymbol{u}=-\frac{1}{\rho}\nabla p+\nu\nabla^2\boldsymbol{u} \tag{1.71}$$

we have

$$-\frac{1}{\rho}\nabla^2 p=\nabla\cdot(\boldsymbol{u}\cdot\nabla\boldsymbol{u})=\nabla\boldsymbol{u}:\boldsymbol{u}\nabla+\boldsymbol{u}\cdot\nabla(\nabla\cdot\boldsymbol{u})=\nabla\boldsymbol{u}:\boldsymbol{u}\nabla \tag{1.72}$$

i.e.

$$\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i \partial x_i}=-\frac{\partial u_i}{\partial x_j}\frac{\partial u_j}{\partial x_i} \tag{1.73}$$

In incompressible CFD, the continuity equation ($\nabla \cdot \boldsymbol{u} = 0$) is responsible for solving for the pressure as each of the three momentum equations are responsible for one velocity component. Note that there is no time in the continuity/Poisson equations, putting challenges on obtaining time-accurate, divergence-free velocity fields. See fractional-step/projection methods.

Chapter 2

Vortex dynamics

Vorticity is defined as the curl of the velocity field:

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{u} \tag{2.1}$$

2.1 Kelvin's theorem

Ref. Kundu and Cohen.

The circulation along a closed line is defined as

$$\Gamma = \oint_l \boldsymbol{u} \cdot d\boldsymbol{x} = \iint_A (\nabla \times \boldsymbol{u}) \cdot d\boldsymbol{A}, \tag{2.2}$$

according to Stokes' theorem.

Consider the momentum equation

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nabla \cdot \boldsymbol{\sigma} + \boldsymbol{f}. \tag{2.3}$$

The material (Lagrangian) derivative of Γ is

$$\frac{D\Gamma}{Dt} = \frac{D}{Dt}\left(\oint_l \boldsymbol{u} \cdot d\boldsymbol{x}\right) \tag{2.4}$$

$$= \oint_l \frac{D\boldsymbol{u}}{Dt} \cdot d\boldsymbol{x} + \oint_l \boldsymbol{u} \cdot \frac{D(d\boldsymbol{x})}{Dt} \tag{2.5}$$

$$= -\oint_l \left(\frac{1}{\rho}\nabla p\right) \cdot d\boldsymbol{x} + \oint_l (\nabla \cdot \boldsymbol{\sigma}) \cdot d\boldsymbol{x} + \oint_l \boldsymbol{f} \cdot d\boldsymbol{x} + \oint_l \boldsymbol{u} \cdot \frac{D(d\boldsymbol{x})}{Dt} \tag{2.6}$$

Assumptions:

1. Inviscid fluid: $\nabla \cdot \boldsymbol{\sigma} = 0$.
2. Conservative body force: $\boldsymbol{f} = -\nabla\Phi, \nabla \times \boldsymbol{f} = 0$ (it is the gradient of a potential field). Conservative means the potential difference does not depend on the path:

$$\int_A^B \boldsymbol{f} \cdot d\boldsymbol{x} = -\int_A^B \nabla\Phi \cdot d\boldsymbol{x} = -\int_A^B d\Phi = \Phi_A - \Phi_B. \tag{2.7}$$

For a close path, $A = B$ hence the integral is zero.

3. Barotropic flow: $\nabla\rho \times \nabla p = 0$.

Moreover, by

$$\boldsymbol{u} + d\boldsymbol{u} = \frac{D}{Dt}(\boldsymbol{x} + d\boldsymbol{x}) = \frac{D\boldsymbol{x}}{Dt} + \frac{D(d\boldsymbol{x})}{Dt}, \tag{2.8}$$

we have

$$\frac{D(d\boldsymbol{x})}{Dt} = d\boldsymbol{u} = d\boldsymbol{x} \cdot \nabla \boldsymbol{u} \tag{2.9}$$

so the last term in (2.6) is

$$\oint_l \boldsymbol{u} \cdot \frac{D(d\boldsymbol{x})}{Dt} = \oint_l \boldsymbol{u} \cdot d\boldsymbol{u} = \oint_l d(\boldsymbol{u}^2) = 0. \tag{2.10}$$

Then we are able to prove all RHS terms in (2.6) are zero hence

$$\frac{D\Gamma}{Dt} = 0 \tag{2.11}$$

along any arbitrary closed curves.

2.1.1 Helmholtz's theorems

2.2 Vorticity transport equation

By Eq. (A.11) we know

$$\nabla \cdot \boldsymbol{\omega} = 0 \tag{2.12}$$

i.e., the continuity of vorticity.

The incompressible Navier–Stokes equation in vector form:

$$\frac{D\boldsymbol{u}}{Dt} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{f} \tag{2.13}$$

Using Eq. (1.61), the rotational form of the Navier–Stokes

$$\frac{\partial \boldsymbol{u}}{\partial t} + \nabla \frac{|\boldsymbol{u}|^2}{2} - \boldsymbol{u} \times \boldsymbol{\omega} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{F} \tag{2.14}$$

from which the gradient of $|\boldsymbol{u}|^2/2$ can be absorbed into the Bernoulli pressure

$$P = p + \frac{|\boldsymbol{u}|^2}{2} \tag{2.15}$$

and $\boldsymbol{H} = \boldsymbol{u} \times \boldsymbol{\omega}$ is the Lamb vector. Using the (A.21) and take the curl of Eq. (2.14),

$$\text{LHS} = \nabla \times (\frac{\partial \boldsymbol{u}}{\partial t} + \nabla \frac{\boldsymbol{u}^2}{2} - \boldsymbol{u} \times \boldsymbol{\omega}) \tag{2.16}$$

$$= \frac{\partial \boldsymbol{\omega}}{\partial t} - \nabla \times (\boldsymbol{u} \times \boldsymbol{\omega}) \tag{2.17}$$

$$= \frac{\partial \boldsymbol{\omega}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u} + \boldsymbol{\omega}(\nabla \cdot \boldsymbol{u}) - \boldsymbol{u}(\nabla \cdot \boldsymbol{\omega}) \tag{2.18}$$

$$= \frac{\partial \boldsymbol{\omega}}{\partial t} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{\omega} - (\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u} \tag{2.19}$$

$$\text{RHS} = \nabla\times(-\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{f}) \tag{2.20}$$

$$= \nu\nabla^2\boldsymbol{\omega} + \nabla\times\boldsymbol{f} + \frac{1}{\rho^2}\nabla\rho\times\nabla p \tag{2.21}$$

Equaling both sides we obtain

$$\underbrace{\frac{\partial \boldsymbol{\omega}}{\partial t}}_{\substack{\text{rate of}\\ \text{change}}} + \underbrace{\boldsymbol{u}\cdot\nabla\boldsymbol{\omega}}_{\text{advection}} = \underbrace{\boldsymbol{\omega}\cdot\nabla\boldsymbol{u}}_{\substack{\text{vortex}\\ \text{stretching}}} + \underbrace{\nu\nabla^2\boldsymbol{\omega}}_{\substack{\text{viscous}\\ \text{diffusion}}} + \underbrace{\nabla\times\boldsymbol{f}}_{\substack{\text{external torque in}\\ \text{a non-conservative field}}} + \underbrace{\frac{1}{\rho^2}\nabla\rho\times\nabla p}_{\substack{\text{baroclinic}\\ \text{torque}}} \tag{2.22}$$

Again, the stretching term $\boldsymbol{\omega}\cdot\nabla\boldsymbol{u}$ comes from the non-linear advection/inertial term in N-S. It is important in the energy cascade in turbulence. Also from (3.18), $\boldsymbol{\omega}\cdot\boldsymbol{\Omega} = \boldsymbol{0}$, we have

$$\boldsymbol{\omega}\cdot\nabla\boldsymbol{u} = \boldsymbol{\omega}\cdot\boldsymbol{S}, \tag{2.23}$$

i.e., vortex stretching happens as a result of its interaction with the strain rate. Hence, the alignment of vorticity with the eigenvectors of the strain rate is important. It was found by Ashurst *et al.* (1987) that, in turbulence, vorticity tend to align to the eigenvector corresponding to the second eigenvalue of $\boldsymbol{S}$, which tend to be positive (indicating positive stretching production). This is called the kinematic alignment effect.

Vorticity describes a lot about the dynamics of the flow, but it is not independent of the velocity. According to the vector identity (A.22), they are related through the Poisson equation

$$\nabla\times\boldsymbol{\omega} = \nabla(\nabla\cdot\boldsymbol{u}) - \nabla^2\boldsymbol{u} = -\nabla^2\boldsymbol{u}, \tag{2.24}$$

which is a relation useful in Fourier spectral methods.

2.2.1 Enstrophy transport equation

Define enstrophy as:

$$\mathcal{E} \triangleq \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{\omega} = \frac{1}{2}\omega_i\omega_i \tag{2.25}$$

Re-write (2.22) into tensor notation we have

$$\frac{\partial\omega_i}{\partial t} + u_j\frac{\partial\omega_i}{\partial x_j} = \omega_j\frac{\partial u_i}{\partial x_j} + \nu\frac{\partial^2\omega_i}{\partial x_j^2} + \epsilon_{ijk}\frac{\partial f_k}{\partial x_j} + \frac{1}{\rho^2}\epsilon_{ijk}\frac{\partial\rho}{\partial x_j}\frac{\partial p}{\partial x_k} \tag{2.26}$$

$\omega_i\times$ (2.26) we have

$$\frac{\partial}{\partial t}(\frac{1}{2}\omega_i\omega_i) + u_j\frac{\partial}{\partial x_j}(\frac{1}{2}\omega_i\omega_i) = \omega_i\omega_j\frac{\partial u_i}{\partial x_j} + \nu\frac{\partial^2}{\partial x_j^2}(\frac{1}{2}\omega_i\omega_i) - \nu\frac{\partial\omega_i}{\partial x_j}\frac{\partial\omega_i}{\partial x_j} \tag{2.27}$$

$$+ \epsilon_{ijk}\omega_i\frac{\partial F_k}{\partial x_j} + \frac{1}{\rho^2}\epsilon_{ijk}\omega_i\frac{\partial\rho}{\partial x_j}\frac{\partial p}{\partial x_k} \tag{2.28}$$

Note that ϵ_{ijk} is the Levi-Civita symbol, not to be confused with the turbulent kinetic energy rate ε or Reynolds stresses dissipation rate ε_{ij}.

Re-write back into vector form:

$$\frac{\partial \mathcal{E}}{\partial t} + \boldsymbol{u}\cdot\nabla\mathcal{E} = \boldsymbol{\omega\omega} : \nabla\boldsymbol{u} + \nu\nabla^2\mathcal{E} \underbrace{-\nu\nabla\boldsymbol{\omega} : \nabla\boldsymbol{\omega}}_{\substack{\text{viscous}\\ \text{dissipation}}} + \boldsymbol{\omega}\cdot(\nabla\times\boldsymbol{F}) + \boldsymbol{\omega}\cdot\frac{\nabla\rho\times\nabla p}{\rho^2} \tag{2.29}$$

Note that we are assuming an incompressible flow, hence $\nabla\cdot\boldsymbol{u}$ related terms are not appearing in Eq. (2.29). A new mechanism compared to (2.22) is the viscous dissipation of enstrophy. This term is always negative and removing enstrophy. Since $\omega_i\omega_j$ is a symmetric tensor and Ω_{ij} is an antisymmetric tennsor, $\omega_i\omega_j\Omega_{ij} = 0$ and we can simplify

$$\boldsymbol{\omega\omega} : \nabla\boldsymbol{u} = \boldsymbol{\omega\omega} : \boldsymbol{S}. \tag{2.30}$$

2.3 Vorticity–streamfunction formulation

According to the Helmholtz decomposition theorem, the velocity field $\boldsymbol{u}$ can be decomposed to the addition of a potential and a curl field as

$$\boldsymbol{u} = \nabla\phi + \nabla\times\boldsymbol{A}, \tag{2.31}$$

with the first part being irrotational and the second solenoidal. The two potentials ϕ and $\boldsymbol{A}$ are called the scalar and vector potentials.

In two-dimensional flows, the vector potential reduces to $\boldsymbol{A} = (0, 0, \psi)$, where ψ is the streamfunction, defined as

$$\psi(x, y) = \int_0^y u(0, \xi)\, d\xi - \int_0^x v(\zeta, y)\, d\zeta + C, \tag{2.32}$$

is constant along the streamlines. A reference point would need to be specified. And it is noted that between two streamlines the volume flow rate is a constant (physical meaning of streamfunctions). For equally spaced streamlines, the locations where they are denser have a faster velocity.

In general, for a divergence-free field ($\nabla\cdot\boldsymbol{u} = 0$) there exist a vector potential. The equations to solve are, $\boldsymbol{u} = \nabla\times\boldsymbol{A}$, or component-wise

$$u = \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \tag{2.33}$$

$$v = \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \tag{2.34}$$

$$w = \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \tag{2.35}$$

whose solution is far from unique. We note that $\nabla\times(\boldsymbol{A} + \nabla f) = \nabla\times\boldsymbol{A}$, for any arbitrary scalar potential ∇f. It can be chosen to $\nabla f = (0, 0, -A_3)$ by integrating $f = -\int A_3\, dz$ such that the above equations are simplifed to

$$u = -\frac{\partial A_2}{\partial z} \tag{2.36}$$

$$v = \frac{\partial A_1}{\partial z} \tag{2.37}$$

$$w = \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \tag{2.38}$$

which are easier to solve.

2.3.1 In two dimensions

As a strategy to solve the N–S, it is common to transform (u, v, p) to (ω, ψ), equivalent to taking the curl of the N–S to eliminate the pressure. Under such transformation, 2D N–S equations are converted to

$$\frac{\partial \omega}{\partial t} + u\frac{\partial \omega}{\partial x} + \frac{\partial \omega}{\partial y} = \nu\left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2}\right) \tag{2.39}$$

$$\omega = -\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2}\right) \tag{2.40}$$

where

$$u = \frac{\partial \psi}{\partial y},\ v = -\frac{\partial \psi}{\partial x}. \tag{2.41}$$

The common procedures in a Fourier spectral method (in a doubly periodic domain) solving the above equations include time-stepping of (2.39), with the streamfunction calculated from the inversion of (2.40) and used to obtained the velocities through (2.41). Such a formulation can be similarly obtain for axisymmetric flows – see section D.1.5. The elliptic character of (2.40) and the parabolic character of (2.39) jointly reflect the mixed elliptic-parabolic nature of the Navier–Stokes. We also note that the elliptic nature is closely associated with incompressibility.

The absence of evolution equation for the pressure is reflected in the Poisson equation (2.40). If the pressure field is desired, it can be recovered by solving the pressure Poisson:

$$\left(\frac{\partial^2 p}{\partial^2 x} + \frac{\partial^2 p}{\partial^2 y}\right) = -\rho\frac{\partial u_\beta}{\partial x_\alpha}\frac{\partial u_\alpha}{\partial x_\beta} \tag{2.42}$$

$$= -\rho\left[\left(\frac{\partial u}{\partial x}\right)^2 + 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + \left(\frac{\partial v}{\partial y}\right)^2\right] \tag{2.43}$$

$$= -2\rho\left(\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial v}{\partial y}\right) \tag{2.44}$$

$$= 2\rho\left[\frac{\partial^2 \psi}{\partial x^2}\frac{\partial^2 \psi}{\partial y^2} - \left(\frac{\partial^2 \psi}{\partial x \partial y}\right)^2\right] \tag{2.45}$$

where the last simplification uses the continuity to reach an expression that requires fewer operation counts. Since the equation is second-order in space, we need two BC's in each direction. At solid walls, we have the no-slip condition $u = v = 0$ such that $\partial_x u = \partial_x v = 0$ at the wall. The N–S equation at the wall reads

$$\frac{\partial p}{\partial x} = \mu\frac{\partial^2 u}{\partial y^2} = \mu\frac{\partial \omega}{\partial y}, \tag{2.46}$$

where in a finite-difference scheme the pressure can be computed from a one-sided FDA of $\partial\omega/\partial y$, given pressure should be known at least for one point on the wall.

The convenience that this method offers comes at the expense of difficulties of imposing the boundary conditions. For streamfunction it is relatively easy – it should be a constant on solid boundaries. For vorticity, neither its value nor its first derivative is known before the

computation. Special treatments are required. For example, at the wall of $y = 0$, the Poisson equation of ψ reduces to

$$\omega = -\frac{\partial^2 \psi}{\partial y^2} \tag{2.47}$$

since $\partial_x v = 0$.

2.3.2 In three dimensions (omega y-Laplacian v)

When deriving the vorticity equation, we took the curl of the Lamb-type equation (2.14), resulting in

$$\frac{\partial \boldsymbol{\omega}}{\partial t} - \nabla \times \boldsymbol{H} = \nu \nabla^2 \boldsymbol{\omega}, \tag{2.48}$$

where $\boldsymbol{H} = \boldsymbol{u} \times \boldsymbol{\omega}$ is the Lamb vector. The components of the vortcity equation are

$$\partial_t \omega_x - (\partial_y H_3 - \partial_z H_2) = \nu \nabla^2 \omega_x \tag{2.49}$$

$$\partial_t \omega_y - (\partial_z H_1 - \partial_x H_3) = \nu \nabla^2 \omega_y \tag{2.50}$$

$$\partial_t \omega_z - (\partial_x H_2 - \partial_y H_1) = \nu \nabla^2 \omega_z \tag{2.51}$$

We take the second equation to be one part of the ω_y–$\triangle v$ formulation. Next, we establish a relation:

$$\partial_x \omega_z - \partial_z \omega_x = \partial_x(\partial_x v - \partial_y u) - \partial_z(\partial_y w - \partial_z v) \tag{2.52}$$

$$= \partial_{xx} v - \partial_y(\partial_x u + \partial_z w) + \partial_{zz} v \tag{2.53}$$

$$= (\partial_{xx} + \partial_{yy} + \partial_{zz}) v \tag{2.54}$$

$$= \nabla^2 v \tag{2.55}$$

Similarly, we take the curl of the vorticity equation (the second curl), $\partial_x \omega_z - \partial_z \omega_x$, which yields the omega y-Laplacian v equations:

$$\partial_t(\nabla^2 v) - (\partial_{xx} + \partial_{zz}) H_2 + \partial_y(\partial_x H_1 + \partial_z H_3) = \nu \nabla^2 (\nabla^2 v) \tag{2.56}$$

$$\partial_t \omega_y - (\partial_z H_1 - \partial_x H_3) = \nu \nabla^2 \omega_y \tag{2.57}$$

which are second- and fourth-order equations. The fourth order equation requires four boundary conditions at $y = 0, 2h$ for a channel flow setting. Naturally, $v = 0$ at the walls are two. The no-slip conditions, $u = w = 0$ at the walls, leads to $\partial_x u = \partial_z w = 0$ at the walls. Hence, we can impose $\partial_y v = 0$ at both walls. These are the equations solved in Kim *et al.* (1987), KMM87, where the way to recover the primitive variables (u, v, w, p) was also described. We first obtain v from solving $\nabla^2 v$, and then u, w are obtained from

$$\partial_x u + \partial_z w = -\partial_y v \tag{2.58}$$

$$\partial_z u - \partial_x w = \omega_y \tag{2.59}$$

where we note that x, z are Fourier directions so it should be straightforward by solving the linear system:

$$\mathrm{i} k_x \hat{u} + \mathrm{i} k_z \hat{w} = -\partial_y \hat{v} \tag{2.60}$$

$$\mathrm{i}k_z\hat{u} - \mathrm{i}k_x\hat{w} = \hat{\omega}_y \tag{2.61}$$

and

$$\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{w}\end{bmatrix} = C\begin{bmatrix}\hat{v}\\ \hat{\omega}_y\end{bmatrix},\ C = \frac{1}{k_x^2+k_z^2}\begin{bmatrix}\mathrm{i}k_x\partial_y & -\mathrm{i}k_z\\ (k_x^2+k_z^2) & 0\\ \mathrm{i}k_z\partial_y & \mathrm{i}k_x\end{bmatrix}. \tag{2.62}$$

Then obtain the pressure if necessary (see KMM87). The inverse of (2.62) is sometimes called the Orr-Sommerfeld transform:

$$\begin{bmatrix}\nabla^2\hat{v}\\ \hat{\omega}_y\end{bmatrix} = \begin{bmatrix}0 & \nabla^2 & 0\\ \mathrm{i}k_z & 0 & -\mathrm{i}k_x\end{bmatrix}\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{w}\end{bmatrix} \tag{2.63}$$

where

$$\nabla^2 = \partial_y^2 - k_x^2 - k_z^2. \tag{2.64}$$

We note that this formulation is common in incompressible CFD and stability problems, due to the advantage of going pressure-less – reduced computational cost.

stability (algebraic growth paper cf.)

2.4 Potential flow

In potential flows, $\boldsymbol{A} = \boldsymbol{0}$ in (2.31) and the velocity field can be represented as the gradient of the scalar potential. The continuity condition, $\nabla \cdot \boldsymbol{u} = 0$, leads to

$$\nabla^2\phi = 0, \tag{2.65}$$

i.e., the scalar potential satisfies the Laplacian equation. Since the Laplacian equation is linear, superposition is enabled and basic solutions can be used as building blocks to construct more complicated flow fields.

Note that the Helmholtz decomposition is not necessarily unique. For example, in 2D potential flows, the velocity can be expressed by solely an irrotational or a solenoidal field:

$$u = \frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y} \tag{2.66}$$

$$v = \frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x} \tag{2.67}$$

where both the velocity potential and the streamfunction satisfy the Cauchy–Riemann (C–R) condition and are harmonic functions that solve Laplacian equation:

$$\nabla^2\phi = 0 \tag{2.68}$$

$$\nabla^2\psi = 0 \tag{2.69}$$

As a consequence,

$$\nabla\phi \cdot \nabla\psi = 0, \tag{2.70}$$

i.e., contours of velocity potential and streamfunctions are perpendicular.

In the language of complex analysis:

$$z = x + iy \tag{2.71}$$

$$\bar{z} = x - iy \tag{2.72}$$

and

$$x = \frac{1}{2}(z + \bar{z}) \tag{2.73}$$

$$y = \frac{1}{2i}(z - \bar{z}). \tag{2.74}$$

We have the derivatives

$$\partial_z = \partial_x \partial_z x + \partial_y \partial_z y = \frac{1}{2}(\partial_x - i\partial_y) \tag{2.75}$$

and similarly

$$\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y). \tag{2.76}$$

Denote the complex potential as $\Phi = \phi + i\psi$, we can show that

$$\frac{d\Phi}{dz} = \frac{1}{2}(\phi_x + i\psi_x) - \frac{i}{2}(\phi_y + i\psi_y) = u - iv, \tag{2.77}$$

which can be sometimes easier to find. We can also show (by the C–R equations) that

$$\frac{d\Phi}{d\bar{z}} = 0. \tag{2.78}$$

Such function Φ is called analytic/homomorphic function whose real and imaginary parts ϕ, ψ are called the harmonic conjugates which satiafy the C–R equation and the Laplacian.

2.4.1 2D potential flow in cylindrical coordinates

The continuity equation in two dimensions reads

$$\frac{\partial(ru)}{\partial r} + \frac{\partial v}{\partial \theta} = 0. \tag{2.79}$$

We can define the streamfunction and scalar potential as

$$u = \frac{1}{r}\frac{\partial \psi}{\partial \theta},\ v = -\frac{\partial \psi}{\partial r} \tag{2.80}$$

$$u = \frac{\partial \phi}{\partial r}, \quad v = \frac{1}{r}\frac{\partial \phi}{\partial \theta}. \tag{2.81}$$

In the limit of 2D axisymmetric flow, we have $\partial_\theta = 0$ and hence

$$\frac{\partial(ru)}{\partial r} = 0,\ u = u(r),\ v = v(r), \tag{2.82}$$

the integration of the former leads to

$$u = \frac{Q}{2\pi r} \tag{2.83}$$

where Q is constant volume flow rate. In an inviscid barotropic potential flow, the conservation of circulation implies $\Gamma = \int_0^{2\pi} vr\, d\theta = 2\pi rv = \text{const}$, and hence

$$v = \frac{\Gamma}{2\pi r}. \tag{2.84}$$

The corresponding streamfunction and scalar potential are

$$\psi = -\frac{\Gamma}{2\pi}\ln r + \frac{Q}{2\pi}\theta + C \tag{2.85}$$

$$\phi = \frac{Q}{2\pi}\ln r + \frac{\Gamma}{2\pi}\theta + C' \tag{2.86}$$

representing the superposition of two typical potential solutions: a line source/sink with

$$u = \frac{Q}{2\pi r},\ v = 0 \tag{2.87}$$

and a line vortex

$$u = 0,\ v = \frac{\Gamma}{2\pi r}. \tag{2.88}$$

The vorticity reduces to

$$\omega_z = \frac{1}{r}\frac{\partial(rv)}{\partial r} = \frac{v}{r} + \frac{\partial v}{\partial r}, \tag{2.89}$$

with the first term called the curvature vorticity and the second called the shear vorticity.

2.5 Stokes flows

In Stokes flows (creeping flows at very low Re), the governing equation reduces to

$$0 = -\frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u}. \tag{2.90}$$

Taking the divergence, curl, and Laplacian respectively, we find that

$$\nabla^2 p = 0 \tag{2.91}$$

$$\nabla^2\boldsymbol{\omega} = \mathbf{0} \tag{2.92}$$

$$\nabla^2\nabla^2\boldsymbol{u} = \mathbf{0} \tag{2.93}$$

$$\nabla^2\nabla^2\psi = 0\,(\text{2D}) \tag{2.94}$$

that all flow variables are harmonic or biharmonic functions.

2.6 Beltrami flows

Recall in (1.62) that the Lamb vector is defined as $\boldsymbol{H} = \boldsymbol{u}\times(\nabla\times\boldsymbol{u}) = \boldsymbol{u}\times\boldsymbol{\omega}$. The Beltrami flows satisfy

$$\boldsymbol{H} = \mathbf{0}, \tag{2.95}$$

i.e., streamlines parallel to vortex lines. Moreover, the generalized Beltrami flows satisfy

$$\nabla \times \boldsymbol{H} = \boldsymbol{0} \tag{2.96}$$

such that the second term in (2.17) is zero and the vorticity equation (2.22), in the absence of external torque and baroclinicity, reduces to

$$\frac{\partial \boldsymbol{\omega}}{\partial t} = \nu \nabla^2 \boldsymbol{\omega}. \tag{2.97}$$

Hence, there is no vorticity generation mechanism in Beltrami and generalized Beltrami flows. Although (2.97) is easy to solve, the boundary conditions for vorticity are usually hard to prescribe.

In two dimensions, we also have

$$\omega = -\nabla^2 \psi, \tag{2.98}$$

and the generalized Beltrami flow condition (2.96) translates to

$$\frac{\partial \omega_z}{\partial x}\frac{\partial \psi}{\partial y} - \frac{\partial \omega_z}{\partial y}\frac{\partial \psi}{\partial x} = 0, \tag{2.99}$$

i.e.,

$$\nabla \omega_z \,//\, \nabla \psi, \tag{2.100}$$

implying that the two rows are linearly dependent and $\omega_z = f(\psi, t)$ that vorticity is a constant along the streamline.

The equation that ψ satisfies is

$$\frac{\partial}{\partial t}(\nabla^2 \psi) = \nu \nabla^2 \nabla^2 \psi, \tag{2.101}$$

with some found exact solutions provided below.

2.6.1 Examples

The ABC (Arnold–Beltrami–Childress) flow

$$u = A \sin z + C \cos y \tag{2.102}$$
$$v = B \sin x + A \cos z \tag{2.103}$$
$$w = C \sin y + B \cos x \tag{2.104}$$
$$\tag{2.105}$$

has $\boldsymbol{\omega} = \boldsymbol{u}$. It is a Beltrami flow in the narrow sense.

Taylor's decaying vortex:

$$\psi(x, y, t) = a \cos(mx) \cos(ny) \exp[-\nu(m^2 + n^2)t], \tag{2.106}$$

with the wavenumbers m, n being free parameters. It is often used for testing numerical algorithms.

Kelvin's cat eye vortex:

$$\psi(x, y, t) = a \cosh(mx) \cos(ny) \exp[\nu(m^2 - n^2)t], \tag{2.107}$$

with the wavenumbers m, n being free parameters.

2.7 Point vortex models

Hamiltonian system, single, dipole, and more. Biot-Savart's law.

2.8 Lamb–Oseen similarity solution, Burgers vortex

We recall that a potential/irrotational vortex has a velocity profile as

$$u_\theta = \frac{\Gamma}{2\pi r} \tag{2.108}$$

where $\Gamma = \int_0^{2\pi} u_\theta r \, \mathrm{d}\theta$ is the circulation. The vorticity is

$$\omega_z = \frac{1}{r}\frac{\partial (r u_\theta)}{\partial r} = 0 \tag{2.109}$$

hence the potential vortex is *irrotational*.

Oseen considered a viscous solution in the form of

$$u_r = 0, \; u_\theta = \frac{\Gamma}{2\pi r} g(r,t), \tag{2.110}$$

where $g(r,t)$ is a non-dimensional similarity function that combines the time-dependent spreading vortex core size $R(t)$ and time t, that collapses when the non-dimensional radius agree.

It is easy to see that the viscous length scale is $\sqrt{\nu t}$ in a diffusion process. So we take

$$g(r,t) = g(\hat{r}^2) = g\left(\frac{r^2}{4\nu t}\right) \tag{2.111}$$

where $\hat{r} = r/2\sqrt{\nu t}$ is the self-similar length scale and $\eta = \hat{r}^2$ is the similarity variable. The same similarity transform converts the heat equation to an ODE from which the heat kernel (2.124) can be solved.

The simplified azimuthal momentum equation is

$$\frac{\partial u_\theta}{\partial t} = \nu\left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{1}{r}\frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r^2}\right), \tag{2.112}$$

which represents a diffusion process, and it reduces to and ODE under the similarity transform (2.111):

$$g' + g'' = 0, \tag{2.113}$$

where $g'(\eta) = \partial_\eta g$. The general solution is

$$g(\eta) = c_1 + c_2 e^{-\eta}. \tag{2.114}$$

Given the boundary conditions

$$g(0) = 0, \; g(\infty) = 1, \tag{2.115}$$

we have

$$g(r,t) = g(\eta) = 1 - \exp\left(\frac{r^2}{4\nu t}\right) \tag{2.116}$$

and the velocity is

$$u_\theta = \frac{\Gamma}{2\pi r}\left(1 - \exp\left(-\frac{r^2}{4\nu t}\right)\right). \tag{2.117}$$

It approaches the limit of potential vortex as $r >> R = 2\sqrt{\nu t}$, with the diffusion speed being $u_d = R/t = 2\sqrt{\nu/t}$. Also, at the inviscid limit ($\nu \to \infty$), it is just the potential vortex solution (with $\omega_z = 0$).

The vorticity, is

$$\omega_z = \frac{1}{r}\frac{\partial(ru_\theta)}{\partial r} = \frac{\Gamma}{4\pi\nu t}\exp\left(-\frac{r^2}{4\nu t}\right). \tag{2.118}$$

It is straightforward to verify that ω_z satisfies the diffusion equation in cylinderical coordinates (with $\partial_\theta = 0$ and no advection term since $u_r = 0$; see (D.83))

$$\frac{\partial \omega_z}{\partial t} = \nu\nabla^2\omega_z \tag{2.119}$$

$$= \frac{\nu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \omega_z}{\partial r}\right), \tag{2.120}$$

The evolution of an initially Gaussian vortex profile with initial radius $a_0^2 = 4\nu t_0$ at t_0 will be

$$\omega_z(r,t) = \omega_0(t)\exp\left(-\frac{r^2}{a_0^2}\right) \tag{2.121}$$

where the vortex core size grows as $\sqrt{\nu t}$

$$a_0^2(t) = a_0^2(t_0) + 4\nu t \tag{2.122}$$

and

$$\omega_0(t) = \omega_0(t_0)\frac{t_0}{t} = \omega_0(t_0)\frac{a_0^2(0)}{a_0^2(t)}, \tag{2.123}$$

all expressed in terms of measured variables such as vortex radius and characteristic vorticity, denoted by subscript '0'. Explicit dependence on t_0 has been removed and the time t is to be interpreted as t after t_0 ($t > t_0$).

Moreover, (2.118) is the just heat kernel (fundamental solution of the heat equation) with two spatial dimensions. In a d-dimensional space, the heat kernel is generally written as

$$K(\boldsymbol{x} - \boldsymbol{x}_0, t - t_0) = \frac{1}{(4\pi\nu(t-t_0))^{d/2}}\exp\left(-\frac{|\boldsymbol{x} - \boldsymbol{x}_0|^2}{4\nu(t-t_0)}\right). \tag{2.124}$$

It is easy to verify that the Lamb vector is $\boldsymbol{L} = L(r)\boldsymbol{e}_r$ so $\nabla \times \boldsymbol{L} = 0$ and it is verified that the Lamb-Oseen solution is generalized Beltrami.

Similar to the Lamb–Oseen solution, the Burgers solution can be established as:

$$u_r = -\alpha r \tag{2.125}$$

$$u_\theta = \frac{\Gamma}{2\pi r}\left(1 - \exp\left(-\frac{\alpha r^2}{2\nu}\right)\right) \tag{2.126}$$

$$u_z = 2\alpha z \tag{2.127}$$

with an axisymmetric stagnation flow and α being the strain rate. It has a vortex stretching mechanism.

2.9 Heat diffusion: a similarity solution

Consider one-dimensional heat equation:

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right), \tag{2.128}$$

when the thermal conductivity k is a constant in space, it reduces to

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}, \tag{2.129}$$

where $\alpha = k/\rho c_p$ is the heat diffusivity. Eqn. (2.129) is also the governing equation for general diffusion processes, for example, of a scalar or chemical species.

Consider the following initial-boundary conditions:

$$T(x, t < 0) = T_s \tag{2.130}$$

$$T(0, t) = T_0,\ T(\infty, t) = T_s, \tag{2.131}$$

corresponding to an infinity long rod with initial temperature T_s subject to an abrupt input at $t = 0$ that brings $T(0, t) = T_0$.

Non-dimensionalization:

$$\theta = \frac{T - T_0}{T_s - T_0} \tag{2.132}$$

such that the PDE is

$$\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial x^2}, \tag{2.133}$$

and the I/BCs are

$$\theta(x, t < 0) = 1 \tag{2.134}$$

$$\theta(0, t) = 0,\ \theta(\infty, t) = 1. \tag{2.135}$$

Consider the non-dimensional similarity variable

$$\eta = \frac{x}{2\sqrt{\alpha t}} \tag{2.136}$$

and $\theta(x, t) = f(\eta)$. Eqn. (2.133) can be converted to an ODE as

$$f'' + 2\eta f' = 0, \tag{2.137}$$

subject to boundary conditions of

$$f(0) = 0,\ f(\infty) = 1. \tag{2.138}$$

Direct integration of

$$\frac{f''}{f'} = -2\eta \tag{2.139}$$

yields

$$\ln f' = -\eta^2 + \ln C_1 \tag{2.140}$$

and

$$f' = C_1 e^{-\eta^2}. \tag{2.141}$$

Integrating again we have

$$f(\eta) = f(\eta) - f(0) = C_1 \int_0^\eta e^{-t^2}\, dt \tag{2.142}$$

where the error function is defined as

$$\mathrm{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-t^2}\, dt \tag{2.143}$$

and

$$\mathrm{erf}(0) = 0,\ \mathrm{erf}(\infty) = 1. \tag{2.144}$$

Hence, we have $C_1 = 2/\sqrt{\pi}$ and

$$f(\eta) = \mathrm{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-t^2}\, dt. \tag{2.145}$$

The solution to the original equation reads

$$T(x,t) = (T_s - T_0)\mathrm{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right) + T_0. \tag{2.146}$$

Moreover, the complementary error function is defined as

$$\mathrm{erfc}(\eta) = 1 - \mathrm{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_\eta^\infty e^{-t^2}\, dt \tag{2.147}$$

and reaches values of

$$\mathrm{erfc}(0) = 1,\ \mathrm{erfc}(\infty) = 0. \tag{2.148}$$

Chapter 3

Velocity gradient tensor, its decomposition and dynamics

Following the previous section, we continue to consider the local flow structures represented by the velocity gradients.

The velocity gradient tensor $\boldsymbol{u}\nabla$ is

$$\boldsymbol{u}\nabla = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial w} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial w} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial w} \end{bmatrix} \tag{3.1}$$

and in entity notation

$$(\boldsymbol{u}\nabla)_{ij} = \frac{\partial u_i}{\partial x_j}, \tag{3.2}$$

and we note that is a Jacobian such that

$$\delta\boldsymbol{u} = (\boldsymbol{u}\nabla)\cdot\delta\boldsymbol{x} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial w} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial w} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial w} \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta z \end{bmatrix}. \tag{3.3}$$

For an arbitrary tensor $\boldsymbol{A}$, it is always possible to decompose it into symmetric and antisymmetric (skew-symmetric) parts:

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{\Omega} \tag{3.4}$$

where

$$\boldsymbol{S} = \frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^T) \tag{3.5}$$

$$\boldsymbol{\Omega} = \frac{1}{2}(\boldsymbol{A} - \boldsymbol{A}^T) \tag{3.6}$$

such that

$$\boldsymbol{S}^T = \boldsymbol{S},\ \boldsymbol{\Omega}^T = -\boldsymbol{\Omega}. \tag{3.7}$$

For the velocity gradient tensor, $\boldsymbol{S} = 1/2(\boldsymbol{u}\nabla + \nabla\boldsymbol{u})$ is called the rate-of-strain tensor and $\boldsymbol{\Omega} = 1/2(\boldsymbol{u}\nabla - \nabla\boldsymbol{u})$ is called the rotation tensor.

3.1 Pseudo-vector and associated antisymmetric rotation tensor

Vorticity

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{u} \tag{3.8}$$

$$= \begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ \partial_x & \partial_y & \partial_z \\ u & v & w \end{vmatrix} \tag{3.9}$$

$$= \begin{bmatrix} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \end{bmatrix} \tag{3.10}$$

is a pseudo-vector ($\omega_i = \epsilon_{ijk}\partial_j u_k$) whose sign depends on the coordinate system (the order of i, j, k; left-hand or right-hand; cyclic or anticyclic), and is related to the antisymmetric part of velocity gradient tensor $\boldsymbol{u}\nabla$ (the rotation rate tensor $\boldsymbol{\Omega}$):

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right). \tag{3.11}$$

or

$$\boldsymbol{\Omega} = \frac{1}{2}\begin{bmatrix} 0 & -\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) & \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) \\ \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) & 0 & -\left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) \\ -\left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right) & \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right) & 0 \end{bmatrix} \tag{3.12}$$

$$= \frac{1}{2}\begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \tag{3.13}$$

$$= \begin{bmatrix} 0 & -\omega_z^* & \omega_y^* \\ \omega_z^* & 0 & -\omega_x^* \\ -\omega_y^* & \omega_x^* & 0 \end{bmatrix} \tag{3.14}$$

where $\boldsymbol{\omega}^*$ is the angular velocity and $\boldsymbol{\omega} = 2\boldsymbol{\omega}^*$ (vorticity is twice of the angular velocity of the local solid-body rotation motion).

Each antisymmetric tensor $\boldsymbol{\Omega}$ can be represented by a pseudo-vector $\boldsymbol{\omega}^*$ (since it just has three independent elements), such that

$$\Omega_{ij} = -\epsilon_{ijk}\omega_k^* \tag{3.15}$$

$$\omega_k^* = -\frac{1}{2}\epsilon_{ijk}\Omega_{ij} \tag{3.16}$$

and according to (3.15)-(3.16) the inner product of the tensor $\boldsymbol{\Omega}$ with and arbitrary vector $\boldsymbol{a}$ can be written as

$$\boldsymbol{\Omega} \cdot \boldsymbol{a} = \boldsymbol{\omega}^* \times \boldsymbol{a}. \tag{3.17}$$

It is easy to verify (3.15) by definition and (3.16) using (A.8). As a corollary, we have

$$\boldsymbol{\Omega} \cdot \boldsymbol{\omega}^* = \boldsymbol{\omega}^* \times \boldsymbol{\omega}^* = \mathbf{0}. \tag{3.18}$$

We can also find that

$$\|\boldsymbol{\Omega}\|^2 = \Omega_{ij}\Omega_{ij} = 2\delta_{kl}\omega_k^*\omega_l^* = \frac{1}{2}\boldsymbol{\omega}^2, \tag{3.19}$$

i.e., the norm of the rotation tensor is just the enstrophy.

Note the vorticity is $\boldsymbol{\omega} = 2\boldsymbol{\omega}^*$. We will further look into the eigenvalue decomposition and principle directions of $\boldsymbol{S}$ to further understand how it describes the geometry of the flow.

3.2 Strain rate tensor

$$\boldsymbol{S} = \begin{bmatrix} \epsilon_1 & \frac{1}{2}\gamma_3 & \frac{1}{2}\gamma_2 \\ \frac{1}{2}\gamma_3 & \epsilon_2 & \frac{1}{2}\gamma_1 \\ \frac{1}{2}\gamma_2 & \frac{1}{2}\gamma_1 & \epsilon_3 \end{bmatrix} \tag{3.20}$$

where the normal strain rate is

$$\epsilon_i = \frac{\partial u_{(i)}}{\partial x_{(i)}}, \tag{3.21}$$

indices inside parentheses don't imply summation, and the shear strain rate is

$$\gamma_i = \left(\frac{\partial u_j}{\partial x_k} + \frac{\partial u_k}{\partial x_j} \right), \; i \neq j \neq k. \tag{3.22}$$

Additionally, it is easy to verify with index notation that

$$\boldsymbol{S} : \boldsymbol{\Omega} = 0, \tag{3.23}$$

where (:) denotes tensor inner-product, which induces the Frobenius norm. But the kinematic alignment between the principal directions of $\boldsymbol{S}$ and the vorticity vector is still an important question in turbulence (Ashurst *et al.*, 1987).

Examples.

1. Consider a pure stretching motion, $\epsilon_1 = \partial_x u \neq 0$ only.

$$\frac{d(\delta x)}{dt} = u = \epsilon_1 \delta x \tag{3.24}$$

hence

$$\epsilon_1 = \frac{1}{\delta x}\frac{d(\delta x)}{dt} \tag{3.25}$$

is the expansion rate (per unit time and per unit length).

2. Consider a pure rotation, $\gamma_1 \neq 0$ only. According to

$$\gamma_1 = \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) \tag{3.26}$$

and

$$\omega_1 = \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} = 0 \tag{3.27}$$

we have

$$v = \frac{1}{2}\gamma_1 \delta z,\ w = \frac{1}{2}\gamma_1 \delta y \tag{3.28}$$

Define α_{23} as the angle between two line elements that are initially aligned with x_2 and x_3, we have (assumingi infinitesimal angle of rotaion during δt)

$$\frac{d\alpha_{23}}{dt} = \frac{w\delta t/\delta y - v\delta t/\delta z}{\delta t} = (\frac{1}{2}\gamma_1 - (-\frac{1}{2}\gamma_1)) = \gamma_1. \tag{3.29}$$

Hence, γ_i can be understood as the rate of change of two perpendicular elements in the plane normal to $\boldsymbol{e}_i$.

Motivation: during the deformation, can we find a set of axes the mutual angle between each pair is not change? That leads to the eigenvalues/principal strains and principal directions of the strain rate tensor.

3.3 Velocity field decomposition examples

1. Consider a plane constant-rate pure solid body rotation with the position vector being

$$\boldsymbol{r} = \cos\theta \boldsymbol{e}_x + \sin\theta \boldsymbol{e}_y, \tag{3.30}$$

and we have

$$\begin{aligned} \boldsymbol{u} &= \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} && (3.31)\\ &= \dot{\theta}(-\sin\theta \boldsymbol{e}_x + \cos\theta \boldsymbol{e}_y) && (3.32)\\ &= \begin{bmatrix} 0 & -\dot{\theta} \\ \dot{\theta} & 0 \end{bmatrix} \begin{bmatrix} \cos\theta \boldsymbol{e}_x \\ \sin\theta \boldsymbol{e}_y \end{bmatrix} && (3.33)\\ &= \boldsymbol{\Omega} \cdot \boldsymbol{r}. && (3.34) \end{aligned}$$

With $\boldsymbol{\omega}^* = \dot{\theta}\boldsymbol{e}_z$ being the angular velocity, we have

$$\boldsymbol{u} = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \boldsymbol{\Omega} \cdot \boldsymbol{r} = \boldsymbol{\omega}^* \times \boldsymbol{r} \tag{3.35}$$

where $\boldsymbol{\omega}^*$ is the angular velocity.

$$\begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \end{bmatrix}, \tag{3.36}$$

In general, the decomposition of velocity gradient tensor can be applied to yield:

$$\begin{aligned}\delta\boldsymbol{u} &= \boldsymbol{u}\nabla\cdot\delta\boldsymbol{x} && (3.37)\\ &= \boldsymbol{S}\cdot\delta\boldsymbol{x}+\boldsymbol{\Omega}\cdot\delta\boldsymbol{x} && (3.38)\\ &= \boldsymbol{S}\cdot\delta\boldsymbol{x}+\boldsymbol{\omega}^*\times\delta\boldsymbol{x} && (3.39)\\ &= \nabla\varphi+\boldsymbol{\omega}^*\times\delta\boldsymbol{x} && (3.40)\end{aligned}$$

where the potential is

$$\begin{aligned}\varphi &= \frac{1}{2}(\epsilon_1\delta x^2+\epsilon_2\delta y^2+\epsilon_3\delta z^2+\gamma_1\delta y\delta z+\gamma_2\delta z\delta x+\gamma_3\delta x\delta y) && (3.41)\\ &= \frac{1}{2}\delta\boldsymbol{x}\cdot\boldsymbol{S}\cdot\delta\boldsymbol{x} && (3.42)\end{aligned}$$

such that

$$\nabla\varphi=\boldsymbol{S}\cdot\delta\boldsymbol{x}. \tag{3.43}$$

According to the Helmholtz decomposition theorem, a vector field can be decomposed into the sum of an irrotational field (gradient field/curl-free) and a solenoidal field (curl field/divergence free). Here we provide a construction for the velocity field.

2. Consider plain shear flow with $U(y)=ay$ and

$$\boldsymbol{u}\nabla=\begin{bmatrix}0&a\\0&0\end{bmatrix}=\begin{bmatrix}0&a/2\\a/2&0\end{bmatrix}+\begin{bmatrix}0&a/2\\-a/2&0\end{bmatrix}=\boldsymbol{S}+\boldsymbol{\Omega}. \tag{3.44}$$

The flow is liearly decomposed into a pure deformation/strain motion with $\partial_y u=\partial_x v=a/2$ and a rotation motion with $\partial_y u=-\partial_x v=a/2$.

3.4 Dynamics of the velocity gradient tensor

3.4.1 Dynamics of A and its powers

Let

$$\boldsymbol{A}=\boldsymbol{u}\nabla=\left[\frac{\partial u_i}{\partial x_j}\right]. \tag{3.45}$$

Taking the gradient of Navier–Stokes we have

$$\frac{\partial\boldsymbol{A}}{\partial t}+u_k\frac{\partial\boldsymbol{A}}{\partial x_k}=-\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i x_j}+\nu\frac{\partial^2\boldsymbol{A}}{\partial x_k^2}-\boldsymbol{A}\cdot\boldsymbol{A} \tag{3.46}$$

Contracting the indices in (3.46) we have

$$-\frac{1}{\rho}\frac{\partial^2 p}{\partial x_k^2}=\operatorname{tr}(\boldsymbol{A}^2)=-u_{i,j}u_{j,i} \tag{3.47}$$

and i.e.,

$$\nabla^2 p = 2\rho Q, \tag{3.48}$$

where

$$Q = \frac{1}{2}[(\operatorname{tr} \boldsymbol{A})^2 - \operatorname{tr}(\boldsymbol{A}^2)] \tag{3.49}$$

is the second invariant of $\boldsymbol{A}$ (see Appendix A.3). Hence, people argue that the sign of Q implies the existence of local pressure minimum and so forth the existence of vortices Hunt *et al.* (1988); Jeong & Hussain (1995).

The next question is the dynamics of its eigenvalues, invariants, and powers. We immediately deal with the latter since $\boldsymbol{A}^2$ appears in the equation of $\boldsymbol{A}$.

Re-write (3.46) as

$$\mathcal{L}(A_{ik}) = 0,\ \mathcal{L}(A_{kj}) = 0 \tag{3.50}$$

and multiply with A_{kj} and A_{ik} we have

$$\frac{\partial \boldsymbol{A}^2}{\partial t} + u_k \frac{\partial \boldsymbol{A}^2}{\partial x_k} = -\frac{1}{\rho}\left(\boldsymbol{A}\cdot\nabla(\nabla p) + \nabla(\nabla p)\cdot\boldsymbol{A}\right) + \nu\frac{\partial^2 \boldsymbol{A}^2}{\partial x_k^2} - 2\boldsymbol{A}^3. \tag{3.51}$$

Here $\nabla(\nabla p)$ is the pressure Hessian. We can see that, similar to the Reynolds average closure problem, the transport equations for $\boldsymbol{A}$ are never closed due to the appearance of even-higher powers.

Contracting the indices in (3.51) and assuming incompressibility ($\operatorname{tr} \boldsymbol{A} = 0$) we have

$$\frac{\partial Q}{\partial t} + u_k \frac{\partial Q}{\partial x_k} = \frac{1}{\rho}(\boldsymbol{A} : \nabla(\nabla p)) + \nu\frac{\partial^2 Q}{\partial x_k^2} + \operatorname{tr}(\boldsymbol{A}^3), \tag{3.52}$$

while we note that $-2Q = A_{ik}A_{ki}$.

3.4.2 Dynamics of S, Ω, and the eigenvalues of $\mathbf{S}^2 + \mathbf{\Omega}^2$

$(3.46)^T \cdot \boldsymbol{A} + (3.46)\cdot\boldsymbol{A}^T$ we have

$$\frac{\partial S_{ij}}{\partial t} + u_k \frac{\partial S_{ij}}{\partial x_k} = -\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i x_j} + \nu\frac{\partial^2 S_{ij}}{\partial x_k^2} - (S_{ik}S_{kj} + \Omega_{ik}\Omega_{kj}) \tag{3.53}$$

Consider a balance between the last term in the RHS and pressure Hessain,

$$\nabla(\nabla p) = -\rho(\boldsymbol{\Omega}^2 + \boldsymbol{S}^2), \tag{3.54}$$

which can be, similarly to (3.48) (which can also be obtained by contracting (3.53)), be used to search for pressure minimum (Jeong & Hussain, 1995).

Assume $\lambda_1, \lambda_2, \lambda_3$ are the eigenvalues of $\boldsymbol{\Omega}^2 + \boldsymbol{S}^2$. Since

$$Q = -\frac{1}{2}\operatorname{tr}(\boldsymbol{A}^2) = -\frac{1}{2}\operatorname{tr}(\boldsymbol{\Omega}^2 + \boldsymbol{S}^2) = -\frac{1}{2}(\lambda_1 + \lambda_2 + \lambda_3), \tag{3.55}$$

a positive Q corresponds to $\lambda_1 + \lambda_2 + \lambda_3 < 0$. Then assume $-\lambda_3 > -\lambda_2 > -\lambda_1$, we have $\lambda_3 < 0, \lambda_1 > 0$. The requirement for p to have a (plane) local minimum (where $\nabla p = 0$ and a sub-matrix of $\nabla(\nabla p)$ is positive definite) is that the second eigenvalue $\lambda < 0$. Hence, a negative

λ_2 is a sufficient condition for a pressure minimum that defines a vortex core (Jeong & Hussain, 1995).

On the other hand, the equation for $\mathbf{\Omega}$,

$$\frac{\partial \Omega_{ij}}{\partial t} + u_k \frac{\partial \Omega_{ij}}{\partial x_k} = \nu \frac{\partial^2 \Omega_{ij}}{\partial x_k^2} - (S_{ik}S_{kj} + \Omega_{ik}\Omega_{kj}), \tag{3.56}$$

does not involve pressure directly. It is equivalent to the vorticity transport equation.

3.4.3 Dynamics of the invariant space

In Euler equations (neglecting viscosity), (3.46) is written as

$$\frac{\partial \boldsymbol{A}}{\partial t} + u_k \frac{\partial \boldsymbol{A}}{\partial x_k} = -\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i x_j} - \boldsymbol{A}\cdot\boldsymbol{A} \tag{3.57}$$

Given

$$-\nabla^2 p = \mathrm{tr}(\boldsymbol{A}) = A_{ik}A_{ki}, \tag{3.58}$$

we have

$$\mathrm{RHS} = -\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i x_j} - A_{ik}A_{kj} \tag{3.59}$$

$$= -\frac{1}{\rho}\frac{\partial^2 p}{\partial x_i x_j} - A_{ik}A_{kj} + \frac{1}{3}\left(\frac{\partial^2 p}{\partial x_k x_k} + A_{mk}A_{km}\right)\delta_{ij} \tag{3.60}$$

$$= -\frac{1}{\rho}\left(\frac{\partial^2 p}{\partial x_i x_j} - \frac{1}{3}\frac{\partial^2 p}{\partial x_k x_k}\delta_{ij}\right) - \left(A_{ik}A_{kj} - \frac{1}{3}A_{mk}A_{km}\delta_{ij}\right) \tag{3.61}$$

Assume the heterogeneous part of the pressure is negligible (at least homogeneous in the small scales), we have

$$\frac{d\boldsymbol{A}}{dt} + \left(A_{ik}A_{kj} - \frac{1}{3}A_{mk}A_{km}\delta_{ij}\right) = 0. \tag{3.62}$$

Multiply (3.62) with A_{ji}, given

$$A_{ji}\frac{dA_{ij}}{dt} = \frac{1}{2}\frac{d}{dt}(A_{ij}A_{ji}) = -\frac{dQ}{dt}, \tag{3.63}$$

and $Q = -1/2\,\mathrm{tr}(\boldsymbol{A})$, $R = 1/3\,\mathrm{tr}(\boldsymbol{A}^3)$, we have

$$\frac{dQ}{dt} = 3R. \tag{3.64}$$

Multiply (3.62) with $A_{jk}A_{ki}$, given

$$A_{jk}A_{ki}\frac{dA_{ij}}{dt} = \frac{1}{3}\frac{d}{dt}(A_{ij}A_{jk}A_{ki}) = \frac{dR}{dt} \tag{3.65}$$

we have

$$\frac{dR}{dt} + \mathrm{tr}(\boldsymbol{A}^4) - \frac{1}{3}[\mathrm{tr}(\boldsymbol{A})^2]^2 = 0 \tag{3.66}$$

From Appendix A.3 we know (in incompressible flows)

$$P = \mathrm{tr}(\boldsymbol{A}) = 0 \tag{3.67}$$

$$Q = \frac{1}{2}[\mathrm{tr}(\boldsymbol{A})^2 - \mathrm{tr}(\boldsymbol{A}^2)] = -\frac{1}{2}\,\mathrm{tr}(\boldsymbol{A}^2) \tag{3.68}$$

$$R = \det(\boldsymbol{A}) = \frac{1}{6}\left(\mathrm{tr}(\boldsymbol{A})^3 - 3\,\mathrm{tr}(\boldsymbol{A})\,\mathrm{tr}(\boldsymbol{A}^2) + 2\,\mathrm{tr}(\boldsymbol{A}^3)\right) = \frac{1}{3}\,\mathrm{tr}(\boldsymbol{A}^3) \tag{3.69}$$

where the last relation from the Newton's identity.

According to Cayley–Hamilton theory, matrix $\boldsymbol{A}$ satisfies its characteristic polynomial as

$$\boldsymbol{A}^3 - P\boldsymbol{A}^2 + Q\boldsymbol{A} - R\boldsymbol{I} = 0 \tag{3.70}$$

so

$$\boldsymbol{A}^4 = \boldsymbol{A}\boldsymbol{A}^3 = -Q\boldsymbol{A}^2 + R\boldsymbol{A} \tag{3.71}$$

and

$$\mathrm{tr}(\boldsymbol{A}^4) = -Q\,\mathrm{tr}(\boldsymbol{A}^2) = \frac{1}{2}\,\mathrm{tr}(\boldsymbol{A}^2)^2. \tag{3.72}$$

Finally, we have

$$\frac{dR}{dt} = -\frac{2}{3}Q^2. \tag{3.73}$$

Equations (3.64)-(3.73) form a autonomous system, which can be further integrated:

$$dt = \frac{dQ}{3R} = \frac{dR}{-2Q/3} \tag{3.74}$$

$$Q^3 + \frac{27}{4}R^2 = \text{const.} \tag{3.75}$$

We note that $\triangle = Q^3 + 27/4R^2$ is just the discriminant of $\boldsymbol{A}$. The approximate inviscid dynamics is just for the discriminant to conserve along the path. But from (3.62) on the characters of the N–S is less seen.

Ref. Meneveau (2011).

3.5 Lagrangian representations

Cauchy-Green tensor etc.

Chapter 4

Laminar wall flows

When the flow is parallel, there is subsitential simplification can be done to the N–S, allowing analytical solutions.

4.1 Plane Poiseuille–Couette

Consider 2D parallel flow at steady state between two planes at $y = 0, 2h$ (h is the channel half height) and the boundary conditions are

$$u(x,0) = v(x,0) = 0, \tag{4.1}$$
$$u(x,2h) = U,\ v(x,2h) = 0. \tag{4.2}$$

The 2D N–S equation reads:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{4.3}$$
$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \tag{4.4}$$
$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right). \tag{4.5}$$

With streamwise invariance, we have $\partial_x u = \partial_x v = 0$. With continuity we also have $\partial_y v = 0$, leading to $v(x,y) = v(x,0) = 0$. Hence the second derivatives $\partial_{xx} u = \partial_{xx} v = \partial_{yy} v = 0$ as well. The momentum equations are reduced to

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\frac{\partial^2 u}{\partial y^2}, \tag{4.6}$$
$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial y}. \tag{4.7}$$

At the wall, (4.6) implies

$$\frac{1}{\rho}\frac{\partial p_w}{\partial x} = \nu\frac{\partial^2 u}{\partial y^2}, \tag{4.8}$$

where the RHS is independent of x (fully developed). Combined with (4.7), we have

$$\frac{\partial p}{\partial x} = \frac{\partial p_w}{\partial x} \triangleq \frac{\partial P}{\partial x}, \tag{4.9}$$

i.e., the pressure gradient is a constant. Then we can solve for the velocity with:

$$\nu \frac{d^2 u}{dy^2} = \frac{1}{\rho} \frac{dP}{dx}. \tag{4.10}$$

Given the two boundary conditions (4.2), the general velocity profile for Poiseuille–Couette flows can be written as

$$u(y) = U \frac{y}{2h} - \frac{2h^2}{\rho\nu} \frac{dP}{dx} \left[\frac{y}{2h} - \left(\frac{y}{2h} \right)^2 \right] \tag{4.11}$$

and it reduces to the Poiseuille solution with $U = 0$

$$u(y) = -\frac{2h^2}{\rho\nu} \frac{dP}{dx} \left[\frac{y}{2h} - \left(\frac{y}{2h} \right)^2 \right] \tag{4.12}$$

and the Couette solution with $dP/dx = 0$

$$u(y) = U \frac{y}{2h}. \tag{4.13}$$

We note the both the Poiseuille and Couette, or the Poiseuille–Couette profiles are Re-independent (but the friction coefficients will be Re-dependent) and have no inflection point. The mechanism for turbulence generation has to be transient. For the balance in a turbulent channel see section 5.6.1.

4.2 Laminar Poiseuille channel

Now lets consider the momentum balance in a control volume of $[0, L_x] \times [0, 2h]$. In plane Poiseuille, it can be shown that the pressure gradient equals twice the wall shear stress (simply because there are two walls; a half-channel balance within $[0, L_x] \times [0, h]$ can be established too, with the centerplane being stress-free) as

$$-2h \frac{dP}{dx} = 2\tau_w = 2\mu \frac{\partial u}{\partial y}. \tag{4.14}$$

That being said, the wall shear stress is balanced by the constant pressure gradient. On one side of the wall,

$$-\frac{dP}{dx} = \frac{\tau_w}{h}, \tag{4.15}$$

which along with (4.14), don't depend on the shape of the profile.

The centerline velocity is

$$U_0 = u(h) = -\frac{h^2}{2\rho\nu} \frac{dP}{dx} \tag{4.16}$$

and the bulk velocity is

$$\bar{U} = \frac{1}{2h} \int_0^{2h} u(y)\, dy = \frac{1}{h} \int_0^{h} u(y)\, dy = -\frac{h^2}{3\rho\nu} \frac{dP}{dx} = \frac{2}{3} U_0 \tag{4.17}$$

and the centerline and bulk Reynolds numbers are

$$Re_0 = U_0 h/\nu, \tag{4.18}$$

$$Re = 2\bar{U}h/\nu = 4/3 Re_0. \tag{4.19}$$

The friction velocity is

$$u_\tau = \sqrt{\tau_w/\rho} = \sqrt{-\frac{h}{\rho}\frac{dP}{dx}} = \sqrt{\frac{2\nu U_0}{h}} = \sqrt{\frac{3\nu\bar{U}}{h}}, \tag{4.20}$$

and the ratios are

$$\frac{u_\tau}{U_0} = \sqrt{\frac{2}{Re_0}} = \sqrt{\frac{8}{3Re}}, \tag{4.21}$$

$$\frac{u_\tau}{\bar{U}} = \sqrt{\frac{6}{Re}}. \tag{4.22}$$

The experimental critical bulk Reynolds number is $Re \approx 1350$, below which a channel can stay laminar, corresponds to

$$\frac{u_\tau}{U_0} = 0.0444, \tag{4.23}$$

leading to a friction Reynolds number of

$$Re_\tau = \frac{u_\tau h}{\nu} = 45, \tag{4.24}$$

which is the theoretical minimum for turbulent channels. The relation between Re_τ and Re in laminar channels is

$$Re_\tau = \sqrt{\frac{3Re}{2}} = \sqrt{2Re_0}. \tag{4.25}$$

The friction Reynolds number based on the normal-mode critical Reynolds number $Re_{0,c} = 5772$ is $Re_\tau \approx 107$. As a reference, the Reynolds numbers in the turbulent channel of Kim *et al.* (1987) are about $Re = 5600$ and $Re_\tau = 180$.

Define the skin friction coefficient as

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho U_0^2} \tag{4.26}$$

it can be shown that

$$c_f = \frac{16}{3Re}, \tag{4.27}$$

which is called the laminar friction law for channels. Equally useful is the skin friction coefficient based on the bulk velocity,

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho\bar{U}^2}, \tag{4.28}$$

with $U_0 = 3\bar{U}/2$ and

$$C_f = \frac{12}{Re}. \tag{4.29}$$

4.3 Hagen–Poiseuille (circular pipe) flow

Hagen–Poiseuille flow has been quite important in experiments, due to the fact that a circular pipe is easy to build and the implications on pipelines. In a circular pipe, the laminar Hagen–Poiseuille flow is governed by

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \frac{\nu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) \tag{4.30}$$

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial r} \tag{4.31}$$

and the solution is (a parabola)

$$u(r) = -\frac{1}{4\mu}\frac{dP}{dx}(R^2 - r^2) \tag{4.32}$$

where the constant pressure gradient $dP/dx < 0$ and R is the pipe radius. It can be shown that the centerline velocity is

$$U_0 = u(0) = -\frac{1}{4\mu}\frac{dP}{dx}R^2 \tag{4.33}$$

and the bulk velocity is

$$\bar{U} = \frac{1}{\pi R^2}\int_0^R u(r)2\pi r\, dr = -\frac{1}{8\mu}\frac{dP}{dx}R^2 = \frac{1}{2}U_0. \tag{4.34}$$

It defines the bulk Reynolds number $Re = \rho\bar{U}D/\mu$. The streamwise pressure gradient is a constant for the same reason as in Poiseuille channels. The volume flow rate

$$Q = \pi R^2\bar{U} = -\frac{\pi R^4}{8\mu}\frac{dP}{dx} \tag{4.35}$$

is proportional to the pressure gradient.

The shear stress,

$$\tau = -\mu\frac{du}{dr} = -\frac{r}{2}\frac{dP}{dx} \tag{4.36}$$

has a linear profile that vanishes at the centerline and peaks at the wall, which leads to the balance between the wall shear stress and the pressure gradient

$$-\frac{dP}{dx} = \frac{2\tau_w}{R}, \tag{4.37}$$

which can also be obtained from a control volume analysis. The latter is more general as it does not depend on the details of the velocity profile (whether it is laminar or turbulent). We also note that the shear stress is independent on viscosity/density.

The skin friction coefficient is

$$C_f = \frac{\tau_w}{\frac{1}{2}\rho\bar{U}^2} = \frac{16}{Re}, \tag{4.38}$$

where the bulk Reynolds number is $Re = \rho\bar{u}D/\mu$, and it can be expressed as a form easier to measure in experiments as the (Darcy) friction factor,

$$f = \frac{\Delta pD}{\frac{1}{2}\rho\bar{U}^2L} = 4C_f = \frac{64}{Re} \tag{4.39}$$

where D is the pipe diameter, L is the pipe length and Δp is the pressure drop in between. Eqn. (4.39) is called the laminar friction law. It also sets that the pressure drop is prortional to the bulk velocity,

$$\Delta p/L \propto \bar{U}, \tag{4.40}$$

where in turbulent flows there is

$$\Delta p/L \propto \bar{U}^{1.75}. \tag{4.41}$$

The corresponding non-dimensional relations, which can alternatively be obtained by Π-group analysis and calibrated/measured from data, are

$$\frac{D\Delta p}{\frac{1}{2}\rho\bar{U}^2 L} = \phi\left(\frac{\rho\bar{U}D}{\mu}\right) = \frac{64}{Re} \tag{4.42}$$

and

$$\frac{\rho D^3 \Delta p}{L\mu^2} = \phi\left(\frac{\rho\bar{U}D}{\mu}\right) \approx 0.155\left(\frac{\rho\bar{U}D}{\mu}\right)^{1.75} \tag{4.43}$$

for laminar and turbulent pipes. We notice that the pressure drop (and hence wall shear stress) depends weakly on μ in turbulent pipes ($\Delta p \sim \mu^{0.25}$) than in laminar pipes ($\Delta p \sim \mu$). The relation (4.43) was first discovered by Blasius by fitting experimental data with insights from non-dimensional groups.

The friction Reynolds number is related to the bulk Reynolds number in laminar pipes as

$$Re_\tau = \frac{\rho u_\tau R}{\mu} = \sqrt{2Re} \tag{4.44}$$

and the critical Reynolds number above which the flow cannot remain laminar is around $Re_{\text{cri}} \approx 2300$ (White, 2008), corresponding to $Re_\tau \approx 68$.

4.4 The boundary layer theory

The boundary layer theory played central roles in the development of 20th century's fluid mechanics and aerodyanmics. Consider a uniform freestream flowing over a flat plat. The effect of viscosity, reflected by the no-slip boundary condition, is confined in a thin layer near the wall that scales with $\sqrt{\nu x/U_0}$ and the external flow is potential. This decomposition of flow regions enables separation of treatments of the lift, which can be computed from the potential theory, and of the drag, which can be obtained from another set of viscous-dependent and outer-flow-dependent boundary-layer equations. The effect of the boundary-layer solution to the potential flow can be approximated as a 'displacement' of the streamlines and the potential solution can be corrected accordingly. The boundary-layer solution, known as the Blasius solution, is self-similar. Similar self-similar solutions are found in corner flow (Falkner–Skan) and plane jets, wakes, and mixing layers, whose governing equations are of boundary-layer type.

4.5 Boundary layer equations

Here the freestreem pressure and velocity are $p_0(x), U_0(x)$ and they are related by the Bernoulli theorem (since the outer flow is inviscid), $p_0 + 1/2\rho U_0^2 =$ constant, hence

$$-\frac{dp_0}{dx} = \rho U_0 \frac{dU_0}{dx} \tag{4.45}$$

and the boundary layer equations are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{4.46}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{dp_0}{dx} + \frac{1}{\rho}\frac{\partial \tau}{\partial y} \tag{4.47}$$

$$= U_0\frac{dU_0}{dx} + \frac{1}{\rho}\frac{\partial \tau}{\partial y} \tag{4.48}$$

$$0 = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{4.49}$$

which can be easily reached by considering the relevant scales (see Schlichting & Gersten (2016) if needed). And we note that since there is no vertical pressure gradient, the streamwise pressure gradient inside the boundary layer is the same as that of the freestreem (determined by the Bernoulli equations). The viscous term is written as the divergence of the shear stress which takes

$$\tau = \mu\frac{\partial u}{\partial y} \tag{4.50}$$

in laminar flows and

$$\tau = \mu\frac{\partial u}{\partial y} - \rho\overline{u'v'} \tag{4.51}$$

in turbulent flows.

4.5.1 von Kármán momentum integral

The von Kármán integrals provides an approach to estimate critical parameters of the boundary layer, such as the boundary layer thickness, the wall shear stress, and the shape factor, without needing to have the solution analytically. In fact, even some velicity profiles that are quite approximate can lead to results close to the Blasius solution.

$(u - U_0)\times$ (4.46) + (4.48) we have

$$\frac{\partial}{\partial x}[u(u-U_0)] + \frac{\partial}{\partial y}[v(u-U_0)] = (U_0 - u)\frac{dU_0}{dx} + \frac{1}{\rho}\frac{\partial \tau}{\partial y}. \tag{4.52}$$

Integrate in y from 0 to ∞ we have

$$\frac{\partial}{\partial x}\left[\int_0^\infty u(U_0 - u)\,dy\right] + \frac{dU_0}{dx}\int_0^\infty (U_0 - u)\,dy = \frac{\tau_w}{\rho} \tag{4.53}$$

So far, we have not assumed whether (u, v) are the laminar velocities or the turbulent mean velocities. The nominal, displacement, and the momentum thicknesses are defined as

$$u(\delta_{99}) = 0.99U_0 \tag{4.54}$$

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U_0}\right)\,dy \tag{4.55}$$

$$\theta = \int_0^\infty \frac{u}{U_0}\left(1 - \frac{u}{U_0}\right)\,dy \tag{4.56}$$

and (4.53) can be turned into the von Kármán momentum integral

$$\frac{1}{2}c_f = \frac{\tau_w}{\rho U_0^2} = \frac{d\theta}{dx} + \frac{(\delta^* + 2\theta)}{U_0}\frac{dU_0}{dx} \tag{4.57}$$

$$= \frac{d\theta}{dx} + \frac{(H+2)\theta}{U_0}\frac{dU_0}{dx} \tag{4.58}$$

where $H = \delta^*/\theta$ is the shape factor and is around 2.6 and 1.3 for laminar and turbulent boundary layers, respectively. By assuming velocity profiles $u/U_0 = f(y\delta_{99})$, the above parameters can be solved.

Historically, von Kármán (1921) performed the momentum integral with a control-volume analysis of the boundary layer (mass and momentum balances). Assume a ZPGBL, such that the freestreem is a constant. Select a control volume of $(0,0), (0,h), (x,\delta), (x,0)$ where a streamline passes $(0,h)$ and (x,δ) (no mass penetrating the streamlines). The length of the plate into the paper is b.

The momentum conservation is

$$-D = \int_0^h b\rho U_0^2\, dy - \int_0^\delta b\rho u^2(y)\ dy \tag{4.59}$$

so

$$D = \rho b U_0^2 h - \rho b \int_0^\delta u^2(y)\, dy. \tag{4.60}$$

The mass conservation is (since the curve passing $(0,h)$ and (x,δ) is a streamline)

$$\int_0^h b\rho U_0\, dy = \int_0^\delta b\rho u(y)\ dy \tag{4.61}$$

which is used to simplify (4.60) to

$$D = \rho b U_0^2 \int_0^\delta \frac{u}{U_0}\left(1 - \frac{u}{U_0}\right) dy = \rho b U_0^2 \theta, \tag{4.62}$$

or in dimensionless form

$$C_D = \frac{D}{\frac{1}{2}\rho U_0^2 bL} = \frac{2\theta(L)}{L}. \tag{4.63}$$

We can see that the momentum integral relates the force on the solid body to the momentum loss in the boundary layer. What is also shown is that the momentum thickness $\theta(x)$ is a measure of the total drag up to x. Such ideas and definition of momentum thickness are also applicable in wakes, where the drag can be computed the same way using momentum integrals. The difference is that in wakes, θ stays as a constant past the body.

Note that the BL starts at $(0,0)$ so the total dray until $(x,0)$ is

$$D = \int_0^x b\tau_w\, dx \tag{4.64}$$

(τ_w is force per unit area) and hence

$$\tau_w = \frac{1}{b}\frac{dD}{dx} = \rho U_0^2 \frac{d\theta}{dx} \tag{4.65}$$

and the friction coefficient is

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho U_0^2} = 2\frac{d\theta}{dx}. \tag{4.66}$$

This is the same as (4.58) without external pressure gradients. Its implication is that, the momentum thickness represents the loss of momentum (deficit) due to the drag/shear stress acting on the bottom wall.

The equations (4.55)-(4.61) can also be interpreted as

$$U_0\delta^* = \int_0^\delta (U_0 - u)\, dy = U_0\delta - \int_0^\delta u\, dy \tag{4.67}$$

$$U_0 h = \int_0^h U_0\, dy = \int_0^\delta u\, dy = U_0(\delta - \delta^*) \tag{4.68}$$

$$\delta_* = \int_0^h \left(1 - \frac{u}{U_0}\right) dy = \delta - h \tag{4.69}$$

Hence, δ^* is interpreted as the displacement of the stremelines (or the bulk of fluid with velocity U_0 and height h) by the boundary layer. The last equality in (4.68) can be re-written as

$$\int_0^H u\, dy = U_0(H - \delta^*) \tag{4.70}$$

where H is an arbitrary height above δ_{99}. Consider two locations x_1, x_2 and a control volume in between with height H. The mass conservation

$$\int_0^H u_1\, dy = \int_0^H u_2\, dy + V_0(x)(x_2 - x_1) \tag{4.71}$$

where $H > \delta_1^*, \delta_2^*$ and V_0 is the vertical velocity of the external flow. Taking (4.70), it can be converted to

$$U_0(H - \delta_1^*) = U(H - \delta_2^*) + V_0(x_2 - x_1), \tag{4.72}$$

that being said,

$$\frac{V_0}{U_0} = \frac{d\delta^*}{dx}, \tag{4.73}$$

the rate at which the displacement thickness grows measures the displacement velocity.

It is now possible to consider the flow near the 'nominal' edge of the boundary layer. The line of $\delta_{99}(x)$ follows $\sqrt{x}$ according to the Blasius solution (see the next section) and its direction is $(1, d\delta_{99}/dx)$. At the edge of the boundary layer (where δ_{99} is), the direction of the velocity is $\boldsymbol{U}_0 = (U_0, V_0) = U_0(1, d\delta^*/dx)$. It can be seen from the Blasius solution that at each location (x), $\delta_{99} \approx 3\delta^*$. That being said, there is flow normal to $y = \delta_{99}(x)$ into the 'boundary layer', which is not a streamline!

Similar to (4.68) that

$$\int_0^\delta u\, dy = U_0(\delta - \delta^*), \tag{4.74}$$

we can also write down

$$\int_0^\delta u^2\, dy = U_0^2(\delta - \delta^* - \theta), \tag{4.75}$$

which interprets θ as the extra displacement (in addition to δ^*) so compensate for the momentum flux so that it matches that of the potential flow. The (x-)momentum conservation in again the control volume of $[x_1, x_2] \times [0, H]$ gives

$$\rho b \int_0^H u_1^2 \, dy = \rho b \int_0^H u_2^2 \, dy + \rho b \int_{x_1}^{x_2} V_0 U_0 \, dx + F_D \tag{4.76}$$

where $F_D = D_2 - D_1$ is the drag force between x_1, x_2 while D counts from the leading edge. With

$$\int_{x_1}^{x_2} V_0 dx = U_0(\delta_2^* - \delta_1^*), \tag{4.77}$$

we have

$$U_0(H - \delta_1^* - \theta_1) = U_0^2(H - \delta_2^* - \theta_2) + U_0^2(\delta_2^* - \delta_1^*) + \frac{F_D}{\rho b} \tag{4.78}$$

and

$$\theta_2 - \theta_1 = \frac{F_D}{\rho b U_0^2}. \tag{4.79}$$

That being said, the change of momentum thickness reflects the drag acting on the fluid during its travel. This is an equivalant statement to (4.65).

NB. The control volume selection of $[x_1, x_2] \times [0, H]$ instead of a CV starts at the origin bypasses the issues with the transient period and takes advantages of the self-similar solution being estabilished and the intial conditions being forgotton.

4.5.2 The Blasius similarity solution

The references are Schlichting & Gersten (2016); Kundu *et al.* (2015); White (2008), with the definition of $\delta(x)$ different by a factor of $\sqrt{2}$. Here we will follow the notations in Kundu *et al.* (2015); White (2008).

The laminar zero-pressure-gradient boundary layer equations are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{4.80}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2} \tag{4.81}$$

The idea of self-similar solutions is that the velocity profile $u(y)$ at different x will be the same under some proper transformation/normalization of u and y. The scale for u is apparently U_0, while the scale for y is δ. From the viscous scaling $v \sim \nu/\delta$ ($\delta \sim \sqrt{\nu T}, T \sim x/U_0$) and the scaling of the continuity equation $v/\delta \sim U_0/x$ we have

$$\delta^2 \sim \nu T \sim \frac{\nu x}{U_0} \tag{4.82}$$

and we accodringly define the boundary layer length scale as

$$\delta_\nu(x) = \sqrt{\frac{\nu x}{U_0}} \tag{4.83}$$

such that the similarity transformation is

$$\eta = \frac{y}{\delta_\nu(x)} = y\sqrt{\frac{U_0}{\nu x}} \tag{4.84}$$

and

$$\frac{u}{U_0} = f(\eta), \tag{4.85}$$

where $f(\eta)$ is the similarity function and η is the similarity coordinate. And we note

$$\frac{\partial \eta}{\partial x} = -\frac{1}{2}\frac{\eta}{x} \tag{4.86}$$

$$\frac{\partial \eta}{\partial y} = \sqrt{\frac{U_0}{\nu x}} \tag{4.87}$$

We note that the streamfunction ψ depends on ν, U_0, x, y and dimensionally

$$\psi(x,y) = U_0\delta_\nu(x)f(\eta) = \sqrt{\nu U_0 x}f(\eta) \tag{4.88}$$

and hence

$$u = \frac{\partial \psi}{\partial y} = U_0 f' \tag{4.89}$$

$$v = -\frac{\partial \psi}{\partial x} = \frac{1}{2}\sqrt{\frac{U_\infty \nu}{x}}(\eta f' - f) \tag{4.90}$$

The derivatives are

$$\frac{\partial u}{\partial x} = -\frac{U_0}{2x}f''\eta \tag{4.91}$$

$$\frac{\partial u}{\partial y} = U_0 f''\sqrt{\frac{U_0}{\nu x}} \tag{4.92}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{U_0^2}{\nu x}f''' \tag{4.93}$$

and then

$$u\frac{\partial u}{\partial x} = -\frac{U_0^2}{2x}f'f''\eta \tag{4.94}$$

$$v\frac{\partial u}{\partial y} = \frac{U_0^2}{2x}f''(\eta f' - f) \tag{4.95}$$

$$\nu\frac{\partial^2 u}{\partial y^2} = \frac{U_0^2}{x}f''' \tag{4.96}$$

and finally we have the ODE

$$f''' + \frac{1}{2}ff'' = 0 \tag{4.97}$$

with the boundary conditions being

$$f(0) = 0,\ f'(0) = 0,\ f'(\infty) = 1, \tag{4.98}$$

corresponding to

$$v(y=0)=0,\ u(y=0)=0,\ u(y=\infty)=U_0. \tag{4.99}$$

Note that $v(y=0)=0$ requires $(\eta f'-f)|_{y=0}=0$, meaning $f(0)=0$.

We also note that in the convention of Schlichting & Gersten (2016), $\delta_\nu=\sqrt{2\nu x/U_0}$ and the Blasius equation reads

$$f'''+ff''=0, \tag{4.100}$$

with the boundary conditions being the same. The choice of $\sqrt{2}$ is arbitrary and only for equation-simplification reasons.

It is common to use a Runge-Kutta shooting method to solve (4.98). The solution is called the Blasius solution, which is 'almost' an exact solution since you can know the functional value at any point to any precision that you want. The resulting relations are (approximately)

$$\frac{\delta_{99}}{x}=\frac{4.91}{\sqrt{Re_x}} \tag{4.101}$$

$$\frac{\delta^*}{x}=\frac{1.721}{\sqrt{Re_x}} \tag{4.102}$$

$$\frac{\theta}{x}=\frac{0.664}{\sqrt{Re_x}} \tag{4.103}$$

$$Re_\theta=0.664\sqrt{Re_x} \tag{4.104}$$

$$H=\frac{\delta^*}{\theta}=2.59 \tag{4.105}$$

$$\tau_w=\rho U_0^2\frac{d\theta}{dx}=\frac{0.332\rho U_0^2}{\sqrt{Re_x}} \tag{4.106}$$

$$c_f=\frac{\tau_w}{\frac{1}{2}\rho U_0^2}=\frac{0.664}{\sqrt{Re_x}}=\frac{0.664^2}{Re_\theta} \tag{4.107}$$

$$C_{Df}=\frac{D}{\frac{1}{2}\rho U_0^2 bL}=\frac{\int_0^L b\tau_w\,dx}{\frac{1}{2}\rho U_0^2 bL}=\frac{1.328}{\sqrt{Re_L}} \tag{4.108}$$

We can see that $\theta<\delta^*<\delta_{99}$. Note that the shape factor obtained from a turbulent profile, typically $U/U_0=(y/\delta)^{1/7}$, is $H=1.3$. It is a measure of the flatness of the profile and can be used as an indicator of boundary-layer transition.

4.5.3 Boundary layer with external pressure gradient, separation

4.5.4 Falkner–Skan Equations

The Falkner–Skan solution to the boundary-layer corner flow is another class of exact solutions of Blasius-kind. The external flow is $U_e=ax^n$ and the streamfunction and similarity variable are

$$\psi(x,y)=\sqrt{\nu U_e(x)x}f(\eta)=\sqrt{\nu a}x^{\frac{n+1}{2}}f(\eta) \tag{4.109}$$

$$\eta=y\sqrt{\frac{U_e}{\nu x}}=\sqrt{\frac{a}{\nu}}yx^{\frac{n-1}{2}} \tag{4.110}$$

The velocities are

$$u = \frac{\partial \psi}{\partial y} = a x^n f' \tag{4.111}$$

$$v = -\frac{\partial \psi}{\partial x} = -\frac{\sqrt{\nu a}}{2} x^{\frac{n-1}{2}} [(n+1)f + (n-1)\eta f']. \tag{4.112}$$

We check that when $n = 0$ (Blasius BL), $v = 1/2\sqrt{U_e \nu / x}(\eta f' - f)$ is what we are familiar with.

Let's do some preparation work,

$$\frac{\partial \eta}{\partial y} = \sqrt{\frac{a}{\nu}} x^{\frac{n-1}{2}} \tag{4.113}$$

$$\frac{\partial \eta}{\partial x} = \frac{n-1}{2} \frac{\eta}{x}, \tag{4.114}$$

and

$$\frac{\partial u}{\partial x} = n a x^{n-1} f' + a x^n f'' \frac{n-1}{2} \frac{\eta}{x} \tag{4.115}$$

$$\frac{\partial u}{\partial y} = a x^n f'' \sqrt{\frac{a}{\nu}} x^{\frac{n-1}{2}} \tag{4.116}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{a^2}{\nu} x^{2n-1} f''' \tag{4.117}$$

$$U_e \frac{dU_e}{dx} = n a^2 x^{2n-1}. \tag{4.118}$$

The x-momentum equation

$$u u_x + v u_y = U_e U_{e,x} + \nu u_{yy} \tag{4.119}$$

then converts to

$$n a^2 x^{2n-1} f' f' - \frac{n+1}{2} a^2 x^{2n-1} f f'' = n a^2 x^{2n-1} + a^2 x^{2n-1} f''' \tag{4.120}$$

which is the Falkner–Skan equation

$$f''' + \frac{n+1}{2} f f'' - n f'^2 + n = 0. \tag{4.121}$$

The BCs are

$$f(0) = 0;\ f'(0) = 0;\ f(+\infty) = 1. \tag{4.122}$$

When $n = 0$ it reduces to the Blasius equation $f''' + 1/2 f f'' = 0$.

4.5.5 von Kármán swirling disk flow

Also shooting method.

4.5.6 Laminar planar mixing layer

Governing equation (BL equations)

4.5.7 Laminar planar jet and wake

Momentum integrals. Momentum thickness. Lift coefficients.

4.5.8 Lift and drag

Consider forces on a 2D cylinder in a uniform freestream. The angle $0 < \theta < 2\pi$ counts clockwise from the front stagnation point. On each surface element at an angle of θ with the horizon, there is a wall shear stress τ_w tangential to the surface and a pressure p normal to it. The lift and drag are

$$D = F_x = \int p \cos\theta \, dA + \int \tau_w \sin\theta \, dA \tag{4.123}$$

$$L = F_y = -\int p \sin\theta \, dA + \int \tau_w \cos\theta \, dA \tag{4.124}$$

where the drag can be decomposed into two parts, the friction drag and the pressure (form) drag

$$D_f = \int \tau_w \sin\theta \, dA, \; D_p = \int p \cos\theta \, dA. \tag{4.125}$$

Non-dimensionally,

$$C_D = \frac{D}{\frac{1}{2}\rho U_0^2 A} = \frac{D_f + D_p}{\frac{1}{2}\rho U_0^2 A} = C_{Df} + C_{Dp} \tag{4.126}$$

where

$$C_{Dp} = \frac{D_p}{\frac{1}{2}\rho U_0^2 A} = \frac{\int p \cos\theta \, dA}{\frac{1}{2}\rho U_0^2 A} = \frac{\int \frac{p - p_0}{1/2 \rho U_0^2} \cos\theta \, dA}{A} = \frac{\int C_p \cos\theta \, dA}{A} \tag{4.127}$$

where $C_p = (p - p_0)/(1/2\rho U_0^2)$ is the pressure coefficient.

Chapter 5

Turbulent flows

5.1 Mean flow and fluctuations

5.1.1 Reynolds average

We denote time average as $\overline{(\cdot)}$, space or ensemble average as $\langle\cdot\rangle$, and sometimes use these notations interchangeably given that they are equivalent under the ergodicity assumption. The properties proved for one definition are expected to hold for another. Although Reynolds decomposition and RANS modelings are not an accurate way of computing turbulence, they consist the foundation of our understanding of turbulence.

Below we give briefly some properties of Reynolds averaging:

(i) (Definition) The time average of a physical variable A is

$$\overline{A} \triangleq \lim_{T\to\infty} \frac{1}{T}\int_0^T A\,\mathrm{d}t \tag{5.1}$$

In practice, the limit is often neglected and the average window is assumed to be long enough.

(ii) (Definition) The fluctuation of a physical variable A is

$$A' \triangleq A - \overline{A} \tag{5.2}$$

(iii) (Proposition) The average of fluctuation is zero.

$$\overline{A'} = \overline{A - \overline{A}} = \overline{A} - \overline{A} = 0 \tag{5.3}$$

5.1.2 Continuity and momentum

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{5.4}$$

$$\frac{Du_i}{Dt} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu\frac{\partial u_i}{\partial x_j}) \tag{5.5}$$

Taking the average of Eq. (5.4) we have

$$\frac{\partial u_i}{\partial x_i} = \frac{\partial \overline{u}_i}{\partial x_i} + \frac{\partial u_i'}{\partial x_i} = 0 \tag{5.6}$$

where

$$\frac{\partial \overline{u}_i}{\partial x_i} = \frac{\partial}{\partial x_i}(\frac{1}{T}\int_0^T u_i \,\mathrm{d}t) = \frac{1}{T}\int_0^T (\frac{\partial u_i}{\partial x_i})\,\mathrm{d}t = \frac{1}{T}\int_0^T 0\,\mathrm{d}t = 0 \tag{5.7}$$

Hence we have the continuity for fluctuating velocity

$$\frac{\partial u_i'}{\partial x_i} = 0 \tag{5.8}$$

Taking the average of Eq. (5.5) we have

$$\mathrm{LHS} = (\frac{1}{T}\int_0^T \mathrm{d}t) * [\frac{\partial}{\partial t}(\overline{u}_i + u_i') + (\overline{u}_j + u_j')\frac{\partial}{\partial x_j}(\bar{u}_i + u_i')] \tag{5.9}$$

$$= \frac{\partial \overline{u}_i}{\partial t} + \overline{\frac{\partial u_i'}{\partial t}} + \overline{\overline{u}_j \frac{\partial}{\partial x_j}\overline{u}_i} + \overline{\overline{u}_j \frac{\partial}{\partial x_j} u_i'} + \overline{u_j' \frac{\partial}{\partial x_j}\overline{u}_i} + \overline{u_j' \frac{\partial}{\partial x_j} u_i'} \tag{5.10}$$

$$= \frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j \frac{\partial}{\partial x_j}\overline{u}_i + \overline{u_j' \frac{\partial}{\partial x_j} u_i'} \tag{5.11}$$

$$= \frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j \frac{\partial}{\partial x_j}\overline{u}_i + \overline{\frac{\partial}{\partial x_j}(u_j' u_i')} - \overline{u_i' \frac{\partial u_j'}{\partial x_j}} = \frac{\partial}{\partial x_j}\overline{u_j' u_i'} \tag{5.12}$$

$$= \frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j \frac{\partial \overline{u}_i}{\partial x_j} + \frac{\partial}{\partial x_j}\overline{u_j' u_i'} \tag{5.13}$$

$$\mathrm{RHS} = (\frac{1}{T}\int_0^T \mathrm{d}t)[-\frac{1}{\rho}\frac{\partial}{\partial x_i}(\overline{p} + p') + \frac{\partial}{\partial x_j}[\nu \frac{\partial}{\partial x_j}(\overline{u}_i + u_i')] \tag{5.14}$$

$$= -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu \frac{\partial \overline{u}_i}{\partial x_j}) \tag{5.15}$$

Equaling both sides yields:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \tag{5.16}$$

$$\frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j \frac{\partial \overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu \frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i' u_j'}) \tag{5.17}$$

where the cross-correlation term having dimension of shear stress

$$\tau_{\mathrm{Rey}} = -\overline{u_i' u_j'} \tag{5.18}$$

is called the Reynolds stress term. It is a rank 2 tensor. It comes from the Reynolds averaging of the non-linear advection term on the LHS of Navier–Stokes, and it distinguishes turbulent flows from laminar ones. It represents the momentum transport due to turbulent motions, in anology to the molecular diffusion.

5.1.3 Transport equation of the fluctuating velocity

Denote the material derivative based on the mean flow advection as

$$\frac{\bar{D}}{Dt} = \frac{\partial}{\partial t} + \overline{u}_k \frac{\partial}{\partial x_k} \tag{5.19}$$

and subtract the Reynolds equation from N-S equation

$$\frac{\bar{D}u_i'}{Dt} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu\frac{\partial u_i'}{\partial x_j} + \overline{u_i'u_j'} - u_i'u_j') - u_j'\frac{\partial \overline{u}_i}{\partial x_j} \tag{5.20}$$

The last term shows the mean-flow stretching of the fluctuation, which is a generation mechanism be shown later related to the shear production of turbulent kinetic energy. In the latter section of 8 we will see the same form of perturbation equations will be used for hydrodynamic stability analysis.

5.1.4 Mean-flow and turbulent kinetic energy

The total kinetic energy of the flow can be devided into the mean kinetic energy (MKE) and the turbulent kinetic energy (TKE)

$$K_{\text{tot}} = \frac{1}{2}\overline{u_i u_i} \tag{5.21}$$

$$= \overline{\frac{1}{2}(\overline{u}_i + u_i')(\overline{u}_i + u_i')} \tag{5.22}$$

$$= \overline{\frac{1}{2}\overline{u}_i\overline{u}_i + \overline{u}_i u_i' + \frac{1}{2}u_i'u_i'} \tag{5.23}$$

$$= \frac{1}{2}\overline{u}_i\overline{u}_i + \frac{1}{2}\overline{u_i'u_i'} \tag{5.24}$$

$$= K + k \tag{5.25}$$

We will show how these two parts are related dynamically.

5.1.5 MKE equation

Multiply the Reynolds equation (5.17) by $\overline{u}_i$ we have

$$\text{LHS} = \overline{u}_i\frac{\bar{D}\overline{u}_i}{Dt} = \frac{\bar{D}K}{Dt} \tag{5.26}$$

$$\overline{u}_i(-\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i}) = -\frac{1}{\rho}\frac{\partial \overline{p}\,\overline{u}_i}{\partial x_i} + \frac{1}{\rho}\frac{\partial \overline{u}_i}{\partial x_i} \tag{5.27}$$

$$= -\frac{1}{\rho}\frac{\partial \overline{p}\,\overline{u}_i}{\partial x_i} \tag{5.28}$$

$$= -\frac{1}{\rho}\frac{\partial \overline{p}\,\overline{u}_j}{\partial x_j} \tag{5.29}$$

$$\overline{u}_i\frac{\partial}{\partial x_j}(\nu\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i'u_j'}) = \frac{\partial}{\partial x_j}[\overline{u}_i(\nu\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i'u_j'})] - \frac{\partial \overline{u}_i}{\partial x_j}(\nu\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i'u_j'}) \tag{5.30}$$

$$= -\nu\frac{\partial\overline{u}_i}{\partial x_j}\frac{\partial\overline{u}_i}{\partial x_j} + \frac{\partial\overline{u}_i}{\partial x_j}\overline{u_i'u_j'} + \frac{\partial}{\partial x_j}(\nu\frac{\partial K}{\partial x_j}) - \frac{\partial\overline{u}_i\,\overline{u_i'u_j'}}{\partial x_j} \tag{5.31}$$

Equaling both sides we have

$$\frac{\bar{D}K}{Dt} = \frac{\partial}{\partial x_j}(\underbrace{-\frac{1}{\rho}\overline{p}\,\overline{u}_j}_{\substack{\text{pressure}\\\text{distortion}}} \underbrace{+\nu\frac{\partial K}{\partial x_j}}_{\substack{\text{molecular}\\\text{diffusion}}} \underbrace{-\overline{u}_i\,\overline{u_i'u_j'}}_{\substack{\text{turbulent}\\\text{diffusion}}}) \underbrace{-\frac{1}{2}P_{kk}}_{\substack{\text{production}\\\text{of TKE}}} \underbrace{-\nu\left(\frac{\partial\overline{u}_i}{\partial x_j}\right)^2}_{\text{dissipation}} \tag{5.32}$$

where the term

$$P_{kk} = -2\overline{u_i'u_j'}\frac{\partial\overline{u}_i}{\partial x_j} \tag{5.33}$$

is the production term of the turbulent kinetic energy, and, on the other hand, is the sink in MKE.

5.1.6 TKE equation

Similarly, multiply (5.20) by u_i' and then take the average

$$\text{LHS} = \overline{u_i'(\frac{\partial u_i'}{\partial t} + \overline{u_k}\frac{\partial u_i'}{\partial x_k})} \tag{5.34}$$

$$= \frac{\partial\frac{1}{2}\overline{u_i'u_i'}}{\partial t} + \overline{u_k}\frac{\partial\frac{1}{2}\overline{u_i'u_i'}}{\partial x_k} \tag{5.35}$$

$$= \frac{\bar{D}k}{Dt} \tag{5.36}$$

$$\overline{u_i'(-\frac{1}{\rho}\frac{\partial p'}{\partial x_i})} = -\frac{1}{\rho}\overline{\frac{\partial p'u_i'}{\partial x_i'}} \tag{5.37}$$

$$= -\frac{1}{\rho}\frac{\partial\overline{p'u_k'}}{\partial x_k'} \tag{5.38}$$

$$\overline{u_i'(\frac{\partial}{\partial x_k}\nu\frac{\partial u_i'}{\partial x_k})} = \overline{\frac{\partial}{\partial x_k}(\nu u_i'\frac{\partial u_i'}{\partial x_k})} - \nu\overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_i'}{\partial x_k}} \tag{5.39}$$

$$= \overline{\frac{\partial}{\partial x_k}(\nu\frac{\partial\frac{1}{2}u_i'u_i'}{\partial x_k})} - \nu\overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_i'}{\partial x_k}} \tag{5.40}$$

$$= \frac{\partial}{\partial x_k}(\nu\frac{\partial k}{\partial x_k}) - \nu\overline{(\frac{\partial u_i'}{\partial x_k})^2} \tag{5.41}$$

$$\overline{u_i'(\frac{\partial}{\partial x_k}\overline{u_i'u_k'})} = 0 \tag{5.42}$$

$$\overline{-u_i'(\frac{\partial}{\partial x_k}u_i'u_k')} = -\overline{\frac{1}{2}\frac{\partial u_i'u_i'u_k'}{\partial x_k}} \tag{5.43}$$

$$= -\frac{1}{2}\frac{\partial\overline{u_i'u_i'u_k'}}{\partial x_k} \tag{5.44}$$

$$\overline{u_i'(-u_k'\frac{\partial\overline{u}_i}{\partial x_k})} = -\overline{u_i'u_k'}\frac{\partial\overline{u}_i}{\partial x_k} \tag{5.45}$$

Equaling both sides we have

$$\frac{\bar{D}k}{Dt} = \frac{\partial}{\partial x_k}\Big(\underbrace{\nu\frac{\partial k}{\partial x_k}}_{\substack{\text{molecular}\\ \text{diffusion}}} \underbrace{-\frac{1}{2}\overline{u_i' u_i' u_k'}}_{\substack{\text{turbulent}\\ \text{diffusion}}} \underbrace{-\frac{1}{\rho}\overline{p' u_k'}}_{\substack{\text{pressure}\\ \text{distortion}}}\Big) + \underbrace{\frac{1}{2}P_{kk}}_{\substack{\text{production}\\ \text{of TKE}}} \underbrace{-\nu\overline{\left(\frac{\partial u_i'}{\partial x_k}\right)\left(\frac{\partial u_i'}{\partial x_k}\right)}}_{\text{(pseudo-) dissipation}} \tag{5.46}$$

Comments:

(1) The turbulent kinetic energy production term

$$P = \frac{1}{2}P_{kk} = -\overline{u_i' u_j'}\frac{\partial \overline{u}_i}{\partial x_j} \tag{5.47}$$

can be expressed in tensor notation as

$$P = \boldsymbol{\tau}_{\text{Rey}} : \nabla\overline{\boldsymbol{u}} = \boldsymbol{\tau}_{\text{Rey}} : \bar{\boldsymbol{S}} \tag{5.48}$$

where the inner product represents the projection of the velocity fluctuation correlation on the mean shear/strain rate. The convention of $P_{kk}/2$ is to be consistent with the notations in (5.74).

(2) The (pseudo-) dissipation term

$$\tilde{\varepsilon} = \nu\overline{\left(\frac{\partial u_i'}{\partial x_j}\right)\left(\frac{\partial u_i'}{\partial x_j}\right)} = \nu\overline{S_{ij}' S_{ij}'} + \nu\overline{\Omega_{ij}'\Omega_{ij}'} \tag{5.49}$$

is always positive, representing the dissipation mechanism of turbulence kinetic energy. We can also see that (perturbation) enstrophy is directly linked to the dissipation rate of TKE/total KE. We note that the relations

$$\boldsymbol{S} : \boldsymbol{\Omega} = 0 \tag{5.50}$$

and

$$\nabla\boldsymbol{u} : \nabla\boldsymbol{u} = \boldsymbol{S} : \boldsymbol{S} + \boldsymbol{\Omega} : \boldsymbol{\Omega} \tag{5.51}$$

also carries for the perturbation quantities. Also note $\|\boldsymbol{\Omega}\|^2 = \|\boldsymbol{\omega}\|^2/2$ – the perturbation enstrophy is largely related to turbulent dissipation.

(3) The pressure redistribution term can be split into pressure diffusion (d) and pressure–strain correlation (s). Consider the equation for TKE components, the pressure redistribution term is

$$-\frac{1}{\rho}\overline{u'\frac{\partial p'}{\partial x}} = -\frac{1}{\rho}\frac{\partial}{\partial x}\overline{p' u'} + \frac{1}{\rho}\overline{p'\frac{\partial u'}{\partial x}} \tag{5.52}$$

$$\Phi_x = \Phi_{x,d} + \Phi_{x,s} \tag{5.53}$$

The spatial integration of $\Phi_{x,d}$ is zero, indicating that it is a spatial transport, whereas the sum of Φ_s over all components is zero (due to continuity), indicating that pressure–strain has not net contribution to the TKE but redistributes energy among components (positive indicating energy input).

(4) Note that eqn. (5.46) can alternatively be written as

$$\frac{\bar{D}k}{Dt} = \frac{\partial}{\partial x_k}(\underbrace{2\nu\overline{u_j' s_{kj}'}}_{\text{molecular diffusion}} - \underbrace{\frac{1}{2}\overline{u_i' u_i' u_k'}}_{\text{turbulent diffusion}} - \underbrace{\frac{1}{\rho}\overline{p' u_k'}}_{\text{pressure distortion}}) + \underbrace{\frac{1}{2}P_{kk}}_{\text{production of TKE}} - \underbrace{2\nu\overline{s_{ij}' s_{ij}'}}_{\text{dissipation}} \tag{5.54}$$

with the viscous term in the original equation being $\tau_{ij,j} = 2\nu s_{ij,j}$ instead of a Laplacian, and the relation between dissipation and pseudo-dissipation being

$$\varepsilon = \tilde{\varepsilon} + \nu\frac{\partial^2 \overline{u_i' u_j'}}{\partial x_i \partial x_j}, \tag{5.55}$$

which is small at vanishing ν or can be absorbed into the transport term otherwise.

5.1.7 Reynolds stress transport equation

The velocity fluctuation transport equation is

$$\frac{\bar{D}u_i'}{Dt} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu\frac{\partial u_i'}{\partial x_j} + \overline{u_i' u_j'} - u_i' u_j') - u_j'\frac{\partial \overline{u}_i}{\partial x_j} \tag{5.56}$$

Or if we exchange the two subscripts we obtain:

$$\frac{\bar{D}u_j'}{Dt} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_j} + \frac{\partial}{\partial x_i}(\nu\frac{\partial u_j'}{\partial x_i} + \overline{u_i' u_j'} - u_i' u_j') - u_i'\frac{\partial \overline{u}_j}{\partial x_i} \tag{5.57}$$

$u_j' \times$ (5.56) $+ u_i' \times$ (5.57) and take the time average:

$$\text{LHS} = \frac{\bar{D}\overline{u_i' u_j'}}{Dt} \tag{5.58}$$

$$\text{RHS}_1 = -\frac{1}{\rho}[-2\overline{p' s_{ij}} + \frac{\partial}{\partial x_i}(\overline{p' u_j'}) + \frac{\partial}{\partial x_j}(\overline{p' u_i'})] \tag{5.59}$$

$$\text{RHS}_2 = \overline{u_j'\frac{\partial}{\partial x_k}(\nu\frac{\partial u_i'}{\partial x_k}) + u_i'\frac{\partial}{\partial x_k}(\nu\frac{\partial u_j'}{\partial x_k})} \tag{5.60}$$

$$= \frac{\partial}{\partial x_k}(\nu\frac{\partial \overline{u_i' u_j'}}{\partial x_k}) - 2\nu\overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_j'}{\partial x_k}} \tag{5.61}$$

$$\text{RHS}_3 = \overline{u_j'\frac{\partial}{\partial x_k}\overline{u_i' u_k'} + u_i'\frac{\partial}{\partial x_k}\overline{u_j' u_k'}} \tag{5.62}$$

$$= 0 \tag{5.63}$$

$$\text{RHS}_4 = -\overline{u_j'\frac{\partial}{\partial x_k}(u_i' u_k') + u_i'\frac{\partial}{\partial x_k}(u_j' u_k')} \tag{5.64}$$

$$= -\overline{u_j' u_k'\frac{\partial}{\partial x_k}(u_i') + u_i' u_k'\frac{\partial}{\partial x_k}(u_j') + u_i' u_j'\frac{\partial}{\partial x_k}(u_k')} \tag{5.65}$$

$$(Continuity, \frac{\partial u_k'}{\partial x_k} = 0, is\, used\, twice\, here.) \tag{5.66}$$

$$= -\frac{\partial}{\partial x_k}\overline{u_i' u_j' u_k'} \tag{5.67}$$

$$\text{RHS}_5 = \overline{-u_k' u_j' \frac{\partial \overline{u}_i}{\partial x_k} - u_k' u_i' \frac{\partial \overline{u}_j}{\partial x_k}} \tag{5.68}$$

$$= -\overline{u_k' u_j'} \frac{\partial \overline{u}_i}{\partial x_k} - \overline{u_k' u_i'} \frac{\partial \overline{u}_j}{\partial x_k} \tag{5.69}$$

$$\tag{5.70}$$

By equalizing both sides we obtain

$$\frac{\bar{D}\overline{u_i' u_j'}}{Dt} = \frac{2}{\rho}\overline{p' s_{ij}} - \frac{1}{\rho}\frac{\partial}{\partial x_k}(\overline{p' u_j'})\delta_{ik} - \frac{1}{\rho}\frac{\partial}{\partial x_k}(\overline{p' u_i'})\delta_{jk} + \frac{\partial}{\partial x_k}(\nu \frac{\partial \overline{u_i' u_j'}}{\partial x_k}) \tag{5.71}$$

$$- 2\nu \overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_j'}{\partial x_k}} - \frac{\partial}{\partial x_k}\overline{u_i' u_j' u_k'} - \overline{u_k' u_j'}\frac{\partial \overline{u}_i}{\partial x_k} - \overline{u_k' u_i'}\frac{\partial \overline{u}_j}{\partial x_k} \tag{5.72}$$

$$= \frac{\partial}{\partial x_k}(\nu \frac{\partial \overline{u_i' u_j'}}{\partial x_k} - \overline{u_i' u_j' u_k'} - \frac{1}{\rho}\overline{p' u_i'}\delta_{jk} - \frac{1}{\rho}\overline{p' u_j'}\delta_{ik}) \tag{5.73}$$

$$- (\overline{u_k' u_j'}\frac{\partial \overline{u}_i}{\partial x_k} + \overline{u_k' u_i'}\frac{\partial \overline{u}_j}{\partial x_k}) + \frac{2}{\rho}\overline{p' s_{ij}} - 2\nu \overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_j'}{\partial x_k}} \tag{5.74}$$

Equaling both sides we have

$$\frac{\bar{D}\overline{u_i' u_j'}}{Dt} = d_{ij} + P_{ij} + \Phi_{ij} - \varepsilon_{ij} \tag{5.75}$$

where

$$d_{ij} = \frac{\partial}{\partial x_k}\left(\nu \frac{\partial \overline{u_i' u_j'}}{\partial x_k} - \overline{u_i' u_j' u_k'} - \frac{1}{\rho}\overline{p' u_i'}\delta_{jk} - \frac{1}{\rho}\overline{p' u_j'}\delta_{ik}\right) \tag{5.76}$$

$$P_{ij} = -\overline{u_k' u_j'}\frac{\partial \overline{u}_i}{\partial x_k} - \overline{u_k' u_i'}\frac{\partial \overline{u}_j}{\partial x_k} \tag{5.77}$$

$$\Phi_{ij} = \frac{2}{\rho}\overline{p' s_{ij}'} = \frac{1}{\rho}\overline{p'\left(\frac{\partial x_i'}{\partial x_j} + \frac{\partial x_j'}{\partial x_i}\right)} \tag{5.78}$$

$$\varepsilon_{ij} = 2\nu \overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_j'}{\partial x_k}} \tag{5.79}$$

$$s_{ij} = \frac{1}{2}\left(\frac{\partial u_i'}{\partial x_j} + \frac{\partial u_j'}{\partial x_i}\right) \tag{5.80}$$

Comments:

(1) The left hand side term $\frac{\bar{D}\overline{u_i' u_j'}}{Dt}$ is the rate of change of the Reynolds stress along the particle line.

(2) The term d_{ij} is the diffusion term in the equation, appearing in the form of gradient. It includes viscous term, Reynolds stress term and pressure-velocity fluctuation coupling term. The diffusion is resulted by the spatial non-uniformity of these properties.

(3) The term P_{ij} is the generation term of Reynolds stress, showed in the form of the product of Reynolds stress and the mean flow strain rate.

(4) The term Φ_{ij} is the redistribution term. We note that the contraction of Reynolds stress transport equation is the transport equation for turbulence kinetic energy. And the contraction of Φ_{ij} is $\Phi_{ii} = \frac{2}{\rho}\overline{p's'_{ii}} = 0$ as continuity holds. So the term contributes nothing to the growth of turbulent kinetic energy. It just takes the kinetic energy from one component of fluid motion to another component. [analyse how it transfers energy from a stronger component to a weaker one and even out the components. also how Reynolds stress is destructed by the pressure–strain correlation.]

(5) The term ε_{ij}, whose contraction is positive forever, representing the dissipation mechanism of kinetic energy.

5.1.8 TKE equations in cylindrical coordinates

The analysis of the TKE budget equations written in cylindrical coordinates is necessarily for the understanding of axisymmetric turbulent flows. Although the scalar terms P, ε, etc will be the same when evaluated in both cylindrical and Cartesian coordinates, the component-wise contributions to them and the broken-down TKE component equations can only be seen clearly when cast in corresponding coordinates. It will be applicable in axisymmetric free-shear flows such as jets and wakes, and axisymmetric wall-bounded flows such as pipes and boundary layers over bodies of revolution.

Here we refer to D.1.2 for the N–S equations, where (u, v, w) correspond to (r, θ, z).

$$\frac{\partial(ru)}{\partial r} + \frac{\partial v}{\partial \theta} + \frac{\partial(rw)}{\partial z} = 0 \tag{5.81}$$

$$\frac{\partial u}{\partial t} + \frac{1}{r}\frac{\partial(ru^2)}{\partial r} + \frac{1}{r}\frac{\partial(uv)}{\partial \theta} + \frac{\partial(uw)}{\partial z} - \frac{v^2}{r} = \text{RHSU} \tag{5.82}$$

$$\frac{\partial v}{\partial t} + \frac{1}{r}\frac{\partial(ruv)}{\partial r} + \frac{1}{r}\frac{\partial v^2}{\partial \theta} + \frac{\partial(vw)}{\partial z} + \frac{uv}{r} = \text{RHSV} \tag{5.83}$$

$$\frac{\partial w}{\partial t} + \frac{1}{r}\frac{\partial(ruw)}{\partial r} + \frac{1}{r}\frac{\partial(vw)}{\partial \theta} + \frac{\partial(w^2)}{\partial z} = \text{RHSW} \tag{5.84}$$

The continuity equation average to be

$$\frac{1}{r}\frac{\partial(r\bar{u})}{\partial r} + \frac{1}{r}\frac{\partial \bar{v}}{\partial \theta} + \frac{\partial \bar{w}}{\partial z} = 0 \tag{5.85}$$

The mean momentum equation, written in tensor form,

$$\partial_t \bar{u}_i + \partial_j(\bar{u}_i\bar{u}_j) = -\frac{1}{\rho}\partial_i \bar{p} + \nu\nabla^2\bar{u}_i - \partial_j\overline{u'_i u'_j}, \tag{5.86}$$

where the convective terms are

$$\partial_j\overline{u_i u_j} = \partial_j(\bar{u}_i\bar{u}_j) + \partial_j\overline{u'_i u'_j}, \tag{5.87}$$

can be expanded in cylindrical coordinates with the following taken care of:
(1) the divergence operator in the cylindrical coordinate is

$$\nabla \cdot \boldsymbol{a} = \frac{1}{r}\frac{\partial(ra_r)}{\partial r} + \frac{1}{r}\frac{\partial a_\theta}{\partial \theta} + \frac{\partial a_z}{\partial z}. \tag{5.88}$$

(2) The geometric terms average to be

$$-\frac{\overline{v^2}}{r} = -\frac{\overline{v}^2}{r} - \frac{\overline{v'^2}}{r} = -\frac{\overline{v}^2}{r} - \frac{R_{\theta\theta}}{r} \tag{5.89}$$

$$\frac{\overline{uv}}{r} = \frac{\bar{u}\bar{v}}{r} + \frac{\overline{u'v'}}{r} = \frac{\bar{u}\bar{v}}{r} + \frac{R_{r\theta}}{r} \tag{5.90}$$

(3) The convective operator

$$\boldsymbol{u}\cdot\nabla = u\frac{\partial}{\partial r} + \frac{v}{r}\frac{\partial}{\partial\theta} + w\frac{\partial}{\partial z} \tag{5.91}$$

(4) The scalar Laplacian

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2} \tag{5.92}$$

Here we pay special attention to the convective (including the geometric) terms. The convective terms in the r-momentum equation average to

$$\frac{1}{r}\frac{\partial(r\overline{u^2})}{\partial r} + \frac{1}{r}\frac{\partial\overline{uv}}{\partial\theta} + \frac{\partial\overline{uw}}{\partial z} - \frac{\overline{v^2}}{r} = \frac{1}{r}\frac{\partial(r\bar{u}^2)}{\partial r} + \frac{1}{r}\frac{\partial\bar{u}\bar{v}}{\partial\theta} + \frac{\partial\bar{u}\bar{w}}{\partial z} - \frac{\bar{v}^2}{r} \tag{5.93}$$

$$+ \frac{1}{r}\frac{\partial(r\overline{u'u'})}{\partial r} + \frac{1}{r}\frac{\partial\overline{u'v'}}{\partial\theta} + \frac{\partial\overline{u'w'}}{\partial z} - \frac{\overline{v'v'}}{r} \tag{5.94}$$

in the θ-momentum equation

$$\frac{1}{r}\frac{\partial(r\overline{uv})}{\partial r} + \frac{1}{r}\frac{\partial\overline{v^2}}{\partial\theta} + \frac{\partial\overline{vw}}{\partial z} + \frac{\overline{uv}}{r} = \frac{1}{r}\frac{\partial(r\bar{u}\bar{v})}{\partial r} + \frac{1}{r}\frac{\partial\bar{v}^2}{\partial\theta} + \frac{\partial\bar{v}\bar{w}}{\partial z} + \frac{\bar{u}\bar{v}}{r} \tag{5.95}$$

$$+ \frac{1}{r}\frac{\partial(r\overline{u'v'})}{\partial r} + \frac{1}{r}\frac{\partial\overline{v'v'}}{\partial\theta} + \frac{\partial\overline{v'w'}}{\partial z} + \frac{\overline{u'v'}}{r} \tag{5.96}$$

and in the z-momentum equation

$$\frac{1}{r}\frac{\partial(r\overline{uw})}{\partial r} + \frac{1}{r}\frac{\partial\overline{vw}}{\partial\theta} + \frac{\partial\overline{w^2}}{\partial z} = \frac{1}{r}\frac{\partial(r\bar{u}\bar{w})}{\partial r} + \frac{1}{r}\frac{\partial\bar{v}\bar{w}}{\partial\theta} + \frac{\partial\bar{w}^2}{\partial z} \tag{5.97}$$

$$+ \frac{1}{r}\frac{\partial(r\overline{u'w'})}{\partial r} + \frac{1}{r}\frac{\partial\overline{v'w'}}{\partial\theta} + \frac{\partial\overline{w'w'}}{\partial z} \tag{5.98}$$

Taking continuity into account,

$$\text{LHSU} = \frac{\partial\bar{u}}{\partial t} + \bar{u}\frac{\partial\bar{u}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial\bar{u}}{\partial\theta} + \bar{w}\frac{\partial\bar{u}}{\partial z} - \frac{\bar{v}^2}{r} + \frac{\partial\overline{u'u'}}{\partial r} + \frac{1}{r}\frac{\partial\overline{u'v'}}{\partial\theta} + \frac{\partial\overline{u'w'}}{\partial z} + \frac{\overline{u'u'}}{r} - \frac{\overline{v'v'}}{r} \tag{5.99}$$

$$= \frac{\partial\bar{u}}{\partial t} + \bar{u}\frac{\partial\bar{u}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial\bar{u}}{\partial\theta} + \bar{w}\frac{\partial\bar{u}}{\partial z} - \frac{\bar{v}^2}{r} + \left(\frac{\partial\overline{u'u'}}{\partial r} + \frac{1}{r}\frac{\partial\overline{u'v'}}{\partial\theta} + \frac{\partial\overline{u'w'}}{\partial z} + \frac{R_{rr} - R_{\theta\theta}}{r}\right) \tag{5.100}$$

$$\text{LHSV} = \frac{\partial\bar{v}}{\partial t} + \bar{u}\frac{\partial\bar{v}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial\bar{v}}{\partial\theta} + \bar{w}\frac{\partial\bar{v}}{\partial z} + \frac{\bar{u}\bar{v}}{r} + \frac{\partial\overline{u'v'}}{\partial r} + \frac{1}{r}\frac{\partial\overline{v'v'}}{\partial\theta} + \frac{\partial\overline{v'w'}}{\partial z} + \frac{2\overline{u'v'}}{r} \tag{5.101}$$

$$= \frac{\partial\bar{v}}{\partial t} + \bar{u}\frac{\partial\bar{v}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial\bar{v}}{\partial\theta} + \bar{w}\frac{\partial\bar{v}}{\partial z} + \frac{\bar{u}\bar{v}}{r} + \left(\frac{\partial\overline{u'v'}}{\partial r} + \frac{1}{r}\frac{\partial\overline{v'v'}}{\partial\theta} + \frac{\partial\overline{v'w'}}{\partial z} + \frac{2R_{r\theta}}{r}\right) \tag{5.102}$$

$$\text{LHSW} = \frac{\partial\bar{w}}{\partial t} + \bar{u}\frac{\partial\bar{w}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial\bar{w}}{\partial\theta} + \bar{w}\frac{\partial\bar{w}}{\partial z} + \frac{\partial\overline{u'w'}}{\partial r} + \frac{1}{r}\frac{\partial\overline{v'w'}}{\partial\theta} + \frac{\partial\overline{w'w'}}{\partial z} + \frac{\overline{u'w'}}{r} \tag{5.103}$$

$$= \frac{\partial \bar{w}}{\partial t} + \bar{u}\frac{\partial \bar{w}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \bar{w}}{\partial \theta} + \bar{w}\frac{\partial \bar{w}}{\partial z} + \left(\frac{\partial \overline{u'w'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{v'w'}}{\partial \theta} + \frac{\partial \overline{w'w'}}{\partial z} + \frac{R_{rz}}{r}\right) \tag{5.104}$$

where the terms in parentheses are to be moved to the RHS as Reynolds stress divergence.

Taking the Laplacian and diffusion terms from D.1.2 (those are linear), the RHS are

$$\text{RHSU} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial r} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{u}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{u}}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial \bar{v}}{\partial \theta} - \frac{\bar{u}}{r^2} + \frac{\partial^2 \bar{u}}{\partial z^2}\right) \tag{5.105}$$

$$\text{RHSV} = -\frac{1}{\rho r}\frac{\partial \bar{p}}{\partial \theta} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{v}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{v}}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial \bar{u}}{\partial \theta} - \frac{\bar{v}}{r^2} + \frac{\partial^2 \bar{v}}{\partial z^2}\right) \tag{5.106}$$

$$\text{RHSW} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial z} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{w}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{w}}{\partial \theta^2} + \frac{\partial^2 \bar{w}}{\partial z^2}\right) \tag{5.107}$$

In all, the mean momentum equations in the cylindrical coordinates are

$$\begin{aligned}
\frac{\partial \bar{u}}{\partial t} + \bar{u}\frac{\partial \bar{u}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \bar{u}}{\partial \theta} + \bar{w}\frac{\partial \bar{u}}{\partial z} - \frac{\bar{v}^2}{r} &= -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial r} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{u}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{u}}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial \bar{v}}{\partial \theta} - \frac{\bar{u}}{r^2} + \frac{\partial^2 \bar{u}}{\partial z^2}\right) \\
&\quad - \left(\frac{\partial \overline{u'u'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{u'v'}}{\partial \theta} + \frac{\partial \overline{u'w'}}{\partial z} + \frac{R_{rr} - R_{\theta\theta}}{r}\right) \\
\frac{\partial \bar{v}}{\partial t} + \bar{u}\frac{\partial \bar{v}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \bar{v}}{\partial \theta} + \bar{w}\frac{\partial \bar{v}}{\partial z} + \frac{\bar{u}\bar{v}}{r} &= -\frac{1}{\rho r}\frac{\partial \bar{p}}{\partial \theta} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{v}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{v}}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial \bar{u}}{\partial \theta} - \frac{\bar{v}}{r^2} + \frac{\partial^2 \bar{v}}{\partial z^2}\right) \\
&\quad - \left(\frac{\partial \overline{u'v'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{v'v'}}{\partial \theta} + \frac{\partial \overline{v'w'}}{\partial z} + \frac{2R_{r\theta}}{r}\right) \\
\frac{\partial \bar{w}}{\partial t} + \bar{u}\frac{\partial \bar{w}}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \bar{w}}{\partial \theta} + \bar{w}\frac{\partial \bar{w}}{\partial z} &= -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial z} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \bar{w}}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \bar{w}}{\partial \theta^2} + \frac{\partial^2 \bar{w}}{\partial z^2}\right) \\
&\quad - \left(\frac{\partial \overline{u'w'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{v'w'}}{\partial \theta} + \frac{\partial \overline{w'w'}}{\partial z} + \frac{R_{rz}}{r}\right)
\end{aligned}$$

Then, the fluctuation equations can be obtained by subtracting the full from the mean:

$$\frac{1}{r}\frac{\partial (ru')}{\partial r} + \frac{1}{r}\frac{\partial v'}{\partial \theta} + \frac{\partial w'}{\partial z} = 0 \tag{5.108}$$

We again only pay attention to the convective terms

$$\begin{aligned}
\text{LHSU}'_c &= \text{LHSU}_c - \overline{\text{LHSU}}_c \\
&= \left(\bar{u}\frac{\partial u'}{\partial r} + u'\frac{\partial \bar{u}}{\partial r} + u'\frac{\partial u'}{\partial r}\right) + \left(\frac{\bar{v}}{r}\frac{\partial u'}{\partial \theta} + \frac{v'}{r}\frac{\partial \bar{u}}{\partial \theta} + \frac{v'}{r}\frac{\partial u'}{\partial \theta}\right) + \left(\bar{w}\frac{\partial u'}{\partial z} + w'\frac{\partial \bar{u}}{\partial z} + w'\frac{\partial u'}{\partial z}\right) \\
&\quad - \frac{2\bar{v}v' + v'^2}{r} - \left(\frac{\partial \overline{u'u'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{u'v'}}{\partial \theta} + \frac{\partial \overline{u'w'}}{\partial z} + \frac{\overline{u'u'} - \overline{v'v'}}{r}\right) \\
\text{LHSV}'_c &= \left(\bar{u}\frac{\partial v'}{\partial r} + u'\frac{\partial \bar{v}}{\partial r} + u'\frac{\partial v'}{\partial r}\right) + \left(\frac{\bar{v}}{r}\frac{\partial v'}{\partial \theta} + \frac{v'}{r}\frac{\partial \bar{v}}{\partial \theta} + \frac{v'}{r}\frac{\partial v'}{\partial \theta}\right) + \left(\bar{w}\frac{\partial v'}{\partial z} + w'\frac{\partial \bar{v}}{\partial z} + w'\frac{\partial v'}{\partial z}\right) \\
&\quad + \frac{\bar{u}v' + \bar{v}u' + u'v'}{r} - \left(\frac{\partial \overline{u'v'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{v'v'}}{\partial \theta} + \frac{\partial \overline{v'w'}}{\partial z} + \frac{2\overline{u'v'}}{r}\right)
\end{aligned}$$

$$\text{LHSW}'_c = \left(\bar{u}\frac{\partial w'}{\partial r} + u'\frac{\partial \bar{w}}{\partial r} + u'\frac{\partial w'}{\partial r}\right) + \left(\frac{\bar{v}}{r}\frac{\partial w'}{\partial \theta} + \frac{v'}{r}\frac{\partial \bar{w}}{\partial \theta} + \frac{v'}{r}\frac{\partial w'}{\partial \theta}\right) + \left(\bar{w}\frac{\partial w'}{\partial z} + w'\frac{\partial \bar{w}}{\partial z} + w'\frac{\partial w'}{\partial z}\right) - \left(\frac{\partial \overline{u'w'}}{\partial r} + \frac{1}{r}\frac{\partial \overline{v'w'}}{\partial \theta} + \frac{\partial \overline{w'w'}}{\partial z} + \frac{\overline{u'w'}}{r}\right)$$

In each LHS group, the terms are arranged in the order of (mean advection + mean production [linear] + nonlinear convection) + curvature effect [linear and nonlinear] + Reynolds stresses.

Define the TKE as

$$k = \frac{1}{2}\overline{u'_i u'_i} = \frac{1}{2}(\overline{u'u'} + \overline{v'v'} + \overline{w'w'}), \tag{5.109}$$

its equation can be derived by multiplying LHS$'$ with respective fluctuations, taking the average, and summing over all three equations. Note that the Reynolds stress terms multiplied by fluctuations and averaged will have zero contribution.

$$\overline{u'\text{LHSU}'_c} = \left(\bar{u}\frac{\partial \overline{u'^2}/2}{\partial r} + \overline{u'u'}\frac{\partial \bar{u}}{\partial r} + \overline{u'u'\frac{\partial u'}{\partial r}}\right) + \left(\frac{\bar{v}}{r}\frac{\partial \overline{u'^2}/2}{\partial \theta} + \frac{\overline{u'v'}}{r}\frac{\partial \bar{u}}{\partial \theta} + \frac{\overline{u'v'}}{r}\frac{\partial u'}{\partial \theta}\right) \tag{5.110}$$

$$+ \left(\bar{w}\frac{\partial \overline{u'^2}/2}{\partial z} + \overline{u'w'}\frac{\partial \bar{u}}{\partial z} + \overline{u'w'\frac{\partial u'}{\partial z}}\right) - \frac{2\bar{v}\overline{u'v'} + \overline{v'^2u'}}{r} \tag{5.111}$$

$$= \left(\bar{u}\frac{\partial \overline{u'^2}/2}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \overline{u'^2}/2}{\partial \theta} + \bar{w}\frac{\partial \overline{u'^2}/2}{\partial z}\right) + \left(\overline{u'u'}\frac{\partial \bar{u}}{\partial r} + \frac{\overline{u'v'}}{r}\frac{\partial \bar{u}}{\partial \theta} + \overline{u'w'}\frac{\partial \bar{u}}{\partial z}\right) \tag{5.112}$$

$$+ \frac{1}{2}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{u'^3}) + \frac{1}{r}\frac{\partial}{\partial \theta}\overline{u'^2v'} + \frac{\partial}{\partial z}\overline{u'^2w'}\right) - \frac{2\bar{v}\overline{u'v'} + \overline{v'^2u'}}{r} \tag{5.113}$$

noting that $\frac{1}{2}u'u'\times$ fluctuation continuity has been added to obtain the last equation. Following a similar routine, we have

$$\overline{v'\text{LHSV}'_c} = \left(\bar{u}\frac{\partial \overline{v'^2}/2}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \overline{v'^2}/2}{\partial \theta} + \bar{w}\frac{\partial \overline{v'^2}/2}{\partial z}\right) + \left(\overline{u'v'}\frac{\partial \bar{v}}{\partial r} + \frac{\overline{v'v'}}{r}\frac{\partial \bar{v}}{\partial \theta} + \overline{v'w'}\frac{\partial \bar{v}}{\partial z}\right) \tag{5.114}$$

$$+ \frac{1}{2}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{v'^2u'}) + \frac{1}{r}\frac{\partial}{\partial \theta}\overline{v'^2v'} + \frac{\partial}{\partial z}\overline{v'^2w'}\right) + \frac{\bar{u}\overline{v'v'} + \bar{v}\overline{u'v'} + \overline{u'v'^2}}{r} \tag{5.115}$$

and

$$\overline{w'\text{LHSW}'_c} = \left(\bar{u}\frac{\partial \overline{w'^2}/2}{\partial r} + \frac{\bar{v}}{r}\frac{\partial \overline{w'^2}/2}{\partial \theta} + \bar{w}\frac{\partial \overline{w'^2}/2}{\partial z}\right) + \left(\overline{u'w'}\frac{\partial \bar{w}}{\partial r} + \frac{\overline{v'w'}}{r}\frac{\partial \bar{w}}{\partial \theta} + \overline{w'w'}\frac{\partial \bar{w}}{\partial z}\right) \tag{5.116}$$

$$+ \frac{1}{2}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{w'^2u'}) + \frac{1}{r}\frac{\partial}{\partial \theta}\overline{w'^2v'} + \frac{\partial}{\partial z}\overline{w'^2w'}\right) \tag{5.117}$$

Now we take care of the RHS. The RHS terms (pressure and viscous) of the fluctuations are trivial since they are all linear.

$$\overline{\text{RHS}}_p = -\frac{1}{\rho}\left(\overline{u'\frac{\partial p'}{\partial r} + \frac{v'}{r}\frac{\partial p'}{\partial \theta} + w'\frac{\partial p'}{\partial z}}\right) \tag{5.118}$$

$$= -\frac{1}{\rho}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{u'p'}) + \frac{1}{r}\frac{\partial}{\partial \theta}\overline{v'p'} + \frac{\partial}{\partial z}\overline{w'p'}\right) \tag{5.119}$$

where we have used continuity and written in a more convenient form – the pressure distortion term can further be split into contributions from pressure fluctuation transport and pressure–strain correlation.

$$\overline{\mathrm{RHSUU}}_v/\nu = \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2}\right)\frac{\overline{u'^2}}{2} - \left[\overline{\left(\frac{\partial u'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial u'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial u'}{\partial z}\right)^2}\right] \tag{5.120}$$

$$- \frac{2}{r^2}\overline{u'\frac{\partial v'}{\partial\theta}} - \frac{\overline{u'^2}}{r^2} \tag{5.121}$$

$$\overline{\mathrm{RHSVV}}_v/\nu = \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2}\right)\frac{\overline{v'^2}}{2} - \left[\overline{\left(\frac{\partial v'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial v'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial v'}{\partial z}\right)^2}\right] \tag{5.122}$$

$$+ \frac{2}{r^2}\overline{v'\frac{\partial u'}{\partial\theta}} - \frac{\overline{v'^2}}{r^2} \tag{5.123}$$

$$\overline{\mathrm{RHSWW}}_v/\nu = \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2}\right)\frac{\overline{w'^2}}{2} - \left[\overline{\left(\frac{\partial w'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial w'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial w'}{\partial z}\right)^2}\right] \tag{5.124}$$

$$\overline{\mathrm{RHSK}}_v/\nu = \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2}\right)k \tag{5.125}$$

$$- \left[\overline{\left(\frac{\partial u'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial u'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial u'}{\partial z}\right)^2}\right] \tag{5.126}$$

$$- \left[\overline{\left(\frac{\partial v'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial v'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial v'}{\partial z}\right)^2}\right] \tag{5.127}$$

$$- \left[\overline{\left(\frac{\partial w'}{\partial r}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial w'}{\partial\theta}\right)^2} + \overline{\left(\frac{\partial w'}{\partial z}\right)^2}\right] \tag{5.128}$$

$$- \frac{\overline{u'^2}}{r^2} - \frac{\overline{v'^2}}{r^2} - \frac{2}{r^2}\overline{u'\frac{\partial v'}{\partial\theta}} + \frac{2}{r^2}\overline{v'\frac{\partial u'}{\partial\theta}} \tag{5.129}$$

In summary, the TKE equation the cylindrical coordinates is

$$\frac{\partial k}{\partial t} + \left(\bar{u}\frac{\partial}{\partial r} + \frac{\bar{v}}{r}\frac{\partial}{\partial\theta} + \bar{w}\frac{\partial}{\partial z}\right)k = \nabla\cdot T + P + \varepsilon \tag{5.130}$$

where T is the transport, P is the production, and ε is the dissipation.

$$T = T_p + T_t + T_v$$

$$\nabla\cdot T_p = -\frac{1}{\rho}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{u'p'}) + \frac{1}{r}\frac{\partial}{\partial\theta}\overline{v'p'} + \frac{\partial}{\partial z}\overline{w'p'}\right)$$

$$\nabla\cdot T_t = -\frac{1}{2}\left(\frac{1}{r}\frac{\partial}{\partial r}(r\overline{(u'^2+v'^2+w'^2)u'}) + \frac{1}{r}\frac{\partial}{\partial\theta}\overline{(u'^2+v'^2+w'^2)v'} + \frac{\partial}{\partial z}\overline{(u'^2+v'^2+w'^2)w'}\right)$$

$$\nabla\cdot T_v = \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial z^2}\right)k$$

$$
\begin{aligned}
P =& P_{uu} + P_{vv} + P_{ww} \\
=& -\left(\overline{u'u'}\frac{\partial \bar{u}}{\partial r} + \frac{\overline{u'v'}}{r}\frac{\partial \bar{u}}{\partial \theta} + \overline{u'w'}\frac{\partial \bar{u}}{\partial z}\right) - \left(\overline{u'v'}\frac{\partial \bar{v}}{\partial r} + \frac{\overline{v'v'}}{r}\frac{\partial \bar{v}}{\partial \theta} + \overline{v'w'}\frac{\partial \bar{v}}{\partial z}\right) \\
& -\left(\overline{u'w'}\frac{\partial \bar{w}}{\partial r} + \frac{\overline{v'w'}}{r}\frac{\partial \bar{w}}{\partial \theta} + \overline{w'w'}\frac{\partial \bar{w}}{\partial z}\right) + \frac{\overline{u'v'}\bar{v}}{r} - \frac{\overline{v'v'}\bar{u}}{r} \\
P_{uu} =& -\left(\overline{u'u'}\frac{\partial \bar{u}}{\partial r} + \frac{\overline{u'v'}}{r}\frac{\partial \bar{u}}{\partial \theta} + \overline{u'w'}\frac{\partial \bar{u}}{\partial z}\right) + \frac{2\overline{u'v'}\bar{v} + \overline{u'v'^2}}{r} \\
P_{vv} =& -\left(\overline{u'v'}\frac{\partial \bar{v}}{\partial r} + \frac{\overline{v'v'}}{r}\frac{\partial \bar{v}}{\partial \theta} + \overline{v'w'}\frac{\partial \bar{v}}{\partial z}\right) - \frac{\overline{v'v'}\bar{u} + \overline{u'v'}\bar{v} + \overline{u'v'^2}}{r} \\
P_{ww} =& -\left(\overline{u'w'}\frac{\partial \bar{w}}{\partial r} + \frac{\overline{v'w'}}{r}\frac{\partial \bar{w}}{\partial \theta} + \overline{w'w'}\frac{\partial \bar{w}}{\partial z}\right) \\
\varepsilon =& -\nu\Bigg[\overline{\left(\frac{\partial u'}{\partial r}\right)^2} + \overline{\left(\frac{\partial v'}{\partial r}\right)^2} + \overline{\left(\frac{\partial w'}{\partial r}\right)^2} + \overline{\left(\frac{\partial u'}{\partial z}\right)^2} + \overline{\left(\frac{\partial v'}{\partial z}\right)^2} + \overline{\left(\frac{\partial w'}{\partial z}\right)^2} \\
& + \frac{1}{r^2}\overline{\left(\frac{\partial u'}{\partial \theta}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial v'}{\partial \theta}\right)^2} + \frac{1}{r^2}\overline{\left(\frac{\partial w'}{\partial \theta}\right)^2} + \frac{\overline{u'^2}}{r^2} + \frac{\overline{v'^2}}{r^2} + \frac{2}{r^2}\overline{u'\frac{\partial v'}{\partial \theta}} - \frac{2}{r^2}\overline{v'\frac{\partial u'}{\partial \theta}}\Bigg]
\end{aligned}
$$

We note that there is a term in P_{uu} and P_{vv}

$$
\frac{\overline{u'v'}\bar{v} + \overline{u'v'^2}}{r} \tag{5.131}
$$

transporting energy between u and v components but does not contribute to the total budget. The geometric effect has also led to additional terms in the production. The net production of $\overline{u'u'}$ and $\overline{v'v'}$ due to curvature is $-\overline{u'v'}\bar{v}/r$ and $\overline{v'v'}\bar{u}/r$ respectively. The former is an analog to $-\overline{u'v'}\partial_r\bar{u}$ and the latter $\overline{u'u'}\partial_r\bar{u}$. The total production can also be computed from the tensor expression:

$$
P = -\overline{u_i u_j}\frac{\partial \bar{u}_j}{\partial x_i} = -\overline{\boldsymbol{u}\boldsymbol{u}} : \nabla \bar{\boldsymbol{u}} \tag{5.132}
$$

where the cylindrical representation of $\nabla \boldsymbol{u}$ is

$$
[\nabla \bar{\boldsymbol{u}}] = \begin{bmatrix} \frac{\partial \bar{u}}{\partial r} & \frac{\partial \bar{v}}{\partial r} & \frac{\partial \bar{w}}{\partial r} \\ \frac{1}{r}\frac{\partial \bar{u}}{\partial \theta} - \frac{\bar{v}}{r} & \frac{1}{r}\frac{\partial \bar{v}}{\partial \theta} + \frac{\bar{u}}{r} & \frac{1}{r}\frac{\partial \bar{w}}{\partial \theta} \\ \frac{\partial \bar{u}}{\partial z} & \frac{\partial \bar{v}}{\partial z} & \frac{\partial \bar{w}}{\partial z} \end{bmatrix} \tag{5.133}
$$

It's easy to see that the additional terms are

$$
\frac{\overline{u'v'}\bar{v}}{r} - \frac{\overline{v'v'}\bar{u}}{r}. \tag{5.134}
$$

Similarly, the perturbation VGT is $\nabla \boldsymbol{u}'$ and the dissipation can be seen as:

$$
\varepsilon = -\nu\overline{\nabla \boldsymbol{u}' : \nabla \boldsymbol{u}'} \tag{5.135}
$$

$$
= -\nu\Bigg[\overline{\left(\frac{\partial u'}{\partial r}\right)^2} + \overline{\left(\frac{\partial v'}{\partial r}\right)^2} + \overline{\left(\frac{\partial w'}{\partial r}\right)^2} + \overline{\left(\frac{\partial u'}{\partial z}\right)^2} + \overline{\left(\frac{\partial v'}{\partial z}\right)^2} + \overline{\left(\frac{\partial w'}{\partial z}\right)^2} \tag{5.136}
$$

$$+\frac{1}{r^2}\overline{\left(\frac{\partial u'}{\partial\theta}\right)^2}+\frac{1}{r^2}\overline{\left(\frac{\partial v'}{\partial\theta}\right)^2}+\frac{1}{r^2}\overline{\left(\frac{\partial w'}{\partial\theta}\right)^2}+\frac{\overline{u'^2}}{r^2}+\frac{\overline{v'^2}}{r^2}+\frac{2}{r^2}\overline{u'\frac{\partial v'}{\partial\theta}}-\frac{2}{r^2}\overline{v'\frac{\partial u'}{\partial\theta}}\Bigg] \tag{5.137}$$

Hence, we have verified that the budget terms, especially the production and dissipation, derived from (1) the component-wise momentum equation and (2) the cylindrical expansion of P and ε in tensor notation are fully consistent. Note we have opted for the pseudo-dissipation instead of the standard dissipation for simplicity. If the standard dissipation $\varepsilon = -\nu\overline{\boldsymbol{S}':\boldsymbol{S}'}$ were used, there will be an additional transport term $\overline{-\partial_j(2\nu u_i' S_{ij}')}$ with a spatial vanishing integration.

Similarly, the Reynolds stress transport equation for $\overline{u'w'}$ can be derived.

$$\overline{w'\mathrm{LHSU}'_c}+\overline{u'\mathrm{LHSW}'_c}= \tag{5.138}$$

Discussion on two limits: (1) $r=\infty$, curvature terms all vanish and reduce to the Cartesian version (noting that $rd\theta = dy$ – rotation around infinity is translation); (2) in the axisymmetric case ($\partial_\theta\overline{(\cdot)}=0$, such as in round jets and wakes), there are further simplifications.

We re-write the TKE production rate components with cylindrical notations:

$$\begin{aligned}
P &= P_{xx}+P_{rr}+P_{\theta\theta}\\
&= -\left(\overline{u_x'u_x'}\frac{\partial\bar{u}_x}{\partial x}+\overline{u_x'u_r'}\frac{\partial\bar{u}_x}{\partial r}+\frac{\overline{u_x'u_\theta'}}{r}\frac{\partial\bar{u}_x}{\partial\theta}\right)-\left(\overline{u_x'u_r'}\frac{\partial\bar{u}_r}{\partial x}+\overline{u_r'u_r'}\frac{\partial\bar{u}_r}{\partial r}+\frac{\overline{u_r'u_\theta'}}{r}\frac{\partial\bar{u}_r}{\partial\theta}\right)\\
&\quad-\left(\overline{u_x'u_\theta'}\frac{\partial\bar{u}_\theta}{\partial x}+\overline{u_r'u_\theta'}\frac{\partial\bar{u}_\theta}{\partial r}+\frac{\overline{u_\theta'u_\theta'}}{r}\frac{\partial\bar{u}_\theta}{\partial\theta}\right)+\frac{\overline{u_r'u_\theta'}\bar{u}_\theta}{r}-\frac{\overline{u_\theta'u_\theta'}\bar{u}_r}{r}\\
P_{xx} &= -\left(\overline{u_x'u_x'}\frac{\partial\bar{u}_x}{\partial x}+\overline{u_x'u_r'}\frac{\partial\bar{u}_x}{\partial r}+\frac{\overline{u_x'u_\theta'}}{r}\frac{\partial\bar{u}_x}{\partial\theta}\right)\\
P_{rr} &= -\left(\overline{u_x'u_r'}\frac{\partial\bar{u}_r}{\partial x}+\overline{u_r'u_r'}\frac{\partial\bar{u}_r}{\partial r}+\frac{\overline{u_r'u_\theta'}}{r}\frac{\partial\bar{u}_r}{\partial\theta}\right)+\frac{2\overline{u_r'u_\theta'}\bar{u}_\theta+\overline{u_r'u_\theta'^2}}{r}\\
P_{\theta\theta} &= -\left(\overline{u_x'u_\theta'}\frac{\partial\bar{u}_\theta}{\partial x}+\overline{u_r'u_\theta'}\frac{\partial\bar{u}_\theta}{\partial r}+\frac{\overline{u_\theta'u_\theta'}}{r}\frac{\partial\bar{u}_\theta}{\partial\theta}\right)-\frac{\overline{u_\theta'u_\theta'}\bar{u}_r+\overline{u_r'u_\theta'}\bar{u}_\theta+\overline{u_r'u_\theta'^2}}{r}.
\end{aligned}$$

We note that there are additional terms compared to the production in Cartesian coordinates. There is a term shared between P_{rr} and $P_{\theta\theta}$, $(\overline{u_r'u_\theta'}\bar{u}_\theta+\overline{u_r'u_\theta'^2})/r$, that re-distributes energy between u_r' and u_θ' components but it does not affect the total production. This is the curvature-related inter-component transport. The net production due to curvature is $\overline{u_r'u_\theta'}(\bar{u}_\theta/r)$ and $-\overline{u_\theta'u_\theta'}(\bar{u}_r/r)$, in P_{rr} and $P_{\theta\theta}$, respectively.

5.1.9 Dissipation rate transport equation

The dissipation term in Reynolds stresses transport equation is defined as

$$\varepsilon_{ij}=2\nu\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_j'}{\partial x_p}} \tag{5.139}$$

Multiply equation (5.20) by $2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}$ and take the time derivative we have:

$$\text{LHS} = 2\nu \frac{\bar{D}}{Dt}\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_i'}{\partial x_p}} = \frac{\bar{D}\varepsilon}{Dt} + 2\nu\overline{\frac{\partial \overline{u_k}}{\partial x_p}\frac{\partial u_i'}{\partial x_p}\frac{\partial u_i'}{\partial x_k}} \tag{5.140}$$

$$\overline{2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}(-\frac{1}{\rho}\frac{\partial p'}{\partial x_i})} = -\frac{2\nu}{\rho}\frac{\partial}{\partial x_k}(\overline{\frac{\partial u_k'}{\partial x_p}\frac{\partial p'}{\partial x_p}}) \tag{5.141}$$

$$\overline{2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}(\frac{\partial}{\partial x_k}(\nu\frac{\partial u_i'}{\partial x_k}))} = \frac{\partial}{\partial x_k}(\nu\frac{\partial \varepsilon}{\partial x_k}) - 2\overline{(\nu\frac{\partial^2 u_i'}{\partial x_p \partial x_k})^2} \tag{5.142}$$

$$\overline{2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}(\frac{\partial}{\partial x_k}\overline{u_i'u_k'})} = 0 \tag{5.143}$$

$$\overline{2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}(\frac{\partial}{\partial x_k} - u_i'u_k')} = -2\nu\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_k'}{\partial x_p}\frac{\partial u_i'}{\partial x_k}} + \frac{\partial}{\partial x_k}\overline{u_k'\nu(\frac{\partial u_i'}{\partial x_p})^2} \tag{5.144}$$

$$= -2\nu\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_k'}{\partial x_p}\frac{\partial u_i'}{\partial x_k}} + \frac{\partial}{\partial x_k}\overline{u_k'\varepsilon'} \tag{5.145}$$

$$\overline{2\nu\frac{\partial u_i'}{\partial x_p}\frac{\partial}{\partial x_p}(-u_k'\frac{\partial \overline{u}_i}{\partial x_k})} = -2\nu\frac{\partial \overline{u}_i}{\partial x_k}\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_k'}{\partial x_p}} - 2\nu\frac{\partial^2 \overline{u}_i}{\partial x_k \partial x_p}\overline{u_k'\frac{\partial u_i'}{\partial x_p}} \tag{5.146}$$

By equalizing both side we yield the transport equation for turbulence dissipation rate

$$\frac{\bar{D}\varepsilon}{Dt} = \frac{\partial}{\partial x_k}\left(-\frac{2\nu}{\rho}\overline{\frac{\partial u_k}{\partial x_p}\frac{\partial p}{\partial x_p}} + \nu\frac{\partial \varepsilon}{\partial x_k} - \overline{u_k'\varepsilon'}\right) - 2\nu\frac{\partial \overline{u}_i}{\partial x_k}\left(\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_k'}{\partial x_p}} + \overline{\frac{\partial u_p'}{\partial x_k}\frac{\partial u_p'}{\partial x_i}}\right) \tag{5.147}$$

$$- 2\nu\overline{u_k'\frac{\partial u_i'}{\partial x_p}\frac{\partial^2 \overline{u}_i}{\partial x_p \partial x_k}} - 2\nu\overline{\frac{\partial u_i'}{\partial x_p}\frac{\partial u_k'}{\partial x_p}\frac{\partial u_i'}{\partial x_k}} - 2\overline{\left(\nu\frac{\partial u_i'}{\partial^2 x_p \partial x_k}\right)^2} \tag{5.148}$$

The final equation of the equation agrees with that given in the turbulence book by Shi (1994). This should be the most complicated RANS equation we attempt here. We can see that the RANS second moment equations are never closed and a closure is needed (Chou, 1945).

5.1.10 Scalar flux, its mean and variance transport equations

Similar to Eq. (5.20) we have the transport equation for the mean and fluctuation of a passive scalar c:

$$\frac{\bar{D}\bar{c}}{\bar{D}t} = \frac{\partial}{\partial x_j}(\nu\frac{\partial \bar{c}}{\partial x_j} - \overline{c'u_j'}) \tag{5.149}$$

and

$$\frac{\bar{D}c'}{Dt} = \frac{\partial}{\partial x_j}(\Gamma\frac{\partial c'}{\partial x_j} + \overline{c'u_j'} - c'u_j') - u_j'\frac{\partial \bar{c}}{\partial x_j} \tag{5.150}$$

where Γ is the molecular diffusion coefficient of c.

Take $c'\times$ (5.20) $+ u_i'\times$ (5.150) and apply the average

$$\text{LHS} = \frac{\bar{D}\overline{c'u_i'}}{\bar{D}t} \tag{5.151}$$

$$\text{RHS}_1 = -\frac{1}{\rho}\overline{c'\frac{\partial p'}{\partial x_i}} = -\frac{1}{\rho}(\frac{\partial}{\partial x_j}\overline{p'c'}\delta_{ij} - \overline{p'\frac{\partial c'}{\partial x_i}}) \tag{5.152}$$

$$\text{RHS}_2 = \frac{\partial}{\partial x_j}(\Gamma\overline{u_i'\frac{\partial c'}{\partial x_j}} + \nu\overline{c'\frac{\partial u_i'}{\partial x_j}}) - (\nu+\Gamma)\overline{\frac{\partial u_i'}{\partial x_j}\frac{\partial c'}{\partial x_j}} \tag{5.153}$$

$$\text{RHS}_3 = -\frac{\partial}{\partial x_j}(\overline{c'u_i'u_j'}) \tag{5.154}$$

$$\text{RHS}_4 = -\overline{c'u_j'}\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i'u_j'}\frac{\partial \overline{c}}{\partial x_j} \tag{5.155}$$

then we obtain the transport equation for scalar flux

$$\frac{\bar{D}\overline{c'u_i'}}{\bar{D}t} = d_{jc} + P_{jc} + \Phi_{jc} - \varepsilon_{jc} \tag{5.156}$$

where

$$d_{ic} = \frac{\partial}{\partial x_j}(\Gamma\overline{u_i'\frac{\partial c'}{\partial x_j}} + \nu\overline{c'\frac{\partial u_i'}{\partial x_j}} - \frac{1}{\rho}\overline{p'c'}\delta_{ij} - \overline{c'u_i'u_j'}) \tag{5.157}$$

$$P_{ic} = -\overline{c'u_j'}\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u_i'u_j'}\frac{\partial \overline{c}}{\partial x_j} \tag{5.158}$$

$$\Phi_{ic} = \frac{1}{\rho}\overline{p'\frac{\partial c'}{\partial x_i}} \tag{5.159}$$

$$\varepsilon_{ic} = (\nu+\Gamma)\overline{\frac{\partial u_i'}{\partial x_j}\frac{\partial c'}{\partial x_j}} \tag{5.160}$$

Comments:

(1) Gradient diffusion: velocity-fluctuation scalar-diffusion correlation, momentum-diffusion scalar-fluctation ccorelation, pressure diffusion, turbulence diffusion.

(2) Production: scalar flux interacting with mean shear, turbulent flux (Reynolds stresses) interacting with mean scalar gradient.

(3) Re-distribution.

(4) Dissipation.

Define scalar mean and variance/fluctuation energy as

$$K_c = \frac{1}{2}\bar{c}^2 \tag{5.161}$$

$$k_c = \frac{1}{2}\overline{c'c'} \tag{5.162}$$

$c'\times$ (5.150) and apply the average

$$\text{LHS} = \frac{\bar{D}k_c}{\bar{D}t} \tag{5.163}$$

$$\text{RHS}_1 = \frac{\partial}{\partial x_j}\Gamma\frac{\partial k_c}{\partial x_j} - \Gamma\overline{\frac{\partial c'}{\partial x_j}\frac{\partial c'}{\partial x_j}} \tag{5.164}$$

$$\text{RHS}_2 = -\frac{1}{2}\frac{\partial}{\partial x_j}\overline{c'c'u_j} \tag{5.165}$$

$$\text{RHS}_3 = -\overline{c'u'_j}\frac{\partial \overline{c}}{\partial x_j} \tag{5.166}$$

then we obtain the transport equation for scalar fluctuation energy

$$\frac{\bar{D}k_c}{\bar{D}t} = \frac{\partial}{\partial x_j}(\Gamma\frac{\partial}{\partial x_j}k_c - \frac{1}{2}\overline{c'c'u'_j}) - \overline{c'u'_j}\frac{\partial \overline{c}}{\partial x_j} - \Gamma\overline{\frac{\partial c'}{\partial x_j}\frac{\partial c'}{\partial x_j}} \tag{5.167}$$

For active scalar (for example, density which appears in the momentum equation as buoyancy force), see section 6.7.

5.1.11 Poisson equation for mean and fluctuation pressure

The Reynolds average equation is

$$\frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j\frac{\partial \overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu\frac{\partial \overline{u}_i}{\partial x_j} - \overline{u'_iu'_j}) \tag{5.168}$$

Take the divergence of the equation:

$$\text{LHS} = \frac{\partial \overline{u}_i}{\partial x_j}\frac{\partial \overline{u}_j}{\partial x_i} \tag{5.169}$$

$$\text{RHS} = -\frac{1}{\rho}\nabla^2\overline{p} - \frac{\partial^2 \overline{u'_iu'_j}}{\partial x_i\partial x_j} \tag{5.170}$$

Poisson equation for mean pressure:

$$-\frac{1}{\rho}\nabla^2\overline{p} = \frac{\partial \overline{u}_i}{\partial x_j}\frac{\partial \overline{u}_j}{\partial x_i} + \frac{\partial^2 \overline{u'_iu'_j}}{\partial x_i\partial x_j} \tag{5.171}$$

The velocity fluctuation transport equation is

$$\frac{\bar{D}u'_i}{Dt} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu\frac{\partial u'_i}{\partial x_j} + \overline{u'_iu'_j} - u'_iu'_j) - u'_j\frac{\partial \overline{u}_i}{\partial x_j} \tag{5.172}$$

Take the divergence of the equation:

$$\text{LHS} = \frac{\partial \overline{u}_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} \tag{5.173}$$

$$\text{RHS} = -\frac{1}{\rho}\nabla^2 p' - \frac{\partial^2 \overline{u'_iu'_j}}{\partial x_i\partial x_j} - \frac{\partial u'_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} - \frac{\partial \overline{u}_i}{\partial x_j}\frac{\partial u'_j}{\partial x_i} \tag{5.174}$$

Poisson equation for fluctuation pressure:

$$\frac{1}{\rho}\nabla^2 p' = -\frac{\partial \overline{u}_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} - \frac{\partial \overline{u}_i}{\partial x_j}\frac{\partial u'_j}{\partial x_i} - \frac{\partial u'_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} + \frac{\partial^2 \overline{u'_iu'_j}}{\partial x_i\partial x_j} \tag{5.175}$$

$$= -\frac{\partial u'_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} + \frac{\partial^2 \overline{u'_iu'_j}}{\partial x_i\partial x_j} - 2\frac{\partial \overline{u}_j}{\partial x_i}\frac{\partial u'_i}{\partial x_j} \tag{5.176}$$

$$= -2\frac{\partial \overline{u}_j}{\partial x_i}\frac{\partial u_i'}{\partial x_j} - \frac{\partial^2}{\partial x_i x_j}(u_i'u_j' - \overline{u_i'u_j'}) \tag{5.177}$$

where the last equation is due to the fact that

$$\frac{\partial u_i'}{\partial x_j}\frac{\partial u_j'}{\partial x_i} = \frac{\partial^2 u_i'u_j'}{\partial x_i x_j}. \tag{5.178}$$

According to the source terms, the pressure can be decomposed into the rapid, slow, and harmonic components

$$p' = p^{(r)} + p^{(s)} + p^{(h)}, \tag{5.179}$$

such that

$$\frac{1}{\rho}\nabla^2 p^{(r)} = -2\frac{\partial \overline{u}_j}{\partial x_i}\frac{\partial u_i'}{\partial x_j} \tag{5.180}$$

$$\frac{1}{\rho}\nabla^2 p^{(s)} = -\frac{\partial^2}{\partial x_i x_j}(u_i'u_j' - \overline{u_i'u_j'}) \tag{5.181}$$

$$\nabla^2 p^{(h)} = 0 \tag{5.182}$$

and modelled separately.

5.1.12 Turbulent vorticity and enstrophy

Similarly, vorticity can be decomposed into the mean and the pertubation. We give the equation of pertuabtion vorticity without derivation:

$$\frac{\bar{D}\omega_i'}{\bar{D}t} = \omega_j'\overline{S}_{ij} + \overline{\omega_j}S_{ij}' + \omega_j'S_{ij}' - \overline{\omega_j'S_{ij}'} - u_j'\frac{\partial \overline{\omega}_i}{\partial x_j} + \frac{\partial}{\partial x_j}(\overline{u_j'\omega_i'} - u_j'\omega_i') + \nu\frac{\partial^2\omega_i'}{\partial x_j^2} \tag{5.183}$$

where $\overline{S}_{ij}$ and S_{ij}' are the mean and the fluctuation shear, respectively.

We define the fluctuating enstrophy as

$$\mathscr{E} = \frac{1}{2}\overline{\omega_i'\omega_i'} \tag{5.184}$$

$\omega_i' \times$ (5.183) and take the time average

$$\text{LHS} = \frac{\bar{D}\mathscr{E}}{\bar{D}t} \tag{5.185}$$

$$\text{RHS}_1 = \overline{\omega_i'\omega_j'}\,\overline{S_{ij}} + \overline{\omega}_j\overline{\omega_i'S_{ij}'} + \overline{\omega_i'\omega_j'S_{ij}'} \tag{5.186}$$

$$\text{RHS}_2 = -\overline{\omega_i'u_j'}\frac{\partial \overline{\omega_i}}{\partial x_j} \tag{5.187}$$

$$\text{RHS}_3 = -\frac{1}{2}\frac{\partial}{\partial x_j}(\overline{u_j'\omega_i'\omega_i'}) \tag{5.188}$$

$$\text{RHS}_4 = \nu\overline{\frac{\partial^2\mathscr{E}}{\partial x_j^2}} - \overline{\frac{\partial\omega_i'}{\partial x_j}\frac{\partial\omega_i'}{\partial x_j}} \tag{5.189}$$

Equaling both sides we obtain

$$\frac{\bar{D}\mathscr{E}}{\bar{D}t} = P_{\mathscr{E}} + D_{\mathscr{E}} - \varepsilon_{\mathscr{E}} \tag{5.190}$$

$$P_{\mathscr{E}} = \overline{\omega_i'\omega_j'}\,\overline{S_{ij}} + \overline{\omega_j}\overline{\omega_i' S_{ij}'} + \overline{\omega_i'\omega_j' S_{ij}'} - \overline{\omega_i' u_j'}\frac{\partial \overline{\omega_i}}{\partial x_j} \tag{5.191}$$

$$D_{\mathscr{E}} = \frac{\partial}{\partial x_j}(\nu\frac{\partial \mathscr{E}}{\partial x_j} - \frac{1}{2}\overline{u_j'\omega_i'\omega_i'}) \tag{5.192}$$

$$\varepsilon_{\mathscr{E}} = \nu\overline{\frac{\partial \omega_i'}{\partial x_j}\frac{\partial \omega_i'}{\partial x_j}} \tag{5.193}$$

Comment: The energy balance process of fluctuation enstrophy obeys four principle processes in nature (Kolmogorov):

$$\text{rate of change} = \text{production} + \text{diffusion} + \text{dissipation}$$

Moreover, in instability problems, it is convenience to consider the linearized inviscid perturbation vorticity equation, which reads

$$\frac{\bar{D}\omega_i'}{\bar{D}t} = \omega_j'\bar{S}_{ij} + \bar{\omega}_j S_{ij}' - u_j'\frac{\partial \bar{\omega}_i}{\partial x_j}, \tag{5.194}$$

since perturbation enstrophy is related to turbulent dissipation (see 5.51). It can be obtained either by linearizing the vorticity equation or taking the curl of the linearized momentum equations. It has a version in rotating frames in (6.109).

5.1.13 Spectral TKE equation

Staring with the fluctuation equations

$$\frac{\partial u_i'}{\partial x_i} = 0 \tag{5.195}$$

$$\frac{\partial u_i'}{\partial t} + \bar{u}_j\frac{\partial u_i'}{\partial x_j} + u_j'\frac{\partial \overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\nu\frac{\partial u_i'}{\partial x_j} - u_i'u_j' + \overline{u_i'u_j'}\right) \tag{5.196}$$

we take the Fourier expansion of the fluctuation in the homogeneous directions

$$\boldsymbol{u}'(x,y,z,t) = \sum_{k_x}\sum_{k_z}\hat{\boldsymbol{u}}(k_x,y,k_z,t)e^{i(k_x x + k_z z)} \tag{5.197}$$

$$p'(x,y,z,t) = \sum_{k_x}\sum_{k_z}\hat{p}(k_x,y,k_z,t)e^{i(k_x x + k_z z)} \tag{5.198}$$

where we assume that there is at least one spatial inhomogeneous direction; otherwise, see section 5.3 for the transport equation of the correlation function.

The energy spectrum is

$$\tilde{e}(k_x,y,k_z,t) = \frac{1}{2}\hat{u}_i\hat{u}_i^* = \frac{1}{2}|\hat{u}|^2 + |\hat{v}|^2 + |\hat{w}|^2 \tag{5.199}$$

and its evolution equation (for each wavenumber pair (k_x, k_z)) can be derived by taking the Fourier transform of the fluctuation equations (5.195)–(5.196) in x, z,

$$ik_x\hat{u} + \partial_y\hat{v} + ik_z\hat{w} = 0 \tag{5.200}$$

$$\frac{\partial\hat{u}_i}{\partial t} + \bar{u}_j\widehat{\frac{\partial u_i'}{\partial x_j}} + \hat{u}_j\frac{\partial\overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\widehat{\frac{\partial p'}{\partial x_i}} + \widehat{\frac{\partial}{\partial x_j}\left(-u_i'u_j'\right)} + \nu\widehat{\frac{\partial^2 u_i'}{\partial x_j^2}} \tag{5.201}$$

where $\partial/\partial x_j = (ik_x, \partial_y, ik_z)$. We note that only the nonlinear term (RHS2) involves scale interaction. Note that the Reynolds stress $\overline{u_i'u_j'}$ involves no k_x, k_z dependence and its FT is Dirac delta (meaning that this term only appears in the 0-0 mode – the standard TKE equation – when integrated over the Fourier space).

Multiply (5.201) with $\hat{u}_i^*(k_x, y, k_z)$ and denote $\boldsymbol{k} = (k_x, y, k_z)$, we have the evolution equation of the Fourier modes:

$$\frac{\partial\tilde{e}}{\partial t} = \Re\left\{-\hat{u}_i^*\hat{u}_j\frac{\partial\overline{u}_i}{\partial x_j}\right\} + \Re\left\{-\frac{1}{\rho}\frac{\partial\hat{p}\hat{v}^*}{\partial y}\right\} + \Re\left\{-\hat{u}_i^*\widehat{\frac{\partial u_i'u_j'}{\partial x_j}}\right\} + \nu\frac{\partial^2\tilde{e}}{\partial y^2} - 2\nu(k_x^2+k_z^2)\tilde{e} - \nu\frac{\partial\hat{u}_i^*}{\partial y}\frac{\partial\hat{u}_i}{\partial y} \tag{5.202}$$

where the scale-dependent spectral TKE terms are

$$\hat{P}(k_x, y, k_z) = \Re\left\{-\hat{u}_j(\boldsymbol{k})\hat{u}_i^*(\boldsymbol{k})\frac{\partial\overline{u}_i}{\partial x_j}\right\} = \Re\left\{-\hat{u}^*\hat{v}\frac{\partial\overline{u}}{\partial y}\right\} \tag{5.203}$$

$$\hat{T}_p(k_x, y, k_z) = \Re\left\{-\frac{1}{\rho}\frac{\partial\hat{p}(\boldsymbol{k})\hat{v}^*(\boldsymbol{k})}{\partial y}\right\} \tag{5.204}$$

$$\hat{T}_{nl}(k_x, y, k_z) = \Re\left\{-\hat{u}_i^*(\boldsymbol{k})\widehat{\frac{\partial u_i'(\boldsymbol{k}_1)u_j'(\boldsymbol{k}_2)}{\partial x_j}}(\boldsymbol{k})\right\} = \Re\left\{-\hat{u}_i^*(\boldsymbol{k})\widehat{\frac{\partial u_i'(\boldsymbol{k}_1)v'(\boldsymbol{k}_2)}{\partial y}}(\boldsymbol{k})\right\} \tag{5.205}$$

$$= \Re\left\{-\hat{u}_i^*(\boldsymbol{k})\widehat{u_j'\frac{\partial u_i'}{\partial x_j}}(\boldsymbol{k})\right\} \tag{5.206}$$

$$= \Re\left\{-\frac{\partial}{\partial y}[\hat{u}_i^*(\boldsymbol{k})\widehat{(u_i'(\boldsymbol{k}_1)v'(\boldsymbol{k}_2))}] + \widehat{u_i'(\boldsymbol{k}_1)v'(\boldsymbol{k}_2)}\frac{\partial\hat{u}_i^*}{\partial y}(\boldsymbol{k})\right\} \tag{5.207}$$

$$\hat{D}(k_x, y, k_z) = \nu\frac{\partial^2\tilde{e}(\boldsymbol{k})}{\partial y^2} \tag{5.208}$$

$$\hat{\varepsilon}(k_x, y, k_z) = -2\nu(k_x^2+k_z^2)\tilde{e}(\boldsymbol{k}) - \nu\frac{\partial\hat{u}_i^*(\boldsymbol{k})}{\partial y}\frac{\partial\hat{u}_i(\boldsymbol{k})}{\partial y} \tag{5.209}$$

are the spectral density of production, pressure fluctuation transport, (nonlinear) turbulent transport, viscous diffusion, and (pseudo-)dissipation at each wavenumber and wall distance, with the last equation in (5.203) giving its representation in a turbulent channel flow with the only non-zero shear being $\partial_y\bar{u}$. These equations are consistent with Mizuno (2016); Cho *et al.* (2018). We note that there are different ways of writing $\hat{T}_{nl}$; note $\boldsymbol{k}_1 + \boldsymbol{k}_2 = \boldsymbol{k}$. Typically, the spectral TKE equation will be averaged in time in practice.

We note that the production term is linear, in the sense that it does not involve scale interactions. Since $\partial_y\bar{u}(y)$ has no spatial dependence in the Fourier direction, it is immune to the FT. The FT of $\overline{uv}$ is the co-spectrum $\Phi_{uv}(\boldsymbol{k}) = \Re\{\hat{u}^*(\boldsymbol{k})\hat{v}(\boldsymbol{k})\}$. As a result, $\hat{P}$ is the production spectrum whose integration over the entire Fourier space recovers $P = -\overline{uv}\partial_y\bar{u}$. The

diffusion of spectral density happens only in the inhomogeneous y-direction while dissipation occurs in all spatial directions. The nonlinear term $\hat{T}_{nl}$, is the only term that involves different wavenumbers (in the FT of $\widehat{u_i'(\boldsymbol{k}_1)u_j'(\boldsymbol{k}_2)}$) and hence scale interactions. The other terms are hence linear/scale-local (e.g., it is said that the production is linear). In statistically stationary flows, $d\tilde{e}/dt = 0$.

Some steps of the derivation are provided below:

$$\frac{\partial \tilde{e}}{\partial t} = \frac{1}{2}\frac{\partial}{\partial t}(\hat{u}_i\hat{u}_i^*) = \frac{1}{2}\left(\hat{u}_i^*\frac{\partial \hat{u}_i}{\partial t} + \hat{u}_i\frac{\partial \hat{u}_i^*}{\partial t}\right) = \Re\{\hat{u}_i^*\frac{\partial \hat{u}_i}{\partial t}\} \tag{5.210}$$

$$\hat{u}_i^*\bar{u}_j\widehat{\frac{\partial u_i'}{\partial x_j}} = \hat{u}_i^*(ik_x\bar{u} + \bar{v}\partial_y + ik_z\bar{w})\hat{u}_i,\ \Re\{\cdot\} = 0,\ \bar{v} = 0 \tag{5.211}$$

$$\hat{u}_i^*\widehat{\frac{\partial p'}{\partial x_i}} = (\hat{u}^*, \hat{v}^*, \hat{w}^*)\cdot(ik_x\hat{p}, \partial_y\hat{p}, ik_z\hat{p}) \tag{5.212}$$

$$= \hat{v}^*\partial_y\hat{p} + \hat{p}(ik_x\hat{u}^* + ik_z\hat{w}^*) \tag{5.213}$$

$$= \hat{v}^*\partial_y\hat{p} + \hat{p}\partial_y\hat{v}^* \tag{5.214}$$

$$= \partial_y(\hat{p}\hat{v}^*) \tag{5.215}$$

$$\overline{\widehat{u_i'u_j'}} = \widehat{\overline{u_i'u_j'}} = 0 \tag{5.216}$$

$$\Re\left\{-\hat{u}_i^*\widehat{\frac{\partial u_i'u_j'}{\partial x_j}}\right\} = \Re\left\{-\hat{u}_i^*\widehat{u_j'\frac{\partial u_i'}{\partial x_j}}\right\}\ \text{(another way of writing } \hat{T}_{nl}) \tag{5.217}$$

$$\widehat{u_i'u_j'}(\boldsymbol{k}) = \int_{-\infty}^{+\infty}\hat{u}_j(\boldsymbol{k}')\hat{u}_i(\boldsymbol{k}-\boldsymbol{k}')\,\mathrm{d}\boldsymbol{k}' \tag{5.218}$$

$$\nu\hat{u}_i^*\widehat{\frac{\partial^2 u_i'}{\partial x_j^2}} = \nu\hat{u}_i^*[-(k_x^2+k_z^2) + \partial_y^2]\hat{u}_i \tag{5.219}$$

$$= -\nu(k_x^2+k_z^2)\hat{u}_i\hat{u}_i^* + \nu\hat{u}_i^*\partial_y(\partial_y\hat{u}_i) \tag{5.220}$$

$$= -\nu(k_x^2+k_z^2)\hat{u}_i\hat{u}_i^* + \nu\partial_y\left(\hat{u}_i^*\partial_y\hat{u}_i\right) - \nu\partial_y\hat{u}_i^*\partial_y\hat{u}_i \tag{5.221}$$

$$\Re\left\{\nu\hat{u}_i^*\widehat{\frac{\partial^2 u_i'}{\partial x_j^2}}\right\} = -\nu(k_x^2+k_z^2)\hat{u}_i\hat{u}_i^* + \nu\partial_y^2(\hat{u}_i^*\hat{u}_i/2) - \nu\partial_y\hat{u}_i^*\partial_y\hat{u}_i \tag{5.222}$$

$$\Re\{\hat{u}_i^*\partial_y\hat{u}_i\} = \Re\{\partial_y[\partial_y(\hat{u}_i^*\hat{u}_i)] - \partial_y[\hat{u}_i\partial_y\hat{u}_i^*]\} = \frac{1}{2}\Re\{\partial_y^2\hat{u}_i^*\hat{u}_i\} = \frac{1}{2}\partial_y^2\hat{u}_i^*\hat{u}_i \tag{5.223}$$

where (5.211) implies that the mean convection does not change the spectrum. We can commute

$$\widehat{\frac{\partial a}{\partial x_j}} = \frac{\partial \hat{a}}{\partial x_j} \tag{5.224}$$

as long as $\partial\hat{a}/\partial x_j$ is interpreted as $(ik_x, \partial_y, ik_z)\hat{a}$. The co-spectrum of two variables u' and v' is

$$\Phi_{uv}(\boldsymbol{k}) = \Re\{\hat{u}^*(\boldsymbol{k})\hat{v}(\boldsymbol{k})\} \tag{5.225}$$

whose physical-space dual is the two-point correlation

$$R_{uv}(\boldsymbol{x}, \boldsymbol{x}') = \overline{u'(\boldsymbol{x})v'(\boldsymbol{x}')} \tag{5.226}$$

and whose integration over the wavenumber space is the cross-correlation $R_{uv}(0) = \overline{u'v'}$ (when $\boldsymbol{x}' = \boldsymbol{x}$).

Integration of (5.202) over the entire wavenumber space will reduce to the standard TKE equation (with x, z being homogeneous directions):

$$0 = -\overline{u'v'}\frac{\partial \overline{u}}{\partial y} - \frac{1}{\rho}\frac{\partial \overline{p'v'}}{\partial y} - \frac{\partial \overline{ev'}}{\partial y} + \nu\frac{\partial^2 \bar{e}}{\partial y^2} - \nu\overline{\frac{\partial u_i'}{\partial x_j}\frac{\partial u_i'}{\partial x_j}} \tag{5.227}$$

where $e = 1/2 u_i' u_i'$ and simplifications for a channel geometry has been made. We note that the integral of the nonlinear interscale transfer term over all wavenumbers is zero

$$\int_{\boldsymbol{k}} \hat{T}_{nl}(\boldsymbol{k}, y)\,\mathrm{d}\boldsymbol{k} = 0, \tag{5.228}$$

hence it does not make any contributions to the TKE budget. The spatial integral any spatial transport terms is also zero, which has some implications on the importance of linear mechanisms in perturbation amplification (see (8.27)).

5.1.14 Error evolution equation

5.2 LES equations

5.2.1 Filtering

Assume an unfiltered quantity $f(x)$ contains all scales and denote the filtered quantity is

$$\tilde{f}(x) = \int g(x, x') f(x')\,\mathrm{d}x', \tag{5.229}$$

and typically in homogeneous directions as the convolution of the function $f(x)$ with the kernel $g(x)$:

$$\tilde{f}(x) = \int g(x - x') f(x')\,\mathrm{d}x'. \tag{5.230}$$

There are many choices for the kernel.

- The box filter. It is given by the following function

$$g(x) = \begin{cases} 1/2\Delta & |x| \le \Delta \\ 0 & \text{otherwise} \end{cases} \tag{5.231}$$

whose Fourier pair is

$$2\pi\hat{g}(k) = \frac{\sin(k\Delta)}{k\Delta}. \tag{5.232}$$

This filter corresponds to discretized measurement or computation in physical space, where filtering is usually not explicitly performed but implied by the truncation of Taylor's series. For example, a second-order approximation of $f'(x)$ at x_i is

$$f'(x_i) \approx \frac{f(x_i + \Delta) - f(x_i - \Delta)}{2\Delta} = \frac{d}{dx}\left(\frac{1}{2\Delta}\int_{x_i-\Delta}^{x_i+\Delta} f(x')\,\mathrm{d}x'\right) = \frac{d\tilde{f}}{dx} \tag{5.233}$$

where the RHS is exactly $d\tilde{f}/dx$ with the convolution kernel being the box filter function.

- The spectral cut-off (sharp Fourier) filter. The filtering kernel

$$g(x) = \frac{\sin(2\pi x/\Delta)}{\pi x} \tag{5.234}$$

corresponds to

$$2\pi\hat{g}(k) = \begin{cases} 1 & |k| \leq 2\pi/\Delta \\ 0 & \text{otherwise} \end{cases} \tag{5.235}$$

whose effect it to kill all modes with $|\boldsymbol{k}| > \boldsymbol{k}_{\max} = 2\pi/\Delta$.

- Gaussian filter. The Gaussian kernel

$$g(x) = \sqrt{\frac{6}{\pi\Delta^2}} e^{-6x^2/\Delta^2} \tag{5.236}$$

has a Fourier counterpart that is also Gaussian

$$2\pi\hat{g}(k) = e^{-(k\Delta)^2/24}. \tag{5.237}$$

Hence it is Gaussian in both physical and Fourier space.

We note that the Fourier counterpart of convolution (filtering) is multiplication in Fourier space, hence the filtered spectrum,

$$\hat{\tilde{f}} = 2\pi\hat{g}\hat{f}, \tag{5.238}$$

is spectral content of the signal f modified by filtering (multiplied by the transfer function $2\pi\hat{g}$). The co-spectrum of two arbitrary variables u, v

$$C_{uv}(k) = \Re\{\hat{u}\hat{v}^*\}, \tag{5.239}$$

which is the Fourier transform pair of

$$R_{uv}(x, x') = \langle u(x)v(x')\rangle, \tag{5.240}$$

will be modified to

$$C_{\tilde{u}\tilde{v}}(k) = 4\pi^2|\hat{g}(k)|^2 C_{uv}. \tag{5.241}$$

Relation to spatial average.

5.2.2 Filtered equations of motion

Adopt the large-small scale decomposition

$$u_i(x_i, t) = \tilde{u}_i(x_i, t) + u_i''(x_i, t), \tag{5.242}$$

where $\tilde{u}_i$ is the filtered velocity and u_i'' is the residual obtained by subtracting $\tilde{u}_i$ from u_i, and note that typically (unless a sharp spectral filter is used)

$$\widetilde{u''} = \widetilde{u - \tilde{u}} = \tilde{u} - \tilde{\tilde{u}} \neq 0, \tag{5.243}$$

unlike $\bar{u'} = 0$ in Reynolds averaging.

Assume filtering and spatial/temporal derivatives commute (which is typically the case with uniform filters), we have the filtered N–S equations

$$\frac{\partial \tilde{u}_i}{\partial x_j} = 0 \tag{5.244}$$

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial \widetilde{u_i u_j}}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \tilde{p}}{\partial x_i} + \nu \frac{\partial^2 \tilde{u}_i}{\partial x_j^2} + \tilde{f}_i \tag{5.245}$$

where the filtered total stress can be decomposed into

$$\widetilde{u_i u_j} = \tilde{u}_i \tilde{u}_j + (\widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j) \tag{5.246}$$

and the second part is called the subgrid-scale stress (generalized small-scale co-variance),

$$\tau_{ij}^{\mathrm{SGS}} = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j, \tag{5.247}$$

analogous to the Reynolds stress tensor $(-\overline{u_i' u_j'})$

$$\tau_{ij}^{\mathrm{Rey}} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j = \overline{u_i' u_j'}. \tag{5.248}$$

τ_{ij}^{SGS} exists when a nonlinear system is coarse-grained, that $\widetilde{u_i u_j} \neq \tilde{u}_i \tilde{u}_j$, and it represents the feedback of the residual scale on the filtered scales.

The momentum equation that the filtered velocity satisfies is

$$\frac{\partial \tilde{u}_i}{\partial t} + \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \tilde{p}}{\partial x_i} - \frac{\partial \tau_{ij}^{\mathrm{SGS}}}{\partial x_j} + \frac{\partial}{\partial x_j}(2\nu \tilde{S}_{ij}) + \tilde{f}_i \tag{5.249}$$

where $\tilde{S}_{ij} = 1/2(\partial_i \tilde{u}_j + \partial_j \tilde{u}_i)$. Note that τ_{ij}^{SGS} is a large-scale quantity and the idea that the effect of small scales to the large scales can be represented by the large scales only leads to LES modellings.

The subgrid scale stress can be decomposed as (Leonard, 1975)

$$\tau_{ij}^{\mathrm{SGS}} = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j = L_{ij} + C_{ij} + R_{ij} \tag{5.250}$$

where

$$L_{ij} = \widetilde{\tilde{u}_i \tilde{u}_j} - \tilde{u}_i \tilde{u}_j \tag{5.251}$$

$$C_{ij} = \widetilde{\tilde{u}_i u_j''} + \widetilde{u_i'' \tilde{u}_j} \tag{5.252}$$

$$R_{ij} = \widetilde{u_i'' u_j''} \tag{5.253}$$

representing the contributions of large-scale self-interaction, cross-scale interaction, and the small-scale interaction to the SGS stress. However, the above decomposition is not Galilean-invariant. The Leonard stress L_{ij} can also be interpreted as the difference between the portions of $\tilde{u}_j \tilde{u}_j$ residing below the cut-off wavenumber and the total.

More generally, the generalized second and third moments (Germano *et al.*, 1991),

$$\tau^l(a, b) = \widetilde{ab} - \tilde{a}\tilde{b} \tag{5.254}$$

and

$$\tau^l(a, b, c) = \widetilde{abc} - \tilde{a}\tau^l(b, c) - \tilde{b}\tau^l(a, c) - \tilde{c}\tau^l(a, c) - \tilde{a}\tilde{b}\tilde{c} \tag{5.255}$$

can be used, where the superscript l implies these operations are with respect to the filter scale l (Johnson, 2021).

5.2.3 LES energy equations

In terms of energy, the (filtered) total energy equals to the energy of the filtered scales and of the sub-filter scale

$$\frac{1}{2}\widetilde{u_i u_i} = \frac{1}{2}\tilde{u}_i\tilde{u}_i + \frac{1}{2}(\widetilde{u_i u_i} - \tilde{u}_i\tilde{u}_i) = \frac{1}{2}\tilde{u}_i\tilde{u}_i + \frac{1}{2}\tau_{ii}^{\mathrm{SGS}}. \tag{5.256}$$

Multiply the filtered momentum equation with $\tilde{u}_i$ and note that

$$-\tilde{u}_i\frac{\partial\tau_{ij}^{\mathrm{SGS}}}{\partial x_j} + \tilde{u}_i\frac{\partial}{\partial x_j}(2\nu\tilde{S}_{ij}) = -\frac{\partial}{\partial x_j}(\tilde{u}_i\tau_{ij}^{\mathrm{SGS}}) + \tilde{S}_{ij}\tau_{ij}^{\mathrm{SGS}} + \frac{\partial}{\partial x_j}(\tilde{u}_i 2\nu\tilde{S}_{ij}) - 2\nu\tilde{S}_{ij}\tilde{S}_{ij} \tag{5.257}$$

we have the large-scale energy equation

$$\left(\frac{\partial}{\partial t} + \tilde{u}_j\frac{\partial}{\partial x_j}\right)\frac{1}{2}\tilde{u}_i\tilde{u}_i = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\tilde{p}\tilde{u}_j + \tilde{u}_i\tau_{ij}^{\mathrm{SGS}} - 2\nu\tilde{u}_i\tilde{S}_{ij}\right) - \Sigma - 2\nu\tilde{S}_{ij}\tilde{S}_{ij} + \tilde{u}_i\tilde{f}_i \tag{5.258}$$

where (positive value of) $\Sigma = -\tau_{ij}^{\mathrm{SGS}}\tilde{S}_{ij}$ represents the rate at which energy transfered between the large scale and the small scales. We note that the above equation can also be written as

$$\left(\frac{\partial}{\partial t} + \tilde{u}_j\frac{\partial}{\partial x_j}\right)\frac{1}{2}\tilde{u}_i\tilde{u}_i = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\tilde{p}\tilde{u}_j - 2\nu\tilde{u}_i\tilde{S}_{ij}\right) - \tilde{u}_i\frac{\partial\tau_{ij}^{\mathrm{SGS}}}{\partial x_j} - 2\nu\tilde{S}_{ij}\tilde{S}_{ij} + \tilde{u}_i\tilde{f}_i \tag{5.259}$$

where the inter-scale transfer term is expressed as $\Psi = \tilde{u}_i\partial_j\tau_{ij}^{\mathrm{SGS}}$. Although both are positive when there is direct energy cascade and they are related as $\Sigma = \Psi - \partial_j(\tilde{u}_i\tau_{ij}^{\mathrm{SGS}})$ and hence share the same spatial average, their pointwise values can be a lot different.

The (filtered) total energy equation can be obtained by multiplying the momentum equation with u_i and then filtering:

$$\frac{\partial}{\partial t}\frac{\widetilde{u_i u_i}}{2} + \frac{\partial}{\partial x_j}\frac{\widetilde{u_i u_i u_j}}{2} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\widetilde{pu_j} - 2\nu\widetilde{u_i S_{ij}}\right) - 2\nu\widetilde{S_{ij}S_{ij}} + \widetilde{u_i f_i} \tag{5.260}$$

and the equation for small-scale energy can be obtained by subtracting the large-scale energy equation (5.258) from above equation:

$$\frac{\partial\tau_{ii}^{\mathrm{SGS}}/2}{\partial t} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}(\widetilde{pu_j} - \tilde{p}\tilde{u}_j) - 2\nu(\widetilde{u_i S_{ij}} - \tilde{u}_i\tilde{S}_{ij})\right) + \Sigma - 2\nu(\widetilde{S_{ij}S_{ij}} - \tilde{S}_{ij}\tilde{S}_{ij}) + (\widetilde{u_i f_i} - \tilde{u}_i\tilde{f}_i) \tag{5.261}$$

$$- \frac{\partial}{\partial x_j}\left(\frac{1}{2}\widetilde{u_i u_i u_j} - \frac{1}{2}\tilde{u}_i\tilde{u}_i\tilde{u}_j - \tilde{u}_i\tau_{ij}^{\mathrm{SGS}}\right) \tag{5.262}$$

where the last term (5.262) can also be written with the generalized moments as

$$-\frac{\partial\phi_j}{\partial x_j},\ \phi_i = \frac{1}{2}\tau_{ii}^{\mathrm{SGS}}\tilde{u}_j + \frac{1}{2}\tau(u_i, u_i, u_j). \tag{5.263}$$

or the entire equation with the generalized moments as (Johnson, 2021)

$$\frac{\partial\tau^l(u_i,u_i)/2}{\partial t} = -\frac{\partial}{\partial x_j}\left(\tau^l(p,u_j) - 2\nu\tau^l(u_i,\tilde{S}_{ij}) + \frac{1}{2}\tau^l(u_i,u_i)\tilde{u}_j + \frac{1}{2}\tau^l(u_i,u_i,u_j)\right) \tag{5.264}$$

$$- \tau^l(u_i, u_j)\tilde{S}_{ij} - 2\nu\tau^l(S_{ij}, S_{ij}) + \tau^l(u_i, f_i) \tag{5.265}$$

It can easily be seen that positive value of $\Sigma = -\tau_{ij}^{\mathrm{SGS}}\tilde{S}_{ij}$ represents direct energy cascade across scale l. Since $\tilde{S}_{ij}$ is symmetric, Σ can be evaluated as

$$\Sigma = -\check{\tau}_{ij}^{\mathrm{SGS}}\tilde{S}_{ij},\ \check{\tau}_{ij}^{\mathrm{SGS}} = \tau_{ij}^{\mathrm{SGS}} - \frac{1}{3}\tau_{kk}^{\mathrm{SGS}}\delta_{ij}. \tag{5.266}$$

Similarly, the equation for the small-scale (sub-filter scale) motions, often overlooked, can be derived as

$$\frac{\partial u_i''}{\partial x_j} = 0 \tag{5.267}$$

$$\frac{\partial u_i''}{\partial t} + \tilde{u}_j\frac{\partial u_i''}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p''}{\partial x_i} + \frac{\partial}{\partial x_j}(\tau_{ij}^{\mathrm{SGS}} - \tilde{u}_i u_j'' - u_i''u_j'') + \frac{\partial}{\partial x_j}(2\nu S_{ij}'') + f_i'' \tag{5.268}$$

and its energy equation can be obtained by multiplying (5.268) with u_i'':

$$\left(\frac{\partial}{\partial t} + \tilde{u}_j\frac{\partial}{\partial x_j}\right)\frac{1}{2}u_i''u_i'' = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}p''u_j'' - u_i''\tau_{ij}^{\mathrm{SGS}} + \frac{1}{2}u_i''u_i''u_j'' - 2\nu u_i''S_{ij}''\right) \tag{5.269}$$

$$- u_i''u_j''\frac{\partial\tilde{u}_i}{\partial x_j} + \Sigma'' - 2\nu S_{ij}''S_{ij}'' + u_i''f_i'' \tag{5.270}$$

where $\Sigma'' = -\tau_{ij}^{\mathrm{SGS}}S_{ij}''$ and both itself and $-u_i''u_j''\partial_j\tilde{u}_i$ are responsible for inter-scale energy transfer, from the small scales' perspective. It can also be alternatively written as

$$\left(\frac{\partial}{\partial t} + \tilde{u}_j\frac{\partial}{\partial x_j}\right)\frac{1}{2}u_i''u_i'' = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}p''u_j'' + \frac{1}{2}u_i''u_i''u_j'' - 2\nu u_i''S_{ij}''\right) \tag{5.271}$$

$$- u_i''u_j''\tilde{S}_{ij} + u_j''\frac{\partial\tau_{ij}^{\mathrm{SGS}}}{\partial x_j} - 2\nu S_{ij}''S_{ij}'' + u_i''f_i'' \tag{5.272}$$

where $T = -u_i''u_j''\tilde{S}_{ij} + u_j''\partial_j\tau_{ij}^{\mathrm{SGS}}$ is interpreted as the transfer term (Cardesa & Lozano-Durán, 2019).

Moreover, since $\widetilde{u''} \neq 0$ in general, there is a part of energy residing in the correlated/cross term such that

$$\frac{1}{2}u_iu_i = \frac{1}{2}(\tilde{u}_i + u_i'')(\tilde{u}_i + u_i'') = \frac{1}{2}\tilde{u}_i\tilde{u}_i + \frac{1}{2}u_i''u_i'' + \tilde{u}_iu_i''. \tag{5.273}$$

The equation of the cross-scale-energy can be similarly derived as

$$\left(\frac{\partial}{\partial t} + \tilde{u}_j\frac{\partial}{\partial x_j}\right)\tilde{u}_iu_i'' = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\tilde{p}u_i'' + p''\tilde{u}_i - (\tilde{u}_j - u_j'')\tau_{ij}^{\mathrm{SGS}} + \frac{1}{2}\tilde{u}_i\tilde{u}_iu_j'' - 2\nu\tilde{u}_iS_{ij}'' - 2\nu u_i''\tilde{S}_{ij}\right) \tag{5.274}$$

$$- \tilde{u}_i\frac{\partial u_i''u_j''}{\partial x_j} + (\Sigma - \Sigma'') - 4\nu\tilde{S}_{ij}S_{ij}'' \tag{5.275}$$

When all three energy equations are added together, we can see the following

- Σ, Σ'' don't contribute to the total energy $1/2u_iu_i$ but just redistributes among components.
- The total dissipation equals to

$$-2\nu\tilde{S}_{ij}\tilde{S}_{ij} - 2\nu S_{ij}''S_{ij}'' - 4\nu\tilde{S}_{ij}S_{ij}'' = -2\nu S_{ij}S_{ij} \tag{5.276}$$

5.2.4 Smagorinsky and Dynamic models

In an eddy-viscosity formulation, $\tau_{ij}^{\mathrm{SGS}} - 1/3\tau_{kk}^{\mathrm{SGS}}\delta_{ij} = -2\nu_T\tilde{S}_{ij}$, the SGS 'dissipation' $\Sigma = 2\nu_T\tilde{S}_{ij}\tilde{S}_{ij}$ is always positive. But in reality, there is typically backscatter (negative Σ). Typical eddy-viscodity models include the Smagorinsky model, $\nu_T = L_c\sqrt{2\tilde{S}_{ij}\tilde{S}_{ij}}$, where $L_c = \min(\kappa d, C_s\sqrt[3]{\delta_x\Delta_y\Delta_z})$ is the characteristic length scale, κ is the Karman constant, and d is the distance from the wall. The model coefficient can be a constant (in static Smagorinsky) or dependent on the large-scale flow (in dynamics Smagorinsky).

Germano *et al.* (1991)

5.2.5 Triple decomposition TKE equation

Similar to the Reynolds decomposition, the flow variables can be triply decomposed as

$$[\boldsymbol{u}(\boldsymbol{x},t), p(\boldsymbol{x},t)] = [\bar{\boldsymbol{u}}(\boldsymbol{x}) + \bar{p}(\boldsymbol{x})] + [\boldsymbol{u}'(\boldsymbol{x},t), p'(\boldsymbol{x},t)] \tag{5.277}$$

$$= [\bar{\boldsymbol{u}}(\boldsymbol{x}) + \bar{p}(\boldsymbol{x})] + [\tilde{\boldsymbol{u}}(\boldsymbol{x},t) + \tilde{p}(\boldsymbol{x},t)] + [\boldsymbol{u}''(\boldsymbol{x},t), p''(\boldsymbol{x},t)] \tag{5.278}$$

where tilded variables denote the so-called 'coherent' part of the fluctuation and double-primed variables are the residual. The 'coherent' part could be conditionally averaged unsteady flow (conditioned to spatial or temporal reference phase), called phase-relevant (Hussain & Reynolds, 1970), or by other definitions. The governing equation for the kinetic energy associated with each part can be similarly established.

Let us start from the decomposition of the kinetic energy into the mean (MKE), the large-scale/coherent (LKE), and the small-scale/residual (SKE) parts:

$$k = \frac{1}{2}\overline{u_i u_i} \triangleq k_M + k_L + k_S \tag{5.279}$$

$$= \frac{1}{2}\bar{u}_i\bar{u}_i + \frac{1}{2}\overline{\tilde{u}_i\tilde{u}_i} + \frac{1}{2}\overline{u_i''u_i''}, \tag{5.280}$$

where the TKE $k_T = k_L + k_S$. In order to eliminate cross kinetic energy (CKE), it requires that

$$\overline{\bar{u}_i\tilde{u}_i} = \overline{\bar{u}_i u_i''} = \overline{\tilde{u}_i u_i''} = 0. \tag{5.281}$$

That being said, the three parts of the flow need to be mutually uncorrelated for successful triple decomposition. Think of the case of turbulence superposed on a perturbation wave in Hussain & Reynolds (1970), the turbulent motion and the 'wave' motion are uncorrelated. But we will see this is not strictly necessary since the CKE equation can be derived and its role is like a bridge between LKE and SKE. The ease of this strict requirement will enable more filter choices.

Here we consider a triple decomposition roughly through cut-off filtering in Fourier space where the average $\overline{(\cdot)}$ denotes averaging over all homogeneous directions and ensembles (the zero Fourier mode), and the filtering is performed in all Fourier directions.

Denote the filtering operation as

$$\widetilde{u_i'} = \tilde{u}_i,\ u_i'' = u_i' - \tilde{u}_i \tag{5.282}$$

and in the case of a sharp Fourier filter (not all filters have these properties),

$$\tilde{\tilde{u}}_i = \tilde{u}_i,\ \widetilde{u_i''} = 0 \tag{5.283}$$

and also

$$\overline{\tilde{u}_i} = \overline{u_i''} = 0,\ \tilde{\bar{u}}_i = \bar{u}_i, \tag{5.284}$$

The triple decomposition here can generally be into three non-overlapping scales: mean, large, and small scales. The first one need not necessarily be the mean flow.

By taking the mean and the Fourier cutoff of the continuity equation we have (up to commutation error)

$$\frac{\partial \bar{u}_i}{\partial x_i} = \frac{\partial u_i'}{\partial x_i} = \frac{\partial \tilde{u}_i}{\partial x_i} = \frac{\partial u_i''}{\partial x_i} = 0. \tag{5.285}$$

The equation for the mean field is almost the same as the RANS equation except that the fluctuations are decomposed as

$$\overline{u_i' u_j'} = \overline{(\tilde{u}_i + u_i'')(\tilde{u}_j + u_j'')} = \overline{\tilde{u}_i \tilde{u}_j} + \overline{u_i'' u_j''}. \tag{5.286}$$

The RANS equation can be re-written in an LES-consistent form as

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} - \frac{\partial}{\partial x_j}(\overline{u_i u_j} - \bar{u}_i \bar{u}_j) + \frac{\partial}{\partial x_j}(2\nu \bar{S}_{ij}) + \bar{f}_i, \tag{5.287}$$

or, in a more familiar form

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} - \frac{\partial}{\partial x_j}(\overline{u_i' u_j'}) + \frac{\partial}{\partial x_j}(2\nu \bar{S}_{ij}) + \bar{f}_i, \tag{5.288}$$

where the Reynolds stress can be interpreted as the sub-filter residual stress where $\overline{(\cdot)}$ is the first filtering operation:

$$\tau_{ij}^{\text{Rey}} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j = \overline{u_i' u_j'}. \tag{5.289}$$

The fluctuation equation can be obtained by subtracting the RANS equation from the total momentum equation:

$$\frac{\partial u_i'}{\partial t} + \bar{u}_j \frac{\partial u_i'}{\partial x_j} + u_j' \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p'}{\partial x_i} - \frac{\partial}{\partial x_j}\left(u_i' u_j' - \overline{u_i' u_j'}\right) + \frac{\partial}{\partial x_j}(2\nu S_{ij}') + f_i'. \tag{5.290}$$

The large-scale fluctuation equation can be obtained by filtering the fluctuation equation (denote $\tilde{u}_i = \widetilde{(u_i')}$ as the filtered velocity and $u_i'' = u_i' - \tilde{u}_i$ as its residual; also assume $\tilde{\bar{u}}_i = \bar{u}_i$), leading to

$$\frac{\partial \tilde{u}_i}{\partial t} + \bar{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} + \tilde{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \tilde{p}}{\partial x_i} - \frac{\partial}{\partial x_j}\left(\widetilde{u_i' u_j'} - \overline{u_i' u_j'}\right) + \frac{\partial}{\partial x_j}(2\nu \tilde{S}_{ij}) + \tilde{f}_i \tag{5.291}$$

It can also be written as

$$\frac{\partial \tilde{u}_i}{\partial t} + \bar{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} + \tilde{u}_j \frac{\partial \bar{u}_i}{\partial x_j} + \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \tilde{p}}{\partial x_i} - \frac{\partial}{\partial x_j}\left((\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j) - \overline{u_i' u_j'}\right) + \frac{\partial}{\partial x_j}(2\nu \tilde{S}_{ij}) + \tilde{f}_i \tag{5.292}$$

for the needs of deriving the energy equation.

If $\widetilde{(\cdot)}$ is the phase average used in Hussain & Reynolds (1970), due to

$$\widetilde{u_i' u_j'} = \tilde{u}_i \tilde{u}_j + \widetilde{u_i'' u_j''} \tag{5.293}$$

we will recover the large-scale equation there. However, with the filtering operator,

$$\widetilde{u_i' u_j'} = \widetilde{\tilde{u}_i \tilde{u}_j} + \widetilde{u_i'' u_j''} + \widetilde{u_i'' \tilde{u}_j} + \widetilde{\tilde{u}_i u_j''}, \tag{5.294}$$

there will be additional terms.

The small-scale equation is obtained by subtracting the large-scale equation from the fluctuation equation:

$$\frac{\partial u_i''}{\partial t} + \bar{u}_j \frac{\partial u_i''}{\partial x_j} + u_j'' \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p''}{\partial x_i} - \frac{\partial}{\partial x_j}(u_i' u_j' - \widetilde{u_i' u_j'}) + \frac{\partial}{\partial x_j}(2\nu S_{ij}'') + f_i'' \tag{5.295}$$

$$\frac{\partial u_i''}{\partial t} + \bar{u}_j \frac{\partial u_i''}{\partial x_j} + u_j'' \frac{\partial \bar{u}_i}{\partial x_j} + \tilde{u}_j \frac{\partial u_i''}{\partial x_j} + u_j'' \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p''}{\partial x_i} - \frac{\partial}{\partial x_j}(u_i'' u_j'' + \tilde{u}_i \tilde{u}_j - \widetilde{u_i' u_j'}) + \frac{\partial}{\partial x_j}(2\nu S_{ij}'') + f_i'' \tag{5.296}$$

$$\frac{\partial u_i''}{\partial t} + \bar{u}_j \frac{\partial u_i''}{\partial x_j} + u_j'' \frac{\partial \bar{u}_i}{\partial x_j} + \tilde{u}_j \frac{\partial u_i''}{\partial x_j} + u_j'' \frac{\partial \tilde{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p''}{\partial x_i} - \frac{\partial}{\partial x_j}(u_i'' u_j'' - \tau_{ij}^{\mathrm{SGS}}) + \frac{\partial}{\partial x_j}(2\nu S_{ij}'') + f_i'' \tag{5.297}$$

$$\tau_{ij}^{\mathrm{SGS}} = \widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j \tag{5.298}$$

which is still consistent with the small-scale equation in Hussain & Reynolds (1970) if (5.293) were right. It also takes the same form as the u_i' equation whose respective large-scale is $\bar{u}_i$ – the stresses include the self-interaction stresses and the sub-filter stresses (that couples the large and the small scales).

Multiply respective equations with respective velocity components, the scale-divided kinetic energy equations (MKE, TKE, LKE, and SKE) can be derived as

$$\left(\frac{\partial}{\partial t} + \bar{u}_j \frac{\partial}{\partial x_j}\right)\frac{\bar{u}_i \bar{u}_i}{2} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\bar{p}\bar{u}_j + \bar{u}_i \overline{u_i' u_j'} - 2\nu \bar{u}_i \bar{S}_{ij}\right) + \overline{u_i' u_j'}\frac{\partial \bar{u}_i}{\partial x_j} - 2\nu \bar{S}_{ij}\bar{S}_{ij} + \bar{u}_i \bar{f}_i$$

$$\left(\frac{\partial}{\partial t} + \bar{u}_j \frac{\partial}{\partial x_j}\right)\frac{\overline{u_i' u_i'}}{2} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\overline{p' u_j'} + \frac{1}{2}\overline{u_i' u_i' u_j'} - 2\nu \overline{u_i' S_{ij}'}\right) - \overline{u_i' u_j'}\frac{\partial \bar{u}_i}{\partial x_j} - 2\nu \overline{S_{ij}' S_{ij}'} + \overline{f_i' u_i'}$$

$$\left(\frac{\partial}{\partial t} + \bar{u}_j \frac{\partial}{\partial x_j}\right)\frac{\overline{\tilde{u}_i \tilde{u}_i}}{2} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\overline{\tilde{p}\tilde{u}_j} + \frac{1}{2}\overline{\tilde{u}_i \tilde{u}_i \tilde{u}_j} + \overline{\tilde{u}_i(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j)} - 2\nu \overline{\tilde{u}_i \tilde{S}_{ij}}\right) - 2\nu \tilde{S}_{ij}\tilde{S}_{ij} + \tilde{u}_i \tilde{f}_i$$
$$- \overline{\tilde{u}_i \tilde{u}_j}\frac{\partial \bar{u}_i}{\partial x_j} + \overline{(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j)\frac{\partial \tilde{u}_i}{\partial x_j}}$$

$$\left(\frac{\partial}{\partial t} + \bar{u}_j \frac{\partial}{\partial x_j}\right)\frac{\overline{u_i'' u_i''}}{2} = -\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\overline{p'' u_j''} + \frac{1}{2}\overline{(\tilde{u}_j + u_j'')u_i'' u_i''} - \overline{u_i''(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j)} - 2\nu \overline{u_i'' S_{ij}''}\right)$$
$$- \overline{u_i'' u_j''}\frac{\partial \bar{u}_i}{\partial x_j} - \overline{u_i'' u_j''\frac{\partial \tilde{u}_i}{\partial x_j}} - \overline{(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j)\frac{\partial u_i''}{\partial x_j}} - 2\nu \overline{S_{ij}'' S_{ij}''} + \overline{u_i'' f_i''}$$

Additionally, even though in the present formulation $\overline{u_i'' \tilde{u}_i} = 0$ due to the fact that the large scale and small scale fluctuations are uncorrelated, its transport equation can still be useful serving as a bridge in the large-small interaction / transfer. Also, it will be useful when filters other than Fourier sharp filter is used.

The cross kinetic energy (CKE) is:

$$\left(\frac{\partial}{\partial t}+\bar{u}_j\frac{\partial}{\partial x_j}\right)\frac{\overline{u_i''\tilde{u}_i}}{2}+\overline{(u_i''\tilde{u}_j+u_j''\tilde{u}_i)}\frac{\partial\bar{u}_i}{\partial x_j}$$
$$=-\frac{\partial}{\partial x_j}\left(\frac{1}{\rho}\overline{(u_j''\tilde{p}+\tilde{u}_jp'')}+\overline{\tilde{u}_iu_i''u_j''}+\overline{u_i''\tilde{u}_i\tilde{u}_j}+\frac{1}{2}\overline{\tilde{u}_i\tilde{u}_iu_j''}+\overline{(u_i''-\tilde{u}_i)(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)}-2\nu\overline{(u_i''\tilde{S}_{ij}+\tilde{u}_iS_{ij}'')}\right)$$
$$+\overline{u_i''u_j''\frac{\partial\tilde{u}_i}{\partial x_j}}+\overline{(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)\frac{\partial u_i''}{\partial x_j}}-\overline{(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)\frac{\partial\tilde{u}_i}{\partial x_j}}-4\nu\overline{S_{ij}''\tilde{S}_{ij}}+\overline{(u_i''\tilde{f}_i+\tilde{u}_if_i'')}$$

and in the case of $\overline{\tilde{a}b''}=\overline{a''\tilde{b}}=0$ for any a',b', it simplifies to the following auxiliary equation

$$0=-\frac{\partial}{\partial x_j}(\overline{\tilde{u}_iu_i''u_j''}+\overline{u_i''\tilde{u}_i\tilde{u}_j}+\frac{1}{2}\overline{\tilde{u}_i\tilde{u}_iu_j''}+\overline{(u_i''-\tilde{u}_i)(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)}) \tag{5.299}$$
$$+\overline{u_i''u_j''\frac{\partial\tilde{u}_i}{\partial x_j}}+\overline{(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)\frac{\partial u_i''}{\partial x_j}}-\overline{(\widetilde{u_i'u_j'}-\tilde{u}_i\tilde{u}_j)\frac{\partial\tilde{u}_i}{\partial x_j}} \tag{5.300}$$

Comments:

- Since we are in a RANS framework, the material derivative is associated the mean flow (sitting on a reference frame moving with the mean).
- In the MKE equation,

$$-\frac{\partial}{\partial x_j}(\bar{u}_i\overline{u_i'u_j'}-2\nu\bar{u}_i\bar{S}_{ij})=\frac{\partial}{\partial x_j}[(-\tau_{ij}^{\text{Rey}}+\tau_{ij}^{\text{vis}})\bar{u}_i] \tag{5.301}$$

  represents the turbulent and molecular diffusion. Both are spatial transports. Note that for consistency with SGS stresses, the Reynolds stress ($\tau_{ij}^{\text{Rey}}=\overline{u_i'u_j'}$) has an opposite sign to its conventional definition. We also note $\overline{(\cdot)}$ does not necessarily need to be the Reynolds average, but can be any filtering operation that has the same correlation/commulation properties with $\widetilde{(\cdot)}$.

- If we add LKE, SKE and CKE together, we will exactly recover TKE. The budgets are closed and the triple decomposition is complete. With some algebra it can be shown that

$$\frac{1}{2}\overline{u_i'u_i'u_j'}=\frac{1}{2}\overline{\tilde{u}_i\tilde{u}_i\tilde{u}_j}+\frac{1}{2}\overline{u_i''u_i''u_j''}+\frac{1}{2}\overline{\tilde{u}_ju_i''u_i''}+\overline{u_i''\tilde{u}_i\tilde{u}_j}+\overline{\tilde{u}_iu_i''u_j''}+\frac{1}{2}\overline{\tilde{u}_i\tilde{u}_iu_j''}, \tag{5.302}$$

  relating the fluctuation-related spatial transport terms (∂_j) in those equations.

- The sink in MKE equation is $-\bar{\Sigma}$, where

$$\bar{\Sigma}=-\overline{u_i'u_j'}\frac{\partial\bar{u}}{\partial x_j}=-\tau_{ij}^{\text{Rey}}\bar{S}_{ij} \tag{5.303}$$

  is the source in the TKE equation and can be splitted into

$$\bar{\Sigma}=-\overline{u_i'u_j'}\frac{\partial\bar{u}_i}{\partial x_j}=-\overline{\tilde{u}_i\tilde{u}_j}\frac{\partial\bar{u}_i}{\partial x_j}-\overline{u_i''u_j''}\frac{\partial\bar{u}_i}{\partial x_j}=\tilde{\Sigma}+\Sigma'' \tag{5.304}$$

$$\tilde{\Sigma} = -\overline{\tilde{u}_i \tilde{u}_j} \frac{\partial \bar{u}_i}{\partial x_j} \tag{5.305}$$

$$\Sigma'' = -\overline{u_i'' u_j''} \frac{\partial \bar{u}_i}{\partial x_j} \tag{5.306}$$

with positive $\bar{\Sigma}$ indicating energy being extracted from the mean flow supplied to the LS and SS fluctuations (sources in LKE and SKE equations).

- The inter-scale transfer term, from the large scales' perspective is $-\Sigma^{\mathrm{SGS}}$, where

$$\Sigma^{\mathrm{SGS}} = -\overline{(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j) \frac{\partial \tilde{u}_i}{\partial x_j}} = -\overline{\tau_{ij}^{\mathrm{SGS}} \tilde{S}_{ij}}, \tag{5.307}$$

with positive Σ^{SGS} indicating forward energy cascade. However, from the small scales' perspective, the transfer term is

$$-\overline{u_i'' u_j'' \frac{\partial \tilde{u}_i}{\partial x_j}} - \overline{(\widetilde{u_i' u_j'} - \tilde{u}_i \tilde{u}_j) \frac{\partial u_i''}{\partial x_j}}, \tag{5.308}$$

which can be shown equal to Σ^{SGS} from the auxiliary equation (CKE). Even though the CKE is eventually zero statistically, the cross-correlation between large and small scale represents the way that energy is transferred: the energy is drained from the LKE, transferred to the CKE, and then from CKE to SKE. CKE is like a catylist who doesn't change at the end, but serves a bridging role. That being said, the LS and SS are coupled during some periods of their lifes.

5.3 Theory of homogeneous isotropic turbulence

5.3.1 Correlation functions

Two-point correlations are defined in a general form

$$R_{ij}(\boldsymbol{x}, \boldsymbol{r}, t) = \overline{u_i(\boldsymbol{x} + \boldsymbol{r}, t) u_j(\boldsymbol{x}, t)}. \tag{5.309}$$

When $\boldsymbol{r} = 0$, it reduces to the covariance $\overline{u_i u_j}(\boldsymbol{x}, t)$. The auto-correlation is, for example, R_{uu}. The continuity of u_i requires that

$$\frac{\partial R_{ij}}{\partial r_i} = \overline{\partial_{r_i} u_i(\boldsymbol{x} + \boldsymbol{r}, t) u_j(\boldsymbol{x}, t)} = \overline{\partial_{x_i + r_i} u_i(\boldsymbol{x} + \boldsymbol{r}, t) u_j(\boldsymbol{x}, t)} = 0. \tag{5.310}$$

The symmetries of R_{ij} are that

$$R_{ji}(\boldsymbol{x}, \boldsymbol{r}, t) = R_{ij}(\boldsymbol{x}', -\boldsymbol{r}, t) \tag{5.311}$$

which reduces to

$$R_{ji}(\boldsymbol{r}, t) = R_{ij}(-\boldsymbol{r}, t) \tag{5.312}$$

when homogeneous. Then,

$$\frac{\partial R_{ij}}{\partial r_i}(\boldsymbol{r}, t) = \frac{\partial R_{ji}}{\partial r_i}(-\boldsymbol{r}, t) = \frac{\partial R_{ij}}{\partial r_j}(-\boldsymbol{r}, t) = -\frac{\partial R_{ij}}{\partial r_j}(\boldsymbol{r}, t) = 0. \tag{5.313}$$

When turbulence is isotropic, $R_{ij}(r,t)$ is an isotropic second-order tensors that depends only on r which must be in the form

$$R_{ij}(r,t) = A(r,t)\delta_{ij} + B(r,t)r_i r_j. \tag{5.314}$$

It only depends on the magnitude of r and whether you are looking parallel or perpendicular to $\boldsymbol{r}$, we need two pieces of information which are the two scalar functions.

The longitudinal and transverse correlations are

$$R_{11}(r,t) = \overline{u_1(\boldsymbol{x}+\boldsymbol{e}_1 r,t)u_1(\boldsymbol{x},t)} \tag{5.315}$$

$$R_{22}(r,t) = \overline{u_2(\boldsymbol{x}+\boldsymbol{e}_1 r,t)u_2(\boldsymbol{x},t)} \tag{5.316}$$

Take $\boldsymbol{r} = (r,0,0)$,

$$R_{11}(r,t) = A(r,t) + B(r,t)r^2 = u'^2(t)f(r,t) \tag{5.317}$$

$$R_{22}(r,t) = R_{33}(r,t) = A(r,t) = u'^2(t)g(r,t) \tag{5.318}$$

where the non-dimensionalized correlations are

$$f(r,t) = \frac{R_{11}(r,t)}{u'^2(t)} \tag{5.319}$$

$$g(r,t) = \frac{R_{22}(r,t)}{u'^2(t)} \tag{5.320}$$

such that $f(0,t) = g(0,t) = 1$. Note that $\overline{u_1'^2} = \overline{u_2'^2} = \overline{u_3'^2} = u'^2$ due to isotropy.

It can be solved that

$$A = u'^2 g,\ B = \frac{u'^2}{r^2}(f-g) \tag{5.321}$$

and hence

$$R_{ij}(r,t) = u'^2\left[g\delta_{ij} + (f-g)\frac{r_i r_j}{r^2}\right]. \tag{5.322}$$

Using the continuity (5.313) $\partial_{r_j} R_{ij} = 0$ it can be shown that

$$g(r,t) = f(r,t) + \frac{r}{2}\frac{\partial f(r,t)}{\partial r}, \tag{5.323}$$

where

$$\frac{\partial r}{\partial r_i} = \frac{r_i}{r} \tag{5.324}$$

$$\frac{\partial}{\partial r_j}\left(\frac{r_i r_j}{r^2}\right) = \frac{\partial}{\partial r_j}(r_i r_j)\frac{1}{r^2} - r_i r_j \frac{2}{r^3}\frac{r_j}{r} = \frac{2r_i}{r^2} \tag{5.325}$$

$$\frac{\partial r_i}{\partial r_j} = \delta_{ij},\ \frac{\partial r_j}{\partial r_j} = 3 \tag{5.326}$$

are used. This relation implies that the two-point correlation tensor is solely determined by the longitudinal correlation. It can further be written as

$$g(r,t) = \frac{1}{2}\left(f(r,t) + \frac{\partial}{\partial r}[rf(r,t)]\right) \tag{5.327}$$

and that

$$\int_0^\infty f(r,t)\mathrm{d}r = 2\int_0^\infty g(r,t)\mathrm{d}r \tag{5.328}$$

such that $L_{11} = 2L_{22}$. The integral length scales are not the only scaled that is determined by the correlation functions. So is the Taylor microscale.

The derivatives of the correlation functions at $r = 0$ are related to the dissipation.

$$-u'^2 f''(0) = -u'^2 \lim_{r\to 0} \frac{\partial^2}{\partial r^2} f(r) \tag{5.329}$$

$$= -\lim_{r\to 0} \frac{\partial^2}{\partial r^2} \overline{u_1(\boldsymbol{x}+\boldsymbol{e}_1 r)u_1(\boldsymbol{x})} \tag{5.330}$$

$$= -\lim_{r\to 0} \overline{\left(\frac{\partial^2 u_1}{\partial x_1^2}\right)_{\boldsymbol{x}+\boldsymbol{e}_1 r} u_1(\boldsymbol{x})} \tag{5.331}$$

$$= -\overline{\frac{\partial^2 u_1}{\partial x_1^2} u_1} \tag{5.332}$$

$$= \overline{\left(\frac{\partial u_1}{\partial x_1}\right)^2} \tag{5.333}$$

Similarly,

$$-u'^2 g''(0) = \overline{\left(\frac{\partial u_1}{\partial x_2}\right)^2}. \tag{5.334}$$

Hence, we have proven

$$\overline{\left(\frac{\partial u_1}{\partial x_2}\right)^2} = 2\overline{\left(\frac{\partial u_1}{\partial x_1}\right)^2} \tag{5.335}$$

in isotropic turbulence and

$$\varepsilon = \nu\overline{\left(\frac{\partial u_i}{\partial x_j}\right)^2} = 15\nu\overline{\left(\frac{\partial u_1}{\partial x_1}\right)^2} = 15\nu\frac{u'^2}{\lambda_g^2}. \tag{5.336}$$

The above equation is more commonly used to obtain the Taylor length scale instead of the second derivative at $r = 0$ (which is more noisy).

5.3.2 Structure function

The structure functions describe the statistics of velocity increments across scales. The second-order structure function tensor is

$$D_{ij}(\boldsymbol{x},\boldsymbol{r},t) = \overline{[u_i(\boldsymbol{x}+\boldsymbol{r},t) - u_i(\boldsymbol{x},t)][u_j(\boldsymbol{x}+\boldsymbol{r},t) - u_j(\boldsymbol{x},t)]} = \overline{\delta u_i(\boldsymbol{x},\boldsymbol{r},t)\delta u_j(\boldsymbol{x},\boldsymbol{r},t)}, \tag{5.337}$$

and third-order structure functions can be similarly defined. Similar to the correlation function, in isotropic turbulence the second-order structure function can be expressed in terms of the longitudinal (D_{LL}) and the transverse (D_{NN}) ones:

$$D_{ij}(\boldsymbol{r},t) = D_{NN}(r,t)\delta_{ij} + [D_{LL}(r,t) - D_{NN}(r,t)]\frac{r_i r_j}{r^2}. \tag{5.338}$$

The structure function tensor has an expression of continuity similar to (5.313) that

$$\frac{\partial D_{ij}(\boldsymbol{r},t)}{\partial r_i}=0, \tag{5.339}$$

leading to

$$\frac{\partial D_{ij}}{\partial r_i}=\frac{r_j}{r^2}\left(r\frac{\partial D_{LL}}{\partial r}+2(D_{LL}-D_{NN})\right)=0 \tag{5.340}$$

and hence

$$D_{NN}(r,t)=D_{LL}(r,t)+\frac{r}{2}\frac{\partial D_{LL}(r,t)}{\partial r}, \tag{5.341}$$

similar to (5.323) that implies D_{ij} is solely determined by D_{LL} (or D_{NN}).

Short proof of (5.339). In homogeneous turbulence, it can be shown

$$D_{ij}(r,t)=2R_{ij}(0)-R_{ij}(r)-R_{ji}(r)=2R_{ij}(0)-R_{ij}(r)-R_{ij}(-r) \tag{5.342}$$

according to the definitions. By $\partial_{r_i}R_{ij}=0$ we also have $\partial_{r_i}D_{ij}=0$.

We note that the structure functions are not independent of correlation functions:

$$D_{LL}(r)=2u'^2(1-f(r)) \tag{5.343}$$

$$D_{NN}(r)=2u'^2(1-g(r)) \tag{5.344}$$

In the inertial range, the structure function only depends on ε and r, leading to the non-dimensionlization

$$D_{LL}=C_2(\varepsilon r)^{2/3} \tag{5.345}$$

where typically $C_2=2.0$ from data. Note the dimension of the structure function is energy, the above equation is equivalent to the more commonly used scaling of the energy spectrum (energy per unit wavenumber)

$$E(k)=C\varepsilon^{2/3}k^{-5/3} \tag{5.346}$$

in isotropic turbulence.

5.3.3 The Kármán–Horwarth equation and the Kolmorogov equation

The evolution equation of the correlation functions is

$$\frac{\partial}{\partial t}(u'^2 f)-\frac{u'^3}{r^4}\frac{\partial}{\partial r}(r^4 S_{11,1})=\frac{2\nu u'^2}{r^4}\frac{\partial}{\partial r}\left(r^4\frac{\partial f}{\partial r}\right), \tag{5.347}$$

which is called the Kármán–Horwarth equation. Here

$$S_{ij,k}(\boldsymbol{r},t)=\frac{\overline{u_i(\boldsymbol{x},t)u_j(\boldsymbol{x},t)u_k(\boldsymbol{x}+\boldsymbol{r},t)}}{u'^3} \tag{5.348}$$

is the two-point triple correlation, that represents an inertial process (cascade). In isotropic turbulence, it is related to the third-order structure function as

$$u'^3 S_{11,1}(r,t)=\overline{u_1^2(\boldsymbol{x},t)u_1(\boldsymbol{x}+\boldsymbol{e}_1 r,t)}=\frac{1}{6}D_{LLL}(r,t). \tag{5.349}$$

The equation for f is subject to closure. In the limit $r = 0$, the above equation reduces to the TKE equation

$$\frac{\partial u'^2}{\partial t} = -\frac{2}{3}\varepsilon, \tag{5.350}$$

with $S_{11,1(0)} = 0$ and $f(0) = 1$.

At the small enough scales where the inertial effect is dominated by the viscous effect, or at low Re, the K–H equation reduces to

$$\frac{\partial}{\partial t}(u'^2 f) = \frac{2\nu u'^2}{r^4}\frac{\partial}{\partial r}\left(r^4\frac{\partial f}{\partial r}\right), \tag{5.351}$$

which has the solution form of Gaussian heat kernel

$$f(r,t) = \exp\left(-\frac{r^2}{8\nu t}\right), \tag{5.352}$$

since the Green's function of ∇^2 is Gaussian. This is a low-Re decay and is not to be confused with the (nearly) exponential tail of $E(k)$.

The Kármán–Horwarth equation can also be written in terms of the structure functions

$$\frac{\partial D_{LL}}{\partial t} + \frac{1}{3r^4}\frac{\partial}{\partial r}(r^4 D_{LLL}) = \frac{2\nu}{r^4}\frac{\partial}{\partial r}\left(r^4\frac{\partial D_{LL}}{\partial r}\right) - \frac{4}{3}\varepsilon, \tag{5.353}$$

which can be integrated to the Kolmogorov equation

$$\frac{3}{r^4}\int_0^r s^4\frac{\partial}{\partial t}D_{LL}(s,t)\,\mathrm{d}s = 6\nu\frac{\partial D_{LL}}{\partial r} - D_{LLL} - \frac{4}{5}\varepsilon r. \tag{5.354}$$

This is the closest point that the -5/3 or the -2/3 laws can be obtained from the Navier–Stokes equations. In statistically stationary flows, after re-writing $D_{LL} = \overline{(\delta u)^2}$ and $D_{LLL} = \overline{(\delta u)^3}$, the Kolmogorov equation becomes

$$-\overline{(\delta u)^3} + 6\nu\frac{d}{dr}\overline{(\delta u)^2} = \frac{4}{5}\varepsilon r, \tag{5.355}$$

where $\delta u = u(x+r) - u(x)$. In the limit of vanishing viscosity,

$$D_{LLL}(r) = -\frac{4}{5}\varepsilon r, \tag{5.356}$$

which is known as the Kolmorogov -4/5 law. If further the structure-function skewness is constant,

$$S' = \frac{D_{LLL}}{D_{LL}^{3/2}}, \tag{5.357}$$

the second-order structure function must follow

$$D_{LL}(r) = C_2(\varepsilon r)^{2/3}, \tag{5.358}$$

which is consistent with $E(k) = C\varepsilon^{2/3}k^{-5/3}$ by taking $k = 2\pi/r$. Such an equilibrium in (5.356) is called the Ricardson–Kolmogorov equilibirum.

The spectral equivalence of the Kármán–Howarth equation,

$$\frac{\partial E(\kappa,t)}{\partial t} = \mathcal{P}(\kappa,t) + \mathcal{T}(\kappa,t) - 2\nu\kappa^2 E(\kappa,t), \tag{5.359}$$

represents the energy evolution in homogeneous isotropic turbulence, where $\mathcal{P}$ is the production spectrum, $\mathcal{T}$ is the transfer spectrum, and $-2\nu\kappa^2 E$ is the dissipation spectrum. Since $\int_0^\infty \mathcal{T}(\kappa,t)\,\mathrm{d}\kappa = 0$, the transfer spectrum is related to the inter-scale transfer fluxes defined via a filtering approach as $\Pi_l = \int_0^{\kappa_{\max}} \mathcal{T}(\kappa,t)\,\mathrm{d}\kappa = -\int_{\kappa_{\max}}^\infty \mathcal{T}(\kappa,t)\,\mathrm{d}\kappa = -\Pi_s$, where $\kappa_{\max}$ is the wavenumber corresponding to the largest Π and is in the inertial range. At sufficiently high Reynolds number, most of the production is in $(0,\kappa_{\max})$ while most of the dissipation is in $(\kappa_{\max},\infty)$. By integrating (5.359) in $(\kappa_{\max},\infty)$ we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{\kappa_{\max}}^\infty E(\kappa,t)\,\mathrm{d}\kappa\right) = \Pi_s(t) + \varepsilon_s(t), \tag{5.360}$$

where $\varepsilon_s \approx \varepsilon$. It can be seen that the instantaneous disequilibrium between $\Pi_s(t)$ and $\varepsilon_s(t)$ is accompanied by non-stationarity at the small scales, represented by the left-hand-side of (5.360).

5.3.4 The generalized Kolmogorov equation, Kármán–Horwarth–Monin–Hill

5.3.5 Energy spectrum and its evolution equation

The velocity and its Fourier pair are

$$u_i(\boldsymbol{x}) = \iiint_{-\infty}^{\infty} \hat{u}_i(\boldsymbol{k}) e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\,\mathrm{d}\boldsymbol{k} \tag{5.361}$$

$$\hat{u}_i(\boldsymbol{k}) = \frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty} u_i(\boldsymbol{x}) e^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x} \tag{5.362}$$

or in the discrete form,

$$u_i(\boldsymbol{x}) = \sum_{\boldsymbol{k}} \hat{u}_i(\boldsymbol{k}) e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} \tag{5.363}$$

$$\hat{u}_i(\boldsymbol{k}) = \frac{1}{(2\pi)^3}\sum_{\boldsymbol{x}} u_i(\boldsymbol{x}) e^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}} \tag{5.364}$$

Note that in this section we choose to put the normalization factor $1/(2\pi)^3$ in front of the forward transform.

The cross-correlation tensor $R_{ij}(\boldsymbol{r}) = \overline{u_i(\boldsymbol{x}+\boldsymbol{r})u_j(\boldsymbol{x})}$ (homogeneous turbulence) and its Fourier pair (the co-spectral tensor) are

$$R_{ij}(\boldsymbol{r}) = \iiint_{-\infty}^{\infty} \Phi_{ij}(\boldsymbol{k}) e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{d}\boldsymbol{k} \tag{5.365}$$

$$\Phi_{ij}(\boldsymbol{k}) = \frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty} R_{ij}(\boldsymbol{r}) e^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{d}\boldsymbol{r} \tag{5.366}$$

Given the ensemble average

$$R_{ij}(\boldsymbol{r}) = \overline{u_i(\boldsymbol{x}+\boldsymbol{r})u_j(\boldsymbol{x})} = \iiint_{-\infty}^{\infty} u_i(\boldsymbol{x}+\boldsymbol{r})u_j(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x}, \tag{5.367}$$

the co-spectral tensor can be written as

$$\Phi_{ij}(\boldsymbol{k}) = \frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty} R_{ij}(\boldsymbol{r})e^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{d}\boldsymbol{r} \tag{5.368}$$

$$= (2\pi)^3\frac{1}{(2\pi)^3}\frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty}\iiint_{-\infty}^{\infty} u_i(\boldsymbol{x}+\boldsymbol{r})u_j(\boldsymbol{x})e^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}\,\mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{r} \tag{5.369}$$

$$= (2\pi)^3\left(\frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty} u_j(\boldsymbol{x})e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\,\mathrm{d}\boldsymbol{x}\right)\left(\frac{1}{(2\pi)^3}\iiint_{-\infty}^{\infty} u_i(\boldsymbol{x}+\boldsymbol{r})e^{-\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{x}+\boldsymbol{r})}\,\mathrm{d}\boldsymbol{r}\right) \tag{5.370}$$

$$= (2\pi)^3\hat{u}_j^*(\boldsymbol{k})\hat{u}_i(\boldsymbol{k}) \tag{5.371}$$

which depends not only on the magnitudes of but the phase difference between $\hat{u}_i$ and $\hat{u}_j$.

Since

$$\overline{u_i(\boldsymbol{x})u_j(\boldsymbol{x})} = R_{ij}(0) = \iiint_{-\infty}^{\infty}\Phi_{ij}(\boldsymbol{k})\,\mathrm{d}\boldsymbol{k}, \tag{5.372}$$

the co-spectral tensor represents scale-by-scale contribution to the Reynolds stresses. Similarly, $\Phi_{ii}(\boldsymbol{k})/2$ to the TKE.

The energy spectrum is the integration of the spectral tensor over spherical shells,

$$E(k) = \oint\frac{1}{2}\Phi_{ii}\,\mathrm{d}S(k) = \iiint\frac{1}{2}\Phi_{ii}\delta(|\boldsymbol{k}|-k)\,\mathrm{d}\boldsymbol{k}, \tag{5.373}$$

where $\oint \mathrm{d}S(k) = 4\pi k^2$. It represents the contribution to the TKE at wavenumbers whose modulus is within $k - dk/2 < |\boldsymbol{k}| < k + dk/2$, such that

$$\frac{1}{2}\overline{u_iu_j} = \int_0^{\infty} E(k)\,\mathrm{d}k. \tag{5.374}$$

In practice, it is easier to measure one-dimensional correlations and spectra. The longitudinal and transverse ones are

$$R_{11}(\boldsymbol{e}_1r_1,t) = \overline{u_1^2}(t)f(r_1,t) \tag{5.375}$$

$$R_{22}(\boldsymbol{e}_1r_1,t) = \overline{u_2^2}(t)g(r_1,t) \tag{5.376}$$

and their Fourier transforms are the 1-D spectra

$$E_{11}(k_1) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{11}(\boldsymbol{e}_1r_1)e^{-\mathrm{i}r_1k_1}\,\mathrm{d}r_1 \tag{5.377}$$

$$E_{22}(k_1) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{22}(\boldsymbol{e}_1r_1)e^{-\mathrm{i}r_1k_1}\,\mathrm{d}r_1 \tag{5.378}$$

Similar to (5.372), the inverse FT gives

$$\overline{u_1^2} = R_{11}(0) = \int_{-\infty}^{\infty} E_{11}(k_1)\,\mathrm{d}k_1 \tag{5.379}$$

$$\overline{u_2^2} = R_{22}(0) = \int_{-\infty}^{\infty} E_{22}(k_1)\,\mathrm{d}k_1 \tag{5.380}$$

Evolution equation of Fourier modes.

The Kolmogorov spectrum. Structure function, Spectrum. dissipation.

The Kolmogorov spectrum can typically be obtained on dimensional grounds, also the 1-D spectra:

$$E(k) = C\varepsilon^{2/3}k^{-5/3} \tag{5.381}$$

$$E_{11}(k_1) = C_1\varepsilon^{2/3}k_1^{-5/3} \tag{5.382}$$

$$E_{22}(k_1) = C_1'\varepsilon^{2/3}k_1^{-5/3} \tag{5.383}$$

where the isotropic relation is ...

Dissipation range spectrum.

Cospectrum.

small scale isotropy and 1-D spectrum (isotropic relation).

$$E(k) = \frac{1}{2}k^3\frac{d}{dk}\left(\frac{1}{k}\frac{dE_{11}(k)}{dk}\right) \tag{5.384}$$

$$\langle \delta u^2 \rangle = 2u'^2[1 - \mathcal{F}^{-1}(E_k)] \tag{5.385}$$

and the relation between the -5/3 and the 2/3 laws.

5.4 Scales of turbulent motions

5.4.1 The integral length scale

The longitudinal and transverse integral scales based on correlation functions are

$$L_{11}(t) = \int_0^\infty f(r,t)\,\mathrm{d}r \tag{5.386}$$

$$L_{22}(t) = \int_0^\infty g(r,t)\,\mathrm{d}r. \tag{5.387}$$

It was shown that $L_{11} = 2L_{22}$.

The large-eddy length scale is

$$L = \frac{k^{3/2}}{\varepsilon}, \tag{5.388}$$

based on an inviscid, equilibrium scaling for dissipation

$$\varepsilon = C_\varepsilon\frac{u'^3}{L}, \tag{5.389}$$

where the r.m.s. velocity is $u' = \sqrt{2k/3}$ (due to the rapid decay of the spectrum, energy is mostly contained in the large scales and u' is a large scale). Empirically, $L_{11} \approx 0.43L$.

5.4.2 The Kolmogorov scale

The Kolmogorov scales are:

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4},\ u_\eta = (\varepsilon\nu)^{1/4},\ \tau_\eta = (\nu/\varepsilon)^{1/2}. \tag{5.390}$$

Note the scales depend only on viscosity and dissipation rate (hence the expressions above can be derived from a dimensional analysis). A resulting Reynolds number

$$Re_\eta = \frac{u_\eta \eta}{\nu} = 1 \tag{5.391}$$

indicates that viscous effect is active at the Kolmogorov scale. Also,

$$\varepsilon = \nu \left(\frac{u_\eta}{\eta}\right)^2 \tag{5.392}$$

indicates that u_η and η are the scales of dissipative eddies. The inviscid scaling for dissipation leads to the scale separation expressed in terms of the large-scale Reynolds number

$$\frac{L}{\eta} = \frac{L}{[\nu^3/(u'^3/L)]^{1/4}} = \left(\frac{Lu'}{\nu}\right)^{3/4} = Re_L^{3/4} \tag{5.393}$$

and similarly

$$\frac{u'}{u_\eta} = Re_L^{1/4} \tag{5.394}$$

$$\frac{T}{\tau_\eta} = Re_L^{1/2} \tag{5.395}$$

where $T = L/u'$.

5.4.3 The Taylor microscale

The Taylor length scale is defined based on the correlation functions. The transverse Taylor length scale is defined based on the transverse correlation as

$$\lambda_g = \left[-\frac{1}{2}g''(0)\right]^{-1/2} = \sqrt{-\frac{2}{g''(0)}}, \tag{5.396}$$

which has a polynomial expantion

$$p(r) = g(0) + g'(0)r + \frac{1}{2}g''(0)r^2 + ... \tag{5.397}$$

where

$$g(r) = \frac{\langle u_2(\boldsymbol{x} + r\boldsymbol{e}_1)u_2(\boldsymbol{x})\rangle}{\langle u_2^2\rangle} \tag{5.398}$$

(think the correlation between two points of opposite locations in a vortex). It is known that $g(0) = 1, g'(0) = 0$ and $g''(0) < 0$ (concave). Then

$$p(r) = 1 + \frac{1}{2}g''(0)r^2, \tag{5.399}$$

is a fitting parabola atthr origin that intersects the horizontal axis at

$$r = \lambda_g = \sqrt{-\frac{2}{g''(0)}}, \tag{5.400}$$

which can be interpreted as an approximate 'de-correlation' length. Think two points having a small correlation coefficient when they are more than an eddy size apart (statistical eddy).

By differentiating (5.323) and taking $r = 0$ it can be shown that $g''(0) = 2f''(0)$ such that $\lambda_f = \sqrt{2}\lambda_g$ where $\lambda_f = \sqrt{-2/f''(0)}$ where $f(r) = \langle u_1(\boldsymbol{x} + r\boldsymbol{e}_1)u_2(\boldsymbol{x})\rangle\langle u_1^2\rangle$.

In isotropic turbulence, the velocity gradient takes the form of

$$\left\langle \left(\frac{\partial u_i}{\partial x_j}\right)^2 \right\rangle = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \left\langle \left(\frac{\partial u_1}{\partial x_1}\right)^2 \right\rangle \tag{5.401}$$

(see exercise 5.28 in Pope (2001)) and dissipation can be expressed as

$$\varepsilon = \nu \left\langle \frac{\partial u_i}{\partial x_j}\frac{\partial u_i}{\partial x_j} \right\rangle = 15\nu \left\langle \left(\frac{\partial u_1}{\partial x_1}\right)^2 \right\rangle. \tag{5.402}$$

xxxx do the differential with $f(r)$

Using λ_g, the dissipation is given by

$$\varepsilon = 15\nu \frac{u'^2}{\lambda_g^2}. \tag{5.403}$$

Practically, the Taylor length scale can be computed as

$$\lambda_g = \sqrt{\frac{15\nu u'^2}{\varepsilon}} \tag{5.404}$$

via measuring u' (the rms velocity) and ε. Then the Taylor Reynolds number is

$$Re_\lambda = \frac{u'\lambda_g}{\nu}, \tag{5.405}$$

which is commonly used to characterize isotropic turbulence. It can be related to the large scale Reynolds number,

$$Re_L = \frac{u'L}{\nu} \tag{5.406}$$

through the inviscid scaling of dissipation as

$$Re_\lambda = (15Re_L)^{1/2}, \tag{5.407}$$

hence

$$\frac{L}{\eta} = (Re_L)^{3/4} \sim Re_\lambda^{3/2}. \tag{5.408}$$

Similarly, in isotropic turbulence,

$$\frac{L}{\lambda_g} = (C_\varepsilon/15)Re_\lambda \tag{5.409}$$

where C_ε is order unity.

5.4.4 Other useful scales

The Corrsin scale (below which the flow doesn't feel the shear and turbulence in different shear flows are similar):

$$L_C = \left(\frac{\varepsilon}{S^3}\right)^{1/2}, \tag{5.410}$$

where S is the background shear. Consider the time scale comparison between dissipation and shear distortion:

$$\frac{T_\varepsilon}{T_S} = \frac{k/\varepsilon}{1/S} = \frac{(SL)^2/\varepsilon}{1/S} = S^3 L/\varepsilon = \frac{L^2}{L_C^2}. \tag{5.411}$$

Here L is the length scale of energy containing eddies. When $L > L_C$ or $T_\varepsilon > T_S$, the eddies will be distroyed by shear before being dissipated.

An analog to the Corrsin scale in stratified turbulence is the Ozmidov scale:

$$L_O = \left(\frac{\varepsilon}{N^3}\right)^{1/2} \tag{5.412}$$

where S is replaced by N and a similar timescale comparison can be made. The Kolmogorov scale is

$$\eta = \left(\frac{\nu^3}{\varepsilon}\right)^{1/4}, \tag{5.413}$$

the ratio between these two gives $L_O/\eta = Re_b^{3/4}$. The buoyancy Reynolds is:

$$Re_b = \frac{\varepsilon}{\nu N^2}. \tag{5.414}$$

It can be regarded as a measure of the separation between scales unaffected by stratification and the dissipation range, and hence an indicator of the extent of small-scale isotropy.

Taking L_O as the vertical length scale for overturns and $q_h = \sqrt{(\langle u'^2 + v'^2 \rangle)}/2$ to be the velocity scale, the Froude number is unity hence the scales greater than L_O are influenced by stratification.

5.5 Turbulent free shear flows

5.5.1 Momentum integral

Similarity solutions (turbulent). Pope (2001).

5.5.2 Similarity solutions

The characteristic velocity and length scales are U_s and δ_s, respectively.

The example of plane jet is the easiest to understand and derive so we are the most detailed in that case and more loosely on the others. The same principles and machinery apply to all cases.

Flow type	$U_s(x)$	$\delta_s(x)$	$U_s \propto x^m$	$\delta_s \propto x^n$	$f(\eta), \eta = r/r_{1/2}, y/y_{1/2}$
Round jet	$U_0(x) = \bar{u}(x,0)$	$r_{1/2}$	-1	1	$1/(1+a\eta^2)^2$
Plane jet	$U_0(x) = \bar{u}(x,0)$	$y_{1/2}$	$-1/2$	1	$\mathrm{sech}^2(\ln(1+\sqrt{2})\,\eta)$
Round wake	$U_d(x) = U_\infty - \bar{u}(x,0)$	$r_{1/2}$	$-2/3$	$1/3$	$\exp(-\ln 2\,\eta^2)$
Plane wake	$U_d(x) = U_\infty - \bar{u}(x,0)$	$y_{1/2}$	$-1/2$	$1/2$	$\exp(-\ln 2\,\eta^2)$
Mixing layer	$\Delta U = U_2 - U_1$	$y_{0.9} - y_{0.1}$	0	1	$1/2\,\mathrm{erf}(\eta/\sigma\sqrt{2})$

Table 5.1: Self-similar solution table. For axisymmetric flows the mean is $\bar{u}_i(x,r)$ and for planar flows the mean is $\bar{u}_i(x,y)$. For the round jet, $a = (\sqrt{2}-1)/S^2$. For the mixing layer, $\sigma = [2\sqrt{2}\mathrm{erf}^{-1}(4/5)]^{-1}$.

5.5.3 Round jet

Characteristic scales:
The centerline velocity is

$$U_s(x) = \bar{u}(x, r=0) \tag{5.415}$$

and the characteristic length is the half width, $\delta_s = r_{1/2}(x)$, such that

$$U_d(x, r_{1/2}) = \bar{u}(x, r_{1/2}(x)) = \frac{1}{2}U_s(x). \tag{5.416}$$

Momentum integral constraint:
The boundary layer equation in cylindrical coordinates reads

$$\bar{u}\frac{\partial \bar{u}}{\partial x} + \bar{v}\frac{\partial \bar{u}}{\partial r} = -\frac{1}{r}\frac{\partial(r\overline{u'v'})}{\partial r}. \tag{5.417}$$

Multiply the continuity equation

$$\frac{\partial \bar{u}}{\partial x} + \frac{1}{r}\frac{\partial(r\bar{v})}{\partial r} = 0 \tag{5.418}$$

by $r\bar{u}$ and add it to (5.417) multiplied by r we obtain

$$\frac{\partial(r\bar{u}\bar{u})}{\partial x} + \frac{\partial(r\bar{u}\bar{v})}{\partial r} = -\frac{\partial(r\overline{u'v'})}{\partial r}. \tag{5.419}$$

Integrate (5.419) in r we obtain

$$\int_0^\infty \frac{\partial(r\bar{u}\bar{u})}{\partial r}\,\mathrm{d}r + r\bar{u}\bar{v}|_0^\infty = -r\overline{u'v'}|_0^\infty \tag{5.420}$$

and since $\overline{u'v'}$ and $\bar{u}$ are zero at infinity (note $\bar{v}$ at infinity is not zero due to entrainment), we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_0^\infty r\bar{u}^2\,\mathrm{d}r\right) = 0 \tag{5.421}$$

which implies the momentum flux

$$\dot{J}(x) = \int_0^\infty \rho \bar{u}^2 2\pi r \, \mathrm{d}r = J_0 \tag{5.422}$$

is conserved (as a result of both mass and momentum conservation), where J_0 is the jet exit strength.

Self-similar assumptions:

$$\bar{u} = U_s(x) f(\eta), \ \overline{u'v'} = U_s^2(x) g(\eta), \ \eta = \frac{r}{r_{1/2}} \tag{5.423}$$

where $\eta = r/\delta_s(x)$ with $\delta_s = r_{1/2}$. Substitute (5.423) into (5.422) we have

$$\dot{J}(x) = (2\pi\rho)(U_s^2 \delta_s^2) \left(\int_0^\infty \eta f^2(\eta) \, \mathrm{d}\eta \right) \tag{5.424}$$

to be a constant and implying

$$\frac{\mathrm{d}}{\mathrm{d}x}(U_s^2 \delta_s^2) = 0 \tag{5.425}$$

and hence

$$\frac{\delta_s}{U_s} \frac{\mathrm{d}U_s}{\mathrm{d}x} = -\frac{\mathrm{d}\delta_s}{\mathrm{d}x}. \tag{5.426}$$

Using the continuity equation we have

$$\bar{v} = -\frac{1}{r} \int_0^r \frac{\partial (r\bar{u})}{\partial x} \, \mathrm{d}y = U_s \frac{\mathrm{d}\delta_s}{\mathrm{d}x} \left(\eta f - \frac{1}{\eta} \int_0^\eta f \eta \, \mathrm{d}\eta \right). \tag{5.427}$$

Next we establish the constant spread rate of the round jet (i.e. $\mathrm{d}\delta_s/\mathrm{d}x$ is a constant). Take $\bar{v}$ into the momentum equation we have

$$\frac{\mathrm{d}\delta_s}{\mathrm{d}x} \left[f^2 \eta + f f' \eta + \left(\frac{f}{\eta} + f' \right) \int_0^\eta f \eta \, \mathrm{d}\eta \right] = g + g' \eta \tag{5.428}$$

and then $\mathrm{d}\delta_s/\mathrm{d}x$ has to be a constant. Combining with momentum integral restriction we have

$$\delta_s \propto x, \, U_s \propto x^{-1}. \tag{5.429}$$

We note that $\bar{v}$ is positive near the centerline and it switches sign from positive to negative as r reaches a certain value. This is the entrainment of ambient fluid. The entrainment is can be shown by the mass flux that

$$\dot{M}(x) = \int_0^\infty \rho \bar{u} 2\pi r \, \mathrm{d}r = 2\pi\rho(U_s \delta_s^2) \left(\int_0^\infty \eta f(\eta) \, \mathrm{d}\eta \right) \sim x, \tag{5.430}$$

i.e., the mass flux increases as x.

Eddy-viscosity model:

5.5.4 Plane jet

Characteristic scales:
The centerline velocity is

$$U_s(x) = \bar{u}(x, y = 0) \tag{5.431}$$

and the characteristic length is the half width, $\delta_s = y_{1/2}(x)$, such that

$$U_d(x, y_{1/2}) = \bar{u}(x, y_{1/2}(x)) = \frac{1}{2}U_s(x). \tag{5.432}$$

Momentum integral constraint:
The boundary layer equation for the mean velocity simplifies to

$$\bar{u}\frac{\partial \bar{u}}{\partial x} + \bar{v}\frac{\partial \bar{u}}{\partial y} = -\frac{\partial \overline{u'v'}}{\partial y}. \tag{5.433}$$

Multiply the continuity equation

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} = 0 \tag{5.434}$$

by $\bar{u}$ and add it to (5.433) we obtain

$$\frac{\partial \bar{u}\bar{u}}{\partial x} + \frac{\partial \bar{u}\bar{v}}{\partial y} = -\frac{\partial \overline{u'v'}}{\partial y}. \tag{5.435}$$

Integrate (5.435) in y we obtain

$$\int_{-\infty}^{\infty} \frac{\partial \bar{u}\bar{u}}{\partial x}\,\mathrm{d}y + \bar{u}\bar{v}|_{-\infty}^{\infty} = -\overline{u'v'}|_{-\infty}^{\infty} \tag{5.436}$$

and since $\overline{u'v'}$ and $\bar{u}$ are zero at infinity (note $\bar{v}$ at infinity is not zero due to entrainment), we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{-\infty}^{\infty} \bar{u}^2\,\mathrm{d}y\right) = 0 \tag{5.437}$$

which implies the momentum flux

$$\dot{J}(x) = \int_{-\infty}^{\infty} \rho\bar{u}^2\,\mathrm{d}y = J_0 \tag{5.438}$$

is conserved (as a result of both mass and momentum conservation), where J_0 is the jet exit strength.
Self-similar assumptions:

$$\bar{u} = U_s(x)f(\eta),\ \overline{u'v'} = U_s^2(x)g(\eta),\ \eta = \frac{y}{y_{1/2}} \tag{5.439}$$

where $\eta = y/\delta_s(x)$ and we have

$$\frac{\partial \eta}{\partial x} = -\frac{\eta}{\delta_s}\frac{\mathrm{d}\delta_s}{\mathrm{d}x} \tag{5.440}$$

$$\frac{\partial \eta}{\partial y} = \frac{1}{\delta_s} \tag{5.441}$$

Substitute (5.439) into (5.438) we have

$$\dot{J}(x) = (U_s^2 \delta_s) \left(\int_{-\infty}^{\infty} f^2(\eta)\, \mathrm{d}\eta \right) = J_0 \tag{5.442}$$

is a constant. So it must be

$$\frac{\mathrm{d}}{\mathrm{d}x}(U_s^2 \delta_s) = 0 \tag{5.443}$$

which gives the momentum flux constraint in terms of characteristic variables, and hence

$$\frac{\delta_s}{U_s} \frac{\mathrm{d}U_s}{\mathrm{d}x} = -\frac{1}{2} \frac{\mathrm{d}\delta_s}{\mathrm{d}x} \tag{5.444}$$

Using the continuity equation we have

$$\bar{v} = -\int_0^y \frac{\partial \bar{u}}{\partial x}\, \mathrm{d}y = U_s \frac{\mathrm{d}\delta_s}{\mathrm{d}x} \left(\eta f - \frac{1}{2} \int_0^\eta f\, \mathrm{d}\eta \right) \tag{5.445}$$

Next we establish the constant spread rate of the plane jet (i.e. $\mathrm{d}\delta_s/\mathrm{d}x$ is a constant). Take $\bar{v}$ into the momentum equation we have

$$\frac{1}{2} \frac{\mathrm{d}\delta_s}{\mathrm{d}x} (f^2 + f' \int_0^\eta f\, \mathrm{d}\eta) = g' \tag{5.446}$$

and then

$$\frac{\mathrm{d}\delta_s}{\mathrm{d}x} = \frac{2g'}{f^2 + f' \int_0^\eta f\, \mathrm{d}\eta} = C \tag{5.447}$$

with the LHS only depend on x and RHS only depend on η. Then both sides have to be constant. Combining (5.447) and (5.443) we have

$$\delta_s \propto x,\ U_s \propto x^{-1/2}. \tag{5.448}$$

Entrainment. The mass flux is

$$\dot{M}(x) = (U_s \delta_s) \left(\int_{-\infty}^{\infty} f(\eta)\, \mathrm{d}\eta \right) \sim x^{1/2}. \tag{5.449}$$

Eddy-viscosity model:

5.5.5 Round wake

Characteristic scales:
The centerline velocity deficit is

$$U_0(x) = U_\infty - \bar{u}(x, r = 0) = U_d(x, r = 0) \tag{5.450}$$

and the characteristic length is the half width, $\delta_s = r_{1/2}(x)$, such that

$$U_d(x, r_{1/2}) = U_\infty - \bar{u}(x, r_{1/2}(x)) = \frac{1}{2} U_0(x). \tag{5.451}$$

Momentum integral constraint:
Here we start from the simplified (see plane wake) momentum equation

$$U_\infty \frac{\partial(\bar{u}-U_\infty)}{\partial x} = -\frac{1}{r}\frac{\partial(r\overline{u'v'})}{\partial r} \tag{5.452}$$

and the corresponding far-wake simplified momentum deficit flux conservation

$$\dot{J}(x) = \int_0^\infty \rho\bar{u}(U_\infty-\bar{u})2\pi r\,\mathrm{d}r \approx \int_0^\infty \rho U_\infty(U_\infty-\bar{u})2\pi r\,\mathrm{d}r = J_0. \tag{5.453}$$

Note that we have already replaced $\bar{u}$ with U_∞, assuming (or by order of magnitude analysis) $\bar{u}/U_\infty$ is small and the convection velocity is U_∞.

We note that the momentum thickness is defined as

$$\theta(x) = \frac{1}{U_\infty^2}\int_0^\infty \bar{u}(U_\infty-\bar{u})r\,\mathrm{d}r = \frac{\dot{J}(x)}{2\pi\rho U_\infty^2} \tag{5.454}$$

whereas its first-order approximation with $\bar{u}\approx U_\infty$ is

$$\tilde{\theta} = \frac{1}{U_\infty}\int_0^\infty (U_\infty-\bar{u})r\,\mathrm{d}r = \frac{J_0}{2\pi\rho U_\infty^2}. \tag{5.455}$$

Self-similar assumptions:

$$U_\infty-\bar{u} = U_s(x)f(\eta),\ \overline{u'v'} = U_s^2(x)g(\eta),\ \eta = \frac{r}{r_{1/2}}, \tag{5.456}$$

We have, to the first-order and in the far wake ($U_s \ll U_\infty$),

$$\dot{J}(x) = (U_s\delta_s^2)(2\pi\rho U_\infty)\int_0^\eta f\,\mathrm{d}\eta \tag{5.457}$$

is a constant and hence,

$$\frac{\mathrm{d}}{\mathrm{d}x}(U_s\delta_s^2) = 0. \tag{5.458}$$

Consider the momentum equation, the other constraint reads

$$-\frac{U_\infty}{U_s}\frac{\mathrm{d}\delta_s}{\mathrm{d}x}(2f+f'\eta)\eta = (g'\eta+g) \tag{5.459}$$

We define the spread rate as

$$S = \frac{U_\infty}{U_s}\frac{\mathrm{d}\delta_s}{\mathrm{d}x}, \tag{5.460}$$

and then

$$-S(2f\eta+f'\eta^2) = (g\eta)'. \tag{5.461}$$

S, as a function of x, has to be a constant.

Include the boundary conditions and after integration we get

$$g = -S\eta f \tag{5.462}$$

same as in plane wakes. Combining (5.458) and (5.460) we have

$$\delta_s \propto x^{1/3},\ U_s \propto x^{-2/3}. \tag{5.463}$$

Eddy-viscosity model:
Invoke the eddy viscosity assumption ($\overline{u'v'} = U_s^2 g \sim \nu_T \partial_y U$, $\nu_T = \hat{\nu}_T r_{1/2} U_s$), we have

$$g = \hat{\nu}_T f', \tag{5.464}$$

and by assuming $\hat{\nu}_T$ is a constant (which is quite good for the core of the wake/jet), it can be solved that

$$f(\eta) = Ce^{-\frac{S}{2\hat{\nu}_T}\eta^2}. \tag{5.465}$$

The boundary conditions are $f(0) = 1, f(1) = 1/2$. Finally, the self-similar function is determined to be

$$f(\eta) = e^{-\ln 2\, \eta^2} \tag{5.466}$$

with $S = 2 \ln 2\, \hat{\nu}_T$ (constant spreading rate).

5.5.6 Plane wake

Characteristic scales:
The centerline velocity deficit is

$$U_s(x) = U_\infty - \bar{u}(x, y = 0) = U_d(x, y = 0), U_d \triangleq U_\infty - \bar{u} \tag{5.467}$$

and the characteristic length is the half width, $\delta_s = y_{1/2}(x)$, such that

$$U_\infty - \bar{u}(x, y_{1/2}(x)) = U_d(x, y_{1/2}) = \frac{1}{2} U_s(x). \tag{5.468}$$

Momentum integral constraint:
The boundary layer equation: (note we subtracted U_∞ without affecting anything)

$$\bar{u}\frac{\partial(\bar{u} - U_\infty)}{\partial x} + \bar{v}\frac{\partial(\bar{u} - U_\infty)}{\partial y} = \bar{u}\frac{\partial \bar{u}}{\partial x} + \bar{v}\frac{\partial \bar{u}}{\partial y} = -\frac{\partial \overline{u'v'}}{\partial y}. \tag{5.469}$$

Multiply the continuity equation

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} = 0 \tag{5.470}$$

by $(\bar{u} - U_\infty)$, add it to (5.469), and use integration by parts to obtain

$$\frac{\partial \bar{u}(\bar{u} - U_\infty)}{\partial x} + \frac{\partial \bar{v}(\bar{u} - U_\infty)}{\partial y} = -\frac{\partial \overline{u'v'}}{\partial y}. \tag{5.471}$$

Integrating (5.471) in y we obtain

$$\int_{-\infty}^{\infty} \frac{\partial \bar{u}(\bar{u} - U_\infty)}{\partial x}\, \mathrm{d}y + \bar{v}(\bar{u} - U_\infty)|_{-\infty}^{\infty} = -\overline{u'v'}|_{-\infty}^{\infty} \tag{5.472}$$

and since $\overline{u'v'}$ and $(\bar{u} - U_\infty)$ are zero at infinity, we have

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{-\infty}^{\infty} \bar{u}(\bar{u} - U_\infty)\,\mathrm{d}y\right) = 0 \tag{5.473}$$

which implies the momentum deficit flux

$$\dot{J}(x) = \int_{-\infty}^{\infty} \rho\bar{u}(U_\infty - \bar{u})\,\mathrm{d}y \tag{5.474}$$

is conserved (we note that we haven't assumed far wake, i.e., $U_s \sim (\bar{u} - U_\infty) \ll U_\infty$, yet).

The momentum thickness is related to $\dot{J}(x)$ as

$$\theta(x) = \frac{1}{U_\infty^2}\int_{-\infty}^{\infty} \bar{u}(U_\infty - \bar{u})\,\mathrm{d}y = \frac{\dot{J}(x)}{\rho U_\infty^2}, \tag{5.475}$$

and it is a constant when the boundary-layer approximations that led to (5.469) can be made, say, in the far wake.

Self-similar assumptions:

Take the functional forms

$$U_\infty - \bar{u} = U_s(x)f(\eta),\ \overline{u'v'} = U_s^2(x)g(\eta),\ \eta = \frac{y}{\delta_s} \tag{5.476}$$

and substitute them into (5.474), and assume the far wake has been reached ($U_s/U_\infty \ll 1$) we have

$$\dot{J}(x) = \int_{-\infty}^{\infty} \rho(U_\infty - U_s f)U_s f\delta_s\,\mathrm{d}\eta \tag{5.477}$$

$$= U_\infty^2\int_{-\infty}^{\infty} \rho(1 - \frac{U_s f}{U_\infty})\frac{U_s}{U_\infty} f\delta_s\,\mathrm{d}\eta \tag{5.478}$$

$$= \rho U_\infty U_s\delta_s\int_{-\infty}^{\infty} f\,\mathrm{d}\eta \tag{5.479}$$

which is a constant. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}(U_s\delta_s) = 0. \tag{5.480}$$

Using the continuity equation we have

$$\frac{\partial\bar{v}}{\partial y} = -\frac{\partial\bar{u}}{\partial x}, \tag{5.481}$$

where due to (5.476),

$$\frac{\partial\bar{u}}{\partial x} = -f\frac{dU_s}{dx} - U_s\frac{df}{dx} = -f\frac{dU_s}{dx} - U_s f'\frac{d\eta}{dx} \tag{5.482}$$

$$= -f\frac{dU_s}{dx} + U_s f'\frac{\eta}{\delta_s}\frac{d\delta_s}{dx} = -\frac{dU_s}{dx}(f + f'\eta) = -\frac{dU_s}{dx}(f\eta)'. \tag{5.483}$$

We note that

$$\frac{\partial\eta}{\partial x} = -\frac{y}{\delta_s^2}\frac{d\delta_s}{dx} = -\frac{\eta}{\delta_s}\frac{d\delta_s}{dx},\ \frac{\partial\eta}{\partial y} = \frac{1}{\delta_s}. \tag{5.484}$$

and the equivalence of (5.480),

$$-\frac{U_s}{\delta_s}\frac{d\delta_s}{dx} = \frac{dU_s}{dx}, \tag{5.485}$$

have been used.

Thus, we have expressed the lateral mean velocity $\bar{v}$ with self-similar functions of $\bar{u}$:

$$\bar{v}(y) = \bar{v}(y) - \bar{v}(0) = \int_0^y \frac{\partial \bar{v}}{\partial y}\,\mathrm{d}y = -\int_0^y \frac{\partial \bar{u}}{\partial x}\,\mathrm{d}y = -U_s\frac{\mathrm{d}\delta_s}{\mathrm{d}x} f\eta. \tag{5.486}$$

Note a negative velocity ($\bar{v} < 0$) corresponds to wake entrainment (of high momentum into low momentum region). Also, $\bar{v} \sim U_s L_y/L_x$ follows an order-of-magnitude analysis of the continuity equation. The expression (5.486) is not explicitly needed in the self-similar analysis of the momentum equation.

Now we consider another constraint, the simplified momentum equation. Since in the far wake, the velocity deficit $U_s/U_\infty \ll 1$ ($\bar{u} \approx U_\infty$), the momentum equation (5.469) simplifies to

$$U_\infty \frac{\partial \bar{u}}{\partial x} = -\frac{\partial \overline{u'v'}}{\partial y}. \tag{5.487}$$

for the following reasons. Since

$$\bar{u}(\bar{u} - U_\infty) = (U_\infty - U_s f)(-U_s f) = U_\infty^2(1 - \frac{U_s f}{U_\infty})(-\frac{U_s}{U_\infty} f) \tag{5.488}$$

$$\approx U_\infty^2(-\frac{U_s}{U_\infty} f) = -U_s U_\infty f = U_\infty(\bar{u} - U_\infty), \tag{5.489}$$

and the scale of the first term $\partial\bar{u}(\bar{u} - U_\infty)/\partial x$ is

$$\frac{U_\infty U_s}{L_x} \tag{5.490}$$

while the scale of the second term $\partial\bar{v}(\bar{u} - U_\infty)/\partial y$ (from (5.486)) is

$$\frac{U_s}{\delta_s}\left(U_s\frac{\delta_s}{L_x}\right) = \frac{U_s^2}{L_x} \ll \frac{U_\infty U_s}{L_x}, \tag{5.491}$$

and the Reynolds stress term on thr RHS scales as ($\overline{u'v'} \sim U_s^2 g$)

$$\frac{U_s^2}{L_y} = \frac{U_s U_\infty}{L_x}. \tag{5.492}$$

The reduction to (5.487) is legitimate.

Substituting the expressions of $\bar{u}$ (5.476) (note that (5.486) for $\bar{v}$ is not needed since the momentum equation is simplified, unlike in jets) into (5.487) we have

$$(f + f'\eta)\frac{U_\infty}{U_s}\frac{\mathrm{d}\delta_s}{\mathrm{d}x} = -g' \tag{5.493}$$

where

$$S = \frac{U_\infty}{U_s}\frac{\mathrm{d}\delta_s}{\mathrm{d}x}. \tag{5.494}$$

is the spreading rate. Since S depends only on x, while the remaining terms depend only on η, S must be constant.

Then (5.493) can be rewritten as

$$g' + S(f + f'\eta) = 0 \tag{5.495}$$

which is to say

$$(g + S\eta f)' = 0. \tag{5.496}$$

Integrate from $\eta = 0$ to η and note that $g(0) = 0$, we have

$$g = -S\eta f. \tag{5.497}$$

Power laws:
Combining two conditions (5.480) and (5.494) we solve the two unknown power laws as

$$\delta_s \propto x^{1/2},\ U_s \propto x^{-1/2}. \tag{5.498}$$

Eddy-viscosity model:
Invoke the constant eddy viscosity assumption (EVM), ($\overline{u'v'} = U_s^2 g \sim \nu_T \partial_y U$, $\nu_T = \hat{\nu}_T \delta_s U_s$)

$$g = \hat{\nu}_T f', \tag{5.499}$$

and (5.497), which is one equation for two unknowns (f and g), is closed using the EVM to

$$\hat{\nu}_T f' = -S\eta f. \tag{5.500}$$

The boundary conditions $f(0) = 1, f(1) = 1/2$ determine the integration constants and the self-similar function is solved to be

$$f(\eta) = e^{-\ln 2\, \eta^2}, \tag{5.501}$$

with the analytical $S = 2 \ln 2\, \hat{\nu}_T$. We note that this self-similar function is the same in planar and axisymmetric wakes, both laminar (more to that later) and turbulent.

5.5.7 Plane mixing layer

Characteristic scales:
The two velocities are $U_2 > U_1$ with U_2 on the top. The mean convection velocity is

$$U_c = \frac{1}{2}(U_1 + U_2) \tag{5.502}$$

and the characteristic velocity scale is (invariant in x)

$$U_s = U_2 - U_1. \tag{5.503}$$

The characteristic length is the mixing layer width,

$$\delta_s(x) = y_{0.9} - y_{0.1} \tag{5.504}$$

with cross-stream location $y_\alpha(x)$ such that

$$\bar{u}(x, y_\alpha(x)) = U_1 + \alpha U_s. \tag{5.505}$$

a reference position is

$$\hat{y} = \frac{1}{2}(y_{0.1} + y_{0.9}) \tag{5.506}$$

such that the self-similar variable is defined as

$$\eta = \frac{y - \hat{y}}{\delta_s(x)} \tag{5.507}$$

Momentum integral constraint:

Self-similar assumptions:

$$\bar{u} - U_c = U_s(x) f(\eta),\ \overline{u'v'} = U_s^2(x) g(\eta),\ \eta = \frac{y - \hat{y}}{\delta_s(x)} \tag{5.508}$$

Eddy-viscosity model:

5.6 Turbulent wall flows

5.6.1 Turbulent channel flow

The mean flow equations are

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} = 0 \tag{5.509}$$

$$0 = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial \overline{u'v'}}{\partial y} \tag{5.510}$$

$$0 = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial y} - \frac{\partial \overline{v'^2}}{\partial y} \tag{5.511}$$

Integrate (5.511) in y from 0 to y we have

$$\frac{\bar{p}}{\rho} + \overline{v'^2} = \frac{p_w}{\rho} \tag{5.512}$$

where p_w is the mean wall pressure. Hence, we have

$$\frac{\partial \bar{p}}{\partial x} = \frac{\partial p_w}{\partial x}, \tag{5.513}$$

implying that the pressure drop over the same distance is the same for different heights, even $\partial \bar{p}/\partial y \neq 0$. The wall shear stress balance (4.15) is still valid (control volume from 0 to h) as

$$-\frac{\partial p_w}{\partial x} = \frac{\tau_w}{h}. \tag{5.514}$$

Substitute (5.512) and (5.514) back to (5.510) we have

$$-\frac{\tau_w}{h} = \nu \frac{\partial^2 \bar{u}}{\partial^2 y} - \frac{\partial \overline{u'v'}}{\partial y} \tag{5.515}$$

the $\int_y^h dy$ of which leads to the channel flow basic equation

$$\tau(y) = \tau_w \left(1 - \frac{y}{h}\right) = \rho u_\tau^2 \left(1 - \frac{y}{h}\right) = \mu \frac{\partial \bar{u}}{\partial y} - \rho \overline{u'v'}, \tag{5.516}$$

or simply

$$\tau(y) = u_\tau^2 \left(1 - \frac{y}{h}\right) = \nu \frac{\partial \bar{u}}{\partial y} - \overline{u'v'} = \tau_{\text{vis}} + \tau_{\text{Rey}}, \tag{5.517}$$

with $\rho = 1$. The total stress has a linear dependence on the distance from the wall, with slope being u_τ^2. This relation is usually used to figure out u_τ from an integral/conservation point of view (instead of from the shear stress at the wall). Near the wall, the viscous shear stress dominates and it decreases as the distance from the wall increases. The linear stress is basically conservation of momentum and it provides a scale u_τ for that. If an eddy viscosity model $\nu_T(y)$ is used, for example the Cess analytical model, the turbulent mean profile $\bar{u}(y)$ can be obtained by integrating $u_\tau^2(1 - y/h)/\nu_T$ from 0 to y.

FIK;

Kolmogorov scale in the log layer in plus units;

Each layer, log laws; characteristic scales;

Chapter 6

Geophysical fluid dynamics

6.1 Basics

6.1.1 Non-inertial frames, centrifugal and Coriolis forces

6.1.2 Absolute velocity

In an inertial frame,

$$\boldsymbol{u}_a = \boldsymbol{u}_{(r)} + \boldsymbol{\Omega} \times \boldsymbol{r}, \tag{6.1}$$

where $\boldsymbol{u}_{(r)}$ is the relative (to the non-inertial frame) velocity. Especially, in a cylindrical frame with a plane-normal rotation,

$$u_{\theta,a} = u_\theta + \Omega r. \tag{6.2}$$

6.1.3 Inertial oscillations: buoyancy and Coriolis frequencies

6.2 Boussinesq approximation

6.3 Balanced flows

6.3.1 Hydrostatic and geostrophic balances

In balanced flow, which is usually the case to the first order, there is a background horizontal pressure gradient that balances the Coriolis forces due to horizontal motions and a vertical pressure gradient that balances the background unperturbed density:

$$0 = -\frac{1}{\rho_0}\frac{\partial p_g}{\partial x} + fV \tag{6.3}$$

$$0 = -\frac{1}{\rho_0}\frac{\partial p_g}{\partial y} - fU \tag{6.4}$$

$$0 = -\frac{1}{\rho_0}\frac{\partial p_g}{\partial z} - \frac{\rho^* g}{\rho_0} \tag{6.5}$$

with

$$p_g = p - p_0,\ \rho^* = \rho - \rho_0 - \rho_b(z) \tag{6.6}$$

and the background balance

$$0 = -\frac{\partial p_0}{\partial z} - (\rho_0 + \rho_b)g \tag{6.7}$$

already subtracted. Note that the Boussinesq and hydrostatic approximations are already applied.

The above equations in vector form:

$$\boldsymbol{f}_{\mathrm{c}} \times \boldsymbol{U} = -\frac{1}{\rho_0}\nabla p_g + \frac{\rho^*}{\rho_0}\boldsymbol{g}. \tag{6.8}$$

We have

$$\boldsymbol{U} = (U, V, 0) = -\frac{1}{\rho_0 f}(\frac{\partial p_g}{\partial y}, -\frac{\partial p_g}{\partial x}, 0). \tag{6.9}$$

And we have

$$\nabla_{\mathrm{h}} \cdot \boldsymbol{U} = 0. \tag{6.10}$$

In the world geostrophic, geo means Coriolis and strophic means cyclone/anticyclone or low-/high-pressure systems.

6.3.2 Thermal wind relations

In hydrostatic Boussinesq flow. Taking the vertical gradient of (6.8) and using the hydrostatic balance, we have

$$0 = \frac{g}{\rho_0}\frac{\partial \rho^*}{\partial x} + f\frac{\partial V}{\partial z} \tag{6.11}$$

$$0 = \frac{g}{\rho_0}\frac{\partial \rho^*}{\partial y} - f\frac{\partial U}{\partial z} \tag{6.12}$$

and hence

$$(\frac{\partial U}{\partial z}, \frac{\partial V}{\partial z}) = \frac{g}{\rho_0 f}(\frac{\partial \rho^*}{\partial y}, -\frac{\partial \rho^*}{\partial x}), \tag{6.13}$$

or in vector form,

$$\frac{\partial \boldsymbol{U}}{\partial z} = \frac{\boldsymbol{g}}{\rho_0 f} \times \nabla \rho^* \tag{6.14}$$

In a more general case. Without introducing the hydrostatic balance and Boussinesq approximation, we write

$$\boldsymbol{f}_c \times \boldsymbol{U} = \frac{1}{\rho}\nabla p + \boldsymbol{g}. \tag{6.15}$$

Taking its curl:

$$\mathrm{LHS} = \nabla \times (\boldsymbol{f}_c \times \boldsymbol{U}) = -\boldsymbol{f}_c \cdot \nabla \boldsymbol{U} = -f\frac{\partial \boldsymbol{U}}{\partial z} \tag{6.16}$$

$$\mathrm{RHS} = -\nabla \times (\frac{1}{\rho}\nabla p) + \nabla \times \boldsymbol{g} = -\frac{1}{\rho^2}(\nabla p \times \nabla \rho) \tag{6.17}$$

Re-introduce hydrostatic ($\partial_z p = -\rho g$) and Boussinesq, we have

$$\nabla p \approx \frac{\partial p}{\partial z}\boldsymbol{e}_z = -\rho g \boldsymbol{e}_z \tag{6.18}$$

and

$$-\frac{1}{\rho^2}(\nabla p \times \nabla \rho) = \frac{\rho g}{\rho_0^2}(\boldsymbol{e}_z \times \nabla \rho^*) \tag{6.19}$$

$$= -\frac{\boldsymbol{g}}{\rho_0} \times \nabla \rho^* \tag{6.20}$$

hence we recover

$$\frac{\partial \boldsymbol{U}}{\partial z} = \frac{\boldsymbol{g}}{\rho_0 f} \times \nabla \rho^*. \tag{6.21}$$

Note:

6.3.3 Cyclostrophic wind relations

In analogy to the thermal wind (Coriolis balances the horizontal pressure gradient), we have similarly the cyclostrophic wind (centrifugal balances the radial pressure gradient):

$$\frac{u_\theta^2}{r} = \frac{1}{\rho_0}\frac{\partial p^*}{\partial r} \tag{6.22}$$

$$\frac{\partial p^*}{\partial z} = -\frac{\rho^* g}{\rho_0} \tag{6.23}$$

$$\frac{\partial \rho^*}{\partial r} = -\frac{\rho_0}{g}\frac{\partial}{\partial z}\left(\frac{u_\theta^2}{r}\right) \tag{6.24}$$

where the mean vertical wind shear is supported with a horizontal density gradient, which provides the centripetal accelaration ($a = u_\theta^2/r$). This has something to do with the Jet Stream. Cyclo means 'cyclone' or low-pressure system and strophic means 'turning'.

6.3.4 Example: Taylor-Proudman theory

Consider steady flow with negligible convective term (in geostrophic balance, $Ro \ll 1$):

$$0 = -\frac{1}{\rho}\nabla p + \boldsymbol{u} \times \boldsymbol{f}_\mathrm{c}. \tag{6.25}$$

Taking the curl of the above equation we have

$$0 = \nabla \times (\boldsymbol{u} \times \boldsymbol{f}_\mathrm{c}). \tag{6.26}$$

Using triple product rules (A.21), in an f-plane, we have

$$\boldsymbol{f}_\mathrm{c} \times \nabla \boldsymbol{u} = 0. \tag{6.27}$$

Assuming the rotation axis is normal to the plane, $\boldsymbol{f}_\mathrm{c} = f\boldsymbol{k}$, we have

$$\frac{\partial \boldsymbol{u}}{\partial z} = \boldsymbol{0}. \tag{6.28}$$

That implies very strong rotation negates vertical gradients.

Stratified Taylor column: consider the thermal wind balance

$$0 = -\frac{1}{\rho_0}\nabla p + \boldsymbol{u} \times \boldsymbol{f}_c - \frac{\rho^* g}{\rho_0}\boldsymbol{k}. \tag{6.29}$$

Taking the curl we have

$$\boldsymbol{f}_c \cdot \nabla \boldsymbol{u} = (\boldsymbol{i}\partial_y - \boldsymbol{j}\partial_x)\frac{\rho^* g}{\rho_0}, \tag{6.30}$$

i.e.,

$$[f_3\frac{\partial u}{\partial z}, f_3\frac{\partial v}{\partial z}, f_3\frac{\partial w}{\partial z}] = [\partial_y\frac{\rho^* g}{\rho_0}, -\partial_x\frac{\rho^* g}{\rho_0}, 0], \tag{6.31}$$

i.e.,

$$f_3\frac{\partial u}{\partial z} = \partial_y\frac{\rho^* g}{\rho_0} \tag{6.32}$$

$$f_3\frac{\partial v}{\partial z} = -\partial_x\frac{\rho^* g}{\rho_0} \tag{6.33}$$

$$\frac{\partial w}{\partial z} = 0 \tag{6.34}$$

which implies Q2D flow with

$$\nabla_H \cdot \boldsymbol{u} = \partial_x u + \partial_y v = -\partial_z w = 0. \tag{6.35}$$

The relations (6.32)-(6.33) are essentially thermal wind relations where the vertical wind shear is balanced by (the pressure gradient created by) horizontal density gradients.

6.3.5 Surface and bottom Ekman layers

In geostrophy, the dominant balance is the Coriolis force and the pressure gradient. Here in the laminar Ekman layer, we consider the steady solution resulting from a balance between viscous shear stress and the Coriolis force:

$$\begin{aligned} 0 &= fv + \nu\frac{\partial^2 u}{\partial z^2} \\ 0 &= -fu + \nu\frac{\partial^2 v}{\partial z^2} \end{aligned} \tag{6.36}$$

where the shear is only in the vertical direction due to horizontally uniform winds blowing over the surface (or uni-directional drag on the ocean bottom).

We denote

$$\boldsymbol{\tau}_a = (\tau_a^X, \tau_a^Y) = \left(\nu\frac{\partial u}{\partial z}\bigg|_{z=0}, \nu\frac{\partial v}{\partial z}\bigg|_{z=0}\right) \tag{6.37}$$

and

$$\mathcal{U}(z) = u(z) + iv(z). \tag{6.38}$$

Hence,

$$0 = f(u+iv) + i\nu\frac{\partial^2}{\partial z^2}(u+iv) \tag{6.39}$$

converts to

$$\frac{\partial^2 \mathcal{U}}{\partial z^2} = \frac{if}{\nu}\mathcal{U}. \tag{6.40}$$

The eigenvalues are

$$\lambda = \pm\sqrt{\frac{if}{\nu}} = \pm\frac{1+i}{\sqrt{2}}\sqrt{\frac{f}{\nu}} \tag{6.41}$$

Define the Ekman depth as

$$d_E = \sqrt{\frac{2\nu}{f}} \tag{6.42}$$

which increases as $\sqrt{\nu}$ and decreases as $\sqrt{1/f}$, and the general solution to (6.40) is

$$\mathcal{U}(z) = C_1 \exp\left[\frac{1+i}{\sqrt{2}}\sqrt{f/\nu}z\right] + C_2 \exp\left[-\frac{1+i}{\sqrt{2}}\sqrt{f/\nu}z\right]. \tag{6.43}$$

By knowing $\mathcal{U}(z=-\infty)$ we have $C_2 = 0$ and

$$\mathcal{U}(z) = \mathcal{U}_0 \exp\left[\frac{1+i}{\sqrt{2}}\sqrt{f/\nu}z\right] = \mathcal{U}_0 e^{z/d_E} e^{iz/d_E} \tag{6.44}$$

which tells us that as you go from the surface (z decreases from zero to $z<0$), the magnitude of the velocity decreases and the velocity vector spirals CW (as you look down), in agreement with the direction of the Coriolis force.

Now we attempt to fix the constant $\mathcal{U}_0$. Given the complex velocity, the (directional) shear stress can be compactly written as

$$\boldsymbol{\tau}_a = \tau_a^X + i\tau_a^Y. \tag{6.45}$$

And the boundary condition is

$$\nu \left.\frac{\partial \mathcal{U}}{\partial z}\right|_{z=0} = \frac{\nu(1+i)\mathcal{U}_0}{d_E}, \tag{6.46}$$

leading to

$$\mathcal{U}_0 = \frac{\boldsymbol{\tau}_a d_E}{(1+i)\rho_0 \nu} \tag{6.47}$$

where

$$\frac{1}{1+i} = \frac{1}{\sqrt{2}} e^{-i\pi/4}, \tag{6.48}$$

which implies that the relative direction of the surface flow is 45 deg to the right (CW) of the wind stress.

Now we turn our attention to the Ekman transport over a finite depth of the mixed layer, $z_E \sim -200$ m, which is a few Ekman depths. The Ekman depth $d_E = \sqrt{2\nu_{\text{eff}}/f} \sim 45$ m for $f \sim 10^{-4} \sim 2\pi \times 24$ hr^{-1}. We note that $e^{-5} = 0.0067$ which might have been sufficient to assume the stress is weak enough to be neglected.

Vertically integrate (6.36) from $-z_E$ to 0, we have

$$\mathcal{U}_E = \int_{-z_E}^{0} u\, dz = \frac{\tau_a^Y}{\rho_0 f} \tag{6.49}$$

$$\mathcal{V}_E = \int_{-z_E}^{0} v\, dz = -\frac{\tau_a^X}{\rho_0 f} \tag{6.50}$$

or in a more compact way,

$$(\mathcal{U}_E, \mathcal{V}_E) = -\frac{1}{\rho_0 f}\boldsymbol{k} \times \boldsymbol{\tau}_a \tag{6.51}$$

where we note that the dimensions for $\mathcal{U}_E, \mathcal{V}_E$ are velocity times length. The direction of the transport, $(\mathcal{U}_E, \mathcal{V}_E)$, is $(\tau_a^Y, -\tau_a^X)$, which is perpendicular to and pointing to the right of the direction of the surface winds (and the wind forcing is balanced by the mean Coriolis force that is 90 deg to the right of the Ekman transport – opposite to the surface forcing). This does not mean that the surface forcing does no work, since the velocity on the surface is at a 45 deg angle to the winds.

We note several useful directions:

- The local velocity direction: $\mathcal{U} = u + iv = (u, v)$;
- The local shear direction: $\boldsymbol{\tau} = \tau^X + i\tau^Y = (\tau^X, \tau^Y) = (\nu\partial_z u, \nu\partial_z v)$;
- And the local Reynolds stress direction: $\boldsymbol{\tau}_{Rey} = (-\overline{u'w'}, -\overline{v'w'})$.

6.4 QGPV theory

The geostrophic balance in a stratified fluid is

$$\frac{Du}{Dt} = -p_x + fv \tag{6.52}$$

$$\frac{Dv}{Dt} = -p_y - fu \tag{6.53}$$

$$\frac{Dw}{Dt} = -p_z + b \tag{6.54}$$

$$\frac{Db}{Dt} + wN^2 = 0 \tag{6.55}$$

$$u_x + v_y + w_z = 0 \tag{6.56}$$

where the horizontal divergence-free condition naturally results from the geostrophy. The vertical divergence w_z is an order ($O(Ro)$) smaller and the (hydrostatically and geostrophically) balanced base flow is 2D.

With a streamfunction ψ, the above system can be written as

$$\begin{pmatrix} p \\ u \\ v \\ w \\ b \end{pmatrix} = \begin{pmatrix} f\psi \\ -\psi_y \\ \psi_x \\ 0 \\ f\psi_z \end{pmatrix} \tag{6.57}$$

where we have chosen the streamfunction $\psi = p/f$ to be proportional to the pressure (as an analog to the sea surface height). The idea of QG theory is to approximate (and then add back)

the terms missing in the balanced flow ($Du_i/Dt, Db/Dt, w_z$, in blue) with the known relations (6.57). This technique is called recursion. For example,

$$\frac{Du}{Dt} = u_t + uu_x + vu_y = -\psi_{yt} + \psi_{yx}\psi_y - \psi_x\psi_{yy} = -\psi_{yt} - J(\psi, \psi_y) \tag{6.58}$$

where we define the Jacobian operator as

$$J(p, q) = p_x q_y - p_y q_x = \begin{vmatrix} p_x & p_y \\ q_x & q_y \end{vmatrix} \tag{6.59}$$

for convenience of expressing the convection terms. We have $J = 0$ when $p = q$. As a consequence,

$$\partial_y J(\psi, \psi_y) = J(\psi_y, \psi_y) + J(\psi, \psi_{yy}) = J(\psi, \psi_{yy}). \tag{6.60}$$

Also, J is a linear operator.

The (quasi-geostrophic) QG equations become

$$-\psi_{yt} - J(\psi, \psi_y) + p_x - fv = 0 \tag{6.61}$$
$$\psi_{xt} + J(\psi, \psi_x) + p_y + fu = 0 \tag{6.62}$$
$$p_z = b \tag{6.63}$$
$$f\psi_{zt} + J(\psi, f\psi_z) + wN^2 = 0 \tag{6.64}$$
$$u_x + v_y + w_z = 0 \tag{6.65}$$

By eliminating the pressure (cross-differentiating the first two momentum equations),

$$(\triangle\psi)_t + J(\psi, \triangle\psi) = fw_z, \tag{6.66}$$

where

$$\triangle\psi = (\partial_{xx} + \partial_{yy})\psi = \zeta \tag{6.67}$$

is the vertical (relative) vorticity. Taking the z-derivative of the buoyancy equation,

$$\frac{f^2}{N^2}\psi_{zzt} + J(\psi, \frac{f^2}{N^2}\psi_{zz}) + fw_z = 0, \tag{6.68}$$

that is,

$$(\mathcal{L}\psi)_t + J(\psi, \mathcal{L}\psi) + fw_z = 0, \tag{6.69}$$

where $\mathcal{L} = \partial_z(f^2/N^2\partial_z)$. Combine the vorticity and the buoyancy equations (6.66)&(6.69), we have

$$q_t + J(\psi, q) = 0, \tag{6.70}$$

where

$$q \triangleq \triangle\psi + \mathcal{L}\psi \tag{6.71}$$

is the potential vorticity (QGPV), which is conserved along the geostrophic motions as dictated by (6.70).

6.5 Governing equations of unbalanced motions

It is reasonable to assume directions of both system rotation and gravity are in $\boldsymbol{z}$.

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{6.72}$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial u_i u_j}{\partial x_j} - f\epsilon_{ij3}(u_j - U_j) = -\frac{1}{\rho_0}\frac{\partial p^*}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} - \frac{\rho^* g}{\rho_0}\delta_{i3}, \tag{6.73}$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_i}{\partial x_i} = \frac{\partial J_{\rho,i}}{\partial x_i}, \tag{6.74}$$

$$\tau_{ij} = \nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right),\ J_{\rho,i} = \kappa\frac{\partial \rho}{\partial x_i}. \tag{6.75}$$

In vector form,

$$\begin{gathered}\nabla \cdot \boldsymbol{u} = 0 \\ \frac{\partial \boldsymbol{u}}{\partial t} + \nabla \cdot (\boldsymbol{u}\boldsymbol{u}) + f\boldsymbol{e}_z \times (\boldsymbol{u} - \boldsymbol{U}) = -\frac{1}{\rho_0}\nabla p^* + \nabla \cdot \boldsymbol{\tau} - \frac{\rho^* g}{\rho_0}\boldsymbol{e}_z \\ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = \nabla \cdot \boldsymbol{J}_\rho\end{gathered} \tag{6.76}$$

where the stress and the scalar flux are

$$\boldsymbol{\tau} = \nu(\nabla \boldsymbol{u} + \boldsymbol{u}\nabla),\ \boldsymbol{J}_\rho = \kappa\nabla\rho. \tag{6.77}$$

The total density ρ is decomposed into the reference density ρ_0, the background density $\rho_b(z)$, and the density perturbation ρ^* due to fluid motion,

$$\rho(x, y, z, t) = \rho_0 + \rho_b(z) + \rho^*(x, y, z, t). \tag{6.78}$$

The total pressure is written as

$$p(x, y, z, t) = p_0 + p_g(x, y) + p_a(z) + p^*(x, y, z, t), \tag{6.79}$$

where the reference pressure p_0 is a constant, the hydrostatic (ambient) pressure p_a has a vertical gradient that balances the ambient density ($\rho_a = \rho_0 + \rho_b(z)$), and the geostrophic pressure p_g has a transverse gradient that balances the Coriolis force due to the geostrophic wind $\boldsymbol{U}$. Only the dynamic pressure p^* appears in the momentum equation (6.73).

Instead of using ρ^*, it is also common to express the buoyancy term as

$$b = -\frac{\rho^* g}{\rho_0}, \tag{6.80}$$

and the 'total' buoyancy

$$\tilde{b} = b + \bar{b} = -\frac{(\rho^* + \bar{\rho}(z))g}{\rho_0}, \tag{6.81}$$

where the background linear stratification is $N^2 = \partial\bar{b}/\partial z$ and we have $\tilde{b} = b + N^2 z$ with the reference value $\bar{\rho}(z = 0) = 0$.

Eqn, (6.74) can also be expressed as

$$\frac{\partial \rho^*}{\partial t} + \frac{\partial \rho^* u_i}{\partial x_i} + w\frac{\partial \bar{\rho}}{\partial z} = \kappa \frac{\partial^2 \rho^*}{\partial x_i^2}, \tag{6.82}$$

and hence we have the buoyancy equation

$$\frac{\partial b}{\partial t} + \frac{\partial b u_i}{\partial x_i} + wN^2 = \kappa \frac{\partial^2 b}{\partial x_i^2}, \tag{6.83}$$

and the equation for the total buoyancy is

$$\frac{\partial \tilde{b}}{\partial t} + \frac{\partial \tilde{b} u_i}{\partial x_i} = \kappa \frac{\partial^2 \tilde{b}}{\partial x_i^2}. \tag{6.84}$$

Roughly, the following independent non-dimensional parameters can be obtained from the comparison of relevant terms:

1. The Reynolds number $Re = UL/\nu$, where L is the horizontal length scale.
2. The Rossby number $Ro = U/fL$.
3. The Froude number $Fr = U/NL$.
4. The aspect ratio $\alpha = H/L$.
5. The Mach number $Ma = U/(L/T)$, where L/T is the group velocity. It is the ratio between the nonlinear term and the tendency term.

6.5.1 Incompressibility

Evan though there is a density transport due to the diffusion (due to the special role that ρ plays; this is not the mass conservation equation)

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_i}{\partial x_i} = \frac{\partial J_{\rho,i}}{\partial x_i} \neq 0, \tag{6.85}$$

we could still establish incompressible condition

$$\frac{\partial u_i}{\partial x_i} = 0 \tag{6.86}$$

with some additional assumtions.

First we review the integration form of the general conservation equation for an arbitraty scalar (per unit mass)

$$\frac{\partial}{\partial t}\left(\iiint_V \rho\psi \,\mathrm{d}V\right) = -\iint_{\Omega=\partial V} (\rho \boldsymbol{u}\psi)\cdot \mathrm{d}\boldsymbol{A} - \iint_{\Omega=\partial V} \rho\kappa(-\nabla\psi)\cdot \mathrm{d}\boldsymbol{A} \tag{6.87}$$

$$= -\iiint_V \nabla\cdot(\rho\boldsymbol{u}\psi)\,\mathrm{d}V - \iiint_V \nabla\cdot(\rho\kappa(-\nabla\psi))\,\mathrm{d}V \tag{6.88}$$

and we have

$$\frac{\partial \rho\psi}{\partial t} + \nabla\cdot(\rho\psi\boldsymbol{u}) = \nabla\cdot(\rho\kappa\nabla\psi). \tag{6.89}$$

It is in a general form of a conservational principle

$$\frac{\partial Q_v}{\partial t} + \nabla \cdot \boldsymbol{F} = 0, \tag{6.90}$$

where $\boldsymbol{F}$ is the flux and $\nabla \cdot \boldsymbol{F}$ is the transport term.

Taking $\psi = 1$ we recover the mass conservation equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = \frac{D\rho}{Dt} + \rho(\nabla \cdot \boldsymbol{u}) = 0. \tag{6.91}$$

Usually, density change along the material lines, is small enough such that $(1/\rho)D\rho/Dt \ll U/L$ and hence $\nabla \cdot \boldsymbol{u} \ll U/L$. That being said, being non-dimensionalised, the velocity field is solenoidal. We note in density-variable flows that

$$\nabla \cdot \boldsymbol{u} = 0 \tag{6.92}$$

is an approximation. See Batchelor (1967), section 3.2, for details as why this is valid.

6.5.2 Scalar transport equation

Taking $\psi = s$ (salinity or temperature) and assume diffusivity κ is constant, we have the scalar transport equation

$$s\frac{\partial \rho}{\partial t} + \rho\frac{\partial s}{\partial t} + u_j\rho\frac{\partial s}{\partial x_j} + u_j s\frac{\partial \rho}{\partial x_j} = \kappa\frac{\partial s}{\partial x_j}\frac{\partial \rho}{\partial x_j} + \rho\kappa\frac{\partial^2 s}{\partial x_j^2}, \tag{6.93}$$

taking into account

$$\frac{\partial \rho}{\partial t} + u_j\frac{\partial \rho}{\partial x_j} = 0 \tag{6.94}$$

we have

$$\rho\frac{\partial s}{\partial t} + u_j\rho\frac{\partial s}{\partial x_j} = \kappa\frac{\partial s}{\partial x_j}\frac{\partial \rho^*}{\partial x_j} + \rho\kappa\frac{\partial^2 s}{\partial x_j^2}. \tag{6.95}$$

We note that under Boussinesq assumption, $\rho^*/\rho = \rho^*/(\rho_0 + \rho*) \ll 1$, we have

$$\frac{\partial s}{\partial t} + u_j\frac{\partial s}{\partial x_j} = \kappa\frac{\partial^2 s}{\partial x_j^2}. \tag{6.96}$$

With some linear equations of state, such as

$$\rho = \rho_0\left(1 - \frac{T - T_0}{T_0}\right), \tag{6.97}$$

we can relate s or T to ρ and get a scalar transport equation for ρ as

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + u_j\frac{\partial \rho}{\partial x_j} = \kappa\frac{\partial^2 \rho}{\partial x_j^2}, \tag{6.98}$$

with the incompressibility being implied by $\nabla \cdot \boldsymbol{u} = 0$ from Eqn. (6.91). We note that Eqns. (6.91) and (6.98) correspond to two different physical principles.

Consider the gas EOS,

$$p = \rho R T \tag{6.99}$$

$$\ln p = ln(\rho) + \ln T + \ln R \tag{6.100}$$

$$\frac{\delta p}{p} = \frac{\delta \rho}{\rho} + \frac{\delta T}{T}. \tag{6.101}$$

In general, for $\rho = \rho(T, p)$,

$$\frac{\delta \rho}{\rho} = \frac{1}{\rho}\frac{\partial \rho}{\partial T}\bigg|_p dT + \frac{1}{\rho}\frac{\partial \rho}{\partial p}\bigg|_T dT dp \tag{6.102}$$

$$= -\alpha dT + \chi_T dp \tag{6.103}$$

where

$$\alpha = -\frac{1}{\rho}\frac{\partial \rho}{\partial T}\bigg|_p, \ \chi_T = \frac{1}{\rho}\frac{\partial \rho}{\partial p}\bigg|_T \tag{6.104}$$

are called isobaric thermal expansion coefficient and isothermal compressibility, respectively.

Assuming an isobaric process, we have

$$\frac{\rho^*}{\rho_0} = \frac{\rho - \rho_0}{\rho_0} = -\alpha(T - T_0) = -\alpha T^* \tag{6.105}$$

$$\frac{1}{\rho_0}\frac{\partial \bar{\rho}}{\partial z} = -\alpha \frac{\partial \bar{T}}{\partial z}, \tag{6.106}$$

and it follows that

$$b = -\frac{\rho^* g}{\rho_0} = \alpha T^* g \tag{6.107}$$

$$N^2 = -\frac{1}{\rho_0}\frac{\partial \bar{\rho}}{\partial z} = \alpha g \frac{\partial \bar{T}}{\partial z}. \tag{6.108}$$

6.6 GFD vorticity equations

6.6.1 Absolute vorticity equation

The 'absolute' vorticity, defined as $\boldsymbol{\omega}_a = \boldsymbol{\omega} + \boldsymbol{f}_c$, is the 'relative' vorticity $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ plus the 'planetary' vorticity $\boldsymbol{f}_c = 2\boldsymbol{\Omega}_c$ ($\Omega_c = \Omega \sin\phi$).

Similar to Eq. (2.22), we can derive the governing equation for $\boldsymbol{\omega}_a$ starting from

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} + \boldsymbol{f}_c \times \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} + \boldsymbol{F} \tag{6.109}$$

without the hydrostatic part separated and/or Boussinesq assumed.

According to identity (A.21) we have, *in an f-plane*,

$$\nabla \times (\boldsymbol{f}_c \times \boldsymbol{u}) = -\boldsymbol{f}_c \cdot \nabla \boldsymbol{u}. \tag{6.110}$$

Similar to Eq. (2.14), by taking the curl of (6.109) and taking $\boldsymbol{F} = b\boldsymbol{e}_z$ we have the vorticity equation in a rotating frame

$$\frac{D\boldsymbol{\omega}}{Dt} = \frac{\partial \boldsymbol{\omega}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{\omega} = \boldsymbol{\omega}_a \cdot \nabla \boldsymbol{u} + \nu \nabla^2 \boldsymbol{\omega} + \nabla \times \boldsymbol{F} + \frac{1}{\rho^2}\nabla \rho \times \nabla p, \tag{6.111}$$

i.e., the absolute vorticity equation:

$$\frac{D\boldsymbol{\omega}_a}{Dt} = \frac{\partial \boldsymbol{\omega}_a}{\partial t} + \boldsymbol{u}\cdot\nabla\boldsymbol{\omega}_a = \frac{\partial \boldsymbol{\omega}}{\partial t} + \boldsymbol{u}\cdot\nabla\boldsymbol{\omega} \tag{6.112}$$

$$= \boldsymbol{\omega}_a\cdot\nabla\boldsymbol{u} + \nu\nabla^2\boldsymbol{\omega} + \nabla\times\boldsymbol{F} + \frac{1}{\rho^2}\nabla\rho\times\nabla p. \tag{6.113}$$

In a Boussinesq fluid,

$$\frac{D\boldsymbol{\omega}}{Dt} = \boldsymbol{\omega}_a\cdot\nabla\boldsymbol{u} + \nu\nabla^2\boldsymbol{\omega} + \nabla\times(b\boldsymbol{e}_z) + \frac{1}{\rho_0^2}\nabla\rho^*\times\nabla p^*, \tag{6.114}$$

where $\nabla\times(b\boldsymbol{e}_z) = \epsilon_{ij3}\partial_j b$.

Additionally, the linearized inviscid evolution equation for the perturbation vorticity, similar to (5.194), reads

$$\frac{\bar{D}\omega_i'}{\bar{D}t} = \omega_j'\bar{S}_{ij} + (\bar{\omega}_j + f\delta_{j3})S_{ij}' - u_j'\frac{\partial\bar{\omega}_i}{\partial x_j} + \frac{1}{2}\epsilon_{ij3}f\omega_j' + \epsilon_{ij3}\frac{\partial b'}{\partial x_j}, \tag{6.115}$$

which is convenient for instability considerations.

Example (Taylor-Proudman theorem, 6.3.4; another proof): Assume inviscid, barotropic fluid acted on by conservative force, and that the rotation rate $\boldsymbol{\Omega}_c = \boldsymbol{f}_c/2$ is much greater than other frequencies. Eqn. (6.113) becomes

$$0 = \boldsymbol{f}_c\cdot\nabla\boldsymbol{u} \tag{6.116}$$

and that completes the proof.

6.6.2 Potential vorticity equation; Ertal's theorem

Ref. Pedlosky (2013).

Assume a conserved scalar λ with a governing operator $D\lambda/Dt = S$ where S is a source term for λ. Consider

$$\frac{D}{Dt}\left(\frac{\partial\lambda}{\partial x_i}\right) = \frac{\partial}{\partial x_i}\left(\frac{\partial\lambda}{\partial t} + u_j\frac{\partial\lambda}{\partial x_j}\right) - \frac{\partial u_j}{\partial x_i}\frac{\partial\lambda}{\partial x_j}, \tag{6.117}$$

i.e.,

$$\frac{D}{Dt}(\nabla\lambda) = \nabla\left(\frac{D\lambda}{Dt}\right) - \nabla\boldsymbol{u}\cdot\nabla\lambda. \tag{6.118}$$

$\nabla\lambda\cdot$ (6.113) + $\boldsymbol{\omega}_a\cdot$ (6.118), with a magic that two opposite-sign $\boldsymbol{\omega}_a\cdot\nabla\boldsymbol{u}\cdot\nabla\lambda$ terms cancel, we have

$$\frac{D}{Dt}(\boldsymbol{\omega}_a\cdot\nabla\lambda) = \boldsymbol{\omega}_a\cdot\nabla S + \nu\nabla^2\boldsymbol{\omega}_a\cdot\nabla\lambda + (\nabla\times\boldsymbol{F})\cdot(\nabla\lambda) + \frac{1}{\rho^2}(\nabla\rho\times\nabla p)\cdot(\nabla\lambda) \tag{6.119}$$

Take $\lambda = \tilde{b}$ which is the total buoyancy, with its governing equation being (6.84), assuming conservative external force $\boldsymbol{F}$ and barotropic flow*, we have the potential vorticity (PV) equation:

$$\frac{D}{Dt}\left(\boldsymbol{\omega}_a\cdot\nabla\tilde{b}\right) = \nu\nabla^2\boldsymbol{\omega}_a\cdot\nabla\tilde{b} + \kappa[\nabla^2(\nabla\tilde{b})]\cdot\boldsymbol{\omega}_a, \tag{6.120}$$

where

$$\Pi = \boldsymbol{\omega}_a\cdot\nabla\tilde{b} \tag{6.121}$$

is called the potential vorticity, which is the component of the absolute vorticity perpendicular to the isosurface (or parallel to the gradient) of $\tilde{b}$. In the absence of dissipation,

$$\frac{D\Pi}{Dt} = 0, \tag{6.122}$$

i.e., PV is conserved along the streamlines. Eq. (6.120) is like a double-diffusion problem with one 'passive' scalar diffuse together with vorticity.

Ertel's theorem: under the follwing assumptions, PV conservation along fluid motion is satisfied (from Eq. (6.119)):

- λ is a conserved quantity that following fluid motion $S = 0$.
- Conservative external force: $\nabla \times \boldsymbol{F} = 0$.
- Either
  1. Baroclinicity absent ($\nabla\rho \times \nabla p = 0$)
  2. λ is only a thermodynamic function of p, ρ, i.e., $\lambda = \lambda(p, \rho)$ so that the last term vanishes when $\cdot(\nabla\lambda)$. For example, $\lambda = s$ (entropy).
- Diffusion-less/inviscid: $\nu = \kappa = 0$.

6.6.3 Relation of PV to Kelvin's theorem in a rotating frame

Similarly, Kelvin's circulation theorem (see (2.6)) in a rotating frame is

$$\frac{D\Gamma_a}{Dt} = \iint_A \frac{\nabla\rho \times \nabla p}{\rho^2} \cdot d\boldsymbol{A}, \tag{6.123}$$

where the absolute circulation is

$$\Gamma_a = \int_A \boldsymbol{\omega}_a \cdot d\boldsymbol{A} = \Gamma + \int_A \boldsymbol{f}_c \cdot d\boldsymbol{A}. \tag{6.124}$$

When the surface A is specifically chosen to be on $\lambda = \lambda(\rho, p))$ = constant (and A is enclosed by a contour l that stays on λ = constant for all time), what follows is that $\nabla\lambda$ must be in the parameter plane spanned by $\nabla\rho$ and ∇p, hence $\nabla\lambda \cdot (\nabla\rho \times \nabla p) = 0$. Here, we note that the normal vector for A is $\boldsymbol{n} = \nabla\lambda/|\nabla\lambda|$ so $(\nabla\rho \times \nabla p) \cdot \boldsymbol{n} = 0$ on the entire plane and

$$\frac{D\Gamma_a}{Dt} = 0. \tag{6.125}$$

The choice of λ is just to choose a surface/contour (of λ = constant) on which $\nabla\rho, \nabla p$ lie in the surface and the baroclinicity term makes zero contribution to the circulation in a baroclinic flow. We can regard PV conservation as a special statement of Kelvin's theorem.

Example: $\lambda(\rho, p) = \rho^2 + p^2$. Think of λ = constant as a cylinderical surface.

Example: For a finite flat area A, $\Gamma_a \approxeq \omega_a A$ and its conservation implies angular momentum conservation, and the change of spacing of λ surfaces (associated with change of area) can change relative vorticity — stretching and compressing.

6.7 Turbulence equations for an active scalar

6.7.1 Mean flow equations

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \tag{6.126}$$

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} - f\epsilon_{ij3}(\bar{u}_j - U_j) = -\frac{1}{\rho_0}\frac{\partial \overline{p^*}}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu \frac{\partial \bar{u}_i}{\partial x_j} - \overline{u_i' u_j'}) - \frac{\overline{\rho^*} g}{\rho_0}\delta_{i3} \tag{6.127}$$

$$\frac{\partial \bar{\rho}}{\partial t} + \bar{u}_j \frac{\partial \bar{\rho}}{\partial x_j} = \frac{\partial}{\partial x_i}\left(\kappa \frac{\partial \bar{\rho}}{\partial x_j} - \overline{\rho' u_j'}\right), \tag{6.128}$$

We note that only vertical variation of density is considered:

$$\bar{\rho} = \overline{\rho^*} + \rho_b(z),\ \rho_b = \rho_0\left(1 - \frac{N^2}{g}z\right),\ \frac{\partial \rho_b}{\partial z} = -\frac{\rho_0 N^2}{g} \tag{6.129}$$

and

$$\rho' = \rho - \bar{\rho} = \rho^* - \overline{\rho^*} = \rho^{*\prime}. \tag{6.130}$$

The equation for $\overline{\rho^*}$ becomes

$$\frac{\partial \overline{\rho^*}}{\partial t} + \bar{u}_j \frac{\partial \overline{\rho^*}}{\partial x_j} - \bar{w}\frac{\rho_0 N^2}{g} = \frac{\partial}{\partial x_i}\left(\kappa \frac{\partial \overline{\rho^*}}{\partial x_j} - \overline{\rho' u_j'}\right). \tag{6.131}$$

6.7.2 Fluctuation equations

$$\frac{\partial u_i'}{\partial x_i} = 0 \tag{6.132}$$

$$\frac{\partial u_i'}{\partial t} + \bar{u}_j \frac{\partial u_i'}{\partial x_j} - f\epsilon_{ij3}u_j' = -\frac{1}{\rho_0}\frac{\partial p^{*\prime}}{\partial x_i} + \frac{\partial}{\partial x_j}(\nu \frac{\partial u_i'}{\partial x_j} + \overline{u_i' u_j'} - u_i' u_j') - u_j'\frac{\partial \bar{u}_i}{\partial x_j} - \frac{\rho^{*\prime} g}{\rho_0}\delta_{i3} \tag{6.133}$$

$$\frac{\partial \rho^{*\prime}}{\partial t} + \bar{u}_j \frac{\partial \rho^{*\prime}}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\kappa \frac{\partial \rho^{*\prime}}{\partial x_j} + \overline{\rho^{*\prime} u_j'} - \rho^{*\prime} u_j'\right) - u_j'\frac{\partial \overline{\rho}}{\partial x_i} \tag{6.134}$$

$$= \frac{\partial}{\partial x_j}\left(\kappa \frac{\partial \rho^{*\prime}}{\partial x_j} + \overline{\rho^{*\prime} u_j'} - \rho^{*\prime} u_j'\right) - u_j'\frac{\partial \overline{\rho^*}}{\partial x_i} + w'\frac{\rho_0 N^2}{g} \tag{6.135}$$

We will see later the Coriolis term won't appear in the transport equations of MKE, TKE, and Reynolds stresses. Coriolis just bends the direction of the velocity. In other words, Coriolis does not have direct influence on turbulence budgets, but indirectly through the change of the mean flow. On the other hand, it does make a difference in the vorticity/enstrophy equation (see section **??**).

The mean and fluctuation density equations are from the scalar transport equations in Section 5.1.10.

6.7.3 MKE, MPE, TKE, TPE, and buoyancy flux equations

Define the mean and turbulent kinetic and potential energy as

$$K = \frac{1}{2}\bar{u}_i\bar{u}_i \tag{6.136}$$

$$K_\rho = \frac{1}{2N^2}\bar{b}^2 \tag{6.137}$$

and

$$k = \frac{1}{2}\overline{u_i'u_i'} \tag{6.138}$$

$$k_\rho = \frac{1}{2N^2}\overline{b'b'} \tag{6.139}$$

where the instantaneous, mean, and fluctuation buoyancy are (with the dimension being acceleration)

$$b = -\frac{\rho^* g}{\rho_0},\ \bar{b} = -\frac{\overline{\rho^*} g}{\rho_0},\ b' = -\frac{\rho^{*\prime} g}{\rho_0}, \tag{6.140}$$

such that k and k_ρ have the same dimension as kinetic energy. We note that

$$\rho^{*\prime} = \rho^* - \overline{\rho^*} = (\rho - \rho_b) - \overline{(\rho - \rho_b)} = \rho - \bar{\rho} = \rho'. \tag{6.141}$$

The **MKE equations** is (repeating (5.32) but with a mean transport) :

$$\frac{\partial K}{\partial t} + \bar{u}_j\frac{\partial K}{\partial x_j} = \frac{\partial}{\partial x_j}\left(-\frac{1}{\rho}\overline{p}\,\overline{u}_j + \nu\frac{\partial K}{\partial x_j} - \overline{u}_i\,\overline{u_i'u_j'}\right) + \overline{u_i'u_j'}\frac{\partial \bar{u}_i}{\partial x_j} - \nu\frac{\partial \overline{u}_i}{\partial x_j}\frac{\partial \overline{u}_i}{\partial x_j} + \bar{b}\bar{w} \tag{6.142}$$

The **MPE equations** is:

$$\frac{\partial K_\rho}{\partial t} + \bar{u}_j\frac{\partial K_\rho}{\partial x_j} = \frac{\partial}{\partial x_j}\left(\kappa\frac{\partial K_\rho}{\partial x_j} - \frac{1}{N^2}\bar{b}\,\overline{b'u_j'}\right) + \frac{1}{N^2}\overline{b'u_j'}\frac{\partial \bar{b}}{\partial x_j} - \frac{\kappa}{N^2}\frac{\partial \bar{b}}{\partial x_j}\frac{\partial \bar{b}}{\partial x_j} - \bar{b}\bar{w} \tag{6.143}$$

We note that the mean buoyancy flux $\bar{b}\bar{w}$ moves energy from MKE to MPE and the turbulent buoyancy flux participates in the production $\overline{b'u_j'}\partial\bar{b}/\partial x_j$, when $\partial\bar{b}/\partial x_j$ is negative, is a sink in the MPE equation and is a source in the TPE equation.

The **TKE equations** is:

$$\frac{\partial k}{\partial t} + \bar{u}_j\frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_k}\Big(\underbrace{\nu\frac{\partial k}{\partial x_k}}_{\substack{\text{molecular}\\ \text{diffusion}}} + \underbrace{\frac{1}{2}\overline{u_i'u_i'u_k'}}_{\substack{\text{turbulent}\\ \text{diffusion}}} - \underbrace{\frac{1}{\rho_0}\overline{p'u_k'}}_{\substack{\text{pressure}\\ \text{distortion}}}\Big) \underbrace{-\overline{u_i'u_j'}\frac{\partial \bar{u}_i}{\partial x_j}}_{\text{production } P} - \underbrace{\nu\overline{\frac{\partial u_i'}{\partial x_k}\frac{\partial u_i'}{\partial x_k}}}_{\text{dissipation } \varepsilon} + \underbrace{\overline{b'w'}}_{\substack{\text{buoyancy}\\ \text{flux } B}} \tag{6.144}$$

$$= \nabla\cdot\boldsymbol{T} + P - \varepsilon + B \tag{6.145}$$

where the turbulent buoyancy flux is

$$B = -\frac{g}{\rho_0}\overline{\rho^{*\prime}w'} = \overline{b'w'}. \tag{6.146}$$

The **TPE equation** is:

$$\frac{\partial k_\rho}{\partial t} + \bar{u}_j\frac{\partial k_\rho}{\partial x_j} = \frac{\partial}{\partial x_j}(\kappa\frac{\partial k_\rho}{\partial x_j} - \frac{1}{2N^2}\overline{b'b'u_j'}) - \frac{1}{N^2}\overline{b'u_j'}\frac{\partial \bar{b}}{\partial x_j} - \frac{\kappa}{N^2}\overline{\frac{\partial b'}{\partial x_j}\frac{\partial b'}{\partial x_j}} - \overline{b'w'} \tag{6.147}$$

We can see that the turbulent buoyancy flux B (negative, think $-\overline{u_i' u_j'}$) moves energy from TKE to TPE (similar to the mean buoyancy flux that moves energy from MKE to MPE) and also participates in TPE's own gradient production through $\overline{b' u_j'} \partial \bar{b} / \partial z$ that moves energy from MPE and TPE. We also note that if we sum over MKE, MPE, TKE, TPE equations the production and buoyancy flux terms will have net zero contribution to the total energy. We can also find the role of buoyancu flux bw by multiplying the w equation with b and the b equation with w in (7.1).

The **buoyancy flux equation** is:

$$\frac{\partial \overline{b' u_i'}}{\partial t} + \bar{u}_j \frac{\partial \overline{b' u_i'}}{\partial x_j} = d_{b,i} + P_{b,i} + \Phi_{b,i} - \varepsilon_{b,i} + B_i \tag{6.148}$$

where

$$d_{b,i} = \frac{\partial}{\partial x_j} (\kappa \overline{u_i' \frac{\partial b'}{\partial x_j}} + \nu \overline{b' \frac{\partial u_i'}{\partial x_j}} - \frac{1}{\rho_0} \overline{p^{*\prime} b'} \delta_{ij} - \overline{b' u_i' u_j'}) \tag{6.149}$$

$$P_{b,i} = -\overline{b' u_j'} \frac{\partial \bar{u}_i}{\partial x_j} - \overline{u_i' u_j'} \frac{\partial \bar{b}}{\partial x_j} \tag{6.150}$$

$$\Phi_{b,i} = -\frac{1}{\rho_0} \overline{p^{*\prime} \frac{\partial b'}{\partial x_i}} \tag{6.151}$$

$$\varepsilon_{b,i} = (\nu + \kappa) \overline{\frac{\partial u_i'}{\partial x_j} \frac{\partial b'}{\partial x_j}} \tag{6.152}$$

$$B_i = -\overline{u_i' w'} N^2 + \overline{b' b'} \delta_{i3} \tag{6.153}$$

We note that we have re-written

$$-\frac{1}{\rho_0} \overline{b' \frac{\partial p^{*\prime}}{\partial x_i}} = -\frac{1}{\rho_0} \left(\frac{\partial}{\partial x_j} \overline{p^{*\prime} b'} \delta_{ij} - \overline{p^{*\prime} \frac{\partial b'}{\partial x_i}} \right), \tag{6.154}$$

and the new term (6.153) did not appear when we derive the passive scalar flux equation.

During $\partial \bar{\rho} / \partial x_3 \neq 0$ we are mostly concerned about $\overline{b' w'}$, which is typically nagative (think about a displaced parcel argument). Its equation is ($i = 3$)

$$\frac{\partial \overline{b' w'}}{\partial t} + \bar{u}_j \frac{\partial \overline{b' w'}}{\partial x_j} = \frac{\partial}{\partial x_j} (\kappa \overline{w' \frac{\partial b'}{\partial x_j}} + \nu \overline{b' \frac{\partial w'}{\partial x_j}} - \frac{1}{\rho_0} \overline{p^{*\prime} b'} \delta_{j3} - \overline{b' w' u_j'}) \tag{6.155}$$

$$- \overline{b' u_j'} \frac{\partial \overline{w}}{\partial x_j} - \overline{w' u_j'} \frac{\partial \bar{b}}{\partial x_j} - \overline{w' w'} N^2 + \overline{b' b'} \tag{6.156}$$

$$- \frac{1}{\rho_0} \overline{p^{*\prime} \frac{\partial b'}{\partial x_3}} + (\nu + \kappa) \overline{\frac{\partial w'}{\partial x_j} \frac{\partial b'}{\partial x_j}} \tag{6.157}$$

Two additional terms (with respect to scalar flux equation, for $\overline{u_i' \phi'}$, where ϕ' is a passive scalar) appear:

$$-\overline{w' w'} N^2 + \overline{b' b'} \tag{6.158}$$

that mark the difference made by an active scalar. They would both be zero if the density is not coupled to buoyancy (simply by setting the gravity g to zero). When $g \neq 0$, density-velocity

coupling plays a role in the dynamics of $\overline{b'w'}$. The effect of buoyancy is restoring (making $B = \overline{b'w'}$ less negative) and the effect of vertical perturbation increases the magnitude of B in a stably stratified medium ($N^2 > 0$).

If we neglect the transport terms, inhomogeneity (think homogeneous stratified turbulence; we have eliminated $\partial_z \bar{w}$, $\partial_z \overline{\rho^*}$ but not $\partial_z \rho_b$), and pressure fluctuations, we can reach a simple set of autonomous equations

$$\frac{\partial(\overline{w'w'}/2)}{\partial t} = \overline{b'w'} \tag{6.159}$$

$$\frac{\partial(\overline{b'b'}/2N^2)}{\partial t} = -\overline{b'w'} \tag{6.160}$$

$$\frac{\partial \overline{b'w'}}{\partial t} = -\overline{w'w'}N^2 + \overline{b'b'} \tag{6.161}$$

which can be written as a linear system that has eigenvalues $\lambda = \pm \mathrm{i}\sqrt{2}N$, $\lambda = 0$ (when $N^2 > 0$, stable stratification), where the purely imaginary EVs can be seen from the reducing the system to

$$\partial_{tt}\overline{b'w'} = -2N^2\overline{b'w'}. \tag{6.162}$$

This implies that the transfer mechanism between TKE, TPE, and buoyancy flux is energy-conserving. The phase relations can be seen from the general solution

$$\overline{b'w'} = \cos(\sqrt{2}Nt) \tag{6.163}$$

$$\overline{w'w'}/2 = \frac{1}{\sqrt{2}N}\sin(\sqrt{2}Nt) \tag{6.164}$$

$$\overline{b'b'}/2N^2 = -\frac{1}{\sqrt{2}N}\sin(\sqrt{2}Nt) \tag{6.165}$$

6.8 Miscellaneous

coriolis frequency;

6.8.1 Derivation of Coriolis force

6.8.2 Boussinesq approximation

Chapter 7

Waves in GFD

7.1 Internal waves: governing equations and dispersion relation

The governing equations for waves can be simplified from (6.76), with the following assumptions:

1. The Ekman number is small (neglect viscosity).
2. The fluid is Boussinesq.
3. The 'Mach' number is small (neglect the nonlinear term/only effects to the first order are retained).
4. The aspect ratio is $\alpha = H/L = W/U \ll 1$.

The set of simplified equations read (with $\rho_0 = 1$)

$$\begin{aligned} \frac{\partial u}{\partial t} &= -\frac{\partial p}{\partial x} + fv \\ \frac{\partial v}{\partial t} &= -\frac{\partial p}{\partial y} - fu \\ \frac{\partial w}{\partial t} &= -\frac{\partial p}{\partial z} + b \\ \frac{\partial b}{\partial t} &= -wN^2 \\ \nabla \cdot \boldsymbol{u} &= 0 \end{aligned} \tag{7.1}$$

and by eliminating pressure and cross-differentiation, an equation for w can be obtained as

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right) + f^2\frac{\partial^2 w}{\partial z^2} + N^2(z)\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}\right) = 0. \tag{7.2}$$

The detailed steps include: (1) eliminate pressure from the momentum equations and obtain a system of three pressure-less equations; (2) cross-differentiate to get the forms of $\partial_{xx}b$ and use the buoyancy equation to replace b with w; we note that we assume $N(z)$ is only a function of the vertical coordinate so its z-dependence can be retained; (3) ∂_{yz} the u-equation and ∂_{xz} the v-equation, to elinimate triple-cross-difference terms as $\partial_{txy}u$; (4) organization.

A 'trivial' solution to (7.2) is simply the geostrophic balance:

$$w = 0,\ fv = p_x,\ fu = -p_y,\ b = p_z. \tag{7.3}$$

Assume non-zero solution in the form of normal modes

$$[u, v, w, p, b] = [\hat{u}, \hat{v}, \hat{w}, \hat{p}, \hat{b}] \exp[i(kx + ly + mz - \omega t)] \tag{7.4}$$

where the wavenumber vector is

$$\boldsymbol{k} = (k, l, m), \tag{7.5}$$

the waves that the equation (7.2) describes should satisfy the dispersion relation

$$\omega^2 = \frac{f^2 m^2 + (k^2 + l^2) N^2}{k^2 + l^2 + m^2} = f^2 \sin^2 \phi + N^2 \cos^2 \phi \tag{7.6}$$

where ϕ is the angle between the wave-vector $\boldsymbol{k}$ and the horizontal plane (k, l). It can be seen that the feasible frequencies lie in the range of

$$f < \omega < N \tag{7.7}$$

for the general case in the ocean that $N/f \gg 1$.

From (7.6) we can get

$$\frac{N^2 - \omega^2}{\omega^2 - f^2} = \frac{(N^2 - f^2) \sin^2 \phi}{(N^2 - f^2) \cos^2 \phi} = \tan^2 \phi = \frac{1}{\alpha^2}, \tag{7.8}$$

where the vertical-to-horizontal aspect ratio is defined as

$$\alpha = \frac{k^2 + l^2}{m^2} = \cot \phi, \tag{7.9}$$

which is close to zero at the hydrostatic limit ($m^2 \ll k^2 + l^2$). In such case, $N \gg \omega$.

We can also see that the continuity equation translates to

$$\boldsymbol{k} \cdot \hat{\boldsymbol{u}} = 0, \tag{7.10}$$

that being said, the wavenumber vector is perpendicular to the particle motions.

The dispersion relation also implies that the frequency only depends on the direction of the wavevector, but not its magnitude. All wavenumbers on a cone of constant ϕ have the same frequency, and the direction of the group velocity, $c_p = d\omega/dk$, is perpendicular to the cone surface. A trivial solution to the equation (7.2) is simply $w = 0$, which is corresponding to a geostrophic+hydrostatic balance.

Examples. (1) For a non-rotationg fluid, $\omega^2 = N^2 \cos^2 \phi$, and the portion of stratification that the motion feels is proportional to the projection of its motion in the vertical direction. Oscillation that feels the full effect of stratification is vertical, and correspond to the largest possible frequency, a horizontal wavenumber, and $\phi = 0$. (2) For $\phi = \pi/2$, the wavevector is perpendicular and there is only inertial oscillations.

7.1.1 General solutions

Assume $N = N_0$ is a constant and without lost of generality the coordinate can be re-oriented such that $\partial_y = 0$ and $\partial_x u + \partial_z w = 0$. The rest of the flow quantities can be recovered from the solution of $\hat{w}$ as

$$\hat{u} = -\frac{m}{k}\hat{w} \tag{7.11}$$

$$\hat{v} = i\frac{fm}{\omega k}\hat{w} = -i\frac{f}{\omega}\hat{u} \tag{7.12}$$

$$\hat{p} = -\left(\frac{N^2 - \omega^2}{m\omega}\right)\hat{w} = \left(\frac{N^2 - \omega^2}{k\omega}\right)\hat{u} \tag{7.13}$$

$$\hat{b} = -i\frac{N^2}{\omega}\hat{w},\ \hat{p} = -i\left(\frac{N^2 - \omega^2}{mN^2}\right)\hat{b} \tag{7.14}$$

The above relations have a number of implications:

1. $\hat{u}$ and $\hat{w}$ are in phase.
2. $\hat{u}$ and $\hat{v}$ are out of phase (think inertial oscillations).
3. Pressure and vertical motion are in phase, but both are out of phase with buoyancy.
4. (TBD) the vertical structures (CW or CCW) of the waves.

7.1.2 Hydrostatic approximation

With the hydrostatic approximation is equivalent to

$$k^2 + l^2 \ll m^2, \tag{7.15}$$

with which the dispersion relation (7.6) reduces to

$$\omega^2 = f^2 + \frac{k^2 + l^2}{m^2}N^2. \tag{7.16}$$

7.1.3 Near-inertial waves

The near-inertialness,

$$\frac{\omega - f}{f} \ll 1, \tag{7.17}$$

implies that

$$\frac{\omega^2 - f^2}{f^2} = \frac{(\omega - f)^2 - 2f^2 + 2\omega f}{f^2} \approx \frac{2(\omega - f)}{f}. \tag{7.18}$$

While (7.6) is re-organized to

$$\frac{\omega^2 - f^2}{f^2} = \left(\frac{N^2}{f^2} - 1\right)\frac{k^2 + l^2}{k^2 + l^2 + m^2}, \tag{7.19}$$

which can only be made very small when the second factor is small, that said,

$$k^2 + l^2 \ll m^2, \tag{7.20}$$

which is the hydrostatic relation so (7.16) applies. With that, the dispersion relation for near-inertial waves is

$$\omega = f + \frac{N^2}{2f}\frac{k^2 + l^2}{m^2}. \tag{7.21}$$

In fact, near-inertial waves are super-hydrostatic. Taking an appropriate orientation of the coordinate frame ($l = 0$,$\partial_y = 0$), we can express the near-inertialness indicator

$$\frac{\omega - f}{f} = \frac{N^2 k^2}{f^2 m^2} = \frac{Bu}{2} \ll 1, \tag{7.22}$$

and the group velocities

$$c_{gv} = \frac{d\omega}{dm} = -\frac{N^2 k^2}{f m^3} = -Bu\frac{f}{m}. \tag{7.23}$$

$$c_{gh} = \frac{d\omega}{dk} = \frac{N^2 k}{f m^2} = Bu\frac{f}{k}. \tag{7.24}$$

Both c_{gv} and c_{gh} are small.

7.1.4 Vertical modes

If we ease the assumption (7.4) to allow a vertical dependence of the modes, since N is typically a function of z, i.e.,

$$w(x, y, z, t) = a(z)\exp[i(kx + ly - \omega t)], \tag{7.25}$$

by substituting into (7.2) we have

$$\frac{d^2 a}{dz^2} + \left(\frac{N^2 - \omega^2}{\omega^2 - f^2}\right)(k^2 + l^2)a = 0, \tag{7.26}$$

with rigid lid BC's $w(0) = w(-h) = 0$ that implies $a(0) = a(-h) = 0$. This is a typical Sturm–Liouville eigenvalue problem. Moreover, the derivative $a' = \partial_z a$ also satisfies another Sturm–Liouville equation. It will be shown later that the set of eigenmodes $\{a_m\}$ are mutually orthogonal under the inner-product xxx, where m is now the mode index.

The solution of (7.26) can be done for each specified freuquency: regard $k^2 + l^2$ as the eigenvalue, solve (7.26) with given $N(z)$ and ω. Under hydrostatic, it can be shown from $\alpha \approx 1$ that $N \gg \omega$. Assume the hydrostatic dispersion relation (for each mode m)

$$\omega_m^2 \approx f^2 + c_m^2(k^2 + l^2) \tag{7.27}$$

we have

$$\frac{d^2 a}{dz^2} + \frac{N^2}{c^2}a = 0, \tag{7.28}$$

where the eigenvalues $\{c_m^{-2}\}$ are the wave speed of each mode m and corresponding Rossby radius of deformation is $\lambda_m = c_m/f$. We note that f is absent in (7.28) and the eigenmodes it defines.

7.2 Inertial and buoyancy oscillations

The two most fundamental pedagogical oscillations in GFD are the inertial and buoyancy oscillations. The inertial oscillation results from the pressure-less version of the horizontal momentum equations (7.1),

$$u_t - fv = 0 \tag{7.29}$$

$$v_t + fu = 0. \tag{7.30}$$

It can be arranged into

$$(u + iv)_t + if(u + iv) = 0 \tag{7.31}$$

which has the solution

$$u + iv = e^{-ift}[u(0) + iv(0)], \tag{7.32}$$

which is rotating clockwise in Northern Hemisphere at an angular frequency f – the Coriolis frequency. Similarly, the pressure-less vertical and buoyancy equations give

$$w_{tt} = -N^2 w, \tag{7.33}$$

leading to the solutions

$$w = \begin{cases} w(0)e^{iNt}, \, N^2 > 0 \\ w(0)e^{\sqrt{-N^2}t}, \, N^2 < 0 \end{cases} \tag{7.34}$$

with the first solution corresponding to buoyancy oscillation in stably stratified fluid and the second corresponding to a gravitationally unstable mode under unstable stratification.

7.3 Shallow water waves

Assumptions:

- Inviscid
- Homogeneous fluid
- Horizontal scale $\gg$ vertical scale ($L \gg H$)
- Small amplitude ($\eta \ll H$)

Scaling analysis. Continuity:

$$\frac{W}{H} \sim \frac{U}{L}, \tag{7.35}$$

momentum:

$$\frac{\partial u}{\partial t} \sim \frac{U}{T} \tag{7.36}$$

$$u\frac{\partial u}{\partial x} \sim v\frac{\partial u}{\partial y} \sim U\frac{U}{L} \tag{7.37}$$

$$w\frac{\partial u}{\partial z} \sim U\frac{W}{H} \tag{7.38}$$

$$-\frac{1}{\rho}\frac{\partial p}{\partial x} \sim \frac{U^2}{L} \tag{7.39}$$

$$\frac{\partial w}{\partial t} \sim \frac{W}{T} \tag{7.40}$$

$$-\frac{1}{\rho}\frac{\partial p}{\partial z} \sim \frac{U^2}{H} \tag{7.41}$$

The incompressibility implies that the speed of motion is much smaller than some sort of wave speed:

$$\frac{U}{L/T} = \frac{\text{inertial}}{\text{acceleration}} \ll 1, \tag{7.42}$$

and the aspect ratio implies

$$\frac{\partial w/\partial t}{-(1/\rho_0)\partial p/\partial z} = \left(\frac{H}{L}\right)^2 \ll 1. \tag{7.43}$$

Hence, we have neglected the non-linear terms in horizontal momentum equations and the vertical momentum equation reduces to a hydrostatic balance. The shallow water equations (SWE) read

$$u_t = -\frac{1}{\rho_0}p_x \tag{7.44}$$

$$v_t = -\frac{1}{\rho_0}p_y \tag{7.45}$$

$$0 = -\frac{1}{\rho_0}p_z - g, \tag{7.46}$$

$$u_x + v_y = 0 \tag{7.47}$$

The hydrostatic pressure balance leads to

$$p(z) = p(H+\eta) + \int_{H+\eta}^{z} -\rho g dz = p_0 + \rho g(H + \eta - z), \tag{7.48}$$

where η is the surface height displacement and H is the depth ($z = 0$ at the bottom). The horizontal pressure gradients are given as $p_x = \rho g \eta_x, p_y = \rho g \eta_y$. The intuition is that when the surface is elevated it also gives the fluid below a higher hydrostatic pressure.

Now let's derive an equation for the surface motions.

$$\frac{\partial \eta}{\partial t} \sim \frac{D\eta}{Dt} = w(H+\eta) = w(H+\eta) - w(0) \tag{7.49}$$

$$= \int_0^{H+\eta} \frac{\partial w}{\partial z} dz = \int_0^{H+\eta} -\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) dz \tag{7.50}$$

$$\approx -(H+\eta)\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) \tag{7.51}$$

$$\approx -H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) \tag{7.52}$$

where we have applied the small amplitude assumption and the slab assumption.

Finally, the SWE become

$$u_t - fv + g\eta_x = 0 \tag{7.53}$$

$$v_t + fu + g\eta_y = 0 \tag{7.54}$$
$$\eta_t + H(u_x + v_y) = 0 \tag{7.55}$$

where the lst equation can be written as

$$(g\eta)_t + c^2(u_x + v_y) = 0 \tag{7.56}$$

where $c^2 = gH$ is the phase/group speed and $p = g\eta$ is the hydrostatic pressure. Shallow water waves are non-dispersive. Cross-differentiate the SWE, then we reach

$$\eta_{tt} - c^2(\eta_{xx} + \eta_{yy}) + Hf\zeta = 0, \tag{7.57}$$

where the relative vorticity $\zeta = v_x - u_y$. The curl of the first two equations of the SWE gives

$$\zeta_t + f(u_x + v_y) = 0. \tag{7.58}$$

Combined with the η_t equation (continuity), we have

$$\frac{\partial}{\partial t}\left(\frac{\zeta}{f} - \frac{\eta}{H}\right) = \frac{\partial q}{\partial t} = 0, \tag{7.59}$$

where $q = \zeta/f - \eta/H$ is the potential vorticity (reduction in SWE). The conservation implies

$$\zeta = \frac{\eta f}{H} + fq_0. \tag{7.60}$$

Substitute this into (7.57) we have (the homogeneous part of the equation):

$$\eta_{tt} - c^2(\eta_{xx} + \eta_{yy}) + f^2\eta = 0. \tag{7.61}$$

Assuming normal modes $\eta = \eta_0 \exp[i(kx + ly - \omega t)]$ with oriented coordinates ($l = 0$), we have the dispersion relation

$$\omega^2 = c^2(k^2 + l^2) + f^2. \tag{7.62}$$

- Large $k^2 + l^2$ (fast waves): $\omega \approx \sqrt{gH}k$. Phase speed $c_p = \omega/k = \sqrt{gH}$. Group speed $c_g = d\omega/dk = \sqrt{gH} = c_p$. They don't feel the rotation of the Earth. We note that the phase is

  $$\Phi = kx + ly - \omega t = \sqrt{k^2 + l^2}(kx + ly - \omega t)/\sqrt{k^2 + l^2} = |\boldsymbol{k}|(\hat{\boldsymbol{k}} \cdot \boldsymbol{x} - c_p t).$$

  Example. (Tsunami speed) Assume $H = 4$ km is the averaged ocean depth and $L = 8000$ km is the width of the ocean basin. With the SWE assumption $H \ll L$ satisfied, $c = \sqrt{gH}$ gives ≈ 200 m/s or 720 km/h of propagation speed.

- Small $k^2 + l^2$ (slow waves, small Ro): $\omega \approx f$. Near-inertial waves.

7.3.1 Travelling wave solutions

Consider non-rotating (or fast) waves. The formal solution to the 1D SWE is

$$\eta = G(x - ct) + F(x + ct) \tag{7.63}$$

with both G and F satisfying ($c^2 = Hg$)

$$\eta_{tt} - c^2\eta_{xx} = (\partial_t + c\partial_x)(\partial_t - c\partial_x)\eta = 0. \tag{7.64}$$

Moreover,

$$u = \int \frac{\partial u}{\partial t} dt = \int -g\frac{\partial \eta}{\partial x} = \frac{g}{c}[G(x - ct) - F(x + ct)]. \tag{7.65}$$

Without the loss of generality, we consider right-propagating waves with $F = 0$. Consider a single wavenumber,

$$\eta = \Re\{A\exp[i(kx - \omega t)]\} = A\cos(kx - \omega t), \tag{7.66}$$

where $\omega = ck$ and c is the phase speed. We have

$$u = \frac{gk}{\omega} A\cos(kx - \omega t), \tag{7.67}$$

which is in phase with η. We check *a posteriori* that in order to make the 'Mach' number very small, we should have

$$\frac{u}{c_p} = \frac{gkA/\omega}{\omega/k} = \frac{gk^2/\omega}{\omega^2} A \ll 1, \tag{7.68}$$

where A is the amplitude.

Here

$$\phi = kx - \omega t = \boldsymbol{k}\cdot\boldsymbol{x} - \omega t \tag{7.69}$$

is called the phase. For 2D waves,

$$\phi = kx + ly - \omega t = \sqrt{k^2 + l^2}\left(\frac{kx + ly}{\sqrt{k^2 + l^2}} - \frac{\omega t}{\sqrt{k^2 + l^2}}\right) = |\boldsymbol{k}|\left(\frac{kx + ly}{|\boldsymbol{k}|} - |c_p|t\right) \tag{7.70}$$

and the phase speed is

$$|c_p| = \frac{\omega}{|\boldsymbol{k}|}. \tag{7.71}$$

The phase 'velocity', by definition,

$$\boldsymbol{c}_p = \frac{\omega}{|\boldsymbol{k}|}\left(\frac{k}{|\boldsymbol{k}|}, \frac{l}{|\boldsymbol{k}|}\right) //\boldsymbol{k}, \tag{7.72}$$

is parallel to the wavevector

$$\boldsymbol{k} = (k, l), \tag{7.73}$$

which is perpendicular to the constant phase lines

$$\phi_0 = kx + ly - \omega t_0 = C \to y = -\frac{k}{l}x + \frac{\phi_0}{l} // \left(1, -\frac{k}{l}\right). \tag{7.74}$$

7.4 Deep water waves

We relax the assumption $H \ll L$ in SWE and allow vertical motions (the fluid is no longer moving like slabs) and reach the deep water equations (DWE)

$$u_t = -\frac{1}{\rho_0} p_x \tag{7.75}$$

$$v_t = -\frac{1}{\rho_0} p_y \tag{7.76}$$

$$w_t = -\frac{1}{\rho_0} p_z \tag{7.77}$$

$$u_x + v_y + w_z = 0 \tag{7.78}$$

where the pressure is its deviation from the background hydrostatic balance

$$-\frac{1}{\rho_0}\bar{p}_z = g \rightarrow \tilde{p}(x, y, z, t) = p(x, y, z, t) - \rho_0 g z + p_0. \tag{7.79}$$

We note that $H \ll L$ is essential in eliminating the w-equation in the SWE.

Chapter 8

Hydrodynamic stability

8.1 Linearized Navier–Stokes

Consider the incompressible N-S equations

$$\nabla \cdot \boldsymbol{u} = 0 \tag{8.1}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \boldsymbol{u} \tag{8.2}$$

and the decomposition of velocity and pressure into the base and perturbation states:

$$\boldsymbol{u} = \boldsymbol{U} + \boldsymbol{u}' \tag{8.3}$$

$$p = P + p' \tag{8.4}$$

We note that the base state also satisfies N-S:

$$\nabla \cdot \boldsymbol{U} = 0 \tag{8.5}$$

$$\frac{\partial \boldsymbol{U}}{\partial t} + \boldsymbol{U} \cdot \nabla \boldsymbol{U} = -\frac{1}{\rho}\nabla P + \nu \nabla^2 \boldsymbol{U} \tag{8.6}$$

hence by plugging in the decomposition to (8.1)-(8.2) we have the perturbation equation:

$$\nabla \cdot \boldsymbol{u}' = 0 \tag{8.7}$$

$$\frac{\partial \boldsymbol{u}'}{\partial t} + \boldsymbol{U} \cdot \nabla \boldsymbol{u}' + \boldsymbol{u}' \cdot \nabla \boldsymbol{U} + \boldsymbol{u}' \cdot \nabla \boldsymbol{u}' = -\frac{1}{\rho}\nabla p' + \nu \nabla^2 \boldsymbol{u}' \tag{8.8}$$

And we note that the boundary conditions that the perturbation $\boldsymbol{u}', p'$ satisfy is homogeneous, such that $\boldsymbol{U}$ and p_b satisfy the same BC's as $\boldsymbol{u}$ and p in the original equation.

In linear stability, with the assumption that

$$O(\boldsymbol{u}') = \epsilon O(\boldsymbol{U}), \tag{8.9}$$

we neglect the nonlinear term $\boldsymbol{u}' \cdot \nabla \boldsymbol{u}'$ and the primes, and have the linearised perturbation equation

$$\nabla \cdot \boldsymbol{u} = 0 \tag{8.10}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{U}\cdot\nabla\boldsymbol{u} - \nu\nabla^2\boldsymbol{u} = -\boldsymbol{u}\cdot\nabla\boldsymbol{U} - \frac{1}{\rho}\nabla p \tag{8.11}$$

or if we define the linear operator as

$$\mathcal{L}_{\boldsymbol{U}} = \frac{\partial}{\partial t} + \boldsymbol{U}\cdot\nabla - \nu\nabla^2 \tag{8.12}$$

there is

$$\mathcal{L}_{\boldsymbol{U}}\boldsymbol{u} = -\boldsymbol{u}\cdot\nabla\boldsymbol{U} - \frac{1}{\rho}\nabla p. \tag{8.13}$$

That being said, the linear mechanism is that the fluctuations extract energy from the mean flow, to the leading order effect, instead of interacting with themselvess.

The linearised equations (8.10)-(8.11), if written in matrix form (Arratia, 2011), is

$$\mathcal{L}_{\mathrm{NS}}\boldsymbol{q} = \begin{bmatrix} \mathcal{L}_{\boldsymbol{U}} + \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} & \frac{\partial}{\partial x} \\ \frac{\partial V}{\partial x} & \mathcal{L}_{\boldsymbol{U}} + \frac{\partial V}{\partial y} & \frac{\partial V}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial W}{\partial x} & \frac{\partial W}{\partial y} & \mathcal{L}_{\boldsymbol{U}} + \frac{\partial W}{\partial z} & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \\ p \end{bmatrix} = 0, \tag{8.14}$$

where $\boldsymbol{q} = [u, v, w, p]^T$. This is in the KKT form that will be described below, where we will see that the same mathematical properties of the operators will be shared in both stability analysis and CFD.

On the other hand, it is sometimes also convenient to define the RHS operator as

$$\mathcal{A}_{\boldsymbol{U}}\boldsymbol{u} = -\boldsymbol{U}\cdot\nabla\boldsymbol{u} - \boldsymbol{u}\cdot\nabla\boldsymbol{U} - \frac{1}{\rho}\nabla p + \nu\nabla^2\boldsymbol{u} \tag{8.15}$$

such that

$$\frac{\partial \boldsymbol{u}}{\partial t} = \mathcal{A}_{\boldsymbol{U}}\boldsymbol{u}. \tag{8.16}$$

In practice, the pressure gradient can be neglected when calculating the operator $\mathcal{A}_{\boldsymbol{U}}$ and the resulting pressure is projected onto a divergence-free space (along with satisfying the boundary conditions in each numerical iteration/time-step), under the framework of projection methods. In general, in theoretical analysis or numerical methods where the pressure p is elimenated (by procedures similar to taking the curl of the N–S), the equations are to some extent equivalent to the vorticity equation.

8.1.1 The role of pressure

A separate short note on the pressure playing the role of Lagrangian multiplier in incompressible system. Consider the Stokes flow (actually that can be the linearized equations as described above)

$$-\nabla^2\boldsymbol{u} + \nabla p = \boldsymbol{f} \tag{8.17}$$

$$\nabla\cdot\boldsymbol{u} = 0 \tag{8.18}$$

and in the matrix form

$$\begin{bmatrix} -\nabla^2 & \nabla \\ -\nabla\cdot & O \end{bmatrix} \begin{bmatrix} \boldsymbol{u} \\ p \end{bmatrix} = \begin{bmatrix} \boldsymbol{f} \\ 0 \end{bmatrix} \tag{8.19}$$

and its discrete version

$$\begin{bmatrix} A & B^T \\ B & O \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}, \tag{8.20}$$

which is a saddle point problem or a KKT (Karush-Kuhn-Tucker) system (Benzi *et al.*, 2005). The Stokes equations can be interpreted as a constrained optimization problem (section 3.15.5 of Gresho & Sani (1998))

$$\min J(\boldsymbol{u}) = \frac{1}{2}\int \|\nabla \boldsymbol{u}\|_2^2 dV - \int \boldsymbol{f}\cdot\boldsymbol{u} dV \tag{8.21}$$

$$\text{subject to } \nabla\cdot\boldsymbol{u} = 0 \tag{8.22}$$

where the variable p, introduced to satisfy an additional constraint, plays the role of a Lagrangian multiplier (Gresho & Sani, 1987). We note that the adjoint of the gradient operator is the (negative) divergence operator

$$(\nabla)^\dagger = -\nabla\cdot(\,) \tag{8.23}$$

We now establish this fact. Consider a scalar f and a vector $\boldsymbol{F}$. Consider

$$\int_V \nabla\cdot(f\boldsymbol{F})\,dV = \int_V f(\nabla\cdot\boldsymbol{F})\,dV + \int_V \nabla f\cdot\boldsymbol{F}\,dV = \iint_{\Omega=\partial V} f\boldsymbol{F}\cdot d\boldsymbol{A} \tag{8.24}$$

and if the boundary integral vanishes we have

$$(\nabla f,\boldsymbol{F}) = \int_V (\nabla f\cdot\boldsymbol{F})\,dV = -\int_V f(\nabla\cdot\boldsymbol{F})\,dV = (f,-\nabla\cdot\boldsymbol{F}) = (f,(\nabla)^\dagger(\boldsymbol{F})) \tag{8.25}$$

and hence

$$(\nabla)^\dagger = -\nabla\cdot(\,) \tag{8.26}$$

8.1.2 The role of linear mechanisms, the Reynolds–Orr equation

Integrate the TKE equation space and note the integration of the spatial transport terms vanish at the boundaries, we have the Reynolds–Orr equation, whose general form is

$$\frac{d}{dt}\int_\Omega \frac{1}{2}u_i'u_i'\,\mathrm{d}V = \int_\Omega -u_i'u_j'\frac{\partial U_i}{\partial x_j}\,\mathrm{d}V - \int_\Omega \nu\frac{\partial u_i'}{\partial x_j}\frac{\partial u_i'}{\partial x_j}\,\mathrm{d}V. \tag{8.27}$$

. Here U_j is the laminar base flow in a stability problem, which is itself a solution to the NSE and u_i' are the respectively defined perturbations. The implication is that the total production balances the total dissipation, and the spatial integral of the nonlinear transport term is zero. An equivalant statement in Fourier space is that the sum of the spectral transport term $\hat{T}_n l$ over all wavenumbers is zero. Recall the derivation (5.202). Also, the production term is linear. That being said, the mechanisms responsible for the global energy amplification have to be linear.

8.2 Parallel shear flows

Here, we start from a more generalized framework with a vertical density profile and a buoyancy equation (linearized N–S under Boussinesq) similar to in Section 6.7.2. We will also restrict our attention on one-dimensional horizontal shear $U(y)$ and vertical stratification ($Ri_g = N^2L^2/U^2$) and rotation ($f = 2\Omega$) where the scales for $(\boldsymbol{u}, \boldsymbol{x}, t, p, \rho)$ are $(U, L, L/U, \rho_0 U^2, (\rho_0/g)N^2 L)$.

The non-dimensional linearized perturbation equations read

$$\begin{aligned}
\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} &= 0 \\
\frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + v\frac{\partial U}{\partial y} - 2\Omega v &= -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u \\
\frac{\partial v}{\partial t} + U\frac{\partial v}{\partial x} + 2\Omega u &= -\frac{\partial p}{\partial y} + \frac{1}{Re}\nabla^2 v \\
\frac{\partial w}{\partial t} + U\frac{\partial w}{\partial x} &= -\frac{\partial p}{\partial z} - Ri_b\rho + \frac{1}{Re}\nabla^2 w \\
\frac{\partial \rho}{\partial t} + U\frac{\partial \rho}{\partial x} - w &= \frac{1}{RePr}\nabla^2 \rho
\end{aligned} \tag{8.28}$$

where the primes for perturbational variables are omitted.

With the normal-mode *ansatz*

$$(\boldsymbol{u}, p, \rho) = (\hat{\boldsymbol{u}}, \hat{p}, \hat{\rho})\exp(ik_1 x + ik_3 z + \sigma t) \tag{8.29}$$

where $\sigma = -ik_1 c = -i\omega$ is the growth rate and $c = \omega/k$ is the wave speed, the equations (8.28) can be turned into the normal-mode equations

$$ik_1\hat{u} + D\hat{v} + ik_3\hat{w} = 0 \tag{8.30}$$

$$(\sigma + ik_1 U - \nu\triangle)\hat{u} + (DU - 2\Omega)\hat{v} = -ik_1\hat{p} \tag{8.31}$$

$$(\sigma + ik_1 U - \nu\triangle)\hat{v} + 2\Omega\hat{u} = -D\hat{p} \tag{8.32}$$

$$(\sigma + ik_1 U - \nu\triangle)\hat{w} = -ik_3\hat{p} - Ri_b\hat{\rho} \tag{8.33}$$

$$(\sigma + ik_1 U - \kappa\triangle)\hat{\rho} = \hat{w} \tag{8.34}$$

where we denote

$$D = \partial_y \tag{8.35}$$

$$\nabla^2 = \partial_{xx} + \partial_{yy} + \partial_{zz} \tag{8.36}$$

$$k^2 = k_1^2 + k_3^2 \tag{8.37}$$

$$\triangle = D^2 - k^2 \tag{8.38}$$

$$\nu = \frac{1}{Re} \tag{8.39}$$

$$\kappa = \frac{1}{RePr} \tag{8.40}$$

following the convention in Arobone & Sarkar (2012).

In a compact form, the normal-mode equations can be expressed as

$$\boldsymbol{A}\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{\rho}\end{bmatrix} = \sigma\boldsymbol{B}\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{\rho}\end{bmatrix} \tag{8.41}$$

where

$$\boldsymbol{A} = \begin{bmatrix} -2\Omega k^2 & ik_1(U\triangle - D^2U + 2\Omega D) - \nu\triangle^2 & -ik_3Ri_bD \\ ik_1(DU + UD - 2\Omega) - \nu\triangle D & D^2U + k_1^2U + DUD - 2\Omega D + \nu ik_1\triangle & 0 \\ -ik_1 & -D & k_1k_3U + \kappa ik_3\triangle \end{bmatrix} \tag{8.42}$$

$$\boldsymbol{B} = \begin{bmatrix} 0 & -\triangle & 0 \\ -D & ik_1 & 0 \\ 0 & 0 & ik_3 \end{bmatrix} \tag{8.43}$$

as a generalized eigenvalue problem in terms of σ (Arobone & Sarkar, 2012). It is noted that both w and p are eliminated and this technique of deriving a pressure-less system is quite common in stability analysis and CFD. The cost is, however, differential equations of higher orders that require more boundary conditions, which are hopefully simple, in this parallel flow case.

The first row is effectively the Laplacian-v ($-\triangle\hat{v}$) equation, obtained by $\partial_y(\partial_z M_3\&C) + \partial_{zz}M_2 + \partial_x(\partial_x M_2 - \partial_y M_1)$, or effectively $\partial_x\omega_z - \partial_z\omega_x = \partial_x(\partial_x M_2 - \partial_y M_1) - \partial_z(\partial_y M_3 - \partial_z M_2)$ (taking the curl twice). Here $M_i (i = 1, 2, 3)$ denote the i-th momentum equation and C denotes the continuity equation, and & means substitute.

$\partial_z M_3\&C$:

$$(\sigma + ik_1U - \nu\triangle)(ik_1\hat{u} + D\hat{v}) = -k_3^2\hat{p} + ik_3Ri_b\hat{\rho} \tag{8.44}$$

$\partial_y(\partial_z M_3\&C)$:

$$ik_1(\sigma + ik_1U - \nu\triangle)D\hat{u} - k_1^2(DU)\hat{u} + (\sigma + ik_1U - \nu\triangle)D^2\hat{v} + ik_1(DU)D\hat{v} = -k_3^2D\hat{p} + ik_3Ri_bD\hat{\rho} \tag{8.45}$$

$\partial_{zz}M_2$:

$$-k_3^2(\sigma + ik_1U - \nu\triangle)\hat{v} - 2k_3^2\Omega\hat{u} = Dk_3^2\hat{p} \tag{8.46}$$

$-\partial_x(\partial_y M_1)$:

$$-ik_1(\sigma + ik_1U - \nu\triangle)D\hat{u} + k_1^2(DU)\hat{u} - ik_1(DU - 2\Omega)D\hat{v} - ik_1(D^2U)\hat{v} = -k_1^2D\hat{p} \tag{8.47}$$

$\partial_x(\partial_x M_2)$:

$$-k_1^2(\sigma + ik_1U - \nu\triangle)\hat{v} - 2k_1^2\Omega\hat{u} = k_1^2D\hat{p} \tag{8.48}$$

$\partial_y(\partial_z M_3\&C) + \partial_{zz}M_2 + \partial_x(\partial_x M_2 - \partial_y M_1)$:

$$\text{RHS} = ik_3Ri_bD\hat{\rho} \tag{8.49}$$

$$\text{LHS}_1 = -(\sigma + ik_1U - \nu\triangle)(k_1^2 + k_3^2)\hat{v} - 2(k_1^2 + k_3^2)\Omega\hat{u} \tag{8.50}$$

$$= -(\sigma + ik_1 U - \nu\triangle)k^2\hat{v} - 2k^2\Omega\hat{u} \tag{8.51}$$

$$\text{LHS}_2 = ik_1(DUD - DUD + 2\Omega D - D^2U)\hat{u} = ik_1(2\Omega D - D^2U)\hat{v} \tag{8.52}$$

$$\text{LHS}_3 = \sigma D^2\hat{v} + ik_1 U D^2\hat{v} - \nu\triangle D^2\hat{v} \tag{8.53}$$

Rearrange:

$$\text{RHS} = -\sigma\triangle\hat{v} \tag{8.54}$$

$$\text{LHS} = -2k^2\Omega\hat{u} + ik_1(2\Omega D - D^2U + U\triangle)\hat{v} - \nu\triangle^2\hat{v} - ik_3 Ri_b D\hat{\rho} \tag{8.55}$$

The second row is effectively an equation for ω_z ($=ik_1\hat{v} - D\hat{u}$), obtained by by cross differentiating u, v momentum equations. Pay attention to stuff like the chain rule: $D((DU)\hat{v}) = D^2U\hat{v} + (DU)D\hat{v}$. The third row is from combining the continuity and the scalar equations. The first two equations represents the time-evolution ($\sigma \sim \partial_t$) of Laplacian-v and ω_z (which is more suitable than ω_y for there is stratification/rotation). It is foruth-order in $\hat{v}$ since it contains $\nu\triangle(\triangle\hat{v})$.

It will be easy to reduce from the general case derived here to the specific cases described later.

8.2.1 Orr–Sommerfeld equations

In what follows, we consider homogeneous parallel shear flows with $U(y) \neq 0, \Omega = 0, Ri_g = 0$. Per Squire's theorem (next section), it is always possible to re-orient the coordinate $x - z$ such that any each unstable 3D perturbation always corresponds to a more unstable 2D one. Hence, it is sufficient to study 2D perturbations with $k_3 = 0, \hat{w} = 0$.

The viscous 2D perturbation equations are

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} &= 0 \\ \frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + v\frac{\partial U}{\partial y} &= -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u \\ \frac{\partial v}{\partial t} + U\frac{\partial v}{\partial x} &= -\frac{\partial p}{\partial y} + \frac{1}{Re}\nabla^2 v \end{aligned} \tag{8.56}$$

and by introducing the streamfunction and normal modes

$$u = \psi_y,\ v = -\psi_x,\ (u, v, \psi) = (\hat{u}, \hat{v}, \hat{\phi})\exp[ik(x - ct)] \tag{8.57}$$

we have to the Orr–Sommerfeld equation

$$(U - c)(\hat{\phi}_{yy} - k^2\hat{\phi}) - U_{yy}\hat{\phi} = \frac{1}{ikRe}(\hat{\phi}_{yyyy} - 2k^2\hat{\phi}_{yy} + k^4\phi) \tag{8.58}$$

representing the viscous instability of shear flows.

We note that the normal mode perturbations (8.57) are in the form of plane waves, where $c_r = \omega/k$ is the phase speed and $\sigma_{2D} = kc_i$ is the temporal growth rate. The expression of the perturbations ($\in \mathbb{R}$) are to be understood as, for example,

$$w = \Re\{\hat{w}[\cos(kx - kc_r t) - i\sin(kx - kc_r t)]\}e^{c_i kt} \tag{8.59}$$

$$= \{\Re(\hat{w})[\cos(kx - kc_r t)] - \Im(\hat{w})[\sin(kx - kc_r t)]\} e^{kc_i t}. \tag{8.60}$$

We also note that since we have taken the curl, the ODE is thrid order in velocity and fourth order in streamfunction. The boundary conditions for two no-slip walls are $\hat{\phi}(0) = \hat{\phi}_y(0) = 0$, $\hat{\phi}(1) = \hat{\phi}_y(1) = 0$.

Example: plane Poiseuille flow

Here we give the famous example of the viscous instability of a plane channel flow, determined by the O–S equation. A laminar plane Poiseuille profile $U(y) = 1 - y^2$ is stable to all inviscid perturbations due to the lack of inflection points, according to Rayleigh's theorem (a necessary but not sufficient condition for instability; see following sections), but is unstable to viscous perturbations once a critical Reynolds number is exceeded, which was first solved numerically by Orszag (1971) using Chebyshev spectral method

The O–S eigenvalue problem, written in terms of wall-normal velocity,

$$(U - c)(\partial_y^2 - k^2)\hat{v} - U''\hat{v} = (\mathrm{i}k_x Re)^{-1}(\partial_y^4 - 2k^2\partial_y^2 + k^4)\hat{v}, \tag{8.61}$$

is a fourth-order equation. It will lead to the well-known critical Reynolds number $Re_c = 5772$ (based on channel half width and the centerline velocity) for the Poisueille channel, above which the base flow loses stability to *viscous*, two-dimensional perturbations (per Squire's theorem). For a Blasius boundary layer, it is also stable at $Re = \inf$ but unstable to viscous perturbations at or above $Re \approx 500$, leading to Tollmein-Schlichting waves. The viscous destabilization is counter-intuitive – it is due to the existence of a critical layer where $c = U$, and the effect of viscosity is to alter the phases of $\hat{u}$ and $\hat{v}$ such that they no longer make zero net contribution to the Reynolds stress like in a Rayleigh's equation (Charru, 2011).

Simplification of the equation can be made by knowing the most unstable perturbations have around $k_x = k = 1.02 \approx 1$, near Re_c:

$$(Re)^{-1}(\hat{v}^{(4)} - 2\hat{v}'' + \hat{v}) - 2\mathrm{i}\hat{v} - \mathrm{i}(1 - y^2)(\hat{v}'' - \hat{v}) = -\mathrm{i}c(\hat{v}'' - \hat{v}) = \lambda(\hat{v}'' - \hat{v}), \tag{8.62}$$

which is an eigenvalue problem with $\lambda = -c\mathrm{i}$ being the eigenvalue. After discretizing with Chebyshev polynomials, the most unstable eigenvalue can be found to have $\Re[\lambda^{(1)}] = -0.0007819$.

8.2.2 Rayleigh's inflection point criterion

The inviscid limit of (8.58) is the Rayleigh equation

$$(U - c)\left(\frac{d^2}{dy^2} - k^2\right)\hat{\phi} - \frac{d^2U}{dy^2}\hat{\phi} = 0 \tag{8.63}$$

with two required BC's being no-penetration, $\hat{\phi}(0) = \hat{\phi}(1) = 0$. It can be shown by taking the complex conjugate of this equation that if c and $\hat{\phi}$ are an eigen pair, so as c^* and $\hat{\phi}^*$. We also note that the temporal growth rate is $\exp(c_i t)$ ($c = c_r + ic_i$). Hence, for each unstable mode there is a corresponding stable mode (and v.v.), and for each (neutrally) stable mode c must be real.

Multiply the following form of (8.63) with $\hat{\phi}^*$ and integrate both sides of

$$\hat{\phi}_{yy} - k^2\hat{\phi} - \frac{U_{yy}}{U - c}\hat{\phi} = 0 \tag{8.64}$$

within $0 < y < 1$ (assume $c_i \neq 0$ so $U - c \neq 0$), as a common technique in Sturm–Liouville problems, we have

$$\int |\hat{\phi}_y|^2 \, dy + k^2 \int |\hat{\phi}|^2 \, dy + \int \frac{U_{yy}}{U-c} |\hat{\phi}|^2 \, dy = 0. \tag{8.65}$$

Since the first two terms are real, the imaginary part of the third term must be zero, that said,

$$\Im \left(\int \frac{U-c^*}{U-c^*} \frac{U_{yy}}{U-c} |\hat{\phi}|^2 \, dy \right) = c_i \int \frac{U_{yy}}{|U-c|^2} |\hat{\phi}|^2 \, dy = 0, \tag{8.66}$$

which is valid when either (1) $c_i = 0$ (stable) or (2) $c_i \neq 0$ (unstable) but the integral must be zero, requiring U_{yy} to change sign (say, $U(y)$ having an inflection point where $U_{yy} = 0$) within the domain. As such, the Rayleigh's inflection point is a necessary condition for instability. But on the other hand, for plane Poisueille, for example, the lack of inflection point indicates that the flow cannot be unstable to inviscid perturbations. There is also no such inflection point criterion (as there is no Squire's theorem either) in axisymmetric flows.

For neutrally stable modes $c_i = 0$, the highest derivative drops out from (8.63) and a critical layer is form near $U = c$. Actually, we can obtain a more sufficient condition (a stronger necessary condition that has a narrower range) than the inflection point criterion. The real part of (8.65) is

$$\int \frac{U_{yy}(U-c_r)}{|U-c|^2} |\hat{\phi}|^2 \, dy = -\int (|\hat{\phi}_y|^2 + k^2 |\hat{\phi}|^2) \, dy \tag{8.67}$$

and for nonzero c_i (8.66) reads

$$\int \frac{U_{yy}}{|U-c|^2} |\hat{\phi}|^2 \, dy = 0. \tag{8.68}$$

Assuming the velocity profile is such that Rayleigh's criterion is satisfied and there exist a unique inflection point y_s, where $U(y_s) = U_s$ and $U''(y_s)$, multiply the above equation with $(c_r - U_s)$ and add it to (8.67) we have

$$\int \frac{U_{yy}(U-U_s)}{|U-c|^2} |\hat{\phi}|^2 \, dy = -\int (|\hat{\phi}_y|^2 + k^2 |\hat{\phi}|^2) \, dy \leq 0 \tag{8.69}$$

We note that both U_{yy} and $U - U_s$ changes sign at y_s so the product cannot change sign there. The only way for the above inequality to hold is for the product to be negative somewhere in the flow, i.e.,

$$U_{yy}(U - U_s) \leq 0, \tag{8.70}$$

with the equality only achieved at y_s. This is referred to as the Fjørtoft's criterion.

Examples:

(1) Shear layer with a velocity profile of a hyperbolic tangent

$$U(y) = U_0 \tanh(y/L), \tag{8.71}$$

where the velocity scale is $U_0 = (U_1 - U_2)/2$ and L is the half-width of the layer. We can see that the shear is the largest at the centerline:

$$S^* = U'(y)L/U_0 = \operatorname{sech}^2(y/L), \tag{8.72}$$

which achives a non-dimensional value of unity at $y = 0$ and decays gradually outwards. The second derivative is

$$U''(y) = -\frac{2U_0}{L^2}\mathrm{sech}^2(y/L)\tanh(y/L) \tag{8.73}$$

and it has one inflection point at $y = 0$ such that $U''(0) = 0$. The instabilities of flows with inflection point usually lead to rolled-up blobs, such as the Kelvin–Helmholtz rollers in shear layers.

(2) A wake or a jet with a Gaussian profile, $U(y) = 1-U_0\exp[-(y/L)^2]$ or $U(y) = U_0\exp[-(y/L)^2]$. The inflection points are $y = \pm\sqrt{L/2}$.

8.2.3 Squire's transformation and theorem

In order to justify the 2D analysis in the previous sections, we consider the inviscid system described by (8.56), but subject to 3D normal mode perturbations:

$$(u, v, w, p) = (\hat{u}, \hat{v}, \hat{w}, \hat{p})\exp[i(k_1x + k_3z - k_1ct)] \tag{8.74}$$

which is a plane wave with phase velocity $c_p = k_1c_r/(k_1^1 + k_2^2)^{1/2}$ (the phase, $\varphi = \boldsymbol{k}\cdot\boldsymbol{x} - c_pkt$ is written in terms of k_1ct instead of $k = (k_1^1 + k_2^2)^{1/2}$ for convenience), propagating along the (k_1, k_3) direction, oblique to the base flow.

We have

$$ik_1\hat{u} + \hat{v}_y + ik_3\hat{w} = 0 \tag{8.75}$$

$$ik_1(U - c)\hat{u} + \hat{v}U_y = -ik_1\hat{p} + \frac{1}{Re}[\hat{u}_{yy} - (k_1^2 + k_3^2)\hat{u}] \tag{8.76}$$

$$ik_1(U - c)\hat{v} = -\hat{p}_y + \frac{1}{Re}[\hat{v}_{yy} - (k_1^2 + k_3^2)\hat{v}] \tag{8.77}$$

$$ik_1(U - c)\hat{w} = -ik_3\hat{p} + \frac{1}{Re}[\hat{v}_{yy} - (k_1^2 + k_3^2)\hat{v}] \tag{8.78}$$

Under the Squire transformation,

$$\bar{k} = (k_1^1 + k_2^2)^{1/2},\ \bar{k}\bar{u} = k_1\hat{u} + k_3\hat{w},\ \bar{p}/\bar{k} = \hat{p}/k_1, \tag{8.79}$$

$$\bar{c} = c,\ \bar{v} = \hat{v},\ \bar{k}\overline{Re} = k_1Re \tag{8.80}$$

the first and third momentum equations can be combined to yield

$$i\bar{k}\bar{u} + \bar{v}_y = 0 \tag{8.81}$$

$$i\bar{k}(U - c)\bar{u} + \bar{v}U_y = -i\bar{k}\bar{p} + \frac{1}{\overline{Re}}[\bar{u}_{yy} - \bar{k}^2\bar{u}] \tag{8.82}$$

$$i\bar{k}(U - c)\bar{v} = -\bar{p}_y + \frac{1}{\overline{Re}}[\bar{v}_{yy} - \bar{k}^2\bar{v}] \tag{8.83}$$

which look exactly the same to the normal mode equations in 2D resulting from (8.56) (not shown). However, the growth rate is $\sigma_{2D} = \bar{k}c_i > k_1c_i = \sigma_{2D}$. That being said, for each 3D perturbation, there exist an equivalent 2D perturbation that has a higher growth rate. Also, since $\overline{Re} < Re$, for each critical Reynolds number in 3D, there is a lower corresponding critical Reynolds number for 2D perturbations.

Hence, we may say that for parallel shear flows the 2D perturbations are more unstable than 3D ones and the previous discussions on the O–S and the Rayleigh equations in 2D are sufficient.

8.2.4 Taylor–Goldstein equation

Similar to (8.28), the dimensional, inviscid, linearized perturbation equations for vertical shear flow $(U(z), 0, 0)$ can be expressed as

$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} &= 0 \\ \frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + w\frac{\partial U}{\partial z} &= -\frac{1}{\rho_0}\frac{\partial p}{\partial x} \\ \frac{\partial w}{\partial t} + U\frac{\partial w}{\partial x} &= -\frac{1}{\rho_0}\frac{\partial p}{\partial z} - \frac{g\rho}{\rho_0} \\ \frac{\partial \rho}{\partial t} + U\frac{\partial \rho}{\partial x} - w\frac{N^2\rho_0}{g} &= 0 \end{aligned} \tag{8.84}$$

where the perturbations are assumed to be 2D (according to Squire's theorem) in the form of (u, v, w, p, ρ).

Introducing the streamfunction

$$u = \psi_z,\ w = -\psi_x \tag{8.85}$$

and the normal modes

$$(\rho, p, \psi) = (\hat{\rho}(z), \hat{p}(z), \hat{\psi}(z)) \exp[ik(x - ct)] \tag{8.86}$$

the equations (8.84) are converted into

$$\begin{aligned} (U - c)\hat{\psi}_z - U_x\hat{\psi} &= -\frac{1}{\rho_0}\hat{p} \\ (U - c)k^2\hat{\psi} &= -\frac{1}{\rho_0}\hat{p}_z - \frac{g}{\rho_0}\hat{\rho} \\ (U - c)\hat{\rho} + \frac{N^2\rho_0}{g}\hat{\psi} &= 0 \end{aligned} \tag{8.87}$$

and by eliminating pressure we have the Taylor–Goldstein equation

$$(U - c)\left(\frac{d^2}{dz^2} - k^2\right)\hat{\psi} + \left(\frac{N^2}{U - c} - \frac{d^2U}{dz^2}\right)\hat{\psi} = 0, \tag{8.88}$$

which is just the Rayleigh equation (8.63) with an additional term from density stratification.

Since

$$\overline{\frac{1}{U - c}} = \overline{\left(\frac{U - c^*}{|U - c|^2}\right)} = \frac{U - c}{|U - c|^2} = \frac{1}{U - c^*}, \tag{8.89}$$

it can be seen that $(c, \hat{\psi})$ is an eigen pair, so is $(c^*, \hat{\psi})$. Hence, for each growing mode, there is a corresponding decaying mode. The no-penetration conditions $w(0) = w(1)$ requires that $\hat{\psi}(0) = \hat{\psi}(1) = 0$ at the walls. We also note that since $w = -\psi_x$, the normal modes of w are related to those of ψ as $\hat{w} = -ik\hat{\psi}$ so the TGE can also be cast as

$$(U - c)\left(\frac{d^2}{dz^2} - k^2\right)\hat{w} + \left(\frac{N^2}{U - c} - \frac{d^2U}{dz^2}\right)\hat{w} = 0. \tag{8.90}$$

If there is no background flow, the T–G that describes the instability triggered by perturbations only reads

$$\frac{d^2\hat{\psi}}{dz^2} + k^2\left(\frac{N^2}{\Omega^2} - 1\right)\hat{\psi} = 0, \tag{8.91}$$

where $\Omega^2 = c^2k^2$ is the angular frequency. Here $N = N(z)$ is an arbitraty density profile.

The T–G equation (8.88) and the Rayleigh equation (8.63) are both second-order ODEs in the Sturm–Liouville form (8.319). They are eigenvalue problems with c or Ω as the eigenvalue and k as a parameter, and are **not** self-adjoint but can be transformed to self-adjoint (refs) via the Howard transformations (ref). What viscosity does on the normality?

8.2.5 Howard's semicircle theorem

Takeing the Howard's transformation (non-singular if $c \neq U$)

$$F = \frac{\hat{\psi}}{U - c} \tag{8.92}$$

the T–G equation is transformed into

$$\frac{d}{dz}[(U-c)^2 F_z] - k^2(U-c)^2 F + N^2 F. \tag{8.93}$$

Multiply by F^*, integrate in the vertical domain, and use the homogeneous boundary conditions, we get

$$\int (U-c)^2 Q\, dz = \int N^2 |F|^2\, dz \tag{8.94}$$

where

$$Q = |F_z|^2 + k^2|F|^2 \tag{8.95}$$

is real positive. Separating the real and imaginary parts of (8.94), we have

$$\int [(U-c_r)^2 - c_i^2] Q\, dz = \int N^2|F|^2\, dz \tag{8.96}$$

$$c_i \int (U - c_r) Q\, dz = 0. \tag{8.97}$$

Since we require $c_i \neq 0$ for instability, the second equation is essentially

$$\int (U - c_r) Q\, dz = 0, \tag{8.98}$$

which can be satisfied only if $U - c_r$ changes sign in the domain, i.e.,

$$U_{\min} < c_r < U_{\max}. \tag{8.99}$$

That implies the propagation direction of the unstable waves is the same as the base flow (assume $U_{\min} > 0$) and provides a bound for their phase speed – could not be faster or slower than the base flow.

Combining (8.96) and (8.97) we have

$$\int [U^2 - c_r^2 - c_i^2] Q\, dz \geq 0, \tag{8.100}$$

also naturally we have

$$\int (U - U\text{min})(U - U_{\max})Q\,dz \le 0 \rightarrow \int [U_{\max}U_{\min} + U^2 - U(U_{\max} + U_{\min})]Q\,dz \le 0. \quad (8.101)$$

Combining (8.100) and (8.101) and realizing (8.98) we finally have

$$\int [U_{\max}U_{\min} + c_r^2 + c_i^2 - c_r(U_{\max} + U_{\min})]Q\,dz \le 0. \quad (8.102)$$

That implies a semicircle ($c_i > 0$)

$$[c_r - \frac{1}{2}(U_{\max} + U_{\min})^2] + c_i^2 \le [\frac{1}{2}(U_{\max} - U_{\min})]^2 \quad (8.103)$$

which limits the possible phase speed (c_r) of the unstable waves and their growth rate ($\sigma = kc_i$). This is a necessary condition. Also, the maximum growth rate is limited by (not necessarily reached)

$$kc_i \le \frac{k(U_{\max} - U_{\min})}{2}. \quad (8.104)$$

In the numerical computation of the complex eigenvalue $c(k)$, the above bounds efficiently narrow the searching range. We also note that $N = 0$ will only zero out the RHS of (8.94) and (8.100) but not that of (8.101), so the semicircle theorem is unaffected and is valid for both unstratified and stratified shear flows.

8.2.6 Miles–Howard sufficient condition

Under the transformation (Howard, 1961)

$$\phi = \frac{\hat{\psi}}{\sqrt{U - c}} \quad (8.105)$$

the T–G equation can be converted to

$$\frac{d}{dz}[(U - c)\phi_z] - \left[k^2(U - c) + \frac{1}{2}U_{zz} + \frac{1}{U - c}\left(\frac{1}{4}U_z^2 - N^2\right)\right]\phi = 0. \quad (8.106)$$

Multiply the equation above with ϕ^*, integrate in z, and use the homogeneous boundary conditions, we get

$$\int \frac{1}{U - c}(N^2 - 1/4U_z^2)\,dz = \int (U - c)(|\phi_z|^2 + k^2|\phi|^2)\,dz + \int \frac{1}{2}\int U_{zz}|\phi|^2\,dz \quad (8.107)$$

and notice that

$$\frac{1}{U - c} = \frac{U - c_r + ic_i}{|U - c|^2} \quad (8.108)$$

we have (separating real and imaginary parts):

$$\int \frac{U - c_r}{|U - c|^2}(N^2 - 1/4U_z^2)\,dz = \int (U - c_r)(|\phi_z|^2 + k^2|\phi|^2)\,dz + \int \frac{1}{2}\int U_{zz}|\phi|^2\,dz \quad (8.109)$$

$$\int \frac{c_i}{|U - c|^2}(N^2 - 1/4U_z^2)\,dz = -c_i \int (|\phi_z|^2 + k^2|\phi|^2)\,dz. \quad (8.110)$$

Since the integrant in the second equation is positive, the necessary condition for instability ($c_i \neq 0$) is for $N^2 - 1/4U_z^2$ to be negative at least in some part of the domain, i.e., at least one point in the domain has

$$Ri_g = \frac{N^2}{U_z^2} < \frac{1}{4}. \tag{8.111}$$

This adverse condition is generally referred to as the Miles–Howard sufficient condition: if the flow has everywhere $Ri_g > 1/4$, inviscid linear stability of the normal modes is guaranteed.

8.2.7 The Orr–Sommerfeld–Squires system

Apart from the primitive variables $[u, v, w]$, it is very common to study hydrodynamic stability in a pressure-less ω_y-$\nabla^2 v$ system after the so-called OSS transform, eliminating pressure. Assume a base flow $U(y)$ and the perturbations are (u, v, w). The linearized perturbation equations for the primitive variables are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{8.112}$$

$$\frac{\partial u}{\partial t} + U\frac{\partial u}{\partial x} + U'v = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u \tag{8.113}$$

$$\frac{\partial v}{\partial t} + U\frac{\partial v}{\partial x} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\nabla^2 v \tag{8.114}$$

$$\frac{\partial w}{\partial t} + U\frac{\partial w}{\partial x} = -\frac{\partial p}{\partial z} + \frac{1}{Re}\nabla^2 w \tag{8.115}$$

and the linearized fluctuation pressure equation is

$$\nabla^2 p = -2U'\frac{\partial v}{\partial x}. \tag{8.116}$$

Taking the divergence of the v-equation and substitute in pressure, we have the Orr–Sommerfeld ($\nabla^2 v$) equation; and taking the curl of the u and w momentum equations we get the Squire ($\eta = \omega_y$) equation:

$$\frac{\partial}{\partial t}(\nabla^2 v) + U\frac{\partial}{\partial x}(\nabla^2 v) = U''\frac{\partial v}{\partial x} + \frac{1}{Re}\nabla^2(\nabla^2 v) \tag{8.117}$$

$$\frac{\partial \eta}{\partial t} + U\frac{\partial \eta}{\partial x} = -U'\frac{\partial v}{\partial z} + \frac{1}{Re}\nabla^2 \eta \tag{8.118}$$

noting that the simplification of the second-order operator is

$$\nabla^2\left(U\frac{\partial v}{\partial x}\right) = U''\frac{\partial v}{\partial x} + 2U'\frac{\partial^2 v}{\partial x \partial y} + U\frac{\partial}{\partial x}(\nabla^2 v). \tag{8.119}$$

These operations are called the Orr–Sommerfeld and the Squires transform, also commonly used in numerics (see section 2.3.2). In the Orr–Sommerfeld–Squires system, the O–S modes are expressed in terms of the v instead of the velocity potential ϕ.

The v-equation is autonomous, although it is coupled to the second equation, in turn driven by v (seemingly one-way coupling), through the continuity constraint. These two equations are called the Orr–Sommerfeld and the Squire equations *in physical space*, respectively. Since they are a fourth-order and a second order equation, we need six boundary conditions in total, being

$$v = \partial_y v = 0,\ \eta = 0 \tag{8.120}$$

at both walls, where the $\partial_y v$ condition comes from continuity applied at the walls. We note that those BCs are the same for plane Poiseuille and Couette flows.

The production terms in these two equations,

$$U''\frac{\partial v}{\partial x},\ -U'\frac{\partial v}{\partial z}, \tag{8.121}$$

represent the Orr and the lift-up mechanisms, that are two of the most important linear, non-normal amplification mechanisms, especially in flows without inflection points, and they are through the interaction between the perturbations and the mean flow.

Assume 2D normal modes

$$v(x,y,z,t) = \hat{v}(y)\exp(\mathrm{i}k_x x + \mathrm{i}k_z z - \mathrm{i}\omega t) \tag{8.122}$$
$$\eta(x,y,z,t) = \hat{\eta}(y)\exp(\mathrm{i}k_x x + \mathrm{i}k_z z - \mathrm{i}\omega t) \tag{8.123}$$

where the wavenumbers are assumed to be real and the frequency ω is assumed to be complex (temporal stability). The boundary conditions are $\hat{v}(y=\pm1) = \partial_y\hat{v}(y=\pm1) = 0 = \hat{\eta}(y=\pm1) = 0$. We denote $k^2 = k_x^2 + k_z^2, c = \omega/k_x$, or by Squires theorem 2D perturbations are more unstable so we can comfortably set $k_z = 0$ if the critical Reynolds number for normal mode stability is the sole interest. We note that α and β are also used to denote the streamwise and spanwise wavenumbers.

The equations can be expressed in a compact, matrix form. Denote $\mathcal{D} = \partial_y$ and $k^2 = k_x^2 + k_z^2$. The OSS system in Fourier space is

$$[(-\mathrm{i}\omega + \mathrm{i}k_x U)(\mathcal{D}^2 - k^2) - \mathrm{i}k_x U'' - Re^{-1}(\mathcal{D}^2 - k^2)^2]\hat{v} = 0 \tag{8.124}$$
$$[(-\mathrm{i}\omega + \mathrm{i}k_x U) - Re^{-1}(\mathcal{D}^2 - k^2)]\hat{\omega}_y = -\mathrm{i}k_z U'\hat{v} \tag{8.125}$$

and can be expressed as

$$\mathcal{L}_{\mathrm{OS}}\hat{v} = -\mathrm{i}\omega\triangle\hat{v} \tag{8.126}$$
$$\mathcal{L}_{\mathrm{SQ}}\hat{\eta} + \mathrm{i}k_z U'\hat{v} = \mathrm{i}\omega\hat{\eta} \tag{8.127}$$

where

$$\mathcal{L}_{\mathrm{OS}} = -\mathrm{i}k_x(U\triangle - U'') + Re^{-1}\triangle^2 \tag{8.128}$$
$$\mathcal{L}_{\mathrm{SQ}} = \mathrm{i}k_x U - Re^{-1}\triangle \tag{8.129}$$
$$\triangle = \mathcal{D}^2 - k^2 \tag{8.130}$$

in a compact form (Schmid & Henningson, 2001):

$$\mathcal{L}\boldsymbol{q} = \mathrm{i}\omega M\boldsymbol{q}, \tag{8.131}$$

where

$$\boldsymbol{q} = \begin{bmatrix}\hat{v}\\ \hat{\eta}\end{bmatrix},\ \mathcal{L} = \begin{bmatrix}\mathcal{L}_{\mathrm{OS}} & 0\\ \mathrm{i}k_z U' & \mathcal{L}_{\mathrm{SQ}}\end{bmatrix},\ M = \begin{bmatrix}-\triangle & 0\\ 0 & 1\end{bmatrix}. \tag{8.132}$$

It can be seen that the coupling operator $\mathrm{i}k_z U'$ is zero for 2D perturbations ($k_z = 0$), whose evolution can be determined by solving the O–S equation only. The term $\mathrm{i}k_z U'$ (or its counterpart in the physical space equation) is also the forcing term in the SQ equation.

This is a typical Sturm–Liouville (generalized) eigenvalue problem, with the inner-product defined as

$$(\boldsymbol{a},\boldsymbol{b})_M = \int \boldsymbol{b}^\dagger M\boldsymbol{a}\,\mathrm{d}y \tag{8.133}$$

and we will show latter that M defines an energy two-norm. The boundary conditions are

$$\hat{v}(\pm 1) = \mathcal{D}\hat{v}(\pm 1) = \hat{\eta}(\pm 1) = 0 \tag{8.134}$$

following their physical space counterparts. When only the eigenvalues of the O–S equation is sought, it is typical to solve

$$\triangle^{-1}\mathcal{L}_{\mathrm{OS}}\hat{v} = \mathrm{i}\omega\hat{v}. \tag{8.135}$$

We note the forward and inverse OSS transforms between the primitive variables and the state variables are

$$\boldsymbol{q} = D\boldsymbol{v} \tag{8.136}$$

$$\begin{bmatrix}\hat{v}\\ \hat{\eta}\end{bmatrix} = D\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{w}\end{bmatrix},\ D = \begin{bmatrix}0 & 1 & 0\\ \mathrm{i}k_z & 0 & -\mathrm{i}k_x\end{bmatrix}, \tag{8.137}$$

and

$$\boldsymbol{v} = C\boldsymbol{q} \tag{8.138}$$

$$\begin{bmatrix}\hat{u}\\ \hat{v}\\ \hat{w}\end{bmatrix} = C\begin{bmatrix}\hat{v}\\ \hat{\eta}\end{bmatrix},\ C = \frac{1}{k^2}\begin{bmatrix}\mathrm{i}k_x\mathcal{D} & -\mathrm{i}k_z\\ k^2 & 0\\ \mathrm{i}k_z\mathcal{D} & \mathrm{i}k_x\end{bmatrix}. \tag{8.139}$$

These transformations are invertible ($CD = I_{3\times 3}, DC = I_{2\times 2}$) so physical-space quantities can easily be recovered, for purposes of, for example, evaluating the energy.

Given

$$\hat{u} = \frac{1}{k^2}(\mathrm{i}k_x\mathcal{D}\hat{v} - \mathrm{i}k_z\hat{\eta}) \tag{8.140}$$

$$\hat{w} = \frac{1}{k^2}(\mathrm{i}k_z\mathcal{D}\hat{v} + \mathrm{i}k_x\hat{\eta}) \tag{8.141}$$

we have

$$\frac{1}{2}(|u|^2 + |w|^2) = \frac{1}{2k^2}(|\mathcal{D}\hat{v}|^2 + |\hat{\eta}|^2) \tag{8.142}$$

and the (scaled) total energy is (note k^2 is the squared wavenumber)

$$E = \frac{1}{2}\int_{-1}^{1}(|u|^2 + |v|^2 + |w|^2)\,\mathrm{d}y = \frac{1}{2k^2}\int_{-1}^{1}(|\mathcal{D}\hat{v}|^2 + |\hat{\eta}|^2 + k^2|v|^2)\,\mathrm{d}y \tag{8.143}$$

$$= \frac{1}{2k^2}\int_{-1}^{1}[\hat{v}^*\hat{\eta}^*]\begin{bmatrix}k^2 - \mathcal{D}^2 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}\hat{v}\\ \hat{\eta}\end{bmatrix}\,\mathrm{d}y = \frac{1}{2k^2}(\boldsymbol{q},\boldsymbol{q})_M, \tag{8.144}$$

where integration by parts and the BCs are used.

8.2.8 Transient growth

The time evolution of the OSS system can be expressed as

$$\frac{d\boldsymbol{q}}{dt} = -M^{-1}\mathcal{L}\boldsymbol{q} = A\boldsymbol{q} \tag{8.145}$$

$$A = -M^{-1}\mathcal{L} = \begin{bmatrix} -\triangle^{-1}\mathcal{L}_{\mathrm{OS}} & 0 \\ -\mathrm{i}k_z U' & -\mathcal{L}_{\mathrm{SQ}} \end{bmatrix} \tag{8.146}$$

which has the solution

$$\boldsymbol{q}(y,t) = e^{At}\boldsymbol{q}_0(y). \tag{8.147}$$

Here e^{At} is the state transition operator/matrix. When the complex operator/matrix is normal (a sufficient condition), according to the spectral theorem, it is unitarily diagonalizable to $e^{At} = X\Lambda X^{-1}$, where the unitary matrix $X = [\boldsymbol{x}_1\boldsymbol{x}_2...\boldsymbol{x}_N]$ contain the orthonormal eigenvectors of e^{At}, and the independent variable $\boldsymbol{q}$ in the ODE (8.145) can be expanded in the eigenspace:

$$\boldsymbol{q} = \sum_{i=1}^{N} \kappa_i \boldsymbol{x}_i = X\boldsymbol{\kappa} \tag{8.148}$$

such that the ODE can be converted to decoupled ODEs solving the expansion coefficients κ_i:

$$\frac{d\boldsymbol{\kappa}}{dt} = \Lambda\boldsymbol{\kappa} \tag{8.149}$$

which has the solution

$$\boldsymbol{\kappa} = e^{\Lambda t}\boldsymbol{\kappa}_0 \tag{8.150}$$

where $\boldsymbol{\kappa}_0$ contain the expansion coefficients of the initial condition $\boldsymbol{q}_0$.

For the system

$$\mathcal{L}\boldsymbol{q} + M\frac{d\boldsymbol{q}}{dt} = 0 \tag{8.151}$$

in which $\mathcal{L}$ is non-normal, the eigenvectors of the Sturm–Liouville EVP (assuming normal modes in time: $\boldsymbol{q}(x,t) = \exp(-i\omega t)\tilde{\boldsymbol{q}}(x)$)

$$\mathcal{L}\tilde{\boldsymbol{q}} = i\omega M\tilde{\boldsymbol{q}}, \tag{8.152}$$

although not necessarily orthogonal, can be used to expand the independent variable. Individually, the linearly independent eigenvectors corresponding to distinct eigenvalues are

$$\mathcal{L}\tilde{\boldsymbol{q}}_i = i\omega_i M\tilde{\boldsymbol{q}}_i, \tag{8.153}$$

and then

$$\boldsymbol{q} = \sum_{i=1}^{N} \kappa_i(t)\tilde{\boldsymbol{q}}_i = Q\boldsymbol{\kappa}(t), \tag{8.154}$$

The IVP (8.151) is converted to

$$\sum_{i=1}^{N}\left[\left(\frac{d\kappa_i}{dt} + i\omega_i\kappa_i\right)\tilde{\boldsymbol{q}}_i\right] = 0 \tag{8.155}$$

and according to the linear independence of $\tilde{\boldsymbol{q}}_i$, it turns out that the equation governing the evolution of expansion coefficients is

$$\frac{d\boldsymbol{\kappa}}{dt} = -i\Lambda\boldsymbol{\kappa} \tag{8.156}$$

where $\Lambda = \text{diag}(\omega_1, \omega_2, ..., \omega_N)$ and we choose to denote the eigenvalue as $i\omega$, with the general solution being

$$\boldsymbol{\kappa} = \exp(-i\Lambda t)\boldsymbol{\kappa}_0. \tag{8.157}$$

Although the ODEs (8.157) that govern the expansion coefficients are decoupled (i.e., so do the evolutions of individual eigenfunctions), that does not imply the resulting energy norm of the perturbation $\boldsymbol{q}$ needs to decay if all eigenvalues ω_i are stable. The energy norm of $\boldsymbol{q}$ is

$$\|\boldsymbol{q}\|_M^2 = (\boldsymbol{q}, \boldsymbol{q})_M = \int \boldsymbol{q}^H M \boldsymbol{q}\, dy = \int \boldsymbol{\kappa}^H Q^H M Q \boldsymbol{\kappa}\, dy = \boldsymbol{\kappa}^H \left(\int Q^H M Q\, dy \right) \boldsymbol{\kappa} \triangleq \boldsymbol{\kappa}^H H \boldsymbol{\kappa}, \tag{8.158}$$

where the Gram matrix

$$H = \int Q^H M Q\, dy,\ H_{ij} = \int \tilde{\boldsymbol{q}}_i^H M \tilde{\boldsymbol{q}}_j\, dy, = (\tilde{\boldsymbol{q}}_j, \tilde{\boldsymbol{q}}_i)_M \tag{8.159}$$

contains important information about orthogonality – the normality/non-normality defined in the sense of energy norm is reflected here. Physically, non-orthogonality manifests as the eigenmode shapes looking similar to one another. Practically, the discretized matrix $\mathcal{M}$ should contain a combination of the weight matrix M and the numerical quadrature weights on the Chebyshev points and be positive-definite, such that the total (integrated) energy is given by the matrix product $q^H \mathcal{M} q$.

Since H is Hermitian, it is positive-definite and has a Cholesky decomposition $H = F^H F$ (where F is invertible) when Q has linearly independent columns (eigenvectors; $Q\boldsymbol{a} = \boldsymbol{0}$ iff $\boldsymbol{a} = \boldsymbol{0}$ such that $\boldsymbol{a}^H H \boldsymbol{a} = \boldsymbol{a}^H Q^H \mathcal{M} Q \boldsymbol{a} > 0$). In practice, it is usually the case after the removal of spurious infinite eigenvalues due to the implementation of boundary conditions (boundary bordering), the success of which will be indicated by the success of the Cholesky decomposition. Hence, eigenspace expansion leading to $\boldsymbol{\kappa}$ is justified.

The energy norms so-evaluated, are, respectively,

$$\|\boldsymbol{q}\|_M^2 = (\boldsymbol{q}, \boldsymbol{q})_M = (F\boldsymbol{\kappa}, F\boldsymbol{\kappa})_2 = \|F\boldsymbol{\kappa}\|_2^2 \triangleq (\boldsymbol{\kappa}, \boldsymbol{\kappa})_F = \|\boldsymbol{\kappa}\|_F^2. \tag{8.160}$$

The vector norms are then used to induce the matrix norm that defines the transient growth problem:

$$G(t) = \max_{\|\boldsymbol{q}_0\| \neq 0} \frac{\|\boldsymbol{q}\|_M^2}{\|\boldsymbol{q}_0\|_M^2} = \max_{\|\boldsymbol{\kappa}_0\| \neq 0} \frac{\|\boldsymbol{\kappa}\|_F^2}{\|\boldsymbol{\kappa}_0\|_F^2} \tag{8.161}$$

$$= \max_{\|\boldsymbol{\kappa}_0\| \neq 0} \frac{\|\exp(-i\Lambda t)\boldsymbol{\kappa}_0\|_F^2}{\|\boldsymbol{\kappa}_0\|_F^2} \triangleq \|\exp(-i\Lambda t)\|_F^2 \tag{8.162}$$

$$= \max_{\|\boldsymbol{\kappa}_0\| \neq 0} \frac{\|F\exp(-i\Lambda t)\boldsymbol{\kappa}_0\|_2^2}{\|F\boldsymbol{\kappa}_0\|_2^2} \triangleq \|F\exp(-i\Lambda t)F^{-1}\|_2^2 \tag{8.163}$$

$$= \sigma_1^2(F\exp(-i\Lambda t)F^{-1}) \tag{8.164}$$

Hence, the maximum growth at time t is given by the energy norm of $\|\exp(-i\Lambda t)\|_F$, which is the matrix 2-norm $\|F\exp(-i\Lambda t)F^{-1}\|_2$, given by its largest singular value.

The optimal perturbation modes and optimal response modes, that achieve the optimal growth, are the leading right and left singular vectors. The SVD

$$F\exp(-i\Lambda t)F^{-1} = U\Sigma V^H, \tag{8.165}$$

where $\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, ...)$ contains the singular values ordered by absolute values, give the mapping

$$F\exp(-i\Lambda t)F^{-1}\boldsymbol{v}_1 = \sigma_1\boldsymbol{u}_1 \tag{8.166}$$

from the optimal perturbation (coefficient) $\boldsymbol{\kappa}_0 = F^{-1}\boldsymbol{v}_1$ (right singular vector) to another vector, the optimal response (coefficient) $\boldsymbol{\kappa}_m = F^{-1}\boldsymbol{u}_1$ (left singular vector) through the evolution operator $\exp(-i\Lambda t)$ and with σ_1^2 being the energy amplification factor. Note that the singular vectors will involve at least a combination of multiple eigenvectors (in terms of projection; note the eigenvectors are not orthogonal in general), unlike normal systems in which the maximum amplification depends only on the leading eigenvalue. We note that the function $G(t)$ is the best growth that can be achieved at each time t, which correspond to different initial conditions, and is rather an envelope than the history of any particular initial condition.

In the case of three-dimensional perturbations ($k_z \neq 0$), the coupling operator $\mathcal{L}_c = \mathrm{i}k_z U' \neq 0$, the evolution operator A is non-normal (both OS and SQ are normal). Even if the eigenvalues of A are all stable, there could be significant transient energy growth due to the non-orthogonality of eigenvectors. Some initial conditions are able to achieve larger growth than others and they are called the optimal perturbations.

Now consider the degeneration to the normal matrix case. If A is normal, it has an eigenvalue decomposition (spectral decomposition) $A = V\Lambda V^{-1}$ where $\{\boldsymbol{v}_i\}$ are the orthonormal eigenvectors, and the weight-quadrature matrix $H = I = F$ (that the decay of all components in $\boldsymbol{\kappa}$ also imply that of the resulting energy). The amplification rate of the system $\partial_t\boldsymbol{x} = A\boldsymbol{x}$ is

$$G(t) = \|e^{At}\|_2^2 = \|Ve^{\Lambda t}V^{-1}\|_2^2 = \|e^{\Lambda t}\|_2^2 = e^{2\Re\{\lambda_1\}t} \tag{8.167}$$

where λ_1 is its eigenvalue with the greatest real part. For non-normal matrices, the above provides a lower bound:

$$G(t) \geq e^{2\Re\{\lambda_1\}t}. \tag{8.168}$$

An easy example: the non-normal matrix $[1,\ 1; 0,\ 1]$ having repeated eigenvalues of unity while it amplifies the vector $(0, 1)$ by a factor of $\sqrt{2}$, greater than the largest eigenvalue.

A note on the relation between the (induced) matrix 2-norm and the leading singular value: Assume $e^{At} = U\Sigma V^H$, we have

$$G(t) = \max_{\boldsymbol{x}\neq 0}\frac{\|e^{At}\boldsymbol{x}\|^2}{\|\boldsymbol{x}\|^2} = \max_{\boldsymbol{x}\neq 0}\frac{(U\Sigma V^H\boldsymbol{x}, U\Sigma V^H\boldsymbol{x})}{(\boldsymbol{x},\boldsymbol{x})} \tag{8.169}$$

$$= \max_{\boldsymbol{x}\neq 0}\frac{(\Sigma V^H\boldsymbol{x}, \Sigma V^H\boldsymbol{x})}{(\boldsymbol{x},\boldsymbol{x})} = \max_{\boldsymbol{y}\neq 0}\frac{(\Sigma\boldsymbol{y}, \Sigma\boldsymbol{y})}{(\boldsymbol{y},\boldsymbol{y})} = \sigma_1^2(t) = \|e^{At}\|_2^2 \tag{8.170}$$

8.2.9 Resolvent analysis

The transient growth problem solved in the previous section concerns the **optimal response to initial conditions**, for which one needs to be very lucky to have exactly such perturbations in the flow. In what follows, we will consider the **optimal response of to forcings**, which replaces

the nonlinear Reynolds stress in the RANS equation and can (hopefully) provide some effects of the background turbulence. This is also called an input-output formulation in the language of linear systems. Here we consider the forcing

$$\boldsymbol{f} = [f_x, f_y, f_z]^{\mathrm{T}} \tag{8.171}$$

in the u, v, w equations, leading to the forced linear system below:

$$\frac{\partial \boldsymbol{q}}{\partial t} = A\boldsymbol{q} + B\boldsymbol{f} \tag{8.172}$$

$$\boldsymbol{v} = C\boldsymbol{q} \tag{8.173}$$

where

$$B = \begin{bmatrix} \triangle^{-1} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -\mathrm{i}k_x\mathcal{D} & -k^2 & -\mathrm{i}k_z\mathcal{D} \\ \mathrm{i}k_z & 0 & -\mathrm{i}k_x \end{bmatrix} \tag{8.174}$$

is the OSS transformation operator such that

$$\begin{bmatrix} \triangle\hat{v} \\ \hat{\eta} \end{bmatrix} = \begin{bmatrix} -\mathrm{i}k_x\mathcal{D} & -k^2 & -\mathrm{i}k_z\mathcal{D} \\ \mathrm{i}k_z & 0 & -\mathrm{i}k_x \end{bmatrix} \begin{bmatrix} \hat{u} \\ \hat{v} \\ \hat{w} \end{bmatrix}. \tag{8.175}$$

In the case of monochromatic harmonic forcing,

$$\boldsymbol{q} = \hat{\boldsymbol{q}}(y)\exp(\mathrm{i}k_x z + \mathrm{i}k_z z - \mathrm{i}\omega t) \tag{8.176}$$

$$\boldsymbol{f} = \hat{\boldsymbol{f}}(y)\exp(\mathrm{i}k_x z + \mathrm{i}k_z z - \mathrm{i}\omega t) \tag{8.177}$$

we have

$$(-\mathrm{i}\omega\boldsymbol{I} - A)\hat{\boldsymbol{q}} = B\hat{\boldsymbol{f}} \tag{8.178}$$

and an input-output relation of

$$\hat{\boldsymbol{v}} = C(-\mathrm{i}\omega\boldsymbol{I} - A)^{-1}B\hat{\boldsymbol{f}} = H_\omega\hat{\boldsymbol{f}}, \tag{8.179}$$

where

$$H_\omega = C(-\mathrm{i}\omega\boldsymbol{I} - A)^{-1}B \tag{8.180}$$

is called the resolvent operator, whose singular values determines the max resolvent gain

$$R(\omega; k_x, k_z, Re) = \max_{\hat{\boldsymbol{f}}\neq 0} \frac{\|\hat{\boldsymbol{v}}(\omega; k_x, k_z)\|^2}{\|\hat{\boldsymbol{f}}(\omega; k_x, k_z)\|^2} = \max_{\hat{\boldsymbol{f}}\neq 0} \frac{\|H_\omega\hat{\boldsymbol{f}}\|^2}{\|\hat{\boldsymbol{f}}\|^2} = \sigma_1^2(H_\omega). \tag{8.181}$$

A singular-value decomposition

$$H_\omega = \Phi\Sigma\Psi^{\mathrm{H}} = \sum_j \sigma_j\phi_j\psi_j^{\mathrm{H}} \tag{8.182}$$

leads to the resolvent expansions

$$\hat{\boldsymbol{f}} = \sum_j (\psi_j, \hat{\boldsymbol{f}})\psi_j \tag{8.183}$$

$$\hat{\boldsymbol{v}} = \sum_j (\psi_j, \hat{\boldsymbol{f}}) \sigma_j \phi_j \tag{8.184}$$

where the right singular vectors $\{\psi_j(\omega; k_x, y, k_z)\}$ are called the forcing modes and the left singular vectors $\{\phi_j(\omega; k_x, y, k_z)\}$ are called the response modes. The projection coefficients $(\psi_j, \hat{\boldsymbol{f}})$ are the modal weights, such that

$$H_\omega \hat{\boldsymbol{f}} = \sum_j \sigma_j (\psi_j, \hat{\boldsymbol{f}}) \phi_j. \tag{8.185}$$

8.2.10 The Orr and lift-up mechanisms

Here we give examples of the two most well known transient mechanisms.

8.2.11 The adjoint OSS operator

Given the inner-product. And Lyapunov equations.

The adjoint eigenvalue problem is

$$\mathcal{L}^\dagger \boldsymbol{q} = -\mathrm{i}\omega M \boldsymbol{q}. \tag{8.186}$$

Non-self-adjointness: The Rayleigh equation is not self-adjoint; but it is after the Howard transformation. So as the Taylor–Goldstein.

8.2.12 Axisymmetric parallel shear flows

It is useful to consider the perturbation equations in cylindrical coordinates (x, r, θ) for applications in fully developed pipe flow and quasi-parallel axisymmetric jets and wakes. The base flow is $(U(r), 0, 0, P)$ and the perturbations are (u_x, u_r, u_θ, p) where the primes are dropped. Axisymmetry base flow implies $\partial_\theta U = 0$ and quasi-parallel implies $\partial_x U = 0$.

The perturbation equations are

$$\frac{\partial u_x}{\partial x} + \frac{1}{r}\frac{\partial}{\partial r}(r u_r) + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} = 0 \tag{8.187}$$

$$\frac{\partial u_x}{\partial t} + U\frac{\partial u_x}{\partial x} + U' u_r = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 u_x \tag{8.188}$$

$$\frac{\partial u_r}{\partial t} + U\frac{\partial u_r}{\partial x} = -\frac{\partial p}{\partial r} + \frac{1}{Re}\left(\nabla^2 u_r - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta} - \frac{u_r}{r^2}\right) \tag{8.189}$$

$$\frac{\partial u_\theta}{\partial t} + U\frac{\partial u_\theta}{\partial x} = -\frac{1}{r}\frac{\partial p}{\partial \theta} + \frac{1}{Re}\left(\nabla^2 u_\theta + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2}\right) \tag{8.190}$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} \tag{8.191}$$

Assuming normal modes

$$u_i(x, r, \theta, t) = \hat{u}_i(r)\exp(\mathrm{i}\alpha x + \mathrm{i}m\theta - \mathrm{i}\omega t) \tag{8.192}$$

we have the cylindrical, normal-mode equations :

$$\mathrm{i}\alpha\hat{u}_x + \frac{1}{r}\partial_r(r\hat{u}_r) + \frac{\mathrm{i}m}{r}\hat{u}_\theta = 0 \tag{8.193}$$

$$-\mathrm{i}\omega\hat{u}_x + \mathrm{i}\alpha U\hat{u}_x + U'\hat{u}_r = -\mathrm{i}k\hat{p} + \frac{1}{Re}\nabla^2\hat{u}_x \tag{8.194}$$

$$-\mathrm{i}\omega\hat{u}_r + \mathrm{i}\alpha U\hat{u}_r = -\partial_r\hat{p} + \frac{1}{Re}\left(\nabla^2\hat{u}_r - \frac{2\mathrm{i}m}{r^2}\hat{u}_\theta - \frac{\hat{u}_r}{r^2}\right) \tag{8.195}$$

$$-\mathrm{i}\omega\hat{u}_\theta + \mathrm{i}\alpha U\hat{u}_\theta = -\frac{\mathrm{i}m}{r}\hat{p} + \frac{1}{Re}\left(\nabla^2\hat{u}_\theta + \frac{2\mathrm{i}m}{r^2}\hat{u}_r - \frac{\hat{u}_\theta}{r^2}\right) \tag{8.196}$$

where the Laplacian is

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) - \alpha^2 - \frac{m^2}{r^2} \tag{8.197}$$

and we define

$$k^2 = \alpha^2 + \frac{m^2}{r^2}. \tag{8.198}$$

Velocity-vorticity formulation.

We denote the velocity components (u, v, w) and (u_x, u_r, u_θ) interchangeably.

The radial vorticity is

$$\omega_r = \frac{1}{r}\frac{\partial u_x}{\partial \theta} - \frac{\partial u_\theta}{\partial x}, \tag{8.199}$$

and in Fourier modes,

$$\hat{\eta} = \hat{\omega}_r = \frac{\mathrm{i}m}{r}\hat{u}_x - \mathrm{i}\alpha\hat{u}_\theta. \tag{8.200}$$

The equation for $\hat{\omega}_r$, after eliminating pressure (taking the curl), is

$$-\mathrm{i}\omega\hat{\omega}_r + \mathrm{i}\alpha U\hat{\omega}_r + \frac{\mathrm{i}m}{r}U'\hat{u}_r = Re^{-1}\left(\nabla^2 - \frac{1}{r^2}\right)\hat{\omega}_r - \frac{2\mathrm{i}m}{r^2}\hat{\omega}_\theta \tag{8.201}$$

with

$$(\nabla^2\boldsymbol{\omega})_r = \left(\nabla^2 - \frac{1}{r^2}\right)\omega_r - \frac{2}{r^2}\frac{\partial \omega_\theta}{\partial \theta}. \tag{8.202}$$

Note that

$$\left(\nabla^2 - \frac{1}{r^2}\right)\hat{\omega}_r = \frac{1}{r}(r\hat{\omega}_r')' - \frac{1}{r^2}\hat{\omega}_r - k^2\hat{\omega}_r. \tag{8.203}$$

With the variable change

$$\Phi = -\mathrm{i}r\hat{u}_r \tag{8.204}$$

$$\Omega = \frac{\hat{\omega}_r}{\mathrm{i}k^2 r} = \frac{\alpha r\hat{u}_\theta - m\hat{u}_x}{k^2 r^2} \tag{8.205}$$

we have the following system (note that our Ω is m times that of Schmid & Henningson (2001))

$$(-\mathrm{i}\omega + \mathrm{i}\alpha U)\mathcal{T}\Phi + \frac{\mathrm{i}\alpha}{r}\left(\frac{U'}{k^2 r}\right)'\Phi = \frac{1}{Re}\mathcal{T}(k^2 r^2\mathcal{T})\Phi + \frac{2}{Re}\alpha m\mathcal{T}\Omega \tag{8.206}$$

$$k^2 r^2(-\mathrm{i}\omega + \mathrm{i}\alpha U)\Omega + \frac{\mathrm{i}mU'}{r}\Phi = \frac{1}{Re}\mathcal{S}\Omega + \frac{2\alpha m}{Re}\mathcal{T}\Phi \tag{8.207}$$

where

$$k^2 = \alpha^2 + \frac{m^2}{r^2} \tag{8.208}$$

$$\mathcal{T} = \frac{1}{r^2} - \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{1}{k^2 r}\frac{\partial}{\partial r}\right) \tag{8.209}$$

$$\mathcal{S} = k^4 r^2 - \frac{1}{r}\frac{\partial}{\partial r}\left(k^2 r^3\frac{\partial}{\partial r}\right) \tag{8.210}$$

The corresponding variable transformations are

$$\begin{bmatrix}\Phi\\ \Omega\end{bmatrix} = \begin{bmatrix}0 & -\mathrm{i}r & 0\\ -\frac{m}{k^2r^2} & 0 & \frac{\alpha r}{k^2r^2}\end{bmatrix}\begin{bmatrix}\hat{u}_x\\ \hat{u}_r\\ \hat{u}_\theta\end{bmatrix},\ \boldsymbol{q} = D\boldsymbol{v}, \tag{8.211}$$

$$\begin{bmatrix}\hat{u}_x\\ \hat{u}_r\\ \hat{u}_\theta\end{bmatrix} = \begin{bmatrix}-\frac{\alpha}{k^2r}\frac{\partial}{\partial r} & -m\\ \frac{\mathrm{i}}{r} & 0\\ -\frac{m}{k^2r^2}\frac{\partial}{\partial r} & \alpha r\end{bmatrix}\begin{bmatrix}\Phi\\ \Omega\end{bmatrix},\ \boldsymbol{v} = C\boldsymbol{q}. \tag{8.212}$$

Finally, we have the cylindrical OSS system

$$\mathcal{L}\boldsymbol{q} = \mathrm{i}\omega M\boldsymbol{q} \tag{8.213}$$

where

$$M = \begin{bmatrix}\mathcal{T} & 0\\ 0 & k^2r^2\end{bmatrix} \tag{8.214}$$

$$\mathcal{L} = \begin{bmatrix}\mathcal{L}_{\mathrm{OS}}^{\mathrm{pipe}} & \frac{1}{Re}2\alpha m\mathcal{T}\\ -\frac{\mathrm{i}mU'}{r} - \frac{1}{Re}2\alpha\mathcal{T} & \mathcal{L}_{\mathrm{SQ}}^{\mathrm{pipe}}\end{bmatrix} \tag{8.215}$$

and

$$\mathcal{L}_{\mathrm{OS}}^{\mathrm{pipe}} = -\mathrm{i}\alpha U\mathcal{T} - \frac{\mathrm{i}\alpha}{r}\left(\frac{U'}{k^2r}\right)' + \frac{1}{Re}\mathcal{T}(k^2r^2\mathcal{T}) \tag{8.216}$$

$$\mathcal{L}_{\mathrm{SQ}}^{\mathrm{pipe}} = -\mathrm{i}\alpha k^2r^2U + \frac{1}{Re}\mathcal{S}. \tag{8.217}$$

Note that the shear coupling operator $\mathcal{L}_c = -\mathrm{i}mU'/r$ takes a similar form to the one in channel OSS. There are additionally viscous coupling operators.

Following the boundary conditions for velocity Fourier modes that will be discussed later in the same section, we are able to impose at $r = 0$

$$\begin{cases} m = 0: & \Phi = \Phi' = \Omega = 0\\ m = 1: & \Phi = \Omega = 0, \Phi' < \infty\\ m > 1: & \Phi = \Phi' = \Omega = 0\end{cases} \tag{8.218}$$

and at $r = L_r$ (where $L_r = 1$ for pipe and sufficiently large for jet/wake; continuity is applied there)

$$\Phi = \Phi' = \Omega = 0, \forall m. \tag{8.219}$$

These BCs comply with the axis regularities

$$\Phi \sim r^{|m|+1},\ \Omega \sim r^{|m|}, \tag{8.220}$$

where $J_m \sim r^m$ ($m \geq 0$,$J_0 \to 1$) is a typical asymptotic behavior of Bessel functions.

- For $m = 0$ (axisymmetric),
$$\Phi(r) = a_1 r + a_2 r^2 + ... \tag{8.221}$$
and smoothness at the origin (no azimuthal preference; invariance to rotation in (x, y) domain) requires $a_1 = a_3 = ... = 0$ (even functions of $r = \sqrt{x^2 + y^2}$; note that $f(x, y) = r = \sqrt{x^2 + y^2}$ is not differentiable at $(0, 0)$ as the derivative is different as the origin is approached from 0^+ or 0^-) so we have
$$\Phi(0) = \Phi'(0) = 0. \tag{8.222}$$
For Ω ($\hat{\omega}_r = -\mathrm{i}\alpha\hat{u}_\theta$ when $m = 0$), it is clear that a non-zero $\hat{u}_\theta$ at $m = 0$ at the centerline is not allowed. So $\Omega(0) = 0$.

- For $m = 1$, finite $\Omega'(0)$ is allowed.

- For $m > 1$, it is clear that $\Phi(0) = \Phi'(0) = \Omega(0) = 0$.

Lastly, we consider the energy norm. Similar to the plane OSS system, the weight/mass matrix M defined an energy norm:

$$(\boldsymbol{a}, \boldsymbol{b})_M = \int_0^{L_r} \boldsymbol{b}^{\mathrm{H}} M \boldsymbol{a} \pi r \,\mathrm{d}r \tag{8.223}$$

such that

$$\|\boldsymbol{q}\|_M^2 = (\boldsymbol{q}, \boldsymbol{q})_M = \int_0^{L_r} \boldsymbol{q}^{\mathrm{H}} M \boldsymbol{q} \pi r \,\mathrm{d}r \tag{8.224}$$

which gives exactly the integrated kinetic energy

$$\frac{1}{2}\int_0^{L_r} (\|\hat{u}_x\|^2 + \|\hat{u}_r\|^2 + \|\hat{u}_\theta\|^2) 2\pi r \,\mathrm{d}r = \frac{1}{2}\int_0^{L_r} \left(\frac{\|\Phi\|^2}{r^2} + \frac{\|\Phi'\|^2}{k^2 r^2} + k^2 r^2 \|\Omega\|^2 \right) 2\pi r \,\mathrm{d}r. \tag{8.225}$$

Note that IBP and radial boundary conditions are applied to relate the above two, so the above relation is only valid in global measure not point-wise. This differs from the channel OSS energy norm and also signifies the operator $\mathcal{T}$ for its connection to the energy norm.

Velocity-velocity formulation.

Instead of the velocity-vorticity formulation we can also choose another set of state variables. First, continuity is used to eliminate $\hat{u}_x$ from the $\hat{u}_x$ equation. Then $(m/r)\hat{u}_x - k\hat{u}_\theta$ and $\partial_r \hat{u}_x - \mathrm{i}k\hat{u}_r$ operations give the reduced $\hat{u}_r$–$\hat{u}_\theta$ equations, respectively, as

$$\frac{\partial}{\partial t}\mathcal{L}\boldsymbol{v} = \mathcal{U}\boldsymbol{v} + \mathcal{R}\boldsymbol{v}, \tag{8.226}$$

where the primitive and state variables are

$$\boldsymbol{u} = \begin{bmatrix} \hat{u}_x \\ \hat{u}_r \\ \hat{u}_\theta \end{bmatrix}, \; \boldsymbol{v} = \begin{bmatrix} \hat{u}_r \\ \hat{u}_\theta \end{bmatrix}, \tag{8.227}$$

linked by

$$\boldsymbol{v} = \mathcal{A}\boldsymbol{u},\ \boldsymbol{u} = \mathcal{A}^\dagger \boldsymbol{v}, \tag{8.228}$$

where († denotes operator/matrix adjoint; $\mathcal{A}$ is an operator)

$$\mathcal{A} = \begin{bmatrix} \frac{\mathrm{i}}{k}\frac{\partial}{\partial r} & 1 & 0 \\ -\frac{m}{kr} & 0 & 1 \end{bmatrix},\ \mathcal{A}^\dagger = \begin{bmatrix} \frac{\mathrm{i}}{kr}\frac{\partial}{\partial r}r & -\frac{m}{kr} \\ 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{8.229}$$

The operators are

$$\mathcal{L} = \begin{bmatrix} \frac{\mathrm{i}m}{kr^2}\frac{\partial}{\partial r}r & -\frac{\triangle}{kr^2} \\ \frac{\mathrm{i}}{k}\frac{\partial}{\partial r}\frac{1}{r}\frac{\partial}{\partial r}r - \mathrm{i}k & -\frac{m}{k}\frac{\partial}{\partial r}\frac{1}{r} \end{bmatrix}, \tag{8.230}$$

$$\mathcal{U} = \begin{bmatrix} \frac{-mU'}{r} + \frac{mU}{r^2}\frac{\partial}{\partial r}r & \frac{\mathrm{i}U\triangle}{r^2} \\ \frac{\partial}{\partial r}\frac{U}{r}\frac{\partial}{\partial r}r - \frac{\partial}{\partial r}U' - k^2U & \mathrm{i}m\frac{\partial}{\partial r}\frac{U}{r} \end{bmatrix} \tag{8.231}$$

$$\mathcal{R} = Re^{-1}\begin{bmatrix} \frac{-2\mathrm{i}mk}{r^2} & -\frac{k}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} + \frac{k(1+\triangle)}{r^2} \\ \frac{\mathrm{i}k(1+\triangle)}{r^2} - \frac{\mathrm{i}k}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} & -\frac{2mk}{r^2} \end{bmatrix} \tag{8.232}$$

$$+ Re^{-1}\begin{bmatrix} \frac{m}{r^2}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} - \frac{m\triangle}{r^3} \\ \frac{\partial}{\partial r}\frac{1}{r}\frac{\partial}{\partial r}r\frac{\partial}{\partial r} - \frac{\partial}{\partial r}\frac{\triangle}{r^2} \end{bmatrix} \times \begin{bmatrix} \frac{\mathrm{i}}{kr}\frac{\partial}{\partial r}r & -\frac{m}{kr} \end{bmatrix} \tag{8.233}$$

where in this case the dimensionless parameter

$$\triangle = m^2 + k^2r^2 \tag{8.234}$$

is related to the Laplacian. These expressions are consistent with Heaton & Peake (2007) and are analog to the channel OSS system. This way, the spurious modes from the null space (since pressure has no time derivative, the evolution operator is singular) need to be handled carefully. However, the order of $\hat{u}_r$ equation is raised to fourth-order through the differential-algebraic operations, additional boundary conditions need to be imposed carefully and accurate spectral methods (such as Chebyshev) are needed.

Boundary conditions.

The radial boundary conditions in a cylindrical coordinate system, especially the axial one due to the singularity, require special attention. Here we consider azimuthal Fourier decomposition only:

$$u_i(x, r, \theta, t) = \sum_{m=0}^{\infty} \hat{u}_i(x, r, m, t)e^{\mathrm{i}m\theta}. \tag{8.235}$$

The velocity vector is required to have a vanishing θ-dependence as the axis is approached (Schmid & Henningson, 2001), i.e.

$$\lim_{r\to 0}\frac{\partial}{\partial\theta}(u_x\boldsymbol{e}_x + u_r\boldsymbol{e}_r + u_\theta\boldsymbol{e}_\theta) = 0. \tag{8.236}$$

Given the derivatives of unit vectors $\partial_\theta \boldsymbol{e}_r = \boldsymbol{e}_\theta, \partial_\theta \boldsymbol{e}_\theta = -\boldsymbol{e}_r$, the above condition is converted into a vector equation

$$\mathrm{i}m\hat{u}_x \boldsymbol{e}_x + (\mathrm{i}m\hat{u}_r - \hat{u}_\theta)\boldsymbol{e}_r + (\hat{u}_r + \mathrm{i}m\hat{u}_\theta)\boldsymbol{e}_\theta = \boldsymbol{0}. \tag{8.237}$$

The linear independence of $\boldsymbol{e}_x, \boldsymbol{e}_r, \boldsymbol{e}_\theta$ requires at $r = 0$

$$\begin{aligned} \mathrm{i}m\hat{u}_x &= 0 \\ \mathrm{i}m\hat{u}_r - \hat{u}_\theta &= 0 \\ \hat{u}_r + \mathrm{i}m\hat{u}_\theta &= 0 \\ \mathrm{i}m\hat{p} &= 0 \end{aligned}$$

with the linear system formed by the last two equations only having non-zero solutions of $\hat{u}_r, \hat{u}_\theta$ when $m = 1$. Provided the Fourier transformed continuity equation

$$\frac{\partial \hat{u}_x}{\partial x} + \frac{\partial \hat{u}_r}{\partial r} + \frac{\hat{u}_r}{r} + \frac{\mathrm{i}m\hat{u}_\theta}{r} = 0, \tag{8.238}$$

the axial boundary conditions can be summarised as

$$\begin{cases} \hat{u}_x < \infty,\, m = 0 \\ \hat{u}_x = 0,\, m \neq 0 \end{cases}, \begin{cases} \hat{u}_r + im\hat{u}_\theta = 0,\, \partial_r \hat{u}_r = 0,\, m = 1 \\ \hat{u}_r = \hat{u}_\theta = 0,\, m \neq 1 \end{cases}, \begin{cases} \hat{p} < \infty,\, m = 0 \\ \hat{p} = 0, m \neq 0 \end{cases}. \tag{8.239}$$

or

$$\begin{cases} m = 0: & \hat{u}_r = \hat{u}_\theta = 0, \hat{p} < \infty, \hat{u}_x < \infty \\ m = 1: & \hat{u}_x = \hat{u}_r + i\hat{u}_\theta = \partial_r \hat{u}_r = \hat{p} = 0 \\ m > 1: & \hat{u}_x = \hat{u}_r = \hat{u}_\theta = \hat{p} = 0 \end{cases} \tag{8.240}$$

The boundary conditions have an azimuthal wavenumber dependence, where eigenmodes of $\hat{u}_r, \hat{u}_\theta$ are only allowed to be nonzero at the axis for $m = 1$. The far-field ($r = L_r$) boundary conditions are

$$\hat{u}_r = \hat{u}_\theta = \hat{u}_x = \hat{p} = 0, \forall m. \tag{8.241}$$

We note that the perturbation equations are second-order in r so two BCs are required for each variable.

Regularities at the axis

The boundary conditions above are consistent with smoothness requirements. For a single azimuthal mode,

$$u_i(r, \theta) = \hat{u}_i(r) e^{\mathrm{i}m\theta}, \tag{8.242}$$

it is in general

$$|\hat{u}_i| \sim r^{|m|} \tag{8.243}$$

such that

$$u_i(r, \theta) = u_i(x, y) = r^{|m|} e^{\mathrm{i}m\theta} = \begin{cases} (x + \mathrm{i}y)^m,\, m \geq 0 \\ (x - \mathrm{i}y)^{-m},\, m < 0 \end{cases} \tag{8.244}$$

can be analytic. This applies to $i = x, r, \theta$. Since

$$\cos\theta = \frac{e^{\mathrm{i}\theta} + e^{-\mathrm{i}\theta}}{2},\ \sin\theta = \frac{e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}}{2\mathrm{i}} \tag{8.245}$$

shifts the powers by ± 1, the regularity of

$$u_y = u_r \cos\theta - u_\theta \sin\theta \tag{8.246}$$

$$u_z = u_r \sin\theta + u_\theta \cos\theta \tag{8.247}$$

requires

$$\hat{u}_{r,\theta} \sim r^{|m|-1}. \tag{8.248}$$

In all,

$$\begin{cases} m = 0: & \hat{u}_r \sim r, \hat{u}_\theta \sim r, \hat{u}_x \sim O(1) \\ m \geq 1: & \hat{u}_r \sim r^{|m|-1}, \hat{u}_\theta \sim r^{|m|-1}, \hat{u}_x \sim r^{|m|} \end{cases} \tag{8.249}$$

8.3 Centrifugal and rotational instabilities

8.3.1 Rayleigh's criterion

Consider cylindrical coordinates (x, r, θ).

The θ-momentum equation reduces to

$$\frac{\partial u_\theta}{\partial t} + u_x \frac{\partial u_\theta}{\partial x} + u_r \frac{\partial u_\theta}{\partial r} + \frac{u_r u_\theta}{r} = 0 \tag{8.250}$$

when inviscid, axisymmetric ($\partial_\theta = 0$). It can be re-written as

$$\frac{1}{r}\left[\frac{\partial}{\partial t}(r u_\theta) + u_x \frac{\partial}{\partial x}(r u_\theta) + u_r \frac{\partial}{\partial r}(r u_\theta)\right] = 0, \tag{8.251}$$

or simply

$$\frac{D(r u_\theta)}{Dt} = \frac{DL}{Dt} = 0, \tag{8.252}$$

which is the conservation of angular momentum,

$$L = r u_\theta, \tag{8.253}$$

in the inciscid limit.

The inviscid r-momentum equation reads

$$\frac{Du_r}{Dt} - \frac{u_\theta^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r}. \tag{8.254}$$

Consider an axisymmetric base flow $(0, 0, U_\theta(r), P)$, satisfying the balance

$$\rho \frac{U_\theta^2}{r} = \frac{\partial P}{\partial r}. \tag{8.255}$$

When a ring of fluid at r_1 is displaced to $r_2 > r_1$ due to a radial perturbation $u_r > 0$, angular momentum is conserved

$$r_1 U_\theta(r_1) = r_2 u_\theta^* \tag{8.256}$$

and the derivative of the radial velocity is

$$\frac{Du_r}{Dt} = \frac{(u_\theta^*)^2}{r_2} - \frac{1}{\rho}\frac{\partial P}{\partial r}\bigg|_{r=r_2} \tag{8.257}$$

$$= \frac{(r_1/r_2)^2 U_\theta^2(r_1)}{r_2} - \frac{U_\theta^2(r_2)}{r_2} \tag{8.258}$$

$$= \frac{1}{r_2^3}[r_1^2 U_\theta^2(r_1) - r_2^2 U_\theta^2(r_2)] \tag{8.259}$$

and the flow is unstable if at one point

$$r_1^2 U_\theta^2(r_1) > r_2^2 U_\theta^2(r_2) \tag{8.260}$$

or stable if everywhere

$$r_1^2 U_\theta^2(r_1) < r_2^2 U_\theta^2(r_2). \tag{8.261}$$

The condition for instability is that the absolute magnitude of the angular momentum decreases radially,

$$\frac{1}{r^3}\frac{dL^2}{dr} < 0. \tag{8.262}$$

This is commonly referred to as the Rayleigh's criterion and it is a necessary and sufficient condition for the instability of inviscid columnar vortices subject to 3-D axisymmetric perturbations (Drazin, 2002). Since the flow is axisymmetric, $\omega_z = \partial_r u_\theta + u_\theta/r - \partial_\theta u_r = \partial_r u_\theta + u_\theta/r$, the stability condition (8.262) is equivalent to

$$\frac{U_\theta}{r}\left(\frac{U_\theta}{r} + \frac{\partial U_\theta}{\partial r}\right) = \frac{U_\theta \omega_z}{r} < 0 \tag{8.263}$$

(velocity and vorticity are of opposite signs) everywhere.

8.3.2 Axisymmetric circular flows: normal mode analysis

The above conclusion can also be reached in a standard normal mode analysis.

Consider cylindrical coordinates (x, r, θ), base flow $(0, 0, U_\theta(r), P)$ and perturbations $(u_x', u_r', u_\theta', p')$. The governing equations for axisymmetric perturbations $(\partial_\theta = 0)$ $(\partial_\theta = 0, u_\theta = 0)$ are

$$\frac{\partial u_x'}{\partial x} + \frac{1}{r}\frac{\partial}{\partial r}(r u_\theta') = 0 \tag{8.264}$$

$$\frac{\partial u_x'}{\partial t} = -\frac{1}{\rho}\frac{\partial p'}{\partial x} \tag{8.265}$$

$$\frac{\partial u_r'}{\partial t} - 2\frac{U_\theta u_\theta'}{r} = -\frac{1}{\rho}\frac{\partial p'}{\partial r} \tag{8.266}$$

$$\frac{\partial u_\theta'}{\partial t} + u_r'\left(\frac{\partial U_\theta}{\partial r} + \frac{U_\theta}{r}\right) = 0 \tag{8.267}$$

The last equation can be expressed as

$$\frac{\partial u_\theta'}{\partial t} = -\frac{u_r'}{r}\frac{\partial}{\partial r}(r u_\theta'). \tag{8.268}$$

8.3.3 Inertial/Coriolis instability

The Rayleigh's criterion can be generalized in a rotating reference frame (non-inertial; with rotation rate Ω) by replacing the angular momentum L with the absolute angular momentum

$$L_a = r(U_\theta + \Omega r) = r(U_\theta + \frac{f}{2}r) \tag{8.269}$$

where $U_\theta + \Omega r$ is interpreted as the absolute velocity. It is conserved along material lines

$$\frac{DL_a}{Dt} = 0. \tag{8.270}$$

The condition for instability

$$\frac{1}{r^3}\frac{dL_a^2}{dr} < 0 \tag{8.271}$$

translates to

$$\chi = \left(\frac{2U_\theta}{r} + f\right)(\omega_z + f) < 0, \tag{8.272}$$

which is called the generalized Rayleigh criterion (Kloosterziel & Van Heijst, 1991). Equation (8.272) can be interpreted as that if the absolute velocity and the absolute vorticity are of opposite signs, the flow is unstable.

It can be estabilished similarly to (8.262) as follows. The θ-momentum equation translates to the absolute angular momentum conservation. The inviscid r-momentum equation reads

$$\frac{Du_r}{Dt} - \frac{u_\theta^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r} + fu_\theta. \tag{8.273}$$

Consider an axisymmetric base flow $(0, 0, U_\theta(r), P)$, satisfying the geostrophic balance (r momentum balance)

$$-\frac{U_\theta^2}{r} = -\frac{1}{\rho}\frac{\partial P}{\partial r} + fU_\theta. \tag{8.274}$$

When a ring of fluid at r_1 is displaced to $r_2 > r_1$ due to a radial perturbation $u_r > 0$, the absolute angular momentum conservation leads to

$$r_1 U_\theta(r_1) + \frac{1}{2}fr_1^2 = r_2 u_\theta^* + \frac{1}{2}fr_2^2 \tag{8.275}$$

and the derivative of the radial velocity is

$$\frac{Du_r}{Dt} = \frac{(u_\theta^*)^2}{r_2} - \frac{1}{\rho}\frac{\partial P}{\partial r}\bigg|_{r=r_2} + fu_\theta^* \tag{8.276}$$

$$= \frac{(u_\theta^*)^2 - U_\theta^2(r_2)}{r_2} + f[u_\theta^* - U_\theta(r_2)] \tag{8.277}$$

$$= \frac{1}{r_2^3}[(r_1^2U_\theta^2(r_1) - r_2^2U_\theta^2(r_2)) + (\frac{1}{4}f^2r_1^4 - \frac{1}{4}f^2r_2^4) + (fU_\theta(r_1)r_1^3 - fU_\theta(r_2)r_2^3)] \tag{8.278}$$

$$= \frac{1}{r_2^3}[L_a^2(r_1) - L_a^2(r_2)] \tag{8.279}$$

and the flow is unstable if at one point

$$\frac{1}{r_3}\frac{dL_a^2}{dr} < 0 \tag{8.280}$$

or stable if everywhere

$$\frac{1}{r_3}\frac{dL_a^2}{dr} > 0. \tag{8.281}$$

Taking $r \to \infty$ (translation = rotation around infinity), we recover the absolute vorticity criterion

$$f(\omega_z + f) < 0 \tag{8.282}$$

for anticyclonic parallel shear flows (Holton, 1972), where ω_z also reduces to $\omega_z = \partial_r u_\theta + u_\theta/r = \partial_r u_\theta = -S$. This marks the similarity between the inertial instability in 2D parallel and axisymmetric base flows. The structures resulting from these two instabilities are quasi-streamwise/azimuthal vortices, respectively. However, 8.282 implies that cyclonic shear layer (f and ω_z same sign) is always stable, but both cyclonic and anticyclic axisymmetric vortices can be unstable.

8.4 Some classical instabilities

8.4.1 Kelvin–Helmholtz and Rayleigh–Taylor instabilities

The KHI and RTI can be analysed together as special cases of a general problem of a vortex sheet separating two streams of fluids with velocity U_1, U_2 and density ρ_1, ρ_2. Here 1 denotes the bottom fluid and 2 denotes the top fluid.

The vortex sheet is an interface $z = \zeta(x, y, t)$ and other than that the flow is inviscid/irrotational – $\nabla^2\phi_1 = 0$, $\phi_1 = U_1 x$; $\nabla^2\phi_2 = 0$, $\phi_2 = U_2 x$, where ϕ is the velocity potential such that $\boldsymbol{u} = \nabla\phi$. The momentum equation

$$\frac{\partial \boldsymbol{u}}{\partial t} + \nabla\left(\frac{\boldsymbol{u}^2}{2}\right) - \boldsymbol{u}\times\boldsymbol{\omega} = -\nabla\left(\frac{p}{\rho}\right) + \nu\nabla^2\boldsymbol{u} - g\boldsymbol{e}_z \tag{8.283}$$

can then be converted into

$$\nabla\left(\frac{\partial\phi}{\partial t} + \frac{(\nabla\phi)^2}{2} + \frac{p}{\rho} + gz\right) = 0, \tag{8.284}$$

neglecting viscosity. Hence, in each region, the governing equations are

$$p_1 = \rho_1\left[C_1 - \frac{\partial\phi_1}{\partial t} - \frac{(\nabla\phi_1)^2}{2} - gz\right] \tag{8.285}$$

$$p_2 = \rho_2\left[C_2 - \frac{\partial\phi_2}{\partial t} - \frac{(\nabla\phi_2)^2}{2} - gz\right] \tag{8.286}$$

Then we consider the boundary conditions. First, the kinematic boundary condition is that a material surface ($f(x, y, z, t) = z - \zeta(x, y, t) = 0$) remains a material surface, so $Df/Dt = \partial f/\partial t + \boldsymbol{u}\cdot\nabla f = 0$, or

$$-\zeta_t - \phi_x\zeta_x - \phi_y\zeta_y + \phi_z = 0, \tag{8.287}$$

which holds for both sides of the interface at $z = \zeta$:

$$\phi_{1,z} = \zeta_t + \phi_{1,x}\zeta_x + \phi_{1,y}\zeta_y \tag{8.288}$$
$$\phi_{2,z} = \zeta_t + \phi_{2,x}\zeta_x + \phi_{2,y}\zeta_y \tag{8.289}$$

Second, the dynamic boundary condition concerns the force balance of the interface. Since the flow is inviscid, at $z = \zeta$ the pressure on both sides should be equal:

$$\rho_1 \left[C_1 - \frac{\partial \phi_1}{\partial t} - \frac{(\nabla \phi_1)^2}{2} - gz \right] = \rho_2 \left[C_2 - \frac{\partial \phi_2}{\partial t} - \frac{(\nabla \phi_2)^2}{2} - gz \right]. \tag{8.290}$$

We want to be systematic on the procedures and will follow the standard steps of base flow – linearization – normal modes.

Base flow.

- Outside the vortex sheet: $\nabla^2 \Phi_1 = \nabla^2 \Phi_2 = 0$ and $\Phi_1(z = -\infty) = U_1 x, \Phi_2(z = +\infty) = U_2 x$.
- Kinematic BC: $\zeta = 0$.
- Dynamic BC: $\rho_1[C_1 - U_1^2/2] = \rho_2[C_2 - U_2^2/2]$.

Linearization/governing equations for perturbations. Where $\phi_i = \Phi_i + \phi_i'$.

- Outside the vortex sheet: $\nabla^2 \phi_1' = \nabla^2 \phi_2' = 0$ and $\phi_1'(z = -\infty) = \phi_2'(z = +\infty) = 0$.
- Linearized kinematic BC at $z = \zeta$ or, approximately, at $z = 0$:

$$\phi_{1,z}' = \zeta_t + U_1 \zeta_x \tag{8.291}$$
$$\phi_{2,z}' = \zeta_t + U_2 \zeta_x \tag{8.292}$$

- Linearized dynamic BC at $z = \zeta$ or, approximately, at $z = 0$:

$$\rho_1 \left[C_1 - \frac{\partial \phi_1}{\partial t} - \frac{(\nabla \phi_1)^2}{2} - gz \right] = \rho_2 \left[C_2 - \frac{\partial \phi_2}{\partial t} - \frac{(\nabla \phi_2)^2}{2} - gz \right]. \tag{8.293}$$

where

$$(\nabla \phi_1)^2 = (\nabla (U_1 x + \phi_1'))^2 = (U_1 + \phi_{1,x}', \phi_{1,y}', \phi_{1,z}')^2 = U_1^2 + 2U_i \phi_{1,x}' + (\nabla \phi_1')^2 \approx U_1^2 + 2U_1 \phi_{1,x}' \tag{8.294}$$

and similar for 2. The replacement of $z = \zeta$ with $z = 0$ is justifies as

$$\phi_{1,x}'(x, y, \zeta, t) = \phi_{1,x}'(x, y, 0, t) + \partial_z(\phi_{1,x}')(x, y, 0, t)\zeta + \partial_{zz}(\phi_{1,x}')(x, y, 0, t)\frac{\zeta^2}{2} + \ldots \approx \phi_{1,x}'(x, y, 0, t). \tag{8.295}$$

Using (8.290), we have

$$\rho_1 \left[\frac{\partial \phi_1'}{\partial t} + U_1 \frac{\partial \phi_1'}{\partial x} + gz \right] = \rho_2 \left[\frac{\partial \phi_2'}{\partial t} + U_2 \frac{\partial \phi_2'}{\partial x} + gz \right] \tag{8.296}$$

- Normal modes.

$$\zeta = \hat{\zeta}(z)e^{st+i(kx+ly)} \tag{8.297}$$
$$\phi_1 = \hat{\phi}_1(z)e^{st+i(kx+ly)} \tag{8.298}$$
$$\phi_2 = \hat{\phi}_2(z)e^{st+i(kx+ly)} \tag{8.299}$$

Substitute into the governing equations $\nabla^2\phi_1' = \nabla\phi_2' = 0$, we have

$$\partial_{zz}\hat{\phi}_1 = (k^2+l^2)\hat{\phi}_1 \tag{8.300}$$
$$\partial_{zz}\hat{\phi}_2 = (k^2+l^2)\hat{\phi}_2 \tag{8.301}$$

leading to

$$\hat{\phi}_1(z) = A_1e^{\tilde{k}z} + B_1e^{-\tilde{k}z} \tag{8.302}$$
$$\hat{\phi}_2(z) = A_2e^{\tilde{k}z} + B_2e^{-\tilde{k}z} \tag{8.303}$$

where $\tilde{k}^2 = k^2+l^2$. Given the respective vanishing conditions at $\pm\infty$, we have

$$\hat{\phi}_1(z) = A_1e^{\tilde{k}z},\ \hat{\phi}_2(z) = B_2e^{-\tilde{k}z}. \tag{8.304}$$

Then the kinematic BCs are used to determine the constants:

$$A_1 = \frac{s+ikU_1}{\tilde{k}}\hat{\zeta} \tag{8.305}$$
$$B_2 = -\frac{s+ikU_2}{\tilde{k}}\hat{\zeta} \tag{8.306}$$

The dynamic BC at $z = \zeta = 0$ leads to the dispersion relation

$$\rho_1[A_1(s+ikU_1)+g] = \rho_2[B_2(s+ikU_2)+g] \tag{8.307}$$

i.e.

$$\rho_1\left[\frac{(s+ikU_1)^2}{\tilde{k}}+g\right] = \rho_2\left[-\frac{(s+ikU_2)^2}{\tilde{k}}+g\right] \tag{8.308}$$

which is quadratic in s and has the solution

$$s = -ik\frac{\rho_1U_1+\rho_2U_2}{\rho_1+\rho_2} \pm \left[\frac{k^2\rho_1\rho_2(U_1-U_2)^2}{(\rho_1+\rho_2)^2} - \frac{\tilde{k}g(\rho_1-\rho_2)}{\rho_1+\rho_2}\right]^{1/2}. \tag{8.309}$$

We can see that for large k the first term in the square root which is positive will be larger than the second, negative term hence all large wavenumbers are unstable. Also, when there is no density difference and only pure shear, any wavenumber is unstable.

There are several special cases that have classical implications.

- The KH instability. Assume $\rho_1 = \rho_2$. The growth rate is

$$s = -ik\frac{U_1+U_2}{2} \pm \frac{k(U_1-U_2)}{2}. \tag{8.310}$$

The KH waves travel at $c = \omega/k = \Re\{is\}/k = (U_1+U_2)/2$, the average speed of two streams and the growth rate is positive for any wavenumber and proportional to the wavenumber. This is generally not true for typical finite-amplitude perturbations where large wavelengths are suppressed by the vertical shear length scale (δ) and small ones suppressed by viscosity/capillarity. The largest inviscid growth rate is achived at $k_{\max} \sim 2\pi/\delta$ with frequency $\sim 2\pi(U/\delta)$ proportional to the shear.

- The RT instability. This can be due to density inversion (bottom light) or due to accelerations (such as in supernovae). Assume $U_1 = U_2 = 0$. The growth rate is

$$s = \pm \left[\frac{\tilde{k} g'(\rho_2 - \rho_1)}{\rho_1 + \rho_2} \right]^{1/2}. \tag{8.311}$$

  where $g' = g + a$ and a is the acceleration of the system/frame.

- Gravity waves. Assume $U_1 = U_2 = 0$ and $\rho_2 < \rho_1$. The growth rate is

$$s = \pm i \left[\frac{\tilde{k} g(\rho_1 - \rho_2)}{\rho_1 + \rho_2} \right]^{1/2}, \tag{8.312}$$

  which gives purely oscillatory motions, which interprets wave motions as a special case of stability. The eigenmodes (z-modes) decay exponentially away from the density interface or they are called trapped modes (in the language of quantum mechanics). When $\rho_2 = 0$ (such as the air-water interface), $s = \pm i\sqrt{\tilde{k}g}$, $\omega = is = \pm\sqrt{\tilde{k}g}$ and the wavespeed is

$$c = \omega/\tilde{k} = \pm\sqrt{g/\tilde{k}}. \tag{8.313}$$

8.4.2 Taylor–Couette instability

8.4.3 Rayleigh–Bénard instability

Consider the flow driven by two parallel plates with a temperature difference (hot bottom). The governing equations are (the Boussinesq equations)

$$\nabla \cdot \boldsymbol{u} = 0 \tag{8.314}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla \left(\frac{p + \rho_0 g z}{\rho_0} \right) + \nu \nabla^2 \boldsymbol{u} + \alpha g (T - T_0) \boldsymbol{e}_z \tag{8.315}$$

$$\frac{\partial \boldsymbol{u}}{\partial T} + \boldsymbol{u} \cdot \nabla T = \kappa \nabla^2 T \tag{8.316}$$

8.4.4 The Lorenz system

Its derivation, linearization, and bifurcations.

8.5 Adjoints

8.5.1 Adjoint matrices, operators, and equations

For a complex matrix $A \in \mathbb{C}^{N\times N}$, define an inner product

$$(u, v)_A = (Au, v) \tag{8.317}$$

where $u, v \in \mathbb{C}^N$ and $(u, v) = v^H u$, $(\cdot)^H$ is the Hermitian transpose. Define the adjoint matrix of A as $A^\dagger$ such that

$$(Au, v) = (u, A^\dagger v). \tag{8.318}$$

We note that if A is Hermitian ($A^H = A$), (8.318) is valid. Such matrix A is also called self-adjoint ($A = A^\dagger$). For operators defined on domains like $\mathbb{C}^N$, Hermitian and self-adjointness imply each other and we don't distinguish these two in what follows.

Consider the following standard Sturm–Liouville eigenvalue problem:

$$\frac{d}{dx}\left[p(x)\frac{d\phi}{dx}\right] + q(x)\phi = -\lambda\sigma(x)\phi, \tag{8.319}$$

where $p(x), w(x)$ are positive, and $\lambda, \phi(x)$ are the eigenvalue and corresponding eigenfunction of the problem. The boundary conditions are

$$\alpha_1\phi(a) + \alpha_2\frac{d\phi}{dx}(a) = 0 \tag{8.320}$$

$$\beta_1\phi(b) + \beta_2\frac{d\phi}{dx}(b) = 0 \tag{8.321}$$

$$\tag{8.322}$$

with $\alpha_1^2 + \alpha_2^2 \neq 0, \beta_1^2 + \beta_2^2 \neq 0$.

The LHS operator is defined as

$$\mathcal{L}(y) = \frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + q(x)y \tag{8.323}$$

and the S–L eigenvalue problem is

$$\mathcal{L}(\phi) + \lambda\sigma(x)\phi = 0. \tag{8.324}$$

The Lagrange identity is

$$u\mathcal{L}(v) - v\mathcal{L}(u) = \frac{d}{dx}\left[p\left(u\frac{dv}{dx} - v\frac{du}{dx}\right)\right] \tag{8.325}$$

and the Green's formula is

$$\int_a^b [u\mathcal{L}(v) - v\mathcal{L}(u)]dx = p\left(u\frac{dv}{dx} - v\frac{du}{dx}\right)\Big|_a^b \tag{8.326}$$

When u, v satisfy the same set of boundary conditions (either homogeneous or periodic), we have the self-adjointness, i.e.,

$$\int_a^b [u\mathcal{L}(v) - v\mathcal{L}(u)]dx = 0 \tag{8.327}$$

or

$$\int_a^b u\mathcal{L}(v)dx = \int_a^b v\mathcal{L}(u)dx. \tag{8.328}$$

$$= \int_a^b v\mathcal{L}^\dagger(u)dx. \tag{8.329}$$

and we note the definition of adjoint operator $\mathcal{L}^\dagger$ of $\mathcal{L}$ is that

$$(u, \mathcal{L}(v)) = (v, \mathcal{L}^\dagger(u)), \tag{8.330}$$

with the inner product defined based on spatial integral and the adjoint is dependent on the inner product.

Examples.

1. The Laplacian operator.

   $$\mathcal{L} = \nabla^2. \tag{8.331}$$

   The multidimensional variation of (8.326), with $\mathcal{L} = \nabla^2$, is

   $$\iiint [u\mathcal{L}(v) - v\mathcal{L}(u)]dV = \iiint \nabla \cdot [u\nabla v - v\nabla u]dV = \iint (u\nabla v - v\nabla u) \cdot d\boldsymbol{A} \tag{8.332}$$

   and if u, v satisfy the same homogeneous BC,

   $$\iiint [u\mathcal{L}(v) - v\mathcal{L}(u)]dV = \iint (u\nabla v - v\nabla u) \cdot d\boldsymbol{A} = 0, \tag{8.333}$$

   ∇^2 is self-adjoint.

2. The wave equation.

   $$\mathcal{L} = \frac{\partial^2}{\partial t^2} - c^2\nabla^2. \tag{8.334}$$

   The Green's formula

   $$\int_{t_i}^{t_f} \iiint [u\mathcal{L}(v) - v\mathcal{L}(u)]dV dt \tag{8.335}$$

   $$= \iiint \left(u\frac{\partial v}{\partial t} - v\frac{\partial u}{\partial t}\right) dV \Big|_{t_i}^{t_f} - c^2 \int_{t_i}^{t_f} \left(\iint (u\nabla v - v\nabla u) \cdot d\boldsymbol{A}\right) dt \tag{8.336}$$

   And we note that the $\mathcal{L} = \partial_{tt}$ operator alone is self-adjoint if the boundary terms vanish, by

   $$\int_{t_i}^{t_f} [u\mathcal{L}(v) - v\mathcal{L}(u)]dt = \left(u\frac{\partial v}{\partial t} - v\frac{\partial u}{\partial t}\right)\Big|_{t_i}^{t_f}. \tag{8.337}$$

3. Heat equation. Given above, we only consider the temporal derivative $\mathcal{L} = \partial_t$ here. Since

   $$\int_{t_i}^{t_f} \left(u\frac{dv}{dt} + v\frac{du}{dt}\right) dt = uv|_{t_i}^{t_f} \tag{8.338}$$

   vanishes if the RHS is zero,

   $$\int_{t_i}^{t_f} u\mathcal{L}v\, dt = \int_{t_i}^{t_f} -v\mathcal{L}u\, dt \tag{8.339}$$

   and the adjoint operator is $\mathcal{L}^\dagger = -\partial_t$. And we note that for the boundary term to vanish, we usually only have one BC for u in a first order problem such as $u(a) = 0$ and need to introduce an adjoint BC as $v(b) = 0$.

8.5.2 Non-self-adjointness and non-normality

A normal matrix $L \in \mathbb{C}^{N\times N}$ is defined as

$$L^H L = L L^H \tag{8.340}$$

and it is unitarily diagonalizable ($L = U\Lambda U^H$). The eigenvectors of L span an orthogonal basis of $\mathbb{C}^N$. More specifically, the eigenvectors corresponding to different eigenvalues are orthogonal, and even for degenerate eigenvalues an orthogonal basis can be found. A normal matrix is Hermitian if and only if all its eigenvalues are real.

The normality of a linear operator $\mathcal{L}$ is defined as

$$\mathcal{L}^\dagger \mathcal{L} = \mathcal{L}\mathcal{L}^\dagger. \tag{8.341}$$

The eigenmodes of $\mathcal{L}$ are normal to each other. A self-adjoint operator is hence (an example of) normal operators and a non-normal operator must be non-self-adjoint.

For two eigenmodes $\boldsymbol{\Phi}_1$ and $\boldsymbol{\Phi}_2$, where $\boldsymbol{\Phi}_i = e^{\lambda_i t}\boldsymbol{\phi}_i$, and $\lambda_i, \boldsymbol{\phi}_i$ are the eigenpair. There difference/cancellation $\boldsymbol{f} = \boldsymbol{\Phi}_1 - \boldsymbol{\Phi}_2$ decays if both decays and $(\boldsymbol{\Phi}_1, \boldsymbol{\Phi}_2) = 0$. That said, if the real part of each eigenvalue is negative, the energy of the perturbation will decay. However, for non-self-adjoint operators, there could be a transient growth of the cancellation $\boldsymbol{f}$ (Schmid, 2007), where the decay of individual eigenmodes does not imply the transient decay of the total energy. The idea of optimal perturbations it to find such a transient mode that grows most within a certain period of time.

8.5.3 Adjoint of the linearised N-S equations (Op's)

Op's: optimal linear perturbation solved as an optimal control problem in an optimization formulation constrained by the PDEs with linear operators.

Similar to (8.10)-(8.11), the linearised perturbation equation with buoyancy is

$$\nabla \cdot \boldsymbol{u} = 0 \tag{8.342}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{U}\cdot\nabla\boldsymbol{u} = -\boldsymbol{u}\cdot\nabla\boldsymbol{U} - \frac{1}{\rho_0}\nabla p + \nu\nabla^2\boldsymbol{u} + b\boldsymbol{e}_z \tag{8.343}$$

$$\frac{\partial b}{\partial t} + \boldsymbol{U}\cdot\nabla b + \boldsymbol{u}\cdot\nabla B = \kappa\nabla^2 b \tag{8.344}$$

where $b = B + b'$ is buoyancy and the primes are dropped from the equations above.

Assume the adjoints of $(\boldsymbol{u}, p, b)$ are $(\boldsymbol{v}, q, \varphi)$, multiplying each term in (8.342) by (v_1, v_2, v_3), (8.343) by $\boldsymbol{v}$, and (8.344) by φ, we can derive the adjoint equations of (8.342)-(8.344) as

$$\nabla \cdot \boldsymbol{v} = 0 \tag{8.345}$$

$$\frac{\partial \boldsymbol{v}}{\partial t} + \boldsymbol{U}\cdot\nabla\boldsymbol{v} = \boldsymbol{v}\cdot\nabla\boldsymbol{U} + \varphi\nabla B - \frac{1}{\rho_0}\nabla q - \nu\nabla^2\boldsymbol{v} \tag{8.346}$$

$$\frac{\partial \varphi}{\partial t} + \boldsymbol{U}\cdot\nabla\varphi = -v_3 - \kappa\nabla^2\varphi \tag{8.347}$$

We note the cross contribution terms $\varphi\nabla B$ and $-v_3$. The same set of equations can also be derived from a Lagrangian multiplier approach (a more morden method, but now classic), with the total perturbation energy being the Lagrangian and the set of governing equations along with

BC's being the constraints enforced as multipliers. Such Lagrangian is in the form of energy gain as (Arratia, 2011; Luchini & Bottaro, 2014; Kaminski *et al.*, 2014)

$$\mathscr{L}(T) = \frac{\langle \boldsymbol{u}(T), \boldsymbol{u}(T)\rangle + \langle b(T), b(T)\rangle/N^2}{\langle \boldsymbol{u}_0, \boldsymbol{u}_0\rangle + \langle b_0, b_0\rangle/N^2} \tag{8.348}$$

$$- [\frac{\partial u_i}{\partial t} + u_j\frac{\partial U_i}{\partial x_j} + U_j\frac{\partial u_i}{\partial x_j} + \frac{1}{\rho_0}\frac{\partial p}{\partial x_i} - \nu\frac{\partial^2 u_i}{\partial x_j^2} - b\delta_{i3}, v_i] \tag{8.349}$$

$$- [\frac{\partial b}{\partial t} + u_j\frac{\partial B}{\partial x_j} + U_j\frac{\partial b}{\partial x_j} - \kappa\frac{\partial^2 b}{\partial x_j^2}, \varphi] - [\frac{\partial u_i}{\partial x_i}, q] \tag{8.350}$$

$$- \langle u_i(0) - u_{0i}, v_{0i}\rangle - \langle b(0) - b_0, \varphi_0\rangle \tag{8.351}$$

constrained by the equations through the multipliers $(\boldsymbol{v}, q, \varphi)$ and similarly the BC's. The inner products

$$\langle \boldsymbol{u}, \boldsymbol{v}\rangle = \int_V \boldsymbol{u}\cdot\boldsymbol{v}dV \tag{8.352}$$

$$[\boldsymbol{u}, \boldsymbol{v}] = \int_0^T \langle \boldsymbol{u}, \boldsymbol{v}\rangle dt \tag{8.353}$$

are defined as respective spatial and spatio-temporal integrations.

In the phraseology of optimal control (with PDE constraints, Ref. section 5 of Manzoni *et al.* (2021)):

- Target functional (finite time transient gain):

$$\mathscr{G}(T) = \frac{\langle \boldsymbol{u}(T), \boldsymbol{u}(T)\rangle + \langle b(T), b(T)\rangle/N^2}{\langle \boldsymbol{u}_0, \boldsymbol{u}_0\rangle + \langle b_0, b_0\rangle/N^2}. \tag{8.354}$$

- State variables: $(\boldsymbol{u}, b)$.
- State equations (PDE constraints): (8.342)-(8.344). We note that (adjoint) pressure only appears as a Lagrangian multiplier that enforces the continuity.
- Control variables: $\boldsymbol{u}(0), b(0)$ and hence $\boldsymbol{u}(\boldsymbol{x}, t), b(\boldsymbol{x}, t)$.
- Admissible constraints on controls: none for now.

Taking the variation of (8.351) w.r.t.:

- The multipliers $(\boldsymbol{v}, q, \varphi)$: we recover the 'direct' equations (8.342)-(8.344).
- The terms $\boldsymbol{v}_0, \varphi_0$: we obtain the definition of IC's $\boldsymbol{u}_0, b_0$.
- The 'direct' variables $(\boldsymbol{u}, p, b)$: we obtain the adjoint equations (8.345)-(8.347). This step will be shown in detail.

Other than deriving from a Lagrangian perspective, the adjoint can also be derived using (multiple) integrations by parts. Starting from (8.14), i.e.,

$$\mathcal{L}_{\mathrm{NS}}\boldsymbol{q} = 0, \tag{8.355}$$

with the direct and adjoint variables being $\boldsymbol{q} = (\boldsymbol{u}, p)$ and $\boldsymbol{q}_\mathrm{d} = (\boldsymbol{v}, q)$, we look for the adjoint $\mathcal{L}^\dagger_\mathrm{NS}$ such that

$$[\boldsymbol{q}_\mathrm{d}, \mathcal{L}_\mathrm{NS}\boldsymbol{q}] - [\mathcal{L}^\dagger_\mathrm{NS}\boldsymbol{q}_\mathrm{d}, \boldsymbol{q}] = \mathrm{BT}(\boldsymbol{q}, \boldsymbol{q}_\mathrm{d}) \tag{8.356}$$

and the boundary conditions that make the RHS boundary terms vanish.

Where the inner product $[\cdot, \cdot]$ is that same as in (8.353) such that

$$[\boldsymbol{q}_\mathrm{d}, \boldsymbol{q}] = \int_T \int_V (\boldsymbol{v} \cdot \boldsymbol{u} + qp) dt dV. \tag{8.357}$$

The weak form of (8.14) is

$$[\mathbb{1}, \mathcal{L}_\mathrm{NS}\boldsymbol{q}] = \int_T \int_V (\partial_t u_i + U_j \partial_j u_i - \nu \partial_j^2 u_i + \partial_i p + u_j \partial_j U_i) dt dV = 0. \tag{8.358}$$

We note that (8.358) should also be valid on any arbitrary test function $\boldsymbol{q}_\mathrm{d} = (\boldsymbol{v}, q)$ for (8.14) to hold, such that

$$[\boldsymbol{q}_\mathrm{d}, \mathcal{L}_\mathrm{NS}\boldsymbol{q}] = \int_T \int_V v_i(\partial_t u_i + U_j \partial_j u_i - \nu \partial_j^2 u_i + u_j \partial_j U_i) + q \partial_i p \, dt dV = 0. \tag{8.359}$$

By integration by parts we have

$$[\boldsymbol{q}_\mathrm{d}, \mathcal{L}_\mathrm{NS}\boldsymbol{q}] = \int_T \int_V u_i(-\partial_t v_i - U_j \partial_j v_i - \nu \partial_j^2 v_i + v_j \partial_j U_i) - p \partial_i q \, dt dV + \mathrm{BT}(\boldsymbol{q}, \boldsymbol{q}_\mathrm{d}) \tag{8.360}$$

$$= \int_T \int_V u_i(-\partial_t v_i - U_j \partial_j v_i - \nu \partial_j^2 v_i + v_j \partial_j U_i) - p \partial_i q \, dt dV \tag{8.361}$$

$$= [\mathcal{L}^\dagger_\mathrm{NS}\boldsymbol{q}_\mathrm{d}, \boldsymbol{q}] \tag{8.362}$$

that defines the adjoint operator of $\mathcal{L}_{\boldsymbol{U}} = \partial_t + \boldsymbol{U} \cdot \nabla - \nu \nabla^2$:

$$\mathcal{L}^\dagger_{\boldsymbol{U}} = -\partial_t - \boldsymbol{U} \cdot \nabla - \nu \nabla^2 \tag{8.363}$$

and the adjoint equation

$$\partial_t v_i + U_j \partial_j v_i + \nu \partial_j^2 v_i = v_j \partial_j U_i - \partial_i q \tag{8.364}$$

and we note that by advancing forward in time, the viscous term is injecting energy into the system. Using the transfrom $\tau = -t$ we have

$$\partial_\tau v_i - U_j \partial_j v_i - \nu \partial_j^2 v_i = -v_j \partial_j U_i + \partial_i q. \tag{8.365}$$

In another form, the adjoint equation (without density) can be expressed similar to (8.16) as

$$\frac{\partial \boldsymbol{v}}{\partial t} = -\mathcal{A}^\dagger_{\boldsymbol{U}} \boldsymbol{v} \tag{8.366}$$

where $\mathcal{A}^\dagger_{\boldsymbol{U}}$ is the adjoint operator of $\mathcal{A}_{\boldsymbol{U}}$ in (8.15)

$$\mathcal{A}^\dagger_{\boldsymbol{U}} \boldsymbol{v} = \boldsymbol{U} \cdot \nabla \boldsymbol{v} - \boldsymbol{v} \cdot \nabla \boldsymbol{U} + \frac{1}{\rho_0} \nabla q + \nu \nabla^2 \boldsymbol{v}. \tag{8.367}$$

Chapter 9

Compressible flows, gas dynamics

9.1 Compressibility

Density change due to the pressure and the thermal effects (EOS). Batchelor (1967) Ref.

9.2 Thermodynamics

9.2.1 Useful relations

The equation of state, internal energy, enthalpy, and specific heats are:

$$p = \rho R T \tag{9.1}$$

$$e = c_v T \tag{9.2}$$

$$h = c_p T \tag{9.3}$$

$$c_p = \frac{\gamma R}{\gamma - 1},\ c_v = \frac{R}{\gamma - 1},\ c_p - c_v = R,\ \gamma = \frac{c_p}{c_v} \tag{9.4}$$

Here R is the gas constant and it is realted to the universal constant Λ as

$$R = \frac{\Lambda}{M} \tag{9.5}$$

where M is the molecular weight (kg/kmol or g/mol). For example, taking $\Lambda = 8.314\,\mathrm{J/(mol \cdot K)} = 8314\,\mathrm{J/(kmol \cdot K)}$ and the molecular weight of air $M = 28.97\,\mathrm{kg/kmol}$, we have $R = 287 \cdot \mathrm{J/(kg \cdot K)} = 287\,\mathrm{m^2/(s^2 \cdot K)}$. The specific heat ratio of air is $\gamma = 1.4$, which does not change too much with T. Consequently, $c_v = 718\,\mathrm{m^2/(s^2 \cdot K)}$ and $c_p = 1005\,\mathrm{m^2/(s^2 \cdot K)}$.

Example. Adiabatic expansion of air (temperature lapse rate).

9.2.2 First and second laws of thermodynamics

The thermodynamics first law states that the increase of the internal energy of the system equals heat added (positive δQ for added to the system) minus work done (positive δW for work done by the system),

$$\delta e = \delta Q - \delta W, \tag{9.6}$$

where the work done by isobaric (constant pressure) expansion and contraction in a frictionless system is $pd\nu$ (positive $d\nu$ when expanding and doing work on the surroundings). Hence, for an infinitesimal reversible process

$$de = dQ - pd\nu, \tag{9.7}$$

or

$$dQ = de + pd\nu \tag{9.8}$$

We define enthalpy as (recall Bernoulli, (1.65))

$$h = e + p\nu = e + \frac{p}{\rho}, \tag{9.9}$$

and then we have also

$$dQ = dh + \nu dp. \tag{9.10}$$

The entropy change is

$$Tds = dQ. \tag{9.11}$$

Using $dQ = de + pd\nu$ from the first law and the definition of h,

$$Tds = de + pd\nu \tag{9.12}$$

$$Tds = dh - \nu dp \tag{9.13}$$

where

$$dh = c_p dT,\ de = c_v dT \tag{9.14}$$

and

$$c_p = \left(\frac{\partial h}{\partial T}\right)_p,\ c_v = \left(\frac{\partial e}{\partial T}\right)_\nu. \tag{9.15}$$

In a constant volume process, $-pd\nu = 0$ and we have

$$\left(\frac{\partial Q}{\partial T}\right)_\nu = \left(\frac{\partial e}{\partial T}\right)_\nu = c_v \tag{9.16}$$

and similarly, in a constant pressure process,

$$\left(\frac{\partial Q}{\partial T}\right)_p = \left(\frac{\partial h}{\partial T}\right)_p = c_p \tag{9.17}$$

9.3 Conservation laws

Apart from deriving the Euler equation by neglecting the viscous term in the Navier–Stokes, we present another way here that is more to the taste of gas dynamics. (AFD notes/Shu).

In compressible flows, it is typical to write the gooverning equations in the following conservative form, especially when doing CFD. It is also noted that the density change can no longer be neglected and it will result in the change of pressure, which is a thermodynamic state variable now.

Mass conservation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j) = 0. \tag{9.18}$$

Momentum conservation:

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho f_i, \tag{9.19}$$

where the viscous stresses are

$$\tau_{ij} = \lambda \frac{\partial u_k}{\partial x_k}\delta_{ij} + 2\mu S_{ij} \tag{9.20}$$

$$= 2\mu \left(S_{ij} - \frac{1}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij} \right) \tag{9.21}$$

and $\lambda = -2/3\mu$ according to the Stokes assumption.

Total energy $E = e + u_k^2/2$ conservation:

$$\frac{\partial \rho(e + u_k^2/2)}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j (e + u_k^2/2)) = \rho f_j u_j - \frac{\partial (u_j p)}{\partial x_j} + \frac{\partial (u_i \tau_{ij})}{\partial x_j} + \frac{\partial}{\partial x_j}\left(k \frac{\partial T}{\partial x_j} \right). \tag{9.22}$$

In a vecterized form where the unknown variable is $U = [\rho, \rho u, \rho v, \rho w, \rho E]^T$ instead of the primitive variables ρ, u, v, w, e, T, p, the conservation laws can be expressed as

$$\frac{\partial}{\partial t}\begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho E \end{bmatrix} + \frac{\partial}{\partial x}\begin{bmatrix} \rho u \\ \rho u^2 + p - \tau_{xx} \\ \rho uv - \tau_{yx} \\ \rho uw - \tau_{zx} \\ (E+p)u - u\tau_{xx} - v\tau_{xy} - w\tau_{xz} + q_x \end{bmatrix} + \frac{\partial}{\partial y}\begin{bmatrix} \rho v \\ \rho uv - \tau_{yx} \\ \rho v^2 + p - \tau_{yy} \\ \rho vw - \tau_{yz} \\ (E+p)v - u\tau_{yx} - v\tau_{yy} - w\tau_{yz} + q_y \end{bmatrix} \tag{9.23}$$

$$+ \frac{\partial}{\partial z}\begin{bmatrix} \rho w \\ \rho uw - \tau_{zx} \\ \rho vw - \tau_{zy} \\ \rho w^2 + p - \tau_{zz} \\ (E+p)w - u\tau_{zx} - v\tau_{zy} - w\tau_{zz} + q_z \end{bmatrix} = \begin{bmatrix} 0 \\ \rho f_x \\ \rho f_y \\ \rho f_z \\ \rho(uf_x + vf_y + wf_x) \end{bmatrix} \tag{9.24}$$

where $\boldsymbol{\tau} =$ and $\boldsymbol{q} = -k\nabla T$. The general consevation law can be written as

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = J, \tag{9.25}$$

where F, G, H are called the fluxes. In order to close the system of five equations and nine unknowns (including μ, k), we need to use the EOS $p = \rho RT$, the calorically perfect gas relation $e = c_v T$, constant Prandtl number ($Pr = \nu/\alpha$, $\alpha = k/c_p\rho$), and the Sutherland's formula ($\mu = \mu(T)$). One of the difference between incompressible and compressible flows is that the energy equation is coupled in compressible flows and is to be solved together with the momentum equations. Another difference is that pressure is a thermodynamic variable.

9.3.1 Speed of sound

Through a control-volume analysis (White, 2008; Kundu *et al.*, 2015), it can be estabilished that the sound speed of a compressible fluid is

$$a = \sqrt{\frac{\partial p}{\partial \rho}} \tag{9.26}$$

and the correct way to calculate this derivative, since both p and ρ are thermodynamic state variables, is adiabatically/isentropically. The isentropic process refers to the propagation of sound waves – the change of fluid parcels as the sound wave (which is a weak wave) passes by, can be described as a isentropic process. It doesn't mean the following derivation can only be valid in isentropic flows. Besides, when the wave is strong, such as a shock wave, the entropy production can no longer be neglected.

Nevertheless,

$$\left.\frac{\partial p}{\partial \rho}\right|_s = \frac{\partial}{\partial \rho}(C\rho^\gamma) = \gamma\frac{p}{\rho} = \gamma RT. \tag{9.27}$$

We also note that

$$\left.\frac{\partial p}{\partial \rho}\right|_T = \frac{\partial}{\partial \rho}(\rho RT) = RT \neq \left.\frac{\partial p}{\partial \rho}\right|_s. \tag{9.28}$$

Hence, the speed of sound is

$$a = \sqrt{\gamma RT}. \tag{9.29}$$

Its value for air in the standard atmosphere at room temperature (288 K/sea level) is about 340.3 m/s (1225 km/h). At 10000 m (223K for standard atmosphere), sound speed is about 299.5 m/s (1078 km/h). In incompressible flows, $\delta\rho \approx 0$ following material, so the sound speed (or the propagation speed of pressure signals) is infinity.

9.4 Adiabatic isentropic flows

In shock-less regions away from the walls, for example, the potential flow region, where heating and entropy production due to shocks or boundary layer friction are absent, the flow can be treated as isentropic and the problem is greatly simplified.

The integration of the Tds relations from state 1 to 2 lead to

$$s_2 - s_1 = c_p \ln\frac{T_2}{T_1} - R\ln\frac{p_2}{p_1} \tag{9.30}$$

$$= c_v \ln\frac{T_2}{T_2} - R\ln\frac{\rho_2}{\rho_1} \tag{9.31}$$

and the isentropic condition implies $s_2 = s_1$ and

$$\frac{T_2}{T_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma-1} = \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma}, \frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1}\right)^{\gamma}. \tag{9.32}$$

These are called the isentropic relations. The last condition is equivalent to

$$\frac{p}{\rho^\gamma} = \text{const.} \tag{9.33}$$

An equation of state like so, $p = p(\rho) = C(s)\rho^\gamma$ is called a barotropic EOS, where the constant $C(s)$ is a function of entropy.

The energy conservation (in terms of Bernoulli, (1.70); along the streamlines) for one-dimensional flow is

$$h_0 = h + \frac{V^2}{2} = c_p T + \frac{V^2}{2} = c_p T_0 = \text{const.} \tag{9.34}$$

The total temperature

$$T_0 = T + \frac{V^2}{2c_p} \tag{9.35}$$

is conserved, following the energy (total enthalpy) conservation. It contains a static part and a dynamic part.

Provided the definition of the Mach number

$$Ma = \frac{V}{a} = \frac{V}{\sqrt{\gamma RT}}, \tag{9.36}$$

the total temperature conservation can be written as

$$T_0 = T\left(1 + \frac{\gamma - 1}{2} Ma^2\right), \tag{9.37}$$

which relates static temperature and the Mach number. This relation requires adiabatic (total temperature/enthalpy conservation) but not necessarily isentropic. Subsequently,

$$a_0 = a\left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{1/2} \tag{9.38}$$

since $a = \sqrt{\gamma RT}$.

With the isentropic relations (γ-relations),

$$p_0 = p\left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{\gamma/(\gamma-1)} \tag{9.39}$$

$$\rho_0 = \rho\left(1 + \frac{\gamma - 1}{2} Ma^2\right)^{1/(\gamma-1)} \tag{9.40}$$

mass conservation ($\rho_1 V_1 A_1 = \rho_2 V_2 A_2$) and ideal gas law, the area ratios for a converging-diverging nozzle is

$$\frac{A_2}{A_1} = \frac{Ma_1}{Ma_2}\left(\frac{1 + \frac{\gamma-1}{2} Ma_2^2}{1 + \frac{\gamma-1}{2} Ma_1^2}\right)^{(k+1)/2(k-1)}. \tag{9.41}$$

Taking $A_1 = A^*$ to be the critical/sonic condition (which might not need to exist in the flow; reference value) at the throat of the nozzle, we obtain the isentropic area-Mach number relation:

$$\frac{A}{A^*} = \frac{1}{Ma}\left(\frac{1 + \frac{\gamma-1}{2} Ma^2}{\frac{\gamma+1}{2}}\right)^{(k+1)/2(k-1)}. \tag{9.42}$$

9.4.1 Area effects

The energy equation

$$h + \frac{V^2}{2} = \text{const.} \tag{9.43}$$

can be taken differential

$$dh + VdV = 0 \tag{9.44}$$

and turn into a momentum equation. In isentropic flows,

$$0 = Tds = dh - \frac{dp}{\rho} \tag{9.45}$$

hence

$$dh = \frac{dp}{\rho} \tag{9.46}$$

and

$$\frac{dp}{\rho} + VdV = 0, \tag{9.47}$$

where $dp = a^2 d\rho$. The continuity equation, $\rho V \rho = \text{const.}$ is differentiated to

$$\frac{d\rho}{\rho} + \frac{dV}{V} + \frac{dA}{A} = 0 \tag{9.48}$$

and hence

$$\frac{dV}{V} = \frac{1}{Ma^2 - 1}\frac{dA}{A}. \tag{9.49}$$

We can see the opposite behavior of the flow when $Ma < 1$ and $Ma < 1$.

9.5 Shock waves, discontinuities, and jump conditions

Shock waves are ideally infinitely thin (order of molecular mean-free-path), discontinuous structures in compressible flows. Across the shock, p, ρ, T, s increase, Ma, p_0 decrease, and T_0, h_0 conserve. The pre- and post-shock variables satisfy normal shock relations that will be derived below. Here we only deal with normal shock waves where the flow is perpendicular to the shock. For oblique shock waves the normal component of the flow satisfies the same relation.

Assume a shock wave moving from right to left. When the reference frame is fixed in the shock, the flow goes from left (left= 1) to right (right= 2).

The conservation laws (for a thin CV across the shock) and thermodynamic relations are:

- Mass conservation:

$$\rho_1 V_1 = \rho_2 V_2. \tag{9.50}$$

- Momentum conservation:

$$\rho_1 V_1^2 - \rho_2 V_2^2 = p_2 - p_1. \tag{9.51}$$

- Energy (total enthalpy/total temperature) conservation:

$$h_1 + \frac{1}{2}V_1^2 = h_2 + \frac{1}{2}V_2^2 = h_0. \tag{9.52}$$

- The enthalpy is
$$h = c_p T \tag{9.53}$$
and c_p and k are constants.

- (Perfect gas law):
$$\frac{p_1}{\rho_1 T_1} = \frac{p_2}{\rho_2 T_2}. \tag{9.54}$$

9.5.1 Jump conditions

We have 5 equations for 5 unknowns, p, V, ρ, h, T. By eliminating density (including multiplying the momentum equation with $1/\rho_1 + 1/\rho_2$), we reach the Rankine–Hugoniot relation

$$h_2 - h_1 = \frac{1}{2}\left(\frac{1}{\rho_1} + \frac{1}{\rho_2}\right)(p_2 - p_1), \tag{9.55}$$

in which only thermodynamic properties are contained and the EOS is not invoked.

In more general cases; Integral form of the equations; jump conditions. Conservation form, weak form, weak continuity, test functions.

9.5.2 Rankine–Hugoniot relations / Mach number relations for perfect gas

In a perfect gas, $h = c_p T = \gamma/(\gamma-1)p/\rho$, the density ratio can be related to the pressure ratio as

$$\frac{\rho_2}{\rho_1} = \frac{1 + \beta\frac{p_2}{p_1}}{\beta + \frac{p_2}{p_1}}, \tag{9.56}$$

where $\beta = (\gamma+1)/(\gamma-1)$. Furthermore, the downstream solution are only functions of upstream Mach number Ma_1 and γ, as the Rankine–Hugoniot relations (Mach number relations):

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}(Ma_1^2 - 1) \tag{9.57}$$

$$\frac{V_1}{V_2} = \frac{\rho_2}{\rho_1} = \frac{(\gamma+1)Ma_1^2}{(\gamma-1)Ma_1^2 + 2} \tag{9.58}$$

$$\frac{T_2}{T_1} = 1 + \frac{2(\gamma-1)}{(\gamma+1)^2}\frac{\gamma Ma_1^2 + 1}{Ma_1^2}(Ma_1^2 - 1) \tag{9.59}$$

$$Ma_2^2 = \frac{(\gamma-1)Ma_1^2 + 2}{2\gamma Ma_1^2 - (\gamma-1)} \tag{9.60}$$

$$\frac{s_2 - s_1}{c_v} = \ln\left[\frac{p_2}{p_1}\left(\frac{\rho_1}{\rho_2}\right)^\gamma\right] \tag{9.61}$$

where in order to reach second to last equation the momentum equation in terms of Mach numbers

$$\frac{p_2}{p_1} = \frac{1 + \gamma Ma_1^2}{1 + \gamma Ma_2^2} \tag{9.62}$$

is used.

It can be shown that when $Ma_1^2 - 1$ is small,

$$\frac{s_2 - s_1}{c_v} \approx \frac{2\gamma(\gamma-1)}{3(\gamma+1)^2}(Ma_1^2 - 1)^3, \tag{9.63}$$

which should be positive according to the second law of thermodynamics. Hence, we must have $Ma_1 > 1$ for the upstream flow. Subsequently,

$$p_2 > p_1 \tag{9.64}$$
$$\rho_2 > \rho_1 \tag{9.65}$$
$$T_2 > T_1 \tag{9.66}$$
$$s_2 > s_1 \tag{9.67}$$
$$Ma_2 < 1 < Ma_1 \tag{9.68}$$
$$V_2 < V_1 \tag{9.69}$$
$$\tag{9.70}$$

post-shock.

9.6 Euler equation

9.6.1 Characteristics

Consider 1D Euler equation. We denote the state vector as $U = [\rho, \rho u, E]$ (total energy $E = \rho e + 1/2\rho u^2$), and the 1D conservation law is

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} = 0, \tag{9.71}$$

where the flux vector is

$$F(U) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E+p) \end{bmatrix}. \tag{9.72}$$

It can be written as

$$\frac{\partial U}{\partial t} + A\frac{\partial U}{\partial x} = 0, \tag{9.73}$$

where $A = \partial F/\partial U$. If A is diagonalizable, $A = R\Lambda R^{-1}$, a change to the characteristic variable $v = R^{-1}u$ will diagonalize the system (9.73) to

$$v_t + \Lambda v_x = 0, \tag{9.74}$$

which are decoupled, and such a system is called hyperbolic. The directions Λ are called the characteristics.

With the equation of state $e = c_v T = (p/\rho)/(\gamma - 1)$, $E = p/(\gamma-1) + 1/2\rho u^2$, the Jacobian of the flux vector is

$$F'(U) = \frac{\partial F}{\partial U} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2}(\gamma-3)u^2 & (3-\gamma)u & (\gamma-1) \\ \frac{1}{2}(\gamma-1)u^3 - u(E+p)/\rho & (E+p)/\rho - (\gamma-1)u^2 & \gamma u \end{bmatrix}, \tag{9.75}$$

which has three real eigenvalues

$$\lambda_1 = u - a,\ \lambda_2 = u,\ \lambda_3 = u + a, \tag{9.76}$$

where

$$a = \sqrt{\gamma p/\rho} \tag{9.77}$$

is just the speed of sound.

9.6.2 Riemann invariants

The 1D Euler equation is

$$\frac{\partial \rho}{\partial t} + u\frac{\partial \rho}{\partial x} = -\rho\frac{\partial u}{\partial x} \tag{9.78}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{9.79}$$

The sound speed is

$$a^2 = \frac{\partial p}{\partial \rho} = \gamma\frac{p}{\rho} \tag{9.80}$$

with an isentropic EOS $p = C\rho^\gamma$. It can be shown that

$$2a\,da = (\gamma - 1)a^2\frac{d\rho}{\rho} \tag{9.81}$$

and hence

$$\frac{\partial \rho}{\partial t} = \frac{2\rho}{a(\gamma-1)}\frac{\partial a}{\partial t} \tag{9.82}$$

$$\frac{\partial \rho}{\partial x} = \frac{2\rho}{a(\gamma-1)}\frac{\partial a}{\partial x} \tag{9.83}$$

$$\frac{1}{\rho}\frac{\partial p}{\partial x} = \frac{2a}{\gamma-1}\frac{\partial a}{\partial x} \tag{9.84}$$

The momentum equation converts to

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -a\frac{\partial}{\partial x}\left(\frac{2a}{\gamma-1}\right) \tag{9.85}$$

and the continuity equation to

$$\frac{\partial}{\partial t}\left(\frac{2a}{\gamma-1}\right) + u\frac{\partial}{\partial x}\left(\frac{2a}{\gamma-1}\right) = -a\frac{\partial u}{\partial x} \tag{9.86}$$

The above two equations can be combined to

$$\frac{\partial}{\partial t}\left(u + \frac{2a}{\gamma-1}\right) + u\frac{\partial}{\partial x}\left(u + \frac{2a}{\gamma-1}\right) = -a\left(u + \frac{2a}{\gamma-1}\right) \tag{9.87}$$

$$\frac{\partial}{\partial t}\left(u - \frac{2a}{\gamma-1}\right) + u\frac{\partial}{\partial x}\left(u - \frac{2a}{\gamma-1}\right) = a\left(u - \frac{2a}{\gamma-1}\right) \tag{9.88}$$

i.e.

$$\left(\frac{\partial}{\partial t}+(u+a)\frac{\partial}{\partial x}\right)\left(u+\frac{2a}{\gamma-1}\right)=0 \tag{9.89}$$

$$\left(\frac{\partial}{\partial t}+(u-a)\frac{\partial}{\partial x}\right)\left(u-\frac{2a}{\gamma-1}\right)=0 \tag{9.90}$$

where the two Riemann invariants

$$R_{\pm}=u\pm\frac{2a}{\gamma-1} \tag{9.91}$$

are conserved in isentropic flows along the two characteristics

$$\frac{dx}{dt}=u\pm a, \tag{9.92}$$

respectively. Once the two information-carrying invariants are solved at any (x,t), the velocity and sound speed can be solved as

$$u=\frac{1}{2}(R_{+}+R_{-}) \tag{9.93}$$

$$a=\frac{\gamma-1}{2}(R_{+}-R_{-}) \tag{9.94}$$

9.6.3 Small perturbation linearized equations; acoustic equation

Linearize the Euler equation around the base state $\rho_0, p_0, \boldsymbol{u}_0$ and decompose the variables as

$$\rho=\rho_0+\rho' \tag{9.95}$$

$$p=p_0+p' \tag{9.96}$$

$$\boldsymbol{u}=\boldsymbol{u}_0+\boldsymbol{u}'=\boldsymbol{u}' \tag{9.97}$$

where we have chosen a uniform ρ_0 such that $\nabla\rho_0=\mathbf{0}$ and $\boldsymbol{u}_0=\mathbf{0}$ (moving with the base flow). The linearized perturbation equations are

$$\frac{\partial\rho}{\partial t}+\rho_0(\nabla\cdot\boldsymbol{u})=0 \tag{9.98}$$

$$\frac{\partial\boldsymbol{u}}{\partial t}=-\frac{1}{\rho_0}\frac{dp}{d\rho}\nabla\rho \tag{9.99}$$

where the barotropic EOS, $p=p(\rho)$, is used and primes are dropped without causing confusions. Combining these two equations and eliminate $\boldsymbol{u}$, we have the acoustic wave equation

$$\frac{\partial^2\rho}{\partial t^2}=a^2\nabla^2\rho, \tag{9.100}$$

where $a=dp/d\rho$ is the (phase) speed of sound.

Alternatively, the density equation can be written with pressure ($\rho_t=(d\rho/dp)p_t=p_t/a^2$) as

$$\frac{\partial}{\partial t}\left(\frac{p}{\rho_0}\right)+a^2(\nabla\cdot\boldsymbol{u})=0, \tag{9.101}$$

or ($\rho_0=1$)

$$p_t+a^2(u_x+v_y+w_z)=0 \tag{9.102}$$

and

$$p_{tt}=a^2\nabla^2 p. \tag{9.103}$$

9.7 Kinetic theory, microscopic view of fluid and flow properties

pressure (molecules bouncing back from a flat wall), viscosity, etc.

kinetic theory. Cf. Frank Shu.

AFD notes; the temperature lapse rate example;

Boltzmann equation, microscopic to macroscopic, Knudsen number;

9.8 Compressible turbulence

9.8.1 Helmholtz velocity decomposition

The Helmholtz decomposition can be used to separate the velocity field into solenoidal/vortical and dilatational/irrotational components:

$$\boldsymbol{u} = \boldsymbol{u}_s + \boldsymbol{u}_c \tag{9.104}$$

$$= \nabla \times \psi + \nabla \phi \tag{9.105}$$

$$\nabla \cdot \boldsymbol{u}_s = 0, \nabla \times \boldsymbol{u}_c = 0. \tag{9.106}$$

The solenoidal part is largely similar to incompressible turbulence which also contains a -5/3 inertial range, while the dilatational part represents expansions, compressions, shocklets, and acoustic waves, and contain a much lower fraction of turbulent energy.

9.8.2 The Farve average

The Farve average is a density-weighted average similar to the Reynolds average in incompressible turbulence.

Chapter 10

Topics in Computational Fluid Dynamics

In the era of computers, a Fluid Mechanics note without any computational methods is incomplete. But to avoid going too unbounded in numerical methods, we present several selected methods that the author believes to be central to the computation of incompressible flows. Some fundamental concepts of numerics can be found in Appendix F.

10.1 The projection method for incompressible flows, primitive variables

In incompressible flows, the momentum equation and energy equations are decoupled, and pressure acts as a Lagrange multiplier to enforce the continuity constraint, while its absolute value makes no sense. It is intuitive to think that three velocity components are solved from three momentum equations while the continuity equation is used to solve for the pressure. But the continuity equation does not contain pressure explicitly. On the other hand, the continuity constraint needs to be satisfied in each time step, while the equation does not involve time either. So the need to obtain time-accurate, divergence-free velocity field motivates the development of velocity-pressure coupling method. In this case, a projection (or fractional-step) method.

10.1.1 The Leray Projection

The Helmholtz decomposition theorem states that for any vector field it can be decomposed into an irrotational and a divergence-free component,

$$\boldsymbol{v} = \nabla\phi + \boldsymbol{u} \tag{10.1}$$

where the scalar gradient $\nabla\phi$ is irrotational and $\boldsymbol{u} \in \mathcal{H}^2$ is the divergence-free such that $\nabla\cdot\boldsymbol{u} = 0$. $\mathcal{H}_2$ is the Hilbert space spanned by twice continuous differentiable functions and equipped with an inner product $(\boldsymbol{u}, \boldsymbol{v}) = \int \boldsymbol{u}\cdot\boldsymbol{v}\,\mathrm{d}V$. In CFD, if given an arbitrary velocity field $\boldsymbol{v}$ (could be in a prediction step of momentum equation stepping, or from numerical experiments), the way to correct it to divergence-free is to find a scalar potential ϕ whose gradient does so, and who typically satisfies a Poisson equation $\nabla^2\phi = \nabla\cdot\boldsymbol{v}$.

The goal of a projection method is to find an (orthogonal) projector P such that

$$P\boldsymbol{v} = \boldsymbol{u} \tag{10.2}$$

that projects the velocity field onto the divergence-free space. We note also $P\boldsymbol{u} = \boldsymbol{u}$ and $P^2\boldsymbol{v} = P\boldsymbol{v}$ so we have shown that P is a projector. In order to show P is orthogonal, we need to prove $(\boldsymbol{u}, \nabla\phi) = 0$. The inner product is

$$\int (\boldsymbol{u} \cdot \nabla\phi)\, \mathrm{d}V = \boldsymbol{n} \cdot (\boldsymbol{u}\phi)|_{\text{boundaries}} - \int \phi(\nabla \cdot \boldsymbol{u})\, \mathrm{d}V = 0 \tag{10.3}$$

for periodic or no-penetration boundaries. Thus, P is orthogonal. The projection that removes the divergence of the velocity field is called the Leray projection.

In the Fourier space, continuity implies velocity Fourier modes being orthogonal to wavenumbers,

$$\boldsymbol{k} \cdot \hat{\boldsymbol{u}} = 0. \tag{10.4}$$

The Leray projector is

$$P_{ij}(\boldsymbol{k}) = \delta_{ij} - \frac{k_i k_j}{|\boldsymbol{k}|^2}. \tag{10.5}$$

For any arbitrary vector v_i, its divergence-free component is

$$u_i = P_{ij} v_j = v_i - \frac{k_i k_j}{|\boldsymbol{k}|^2} v_j \tag{10.6}$$

such that $k_i u_i = k_i(P_{ij} v_j) = 0$. It is straightforward to verify that $P^2 = P$ and that $v_i - (k_i k_j/|\boldsymbol{k}|^2)v_j \perp k_i k_j/|\boldsymbol{k}|^2 v_j$.

10.1.2 The projection method

As originally proposed in Chorin (1968), the projection method employs a two-step approach. Instead of solving pressure and velocity simultaneously, the momentum equations are first solved without correct pressure (typically the pressure field from a previous time step). Then, a Poisson equation for pressure correction is formed, where the gradient of the pressure correction is used to enforce velocity continuity.

In a simple forward Euler time-stepping scheme (can easily be extended to more complicated schemes such as RK), the projection approach consists of

$$\frac{\hat{\boldsymbol{u}}^n - \boldsymbol{u}^{n-1}}{\Delta t} = -\nabla p^{n-1} - \boldsymbol{u}^{n-1} \cdot \nabla \boldsymbol{u}^{n-1} + \frac{1}{Re}\nabla^2 \boldsymbol{u}^{n-1} \tag{10.7}$$

$$\nabla^2 \phi^n = \frac{1}{\Delta t}(\nabla \cdot \hat{\boldsymbol{u}}^n) \tag{10.8}$$

$$\boldsymbol{u}^n = \hat{\boldsymbol{u}}^n - \Delta t \nabla \phi^n \tag{10.9}$$

$$p^n = p^{n-1} + \phi^n \tag{10.10}$$

The projection step (10.9) removes the divergence from the provisional velocity $\hat{\boldsymbol{u}}^n$ ($P\hat{\boldsymbol{u}}^n = \boldsymbol{u}^n$) where the required scalar gradient is solved from the Poisson equation (10.8).

The equation that $\boldsymbol{u}^n$ and p^n satisfies simultaneously is

$$\frac{\boldsymbol{u}^n - \boldsymbol{u}^{n-1}}{\Delta t} = -\nabla p^n - \boldsymbol{u}^{n-1} \cdot \nabla \boldsymbol{u}^{n-1} + \frac{1}{Re}\nabla^2 \boldsymbol{u}^{n-1} \tag{10.11}$$

while subject to the constraint

$$\nabla \cdot \boldsymbol{u}^n = 0. \tag{10.12}$$

The projection step (10.9) solves the above constrained optimization problem, via finding a vector in the divergence-free space closest to the provisional velocity vector

$$\boldsymbol{u}^n = \underset{\nabla\cdot\boldsymbol{w}=0}{\arg\min} \frac{1}{2}\|\boldsymbol{w} - \hat{\boldsymbol{u}}^n\|_2^2, \tag{10.13}$$

which is an orthogonal projection.

The Lagrangian of the optimization problem is

$$L(\boldsymbol{w}, p) = \int \frac{1}{2}|\boldsymbol{w} - \hat{\boldsymbol{u}}^n|^2 \,\mathrm{dV} + \int \tilde{p}(\nabla \cdot \boldsymbol{w}) \,\mathrm{dV} \tag{10.14}$$

where $\tilde{p}$ is a Lagrange multiplier, introduced along with the constraint. The integrated constraint is because the constraint is to be satisfied at infinite points, and it is the functional analog to $\lambda^T g$ ($\lambda = [\lambda_1, \lambda_2, ...]^T$). As a result, L is a scalar functional. The optimization problem (10.14) is solved by taking the variations w.r.t. $\boldsymbol{w}$ to be zero:

$$\delta_w L = \int (\boldsymbol{w} - \hat{\boldsymbol{u}}^n) \cdot \delta\boldsymbol{w} \,\mathrm{dV} + \int \tilde{p}(\nabla \cdot \delta\boldsymbol{w}) \,\mathrm{dV} \tag{10.15}$$

$$= \int (\boldsymbol{w} - \hat{\boldsymbol{u}}^n - \nabla\tilde{p}) \cdot \delta\boldsymbol{w} \,\mathrm{dV} \tag{10.16}$$

$$= 0, \tag{10.17}$$

for all variations $\delta\boldsymbol{w}$, provided vanishing boundary terms. Then we have (almost everywhere) the Euler-Lagrange condition

$$\boldsymbol{w} = \hat{\boldsymbol{u}}^n + \nabla\tilde{p}. \tag{10.18}$$

Taking the divergence, the above becomes

$$\nabla^2\tilde{p} = -(\nabla \cdot \hat{\boldsymbol{u}}^n), \tag{10.19}$$

which is the Poisson equation (10.8) up to a scaling factor ($\tilde{p} = -\Delta t\phi^n$). From this we can see that Δt is not explicitly necessary in this process.

The optimization can also be written in a KKT structure

$$\begin{bmatrix} I & -\nabla \\ \nabla\cdot & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{w} \\ \tilde{p} \end{bmatrix} = \begin{bmatrix} \hat{\boldsymbol{u}}^n \\ 0 \end{bmatrix}. \tag{10.20}$$

The discrete, matrix representation of the original N–S system is

$$\begin{bmatrix} M & G \\ D & O \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}, \tag{10.21}$$

where M, G, T are the momentum, gradient, divergence operators and $G^T = D$. This is a saddle-point problem (Benzi *et al.*, 2005). From

$$Mu + Gp = f \tag{10.22}$$

$$Du = 0. \tag{10.23}$$

we can solve formally

$$u = M^{-1}f - M^{-1}Gp. \tag{10.24}$$

$$DM^{-1}Gp = DM^{-1}f \tag{10.25}$$

Here $S = DM^{-1}G$ is the Schur complement, which is dense and expensive. Hence, solving the system (10.21) directly is difficult since it is indefinite and poorly conditioned.

An approximate / split approach to solve (10.21) is the projection method. First solve

$$M\hat{u} = f \tag{10.26}$$

ignoring pressure, then solve an approximate system for pressure ($S = DM^{-1}G \approx DG = \nabla^2$)

$$\nabla^2 p \approx DM^{-1}Gp = DM^{-1}f = DM^{-1}M\hat{u} = D\hat{u}. \tag{10.27}$$

Then from $Mu + Gp = f = M\hat{u}$ we can solve

$$u = \hat{u} - M^{-1}Gp. \tag{10.28}$$

These three steps complete the projection method.

The exact block LU decomposition is

$$\begin{bmatrix} M & G \\ D & O \end{bmatrix} = \begin{bmatrix} I & O \\ DM^{-1} & I \end{bmatrix} \begin{bmatrix} M & G \\ O & -DM^{-1}G \end{bmatrix}, \tag{10.29}$$

and the solution of (10.21) is reduced to solving

$$\begin{bmatrix} M & G \\ O & -DM^{-1}G \end{bmatrix} \begin{bmatrix} u \\ p \end{bmatrix} = \begin{bmatrix} f \\ -DM^{-1}f \end{bmatrix}, \tag{10.30}$$

where the first equation gives (10.28) and the second equation is approximated as (10.27).

10.2 Vorticity–Streamfunction and vorticity–Laplacian formulations, state variables

10.3 Spectral collocation method

10.3.1 Initial perturbation generation

Rogallo

Leray projection in Fourier space.

Acknowledgements

During the course of his studies in fluid mechanics, the author is fortunate enough to take courses from and work under the guidance of many distinguished professors and collaborators, and to interact with fellow students and friends. This note is not possible without their influences.

Appendix A

Vectors, tensors, and their calculus

A.1 Levi-Civita symbol

A.1.1 Determinant representation

The matrix determinants can be expressed in terms of the Levi-Civita symbol. Assume A is a matrix

$$\det(A) = \boldsymbol{a}_1 \cdot (\boldsymbol{a}_2 \times \boldsymbol{a}_3) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \epsilon_{ijk} a_{1i} a_{2j} a_{3k} \tag{A.1}$$

where

$$\boldsymbol{a}_1 = (a_{11}, a_{12}, a_{13})^\top,\ \boldsymbol{a}_2 = (a_{21}, a_{22}, a_{23})^\top,\ \boldsymbol{a}_3 = (a_{31}, a_{32}, a_{33})^\top \tag{A.2}$$

Therefore the Levi-Civita symbol can be expressed as

$$\epsilon_{ijk} = \det(\boldsymbol{e}_i, \boldsymbol{e}_j, \boldsymbol{e}_k) = \boldsymbol{e}_i \cdot (\boldsymbol{e}_j \times \boldsymbol{e}_k) \tag{A.3}$$

Similarly, the outer product of vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ can be written as

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{vmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \epsilon_{ijk} a_j b_k \boldsymbol{e}_i \tag{A.4}$$

Example: $\boldsymbol{\omega}$.

A.1.2 Epsilon identity

$$\epsilon_{ijk}\epsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix} \tag{A.5}$$

$$= \delta_{il}(\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) + \delta_{jl}(\delta_{in}\delta_{km} - \delta_{im}\delta_{kn}) + \delta_{kl}(\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}) \tag{A.6}$$

$$= \delta_{il}\delta_{jm}\delta_{kn} + \delta_{im}\delta_{jn}\delta_{kl} + \delta_{in}\delta_{jl}\delta_{km} - \delta_{il}\delta_{jn}\delta_{km} - \delta_{in}\delta_{jm}\delta_{kl} - \delta_{im}\delta_{jl}\delta_{kn} \tag{A.7}$$

A.1.3 Contracted epsilon identity

Let $i = l$ and notice $\delta_{ii} = 3$

$$\epsilon_{ijk}\epsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km} \tag{A.8}$$

Futhur let $k = m$

$$\epsilon_{ijk}\epsilon_{ijn} = 2\delta_{kn} \tag{A.9}$$

Futhermore

$$\epsilon_{ijk}\epsilon_{ijk} = 6 \tag{A.10}$$

A.2 Vector identities

Assume λ is a scalar and $\boldsymbol{a}$,$\boldsymbol{b}$,$\boldsymbol{c}$,$\boldsymbol{d}$ are vectors in $\mathbb{R}^3$. The identities below might be useful in fluids, some of which have geometric implecations.

$$\nabla \cdot (\nabla \times \boldsymbol{b}) = 0 \tag{A.11}$$
$$\nabla \times (\nabla\lambda) = 0 \tag{A.12}$$
$$\nabla \cdot (\lambda\boldsymbol{b}) = \nabla\lambda \cdot \boldsymbol{b} + \lambda(\nabla \cdot \boldsymbol{b}) \tag{A.13}$$
$$\nabla \times (\lambda\boldsymbol{b}) = \lambda(\nabla \times \boldsymbol{b}) - \boldsymbol{b} \times \nabla\lambda \tag{A.14}$$
$$\nabla(\boldsymbol{a} \cdot \boldsymbol{b}) = \boldsymbol{b} \cdot \nabla\boldsymbol{a} + \boldsymbol{a} \cdot \nabla\boldsymbol{b} + \boldsymbol{b} \times (\nabla \times \boldsymbol{a}) + \boldsymbol{a} \times (\nabla \times \boldsymbol{b}) \tag{A.15}$$
$$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b} \cdot (\boldsymbol{c} \times \boldsymbol{a}) = \boldsymbol{c} \cdot (\boldsymbol{a} \times \boldsymbol{b}) \tag{A.16}$$
$$\nabla \cdot (\boldsymbol{a} \times \boldsymbol{b}) = (\nabla \times \boldsymbol{a}) \cdot \boldsymbol{b} - (\nabla \times \boldsymbol{b}) \cdot \boldsymbol{a} \tag{A.17}$$
$$(\boldsymbol{a} \times \boldsymbol{b}) \cdot (\boldsymbol{c} \times \boldsymbol{d}) = (\boldsymbol{a} \cdot \boldsymbol{c})(\boldsymbol{b} \cdot \boldsymbol{d}) - (\boldsymbol{a} \cdot \boldsymbol{d})(\boldsymbol{b} \cdot \boldsymbol{c}) \tag{A.18}$$
$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b}) \tag{A.19}$$
$$\boldsymbol{b} \times (\nabla \times \boldsymbol{b}) = \nabla(\frac{1}{2}\boldsymbol{b} \cdot \boldsymbol{b}) - \boldsymbol{b} \cdot \nabla\boldsymbol{b} \tag{A.20}$$
$$\nabla \times (\boldsymbol{a} \times \boldsymbol{b}) = (\boldsymbol{b} \cdot \nabla)\boldsymbol{a} - (\boldsymbol{a} \cdot \nabla)\boldsymbol{b} + \boldsymbol{a}(\nabla \cdot \boldsymbol{b}) - \boldsymbol{b}(\nabla \cdot \boldsymbol{a}) \tag{A.21}$$
$$\nabla \times (\nabla \times \boldsymbol{a}) = \nabla(\nabla \cdot \boldsymbol{a}) - \nabla^2\boldsymbol{a} \tag{A.22}$$

Their proofs are left as exercises.
Comments:

(1) Eq. (A.11): A curl field is solenoidal (divergence-free).

(2) Eq. (A.12): A gradient field is irrotational (curl-free).

(3) Eq. (A.21): $\boldsymbol{a} \times \boldsymbol{b}$ is perpendicular to $\boldsymbol{a}$ and $\boldsymbol{b}$, so its curl is in the space spaned by $\boldsymbol{a}$ and $\boldsymbol{b}$.

(4) Eq. (A.19): $\boldsymbol{a} \times (\cdot)$ is perpendicular to $\boldsymbol{a}$ and $(\cdot) \times (\boldsymbol{b} \times \boldsymbol{c})$ is in the space spaned by $\boldsymbol{b}$ and $\boldsymbol{c}$. This two facts in combinition gives the bases of $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$.

(5) Eq. (A.16): This is the volume spaned by $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c})$, and the identity is basically the invariance of a determinant with respect to row/column permutation.

(6) Eq. (A.18): By letting $\boldsymbol{a} = \boldsymbol{c}$ and $\boldsymbol{b} = \boldsymbol{d}$ and noticing the inner product with itself is non-negative, we re-discover the Cauchy-Schwartz inequality.

(7) From (A.15) it can be immediately seen that

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = \nabla(\boldsymbol{u}^2/2) - \boldsymbol{u} \times (\nabla \times \boldsymbol{u}). \tag{A.23}$$

A.3 Tensor eigenvalues, invariants, and its application in fluids

Consider a tensor $\boldsymbol{A}$ in Cartesian coordinate

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \tag{A.24}$$

Its eigenvalues are roots of the characteristic polynomial

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = \begin{vmatrix} \lambda - a_{11} & -a_{12} & -a_{13} \\ -a_{21} & \lambda - a_{22} & -a_{23} \\ -a_{31} & -a_{32} & \lambda - a_{33} \end{vmatrix} = \lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0 \tag{A.25}$$

with the three coefficients being the three principle invariants of $\boldsymbol{A}$

$$\begin{aligned} I_1 &= a_{11} + a_{22} + a_{33} && \text{(A.26)} \\ &= \mathrm{tr}(\boldsymbol{A}) && \text{(A.27)} \\ &= a_{ii} && \text{(A.28)} \\ I_2 &= a_{11}a_{22} + a_{22}a_{33} + a_{33}a_{11} - a_{12}a_{21} - a_{23}a_{32} - a_{13}a_{31} && \text{(A.29)} \\ &= \frac{\mathrm{tr}(\boldsymbol{A})^2 - \mathrm{tr}(\boldsymbol{A}^2)}{2} && \text{(A.30)} \\ &= \frac{1}{2}((a_{ii})^2 - a_{ij}a_{ji}) && \text{(A.31)} \\ I_3 &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) && \text{(A.32)} \\ &= \det(\boldsymbol{A}) && \text{(A.33)} \end{aligned}$$

in both element-wise and coordinate-independent expression.

Now we consider the factorization of the characteristic polynomial as

$$(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = \lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1)\lambda - \lambda_1\lambda_2\lambda_3 = 0, \tag{A.34}$$

and obtain the Vieta's theorem for cubic equations as

$$\begin{aligned} I_1 &= \lambda_1 + \lambda_2 + \lambda_3 && \text{(A.35)} \\ I_2 &= \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 && \text{(A.36)} \\ I_3 &= \lambda_1\lambda_2\lambda_3 && \text{(A.37)} \end{aligned}$$

which are the three principle invariants of tensor $\boldsymbol{A}$.

Additionally, there are more invariants (although not independent) of $\boldsymbol{A}$, such as the main invariants

$$J_1 = \lambda_1 + \lambda_2 + \lambda_3 = I_1 = \mathrm{tr}(\boldsymbol{A}) \tag{A.38}$$
$$J_2 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 = I_1^2 - 2I_2 = \mathrm{tr}(\boldsymbol{A} \cdot \boldsymbol{A}) \tag{A.39}$$
$$J_1 = \lambda_1^3 + \lambda_2^3 + \lambda_3^3 = I_1^3 - 3I_1 I_2 + 3I_3 = \mathrm{tr}(\boldsymbol{A} \cdot \boldsymbol{A} \cdot \boldsymbol{A}) \tag{A.40}$$

which are the coefficients of the characteristic polynomial of the deviatoric part of $\boldsymbol{A}$:

$$\boldsymbol{A} - \frac{\mathrm{tr}(\boldsymbol{A})}{3}\boldsymbol{I}, \tag{A.41}$$

which is traceless and has eigenvalues

$$\lambda_i - \frac{1}{3}. \tag{A.42}$$

A.3.1 Discriminant of a cubic equation

Consider

$$ax^3 + bx^2 + cx + d = 0, \tag{A.43}$$

its determinant is

$$\Delta = (x_1 - x_2)^2(x_2 - x_3)^2(x_3 - x_1)^2 \tag{A.44}$$
$$= 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2 \tag{A.45}$$

with x_1, x_2, x_3 being the three roots.

1. $\Delta > 0$: Three distinct real roots.
2. $\Delta = 0$: All roots are real with at least two identical.
3. $\Delta < 0$: One real and a pair of complex conjugate roots (proof: assume complex roots are $x \pm iy$).

Proof. The Vieta's theorem for (A.43) and the invariant relations can be used to simplify (A.43) to obtain (A.45).

Note: Eq. (A.45) can also be obtained as follows (with some reasons/meanings in algebraic geometry). Consider a cubic equation in canonical form

$$f(x, w) = Ax^3 + 3Bx^2w + 3Cxw^2 + Dw^3 = 0. \tag{A.46}$$

The Hessain matrix is

$$H(f) = \begin{bmatrix} 6Ax + 6Bw & 6Bx + 6Cw \\ 6Bx + 6Cw & 6Cx + 6Dw \end{bmatrix} \tag{A.47}$$

and the Hessain

$$\det(H) = 36[(AC - B^2)x^2 + (AD - BC)xw + (BD - C^2)w^2] \tag{A.48}$$

$$= 18[x, w] \begin{bmatrix} 2(AC - B^2) & (AD - BC) \\ (AD - BC) & 2(BD - C^2) \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix}, \tag{A.49}$$

in quadratic form. Define the Hessain

$$\boldsymbol{H} = \begin{bmatrix} 2(AC - B^2) & (AD - BC) \\ (AD - BC) & 2(BD - C^2) \end{bmatrix} \tag{A.50}$$

The discriminant of the cubic is just the determinant of the Hessain $\boldsymbol{H}$:

$$\Delta = \det(\boldsymbol{H}) = -A^2D^2 + 6ABCD - 4AC^3 - 4B^3D + 3B^2C^2, \tag{A.51}$$

and $\Delta > 0$ for three real roots, $\Delta = 0$ for double or triple real root, and $\Delta < 0$ for single real root.

A.3.2 Application to the velocity gradient tensor

The cubic curve theory, especially root finding, has relation to the characteristic polynomial of the velocity gradient tensor and so forth the local flow geometry (basically the number of real/complex eigenvalues).

The characteristic polynomial for $\boldsymbol{u}\nabla$ is

$$\lambda^3 - P\lambda^2 + Q\lambda - R = 0, \tag{A.52}$$

where $(P, Q, R) = (I_1, I_2, I_3)$ are the three invariants. In incompressible flow, $P = u_{i,i} = 0$, and the equation above degenerates to

$$\lambda^3 + Q\lambda - R = 0. \tag{A.53}$$

It is in the so-called 'depressed' form (comparatively, an elliptic curve is called in the Weierstrass form if it satisfies the Weierstrass equation $y^2 = x^3 + ax + b$)

Positive second invariant $Q = -1/2u_{i,j}u_{j,i} = 1/2(\|\boldsymbol{\Omega}\|^2 - \|\boldsymbol{S}\|^2)$ is used for identifying vortical motions, or 'eddies' Hunt *et al.* (1988); Jeong & Hussain (1995). Consider the Poisson equation,

$$\frac{1}{\rho}\nabla^2 p = -u_{i,j}u_{j,i}, \tag{A.54}$$

we have positive Q corresponding to a local pressure minimum.

The discriminant for depressed cubic equation

$$x^3 + px + q = 0 \tag{A.55}$$

reduces to

$$\Delta = -4p^3 - 27q^2. \tag{A.56}$$

So we have the discriminant for the gradient of a solenoidal field (with renormalized coefficients; note the flipped sign)

$$\triangle = \left(\frac{1}{3}Q\right)^3 + \left(\frac{1}{2}R\right)^2 \tag{A.57}$$

and if $\triangle > 0$ there will be complex eigenvalues (in complex conjugate pair according to the algebra basic theorem) and so-defined vortical motions. Hence, we can see that the invariants of the velocity gradient tensor is largely related to the local geometry (Chong *et al.*, 1990) of the flow.

Moreover, when $\triangle > 0$ and there exist a pair of complex EVs, the complex diagonalization of the VGT is

$$A = P \begin{bmatrix} \lambda_r & & \\ & \lambda_{cr} & \lambda_{ci} \\ & -\lambda_{ci} & \lambda_{cr} \end{bmatrix} P^{-1}, \tag{A.58}$$

where P is an invertible matrix (not necessarily unitary unless A is normal which the VGT is generally not) and this transform is a similarity transform. The three EVs of the VGT become

$$\lambda_1 = \lambda_r,\ \lambda_{2,3} = \lambda_r \pm \lambda_{ci}. \tag{A.59}$$

This leads to the use of λ_{ci}, the imaginary part of the complex eigenvalue(s) of the VGT, to identify vortical regions (Zhou *et al.*, 1999).

A.3.3 Application to the strain rate tensor

We note that both the rate-of-strain tensor $\boldsymbol{S}$ and the Reynolds stress tensor $-\overline{u_i' u_j'}$ are real symmetric, hence they have three real eigenvalues and three orthogonal eigenvectors (principle axes), or, in another word, they are unitarily similar to a diagonal matrix. The rate of strain tensor is diagonal and all strains are normal strains in the principle coordinate.

This part is largely related to eigenvalue decomposition (see section B).

Example. Consider a strain rate tensor

$$\boldsymbol{S} = \begin{bmatrix} 0 & \gamma/2 \\ \gamma/2 & 0 \end{bmatrix}. \tag{A.60}$$

Its eigenvalues are $\lambda_1 = -\gamma/2$, $\lambda_2 = \gamma/2$ and associated eigenvectors are $\boldsymbol{x}_1 = 1/\sqrt{2}[1, -1]^T$ and $\boldsymbol{x}_2 = 1/\sqrt{2}[1, 1]^T$. The two principle directions are 45 deg (stretching, the direction that receives the most amplification) and -45 deg (compressing). The similarity transform is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & \gamma/2 \\ \gamma/2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \frac{1}{\sqrt{2}} = \begin{bmatrix} -\gamma/2 & 0 \\ 0 & \gamma/2 \end{bmatrix} \tag{A.61}$$

where

$$\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2] \tag{A.62}$$

is a unitary matrix such that $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{X}\boldsymbol{\Lambda}$.

Take $\gamma = \partial_y U$, this is an importan example of plane shear flow. Think of the deformation of a rectangle to a diamond as it moves with the shear.

A.3.4 Application to the Reynolds stress tensor, the invariant map, and the Lumley triangle,

Consider the anisotropic (deviatoric) tensor of Reynolds stress

$$a_{ij} = \frac{\overline{u_i' u_j'}}{2k} - \frac{1}{3}\delta_{ij} \tag{A.63}$$

and its three principle invariants

$$I = \sigma_1 + \sigma_2 + \sigma_3 \tag{A.64}$$

$$II = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1 \tag{A.65}$$

$$III = \sigma_1\sigma_2\sigma_3 \tag{A.66}$$

along with its three eigenvalues

$$\sigma_1,\ \sigma_2,\ \sigma_3. \tag{A.67}$$

Since a_{ij} is a deviator, it is traceless and

$$I = a_{ii} = 0. \tag{A.68}$$

Consider turbulence. and has zero determinant

$$\det\left(\frac{\overline{u_i' u_j'}}{2k}\right) = (\sigma_1 + \frac{1}{3})(\sigma_2 + \frac{1}{3})(\sigma_3 + \frac{1}{3}) \tag{A.69}$$

$$= \sigma_1\sigma_2\sigma_3 + \frac{1}{3}(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) + \frac{1}{9}(\sigma_1 + \sigma_2 + \sigma_3) + \frac{1}{27}, \tag{A.70}$$

and we define

$$F = 27III + 9II + 1 \tag{A.71}$$

since $I = 0$.

1. Two-dimensional turbulence: the Reynolds stress tensor $\overline{u_i' u_j'}$ can be diagonalized to

   $$diag(a, k - a, 0)$$

   and has zero determinant (there's a direction that has no turbulence). $F = 0$.

2. Three-dimensional isotropic turbulence: the Reynolds stress tensor $\overline{u_i' u_j'}$ is

   $$diag(k/3, k/3, k/3)$$

   and we have $F = 1$.

3. Axisymmetric turbulence. Similarly, the characteristic polynomial of a_{ij} is in Weierstrass form and the condition for repeated eigenvalues (same energy in two principle directions) is

   $$\triangle = \left(\frac{1}{3}II\right)^3 + \left(\frac{1}{2}III\right)^2 = 0 \tag{A.72}$$

   and hence

   $$III = \pm 2\left(-\frac{II}{3}\right)^3, \tag{A.73}$$

   corresponding to the negative/left (pancake) and positive/right (cigar) limit curves of the Lumley triangle (Lumley & Newman, 1977; Choi & Lumley, 2001).

Appendix B

Matrix Analysis

B.1 Unitary matrix

Unitary transformations preserve inner products (and hence length and angle). Rotation and reflection.

B.2 Singular value decomposition and eigenvalue decomposition

also bilateral relations between SVD/Schur and polar decomposition.

B.3 Conformal mapping

Analytic functions, homomorphism, and conformal mapping.

Angle preserving, Symplectic (area preserving), implications of the latter on numerical schemes.

Appendix C

Fourier analysis

Fourier transform is a linear transformation to a space in which the function can be represented by a set of complete basis where the physics might be better understood. Fourier basis functions (sines and cosines) are the eigenfunction of the Laplacian operator, the matrix of which can be diagonalized by the Fourier transform, and the corresponding eigenvalues are frequencies.

C.1 Fourier series

Assume a function $f(t) \in \mathcal{L}_2([-\pi, \pi])$ (hence $\|f(t)\|$ in (C.4) is bnounded) is periodic on the interval $[-\pi, \pi]$. The Fourier expansion of $f(t)$ in terms of sin's and cos's is

$$f(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} (A_k \cos kt + B_k \sin kt), \tag{C.1}$$

$$A_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos kt \, \mathrm{d}t = \frac{1}{\| \cos kt \|^2} (f(t), \cos kt) \tag{C.2}$$

$$B_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin kt \, \mathrm{d}t = \frac{1}{\| \sin kt \|^2} (f(t), \sin kt) \tag{C.3}$$

where $(\cdot, \cdot)$ denotes the inner product in $\mathcal{L}_2$ space (which is a Hilbert space) and the norm $\| \cdot \|$ is defined as

$$\|f(t)\| = (f(t), f(t))^{1/2} \tag{C.4}$$

and the inner-product therein as

$$(f(t), g(t)) = \int_{-\pi}^{\pi} f(t) \overline{g}(t) \, \mathrm{d}t, \tag{C.5}$$

where $\overline{(\cdot)}$ is the complex conjugate.

If the function is periodic in $[0, L]$, after domain transformation we will get

$$f(t) = \frac{A_0}{2} + \sum_{k=1}^{\infty} \left(A_k \cos\left(\frac{2\pi k}{L} t\right) + B_k \sin\left(\frac{2\pi k}{L} t\right) \right), \tag{C.6}$$

$$A_k = \frac{2}{L} \int_{-\pi}^{\pi} f(t) \cos\left(\frac{2\pi k}{L} t\right) \, \mathrm{d}t \tag{C.7}$$

$$B_k = \frac{2}{L}\int_{-\pi}^{\pi} f(t)\sin\left(\frac{2\pi k}{L}t\right)\,\mathrm{d}t, \tag{C.8}$$

and we note that the normalizing factor just comes from the norms of the trigonometric functions on prescribed intervals and the pre-factor before t is to make the sin's and cos's periodic on this interval.

Now we can think of combining the sin's and cos's into a form similar to $e^{\mathrm{i}kt}$ by leveraging

$$e^{\mathrm{i}kx} = \cos kt + \mathrm{i}\sin kt = \psi_k. \tag{C.9}$$

This will give us the complex Fourier series. Assume $f(t) \in \mathbb{C}$. The series (C.1) on domain $[-\pi, \pi]$ can be re-organized as

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{i}kt} = \sum_{k=-\infty}^{\infty} (\alpha_k + i\beta_k)(\cos kt + \mathrm{i}\sin kt). \tag{C.10}$$

If $f(t) \in \mathbb{R}$, and given $\cos kt = (e^{\mathrm{i}kx} + e^{-\mathrm{i}kx})/2$ and $\sin kt = (e^{\mathrm{i}kx} - e^{-\mathrm{i}kx})/2i$ we have

$$c_k = \frac{\alpha_k + i\beta_k}{2},\ c_{-k} = \frac{\alpha_k - i\beta_k}{2}, \tag{C.11}$$

and the symmetry

$$c_k = \overline{c}_{-k}. \tag{C.12}$$

The relation of the coefficients c_k to those α_k, β_k in (C.1) can be obtained by expanding. For example, $k = 0$ and

$$\frac{A_0}{2} = c_0, \tag{C.13}$$

$k = \pm 1$ and

$$(\alpha_1 + i\beta_1)(\cos t + \mathrm{i}\sin t) + (\alpha_{-1} + i\beta_{-1})(\cos t - \mathrm{i}\sin t) = A_1\cos t + B_1\sin t, \tag{C.14}$$

where we have four conditions and four unknowns.

Assume a function $f(t)$ on the interval $[-L, L]$ is continuous and square-integrable. Its Fourier expansion that decompose $f(t)$ into infinite sum of sin's and cos's is

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{i}k\pi t/L} = \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{i}\omega_k t} = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} (f(t), \psi_k)\psi_k \tag{C.15}$$

$$c_k = \frac{1}{2\pi}(f(t), \psi_k) = \frac{1}{2L}\int_{-L}^{L} f(t)e^{-\mathrm{i}k\pi t/L}\,\mathrm{d}t, \tag{C.16}$$

where the basis functions are called $\{\psi_k\}$ and the corresponding expansion coefficients on this basis are $\{c_k\}$. The angular frequency is

$$\omega_k = \frac{k\pi}{L} = k\Delta\omega, \tag{C.17}$$

that has a dimension of inverse t, where $\Delta\omega$ is the frequency resolution.

C.2 Fourier transform

In the Fourier series, the corresponding function on $[-L, L]$ is the periodic extension of $f(t)$. Assume we don't want to have this periodic extension and take the limit of $L \to \infty$. Then we get the Fourier transform as:

$$f(t) = \lim_{\Delta\omega\to\infty} \sum_{k=-\infty}^{\infty} c_k e^{\mathrm{i}\omega_k t} \tag{C.18}$$

$$= \lim_{\Delta\omega\to\infty} \sum_{k=-\infty}^{\infty} (\frac{\Delta\omega}{2\pi} \int_{-\pi/\Delta\omega}^{\pi/\Delta\omega} f(\xi) e^{-\mathrm{i}k\pi\xi/L}\,\mathrm{d}\xi) e^{\mathrm{i}\omega_k t} \tag{C.19}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} (\int_{-\infty}^{\infty} f(\xi) e^{-\mathrm{i}k\pi\xi/L}\,\mathrm{d}\xi) e^{\mathrm{i}\omega t}\,\mathrm{d}\omega \tag{C.20}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{\mathrm{i}\omega t}\,\mathrm{d}\omega \tag{C.21}$$

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(\xi) e^{-\mathrm{i}\omega\xi}\,\mathrm{d}\xi \tag{C.22}$$

where

$$\hat{f}(\omega) = \mathcal{F}(f(t)) \tag{C.23}$$

$$f(t) = \mathcal{F}^{-1}(\hat{f}(\omega)) \tag{C.24}$$

are the direct and inverse Fourier transforms. We call

$$f(t) \leftrightarrow \hat{f}(\omega) \tag{C.25}$$

a Fourier transform pair. Here the spectral density $(1/2\pi)\hat{f}(\omega)\mathrm{d}\omega$ is the analogy of the amplitude c_k. The (C.21) is can also be written as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\xi) e^{-\mathrm{i}\omega(\xi-t)}\,\mathrm{d}\xi\,\mathrm{d}\omega \tag{C.26}$$

and (C.22) called the Fourier inversion theorem. In many cases, for convenience, (for example, Fourier expansion in spectral methods) the factor of $1/2\pi$ is often placed in front of the forward FT (C.22). Another choice is placing a factor of $1/\sqrt{2\pi}$ in front of both.

C.2.1 Orthogonality, completeness, and linear independence

Up to now, we have probably reached a point where the properties of the Fourier basis should be discussed. The bases are orthogonal and form an infinite-dimensional function space.

First we take a look at $\{\psi_k\}$ where $\psi_k = e^{\mathrm{i}kx} \in [-\pi, \pi]$. We have

$$(\psi_j, \psi_k) = \int_{-\pi}^{\pi} e^{\mathrm{i}jx} e^{-\mathrm{i}kx}\,\mathrm{d}t = \int_{-\pi}^{\pi} e^{\mathrm{i}(j-k)t}\,\mathrm{d}t = \frac{1}{\mathrm{i}(j-k)} e^{\mathrm{i}(j-k)t}|_{-\pi}^{\pi} = \begin{cases} 0, & j \neq k \\ 2\pi, & j = k \end{cases}. \tag{C.27}$$

C.2.2 Derivatives

It is easy to see that

$$f'(t) = \frac{d}{dt}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)e^{\mathrm{i}\omega t}\,\mathrm{d}\omega\right) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{i}\omega\hat{f}(\omega)e^{\mathrm{i}\omega t}\,\mathrm{d}\omega \tag{C.28}$$

and hence

$$f'(t) \leftrightarrow \mathrm{i}\omega\hat{f}(\omega) \tag{C.29}$$

are a Fourier pair.

C.2.3 The inversion theorem; duality

As preparation, we introduce the flip operator

$$\mathcal{R}f(t) = f(-t) \tag{C.30}$$

and it can be shown that if $\hat{f}(\omega) = \mathcal{F}[f(t)]$

$$\mathcal{F}\mathcal{R}f(t) = \mathcal{F}f(-t) = \int_{-\infty}^{\infty} f(-\xi)e^{-\mathrm{i}\omega\xi}\,\mathrm{d}\xi = \int_{-\infty}^{\infty} f(\xi)e^{\mathrm{i}\omega\xi}\,\mathrm{d}\xi = \hat{f}(-\omega) = \mathcal{R}\mathcal{F}f(t) \tag{C.31}$$

$$= \int_{-\infty}^{\infty} f(t)e^{\mathrm{i}\omega t}\,\mathrm{d}t = 2\pi\mathcal{F}^{-1}[f(t)] \tag{C.32}$$

that being said,

$$2\pi\mathcal{F}^{-1}f = \mathcal{F}\mathcal{R}f = \mathcal{R}\mathcal{F}f \tag{C.33}$$

and it can be deduced that

$$f(-t) = \mathcal{R}(t) = \mathcal{F}^{-1}\mathcal{R}\hat{f}(\omega) = \mathcal{F}^{-1}\hat{f}(-\omega) = \frac{1}{2\pi}\mathcal{F}\mathcal{R}\hat{f}(-\omega) = \frac{1}{2\pi}\mathcal{F}\hat{f}(\omega) \tag{C.34}$$

that the duality

$$\hat{f}(\omega) = \mathcal{F}[f(t)] \leftrightarrow f(-t) = \frac{1}{2\pi}\mathcal{F}[\hat{f}(\omega)] \tag{C.35}$$

is true. Thai being said, if you know a function g is the Fourier transform of another function f, you can obtain the Fourier transform of g in terms of f.

C.2.4 Convolution theorem

The convolution of two functions f and g is defined as

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,\mathrm{d}\tau = \int_{-\infty}^{\infty} f(t-\tau)g(\tau)\,\mathrm{d}\tau = (g * f)(t), \tag{C.36}$$

and we will show how it is related to the Fourier transform of the product of two signals.

The convolution theorem is that

$$\mathcal{F}((f * g)(t)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,\mathrm{d}\tau e^{-\mathrm{i}\omega t}\,\mathrm{d}t \tag{C.37}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\tau)g(t-\tau)\,\mathrm{d}\tau e^{-\mathrm{i}\omega\tau}e^{-\mathrm{i}\omega(t-\tau)}\,\mathrm{d}t \tag{C.38}$$

$$= \int_{-\infty}^{\infty} g(t-\tau) e^{-\mathrm{i}\omega(t-\tau)} \,\mathrm{d}(t-\tau) \int_{-\infty}^{\infty} f(\tau) e^{-\mathrm{i}\omega\tau} \,\mathrm{d}\tau \tag{C.39}$$

$$= \hat{f}(\omega)\hat{g}(\omega) \tag{C.40}$$

$$= \mathcal{F}(f(t))\mathcal{F}(g(t)) \tag{C.41}$$

the Fourier transform of the convolution of two functions is the product of the Fourier transforms of these two functions. This provides another way of computing the convolution of two functions.

$$\begin{array}{ccc} f & \xrightarrow{*g} & (f*g)(t) \\ {\scriptstyle\mathcal{F}}\big\downarrow & & \big\downarrow{\scriptstyle\mathcal{F}^{-1}} \\ \hat{f}(\omega) & \xrightarrow[\times\hat{g}(\omega)]{} & \hat{f}(\omega)\hat{g}(\omega) \end{array}$$

We will see that the other way around is also true. Assume

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-\mathrm{i}\omega t} \,\mathrm{d}t \tag{C.42}$$

$$\hat{g}(\omega) = \int_{-\infty}^{\infty} g(t) e^{-\mathrm{i}\omega t} \,\mathrm{d}t \tag{C.43}$$

and the Fourier transform of the product of two functions is

$$\mathcal{F}[f(t)g(t)] = \int_{-\infty}^{\infty} f(t)g(t) e^{-\mathrm{i}\omega t} \,\mathrm{d}t \tag{C.44}$$

$$= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\alpha) e^{\mathrm{i}\alpha t} \,\mathrm{d}\alpha \right) g(t) e^{-\mathrm{i}\omega t} \,\mathrm{d}t \tag{C.45}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\alpha) \left(\int_{-\infty}^{\infty} g(t) e^{-\mathrm{i}(\omega-\alpha)t} \,\mathrm{d}t \right) \mathrm{d}\alpha \tag{C.46}$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\alpha)\hat{g}(\omega-\alpha) \,\mathrm{d}\alpha \tag{C.47}$$

$$= \frac{1}{2\pi} \hat{f} * \hat{g} \tag{C.48}$$

that is the convolution of the Fourier transforms of the two functions.

C.2.5 Correlation functions and spectra; Parseval's theorem

Define the cross-correlation of two complex-value functions u and v as

$$C_{uv}(\tau) = \int_{-\infty}^{\infty} u^*(t) v(t+\tau) \,\mathrm{d}t \tag{C.49}$$

$$= \int_{-\infty}^{\infty} u^*(t-\tau) v(t) \,\mathrm{d}t \tag{C.50}$$

$$= C_{vu}^*(-\tau) \tag{C.51}$$

Its Fourier counterpart is the cross-spectrum

$$\Phi_{uv}(\omega) = \mathcal{F}[C_{uv}(\tau)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u^*(t) v(t+\tau) e^{-\mathrm{i}\omega\tau} \,\mathrm{d}t\mathrm{d}\tau \tag{C.52}$$

$$= 2\pi \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} u^*(t) e^{\mathrm{i}\omega t} \, \mathrm{d}t \right) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} v(t+\tau) e^{-\mathrm{i}\omega(t+\tau)} \, \mathrm{d}\tau \right) \tag{C.53}$$

$$= 2\pi \hat{u}^*(\omega) \hat{v}(\omega) \tag{C.54}$$

whose real part is called the co-spectrum of uv. The phase relation between u, v is $\phi = \tan^{-1}(\Im(\Phi_{uv})/\Re(\Phi_{uv}))$. Note we have chosen to put the $\frac{1}{2\pi}$ factor in the forward FT here. In the special case of $u = v$,

$$\Phi_{uu}(\omega) = 2\pi \hat{u}^*(\omega) \hat{u}(\omega) = 2\pi |\hat{u}(\omega)|^2 \tag{C.55}$$

is the energy spectrum of u, which is the Fourier transform of the two-point auto-correlation C_{uu} (Wiener–Khinchin theorem). We note that $\Phi_{uu}(\omega)$ is real while Φ_{uv} is complex.

From the inverse FT,

$$C_{uv}(\tau) = \int_{-\infty}^{\infty} \Phi_{uv}(\omega) e^{\mathrm{i}\omega\tau} \, \mathrm{d}\omega = 2\pi \int_{-\infty}^{\infty} \hat{u}^*(\omega) \hat{v}(\omega) e^{\mathrm{i}\omega\tau} \, \mathrm{d}\omega. \tag{C.56}$$

Taking $\tau = 0$ we have

$$\int_{-\infty}^{\infty} u^*(t) v(t) \, \mathrm{d}t = C_{uv}(0) = 2\pi \int_{-\infty}^{\infty} \hat{u}^*(\omega) \hat{v}(\omega) \, \mathrm{d}\omega \tag{C.57}$$

or simply, the correlation between u, v being the average/integral in physical/Fourier space:

$$\langle u(t) v(t) \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} u^*(t) v(t) \, \mathrm{d}t = \int_{-\infty}^{\infty} \hat{u}^*(\omega) \hat{v}(\omega) \, \mathrm{d}\omega. \tag{C.58}$$

With $u = v$,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} |u(t)|^2 \, \mathrm{d}t = \int_{-\infty}^{\infty} |\hat{u}(\omega)|^2 \, \mathrm{d}\omega. \tag{C.59}$$

This is called the Parseval's theorem. In the context of DFT, it becomes

$$\frac{1}{N} \sum_{j=0}^{N-1} u_j^* u_j = \sum_{k=0}^{N-1} \hat{u}_k^* \hat{u}_k. \tag{C.60}$$

C.2.6 Symmetries of real FT

A real function $f(t) \in \mathbb{R}$ being Fourier transformed can be decomposed into odd and even parts:

$$f(t) = \frac{1}{2}(f(t) + f(-t)) + \frac{1}{2}(f(t) - f(-t)) \tag{C.61}$$

$$= f_e(t) + f_o(t) \tag{C.62}$$

where

$$f_e(t) = f_e(-t), \; f_o(t) = -f_o(-t). \tag{C.63}$$

The Fourier transform can be so decomposed as

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(\xi) e^{-\mathrm{i}\omega\xi} d\xi \tag{C.64}$$

$$= \int_{-\infty}^{\infty} (f_e(\xi) + f_o(\xi))(\cos(\omega\xi) - \mathrm{i}\sin(\omega\xi))d\xi \tag{C.65}$$

$$= \int_{-\infty}^{\infty} f_e(\xi)\cos(\omega\xi)d\xi - i\int_{-\infty}^{\infty} f_o(\xi)\sin(\omega\xi)d\xi \tag{C.66}$$

$$= \Re\{\hat{f}(\omega)\} + \Im\{\hat{f}(\omega)\} \tag{C.67}$$

$$= \hat{f}_e(\omega) + \hat{f}_o(\omega) \tag{C.68}$$

where we note that the $\int_{-\infty}^{\infty}$ of the product of an even and an odd function is zero.

This has several implications:

- The FT of an even function is real and even ($\hat{f}(\omega) = \hat{f}(-\omega)$). The FT of an odd function is imaginary and odd ($\hat{f}(\omega) = -\hat{f}(-\omega)$).
- The real part of the FT is the FT of the even part of the signal. The imaginary part of the FT is the FT of the odd part of the signal.

These properties can be summarized into

$$\hat{f}(\omega) = \hat{f}_e(\omega) + \hat{f}_o(\omega) \tag{C.69}$$

$$= \hat{f}_e(-\omega) - \hat{f}_o(-\omega) \tag{C.70}$$

$$= \overline{\hat{f}_e(-\omega) + \hat{f}_o(-\omega)} \tag{C.71}$$

$$= \overline{\hat{f}(-\omega)} \tag{C.72}$$

which is called the Hermitian symmetry of the real FT. It can also be seen from replacing ω with $-\omega$ in (C.64). This Hermitian symmetry leads to $\hat{f}(0) = \hat{f}^*(0) \in \mathbb{R}$. In other words, the ω mode and the $-\omega$ mode have equal amplitude but opposite phases:

$$\hat{f}(\omega) = |\hat{f}(\omega)|e^{\mathrm{i}\phi(\omega)} \tag{C.73}$$

$$\hat{f}(-\omega) = |\hat{f}(-\omega)|e^{\mathrm{i}\phi(-\omega)} = \overline{\hat{f}(\omega)} = |\hat{f}(\omega)|e^{-\mathrm{i}\phi(\omega)} \to \phi(-\omega) = -\phi(\omega) \tag{C.74}$$

C.2.7 Fourier transform in multiple dimensions

FT can be straightforwardly extended to higher dimensions as

$$f(x,y,z) = \frac{1}{(2\pi)^3}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \hat{f}(k_x,k_y,k_z)e^{\mathrm{i}k_x x}e^{\mathrm{i}k_y y}e^{\mathrm{i}k_z z}\,\mathrm{d}k_x\mathrm{d}k_y\mathrm{d}k_z \tag{C.75}$$

$$\hat{f}(k_x,k_y,k_z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y,z)e^{-\mathrm{i}k_x x}e^{-\mathrm{i}k_y y}e^{-\mathrm{i}k_z z}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z \tag{C.76}$$

The Hermitian symmetry of real FT ($f \in \mathbb{R}$) is

$$\hat{f}(k_x,k_y,k_z) = \overline{\hat{f}(-k_x,-k_y,-k_z)}. \tag{C.77}$$

C.2.8 Examples

- Dirac delta and unity.

$$\mathcal{F}[\delta(t-t_0)] = \int_{-\infty}^{\infty} \delta(t-t_0)e^{-\mathrm{i}\omega t}\,\mathrm{d}t = e^{-\mathrm{i}\omega t_0} = \cos(\omega t_0) - \mathrm{i}\sin(\omega t_0), \tag{C.78}$$

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t)e^{-\mathrm{i}\omega t}\,\mathrm{d}t = 1, \tag{C.79}$$

$$\delta(-t) = \delta(t) = \mathcal{F}^{-1}[1] = \frac{1}{2\pi}\int_{-\infty}^{\infty} 1 \cdot e^{\mathrm{i}\omega t}\,\mathrm{d}\omega, \tag{C.80}$$

$$2\pi\delta(\omega) = \int_{-\infty}^{\infty} e^{-\mathrm{i}\omega t}\,\mathrm{d}t = \mathcal{F}[1]. \tag{C.81}$$

  that the Dirac $\delta(t)$ and unity are a Fourier transform pair. The Fourier dual of a constant signal is $\mathcal{F}(C) = C \cdot 2\pi\delta(\omega)$.

- Sine and cosine. First we establish

$$\mathcal{F}[e^{\mathrm{i}\alpha t}] = \int_{-\infty}^{\infty} e^{\mathrm{i}\alpha t}e^{-\mathrm{i}\omega t}\,\mathrm{d}t = 2\pi\delta(\omega-\alpha). \tag{C.82}$$

  With

$$\cos(\alpha x) = \frac{e^{\mathrm{i}\alpha x} + e^{-\mathrm{i}\alpha x}}{2} \tag{C.83}$$

$$\sin(\alpha x) = \frac{e^{\mathrm{i}\alpha x} - e^{-\mathrm{i}\alpha x}}{2i} \tag{C.84}$$

  we have

$$\mathcal{F}[\cos(\alpha x)] = \pi[\delta(\omega+\alpha) + \delta(\omega-\alpha)] \tag{C.85}$$

$$\mathcal{F}[\sin(\alpha x)] = i\pi[\delta(\omega+\alpha) - \delta(\omega-\alpha)] \tag{C.86}$$

  that being said, the frequency content of these functions are only at isolated frequency(ies) even though they are widespread in space. On the other hand, the FT of highly spatially localized function $\delta(t-t_0)$ is widespread in Fourier space (see (C.78)).

- Box and sinc. Consider a box function with unit area

$$g(t) = \begin{cases} 1/2\Delta & |t| \le \Delta \\ 0 & \text{otherwise} \end{cases} \tag{C.87}$$

  whose Fourier pair is

$$\hat{g}(\omega) = \frac{1}{2\Delta}\int_{-\Delta}^{\Delta} e^{-\mathrm{i}\omega t}\,\mathrm{d}t \tag{C.88}$$

$$= \frac{1}{2\Delta}\frac{1}{-i\omega}(e^{-\mathrm{i}\omega\Delta} - e^{\mathrm{i}\omega\Delta}) \tag{C.89}$$

$$= \frac{\sin(\omega\Delta)}{\omega\Delta} = \mathrm{sinc}(\omega\Delta). \tag{C.90}$$

We note the sinc (cardinal sine) function has a finite integration (computed using the Laplace transform) over the real domain

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\,\mathrm{d}x = \pi \tag{C.91}$$

and its normalized version is $\mathrm{sinc}_\pi x = \sin(\pi x)/\pi x$ whose integral over the real axis is unity. Thus, spatially compact function square box has an oscillatory spectral counterpart.

On the other hand, spatially oscillatory function

$$h(t) = \frac{\sin(2\pi t/\Delta)}{\pi t} \tag{C.92}$$

corresponds to a spectrally sharp counterpart

$$\hat{h}(\omega) = \begin{cases} 1, & |\omega| \le 2\pi/\Delta \\ 0, & \text{otherwise} \end{cases} \tag{C.93}$$

which can be shown from the inverse transform of the sinc function

$$\left.\begin{array}{ll} 1/2\Delta, & |t| \le \Delta \\ 0, & \text{otherwise} \end{array}\right\} = g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\sin(\omega\Delta)}{\omega\Delta} e^{\mathrm{i}\omega t}\,\mathrm{d}\omega \tag{C.94}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\sin(\Delta t)}{\Delta t} e^{\mathrm{i}\omega t}\,\mathrm{d}t \tag{C.95}$$

equivalently

$$\left.\begin{array}{ll} 1, & |\alpha| \le 2\pi/\Delta \\ 0, & \text{otherwise} \end{array}\right\} = g(\alpha) = \int_{-\infty}^{\infty} \frac{\sin(2\pi\tau/\Delta)}{\pi\tau} e^{-\mathrm{i}\alpha\tau}\,\mathrm{d}\tau \tag{C.96}$$

with COVs $\alpha = -2\pi\omega/\Delta^2$ and $\tau = \Delta^2 t/2\pi$. Taking $\hat{h}(\omega) = g(\alpha)$ completes the proof.

- Gaussian. The Fourier transform of a normalized Gaussian function $f(t) = \sqrt{a/\pi}e^{-at^2}$ is

$$\mathcal{F}[f(t)] = \sqrt{\frac{a}{\pi}}\int_{-\infty}^{\infty} e^{-at^2} e^{-\mathrm{i}\omega t}\,\mathrm{d}t = \sqrt{\frac{a}{\pi}}\int_{-\infty}^{\infty} \exp\left[-a\left(t + \frac{\mathrm{i}\omega}{2a}\right)^2 - \frac{\omega^2}{4a}\right]\,\mathrm{d}t \tag{C.97}$$

$$= \sqrt{\frac{a}{\pi}} e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} \exp\left[-a\left(t + \frac{\mathrm{i}\omega}{2a}\right)^2\right]\,\mathrm{d}t = e^{-\frac{\omega^2}{4a}} \tag{C.98}$$

where we have used the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-at^2}\,\mathrm{d}t = \sqrt{\frac{\pi}{a}}, \tag{C.99}$$

provided

$$\left(\int_{-\infty}^{\infty} e^{-at^2}\,\mathrm{d}t\right)^2 = \left(\int_{-\infty}^{\infty} e^{-ax^2}\,\mathrm{d}x\right)\left(\int_{-\infty}^{\infty} e^{-ay^2}\,\mathrm{d}y\right) \tag{C.100}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-a(x^2+y^2)}\,\mathrm{d}x\mathrm{d}y \tag{C.101}$$

$$= \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} e^{-ar^2} r \, \mathrm{d}r \tag{C.102}$$

$$= \frac{\pi}{a}. \tag{C.103}$$

Take $a = 6/\Delta^2$ in a typical choice of filtering,

$$f(t) = \frac{6}{\pi\Delta^2} e^{-\frac{6t^2}{\Delta^2}} \tag{C.104}$$

has a Fourier transform pair

$$\hat{f}(\omega) = e^{-\frac{(\omega\Delta)^2}{24}}. \tag{C.105}$$

Note that the FT of Gaussian is still Gaussian. Take $a = 1/2\sigma$ we recover the standard Gaussian distribution

$$f(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}. \tag{C.106}$$

Gaussian is compact in both physical and Fourier space.

C.3 Discrete Fourier transform

Consider the discretization of physical domain (sampling, denoted by a subscript 's') first. Assume the sampling period is $L_s = L_0/N$ (or sampling frequency $\omega_s = 2\pi/L_s$) and the sampled function f_s is given by

$$f_s(t) = \sum_{j=-\infty}^{\infty} f(t)\delta(t - jL_s), \tag{C.107}$$

where $\delta(t)$ is the Dirac delta and

$$\int_{-\infty}^{\infty} \delta(t) \, \mathrm{d}t = 1; \tag{C.108}$$

$$\int_{-\infty}^{\infty} \delta(t - t_0) f(t) \, \mathrm{d}t = f(t_0). \tag{C.109}$$

Given (C.22), we have

$$\hat{f}_s(\omega) = \int_{-\infty}^{\infty} \sum_{j=-\infty}^{\infty} f(t)\delta(t - jL_s) e^{-\mathrm{i}\omega t} \, \mathrm{d}t \tag{C.110}$$

$$= \sum_{j=-\infty}^{\infty} f(jL_s) e^{-\mathrm{i}\omega jL_s}. \tag{C.111}$$

Now we also need to discretize the frequency domain. Assume there are $N+1$ samples with indices $0, 1, ..., N$, $f_N = f_0$ and the sampling period being L_s and the length of sampling, or the period of extension is $L_0 = NL_s$. The lowest frequency that can be resolved is $\omega_0 = 2\pi/L_0$ and the discrete frequencies are $\omega_k = k\omega_0 = 2\pi k/L_0$. Replacing ω in the spectral density function $\hat{f}_s(\omega)$ we get

$$\hat{f}_s(k\omega_0) = \int_{-L_0/2}^{L_0/2} \sum_{j=0}^{N-1} f(jL_s) e^{-\mathrm{i}k\omega_0 jL_s} \, \mathrm{d}t \tag{C.112}$$

$$= \sum_{j=0}^{N-1} f(-\frac{L_0}{2} + jL_s)e^{-\mathrm{i}kj\frac{2\pi}{N}}, \tag{C.113}$$

denoted as

$$\hat{f}_k = \sum_{j=0}^{N-1} f_j e^{-\mathrm{i}2\pi kj/N}. \tag{C.114}$$

We also note that, as commonly used, the k-th frequency $\tilde{f}_k$ (the tilde distinguishes 'f'requency from 'f'unction) corresponding to the spectral density (Fourier coefficient) $\hat{f}_k$ is given as

$$\tilde{f}_k = \frac{\tilde{\omega}_k}{2\pi} = \frac{k\tilde{\omega}_0}{2\pi} = \frac{k}{2\pi}\frac{2\pi}{L_0} = \frac{k}{2\pi}\frac{2\pi}{NL_s} = \frac{k}{N}\tilde{f}_s = k\Delta\tilde{f}, \tag{C.115}$$

where $\tilde{f}_s = 1/L_s$ is the sampling frequency and $\Delta\tilde{f} = \tilde{f}_s/N = 1/L_0$ is the frequency resolution (the smallest/lowest resolved frequency). The angular frequency / wavenumber resolution is $\Delta\omega = 2\pi\Delta\tilde{f} = 2\pi/L_0$. Here tildes denotes frequency (not to be confused with the function f). When plotting the power spectral density, $\tilde{f}_k$ will be the discrete frequencies on the horizontal axis (you just multiply k/N, $k = 0, 1, ..., N-1$ with $\tilde{f}_s = 1/L_s$), and the Fourier coefficient $\hat{f}_k$ will be the vertical axis.

In a general case, assume on the interval $[t_0, t_{N-1}]$ the function f is periodic and take the values

$$f_i = f(t_i),\ 0 \le i \le N-1, \tag{C.116}$$

at N equispaced, collocation points $t_0, t_1, ..., t_{N-1}$ which are typically the measurement data, where

$$t_i = \frac{L_0}{N}i,\ i = 0, 1, ..., N-1. \tag{C.117}$$

The discrete Fourier transform (DFT) and its inverse are

$$\hat{f}_k = \sum_{j=0}^{N-1} f_j e^{-\mathrm{i}2\pi jk/N} = \sum_{j=0}^{N-1} f_j(\omega_N^k)^j \tag{C.118}$$

$$f_j = \frac{1}{N}\sum_{k=0}^{N-1} \hat{f}_k e^{\mathrm{i}2\pi jk/N} = \frac{1}{N}\sum_{k=0}^{N-1} \hat{f}_k(\bar{\omega}_N^j)^k \tag{C.119}$$

where

$$\omega_N = e^{-\mathrm{i}2\pi/N} \in \mathbb{C} \tag{C.120}$$

or its complex conjugate $\bar{\omega}_N = e^{\mathrm{i}2\pi/N}$ is the primitive N-th complex root of unity since $(\omega_N)^N = (\bar{\omega}_N)^N = 1$. The FT linearly maps the data in physical space to spectral space or vice versa as

$$(f_0, f_1, ..., f_{N-1}) \leftrightarrow (\hat{f}_0, \hat{f}_1, ..., \hat{f}_{N-1}), \tag{C.121}$$

where both sides are linear combinations of one another and the weights are given by appropriate powers of ω_N.

Note the normalizing factor $1/N$ is similar to the factor of $1/2\pi$ in the inverse Fourier transform (C.21) that preserves the energy after applying the forward and inverse FTs. It can also be seen that the zero mode in (C.119) corresponds to the mean

$$\langle f \rangle = \frac{1}{N}\hat{f}_0 = \frac{1}{N}\sum_{j=0}^{N-1} f_j. \tag{C.122}$$

C.3.1 Discrete orthogonality

We prove that for each k in (C.118), basis functions $\{(\omega_N^k)^j\}$ are orthogonal. That also being said, the columns in the DFT matrix (C.152) form an orthogonal basis.

Define the inner product of two arbitrary powers i, j as

$$[(\omega_N^k)^i, (\omega_N^k)^j] = \sum_{k=0}^{N-1} \overline{(\omega_N^k)^i}(\omega_N^k)^j = \sum_{k=0}^{N-1} e^{\mathrm{i}2\pi k(i-j)/N}. \tag{C.123}$$

Due to the geometric sum formula

$$\sum_{k=0}^{N-1} \alpha^k = \frac{1-\alpha^N}{1-\alpha}, \tag{C.124}$$

and $1 - \omega_N^N = 0$, we have

$$[(\omega_N^k)^i, (\omega_N^k)^j] = \begin{cases} 0, & i \neq j \\ N, & i = j \end{cases}. \tag{C.125}$$

Hence, it is also proved that the DFT matrix is full-rank.

C.3.2 Conjugate symmetry of the real DFT

Since $\hat{f}_k = \hat{f}_{-k}^*$ (here * is the complex conjugate), similar to (C.72), we have

$$\hat{f}_{N-k} = \hat{f}_{-k} = \hat{f}_k^* \tag{C.126}$$

if $f(t) \in \mathbb{R}$. This symmetry implies redundancy in the coefficients in real DFT that we only need to store half of the elements (N is even)

$$k = \{0, 1, ..., \frac{N}{2} - 1\}, \tag{C.127}$$

and the rest can be recovered by complex conjugation. Frequencies $k > N/2 - 1$ are aliased back to $k - N$. For example, $k = 0$ in (C.126) correspond to

$$\hat{f}_0 = \hat{f}_{N/2} \in \mathbb{R}, \tag{C.128}$$

as the DC component.

C.3.3 Frequency ordering and the Nyquist frequency

It is observed that according to the 2π periodicity of $e^{-2\pi ikj/N}$, the DFT (C.118) can be shifted to

$$\hat{f}_k = \sum_{j=-N/2}^{N/2-1} f_j e^{-\mathrm{i}2\pi kj/N}, \tag{C.129}$$

with the same number of terms. Accordingly, the discrete wavenumbers can be arranged as

$$k = \{0, 1, 2, ..., \frac{N}{2} - 1, \frac{N}{2}, \frac{N}{2} + 1, ..., N-2, N-1\} \tag{C.130}$$

$$= \{0, 1, 2, ..., \frac{N}{2} - 1, -\frac{N}{2}, -\frac{N}{2} + 1, ..., -2, -1\}, \tag{C.131}$$

according to the conjugate symmetry of (C.118) and (C.129) that

$$\hat{f}_{N-k} = \hat{f}_{-k}, \tag{C.132}$$

where the second half in (C.130) are interpreted as negative frequencies.

It can be seen that with (C.131), the k-th power of $(\bar{\omega}_N^k)^j$ that you multiply with / corresponds to $\hat{f}_k$ in the Fourier expansion (C.119) becomes

$$\{\bar{\omega}_N^k\} = \exp\left(\mathrm{i}2\pi\{0, 1, 2, ..., \frac{N}{2} - 1, -\frac{N}{2}, -\frac{N}{2} + 1, ..., -2, -1\}/N\right) = \exp(\mathrm{i}\{\tilde{\omega}_k^*\}/N), \tag{C.133}$$

where the corresponding discrete, non-dimensional (in order for the exponential to make sense), angular frequencies are

$$\tilde{\omega}_k^* = 2\pi k,\ k = \{0, 1, 2, ..., \frac{N}{2} - 1, -\frac{N}{2}, -\frac{N}{2} + 1, ..., -2, -1\}, \tag{C.134}$$

such that $\{\tilde{\omega}_k^*\}/N$ uniformly distribute within $[-\pi, \pi)$ on the unit circle in a counter-clockwise order, starting from 0 and having a jump from π to $-\pi$ when $k = N/2$. When multiplied by the frequency scale $1/L_0 = \Delta\tilde{f}$, the angular frequency becomes dimensional:

$$\tilde{\omega}_k = 2\pi\tilde{f}_k = \frac{2\pi k}{L_0},\ k = \{0, 1, 2, ..., \frac{N}{2} - 1, -\frac{N}{2}, -\frac{N}{2} + 1, ..., -2, -1\}, \tag{C.135}$$

or

$$\tilde{f}_k = \frac{k}{L_0} = k\Delta\tilde{f}. \tag{C.136}$$

It is easy to see that the highest frequency is $\tilde{f}_{N/2-1} = (N-2)/(2L_0)$. Given the interval size $L_s = L_0/N = 1/\tilde{f}_s$, $\tilde{f}_{N/2-1} \approx 1/2\tilde{f}_s$ given N is typically large. That being said, the resolvable frequency is at most nearly half of the sampling frequency. The frequency $\tilde{f}_s/2$ is called the Nyquist frequency and the frequencies above this frequency are subject to aliasing (folding). For example, $\{0.6, 1.4, 1.6\}\tilde{f}_s$ will all be aliased to $0.4\tilde{f}_s$, due to the symmetry (about $N/2$ or $\tilde{f}_s/2$) and the N-periodicity of (C.135). In other words, since Fourier modes contain real and complex parts, with the same amount of information from N points in physical space we can only obtain half the number of independent Fourier modes, consistent with the conjugate symmetry (C.126).

C.3.4 Aliasing and de-aliasing

It is common to involve new frequencies above the Nyquist frequency and hence aliasing errors during the multiplications in the physical space, especially in pseudo-spectral treatments of nonlinear PDEs.

For example, consider the quadratic nonlinear term f^2 and its Fourier transform. We first denote

$$[\mathcal{F}_N(f)]_k = \hat{f}_k = \sum_{j=-N/2}^{N/2-1} f_j e^{-\mathrm{i}2\pi kj/N} \tag{C.137}$$

$$[\mathcal{F}_N^{-1}(f)]_j = f_j = \sum_{k=-N/2}^{N/2-1} \hat{f}_k e^{\mathrm{i}2\pi jk/N} \tag{C.138}$$

as the DFTs with N points. The pseudo-spectral treatment of f^2 involves an inverse transform of $\hat{f}$, a multiplication in physical space, and a direct transform to Fourier space. This way, the cost of convolution in Fourier space as in a normal FT of a product in physical space can be avoided.

We have:

$$[\mathcal{F}_N(f^2)]_k = [\mathcal{F}_N([\mathcal{F}_N^{-1}(\hat{f})]^2)]_k \tag{C.139}$$

$$= \sum_{j=-N/2}^{N/2-1} [\mathcal{F}_N^{-1}(\hat{f})]_j^2 e^{-\mathrm{i}2\pi kj/N} \tag{C.140}$$

$$= \sum_{j=-N/2}^{N/2-1} \left(\frac{1}{N^2} \sum_{m=-N/2}^{N/2-1} \sum_{l=-N/2}^{N/2-1} \hat{f}_m \hat{f}_l e^{\mathrm{i}2\pi j(m+l)/N} \right) e^{-\mathrm{i}2\pi kj/N} \tag{C.141}$$

$$= \frac{1}{N^2} \sum_{m=-N/2}^{N/2-1} \sum_{l=-N/2}^{N/2-1} \hat{f}_m \hat{f}_l \sum_{j=-N/2}^{N/2-1} e^{\mathrm{i}2\pi j(m+l-k)/N} \tag{C.142}$$

$$= \frac{1}{N} \sum_{l=-N/2}^{N/2-1} \hat{f}_{k-l} \hat{f}_l \tag{C.143}$$

where the discrete orthogonality condition is used in the simplification to the last equation. Aliasing error occurs as $k-l$ exceeds the frequency range, as can be seen from the decomposition of the following cyclic convolution

$$[\mathcal{F}_N(f^2)]_k = \frac{1}{N} \sum_{l=-N/2}^{N/2-1} \hat{f}_{k-l} \hat{f}_l \tag{C.144}$$

$$= \frac{1}{N} \sum_{-N/2 \le l, k-l \le N/2-1} \hat{f}_{k-l} \hat{f}_l + \frac{1}{N} \sum_{\substack{-N/2 \le l \le N/2-1 \\ k-l > N/2-1 \text{ or } k-l < -N/2}} \hat{f}_{k-l} \hat{f}_l \tag{C.145}$$

where the second term involves aliasing errors.

The idea of the zero-padding method is to choose to perform the DFT in a large space, with K points, with the goal being keeping $[-\frac{N}{2}, \frac{N}{2} - 1]$ aliasing-free. In such a case,

- Wavenumbers with $k-l > K/2-1$ will be aliased to $k-l-K$, requiring $k-l-K < -N/2$ for all possible $k, l \in [-\frac{N}{2}, \frac{N}{2} - 1]$. The maximum of $k - l$ is $N - 1$, achieved when $k = N/2 - 1, l = -N/2$. Hence, $K > 3N/2 - 1$.

- Wavenumbers with $k-l < -K/2$ will be aliased to $k-l+K$, requiring $k-l+K > N/2-1$ for all $k, l \in [-\frac{N}{2}, \frac{N}{2} - 1]$. The minimum of $k - l$ is $-N + 1$, when $k = -N/2, l = N/2 - 1$. Hence, $K > 3N/2 - 2$.

- In all, a choice of $K = 3N/2$ is typically made.

- The computational procedures are: (1) zeros are padded to extend the wavenumber range and $\mathcal{F}_K^{-1}(\hat{f})$ is computed; (2) it is squared in physical space and Fourier transformed to $\mathcal{F}_K([\mathcal{F}_K^{-1}(\hat{f})]^2)$; (3) 1/3 of the wavenumbers between $N/2$ and $K/2$ are dropped and the FT is scaled back to $\mathcal{F}_N$ by multiplying K/N.

C.3.5 MATLAB `fft()`, and `FFTW`

We note that the order (C.131),

$$k = \{0, 1, 2, ..., \frac{N}{2} - 1, -\frac{N}{2}, -\frac{N}{2} + 1, ..., -2, -1\},$$

is used in `FFTW3` (Frigo & Johnson, 2020) and in MATLAB `fft()`. In MATLAB, `fftshift()` can be used to shift (C.131) to

$$k = \{-\frac{N}{2}, -\frac{N}{2} + 1, ..., -1, 0, 1, ..., \frac{N}{2} - 2, \frac{N}{2} - 1\} \tag{C.146}$$

such that it is symmetric about zero and only the positive semi-axis of $\tilde{\omega}_k$ (assuming N is even) needs to be plotted (see (C.135)).

C.3.6 Examples

- Consider the azimuthal decomposition of $f(\theta)$ into its Fourier modes. The IFT is

$$f_k = \sum_{m=0}^{N-1} \hat{f}_m e^{\mathrm{i}2\pi mk/N} = \sum_{m=0}^{N-1} \hat{f}_m e^{\mathrm{i}(m2\pi k/N)} = \sum_{m=0}^{N-1} \hat{f}_m e^{\mathrm{i}(m\theta_k)} \tag{C.147}$$

  where we have absorbed $1/N$ into $\hat{f}_k$, integers $m = 0, 1, ..., N-1$ are the mode indices, and $\theta_k = (2\pi/N)k$, $k = 0, 1, ..., N-1$ are the discrete azimuth angles (collocation points). Domain $L_0 = 2\pi$.

- The DFTs in multiple dimensions are

$$\hat{f}_{k_1,k_2,k_3} = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \sum_{l=0}^{L-1} f_{m,n,l} e^{-\mathrm{i}2\pi(k_1 m/M + k_2 n/N + k_3 l/L)} \tag{C.148}$$

$$f_{m,n,l} = \frac{1}{MNL} \sum_{k_1=0}^{M-1} \sum_{k_2=0}^{N-1} \sum_{k_3=0}^{L-1} \hat{f}_{k_1,k_2,k_3} e^{\mathrm{i}2\pi(k_1 m/M + k_2 n/N + k_3 l/L)} \tag{C.149}$$

  and it is seen that the mean (zero mode) is computed as

$$\langle f \rangle = \frac{1}{MNL} \hat{f}_{0,0,0} = \frac{1}{MNL} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \sum_{l=0}^{L-1} f_{m,n,l}. \tag{C.150}$$

  The conjugate symmetry condition for real DFT ($f \in \mathbb{R}$) is

$$\overline{\hat{f}}(k_1, k_2, k_3) = \hat{f}(-k_1, -k_2, -k_3). \tag{C.151}$$

C.4 The Fast Fourier Transform

C.4.1 The matrix form of the DFT

The DFT (C.118) is often written in matrix form as (following a slightly different normalizing convention that equally places a factor $1/\sqrt{N}$ in direct and inverse FT)

$$\begin{bmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \vdots \\ \hat{f}_{N-1} \end{bmatrix} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & ... & 1 \\ 1 & \omega_N & \omega_N^2 & ... & \omega_N^{N-1} \\ 1 & \omega_N^2 & \omega_N^4 & ... & \omega_N^{2(N-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \omega_N^{(N-1)} & \omega_N^{2(N-1)} & ... & \omega_N^{(N-1)^2} \end{bmatrix} \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_{N-1} \end{bmatrix}, \tag{C.152}$$

or

$$\hat{\boldsymbol{f}} = \boldsymbol{F}\boldsymbol{f},\ \hat{f}_k = F_{kj} f_j,\ 0 \le k, j \le N-1 \tag{C.153}$$

where

$$\boldsymbol{F} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & ... & 1 \\ 1 & \omega_N & \omega_N^2 & ... & \omega_N^{N-1} \\ 1 & \omega_N^2 & \omega_N^4 & ... & \omega_N^{2(N-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \omega_N^{(N-1)} & \omega_N^{2(N-1)} & ... & \omega_N^{(N-1)^2} \end{bmatrix},\ F_{kj} = \omega_N^{kj},\ 0 \le k, j \le N-1 \tag{C.154}$$

is the Fourier transformation matrix. In terms of a special case of a Vandermonde matrix,

$$\boldsymbol{F} = \frac{1}{\sqrt{N}} \boldsymbol{V}(1, \omega_N, \omega_N^2, ..., \omega_N^{(N-1)}), \tag{C.155}$$

where the factor $1/\sqrt{N}$ is to make $\boldsymbol{F}$ unitary.

The DFT matrix in (C.152) is fully dense due to Fourier transform's global property. And it is a Vandermonde matrix that has many nice properties and such as symmetry ($\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}$) and unitary ($\boldsymbol{F}\boldsymbol{F}^{\mathrm{H}} = \boldsymbol{I}$).

As a result, the inverse Fourier transform can be directly written down as

$$\begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_{N-1} \end{bmatrix} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & 1 & ... & 1 \\ 1 & w_N & w_N^2 & ... & w_N^{N-1} \\ 1 & w_N^2 & w_N^4 & ... & w_N^{2(N-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & w_N^{(N-1)} & w_N^{2(N-1)} & ... & w_N^{(N-1)^2} \end{bmatrix} \begin{bmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \vdots \\ \hat{f}_{N-1} \end{bmatrix}, \tag{C.156}$$

or

$$\boldsymbol{f} = \boldsymbol{F}^{-1}\hat{\boldsymbol{f}} \tag{C.157}$$

where

$$\boldsymbol{F}^{-1} = \boldsymbol{F}^{\mathrm{H}} = \overline{\boldsymbol{F}} = \frac{1}{\sqrt{N}}\boldsymbol{V}(1, w_N, w_N^2, ..., w_N^{(N-1)}) \tag{C.158}$$

with $w_N = \bar{\omega}_n = e^{\mathrm{i}2\pi/N}$ being also the primitive N-th root of unity.

In the language of linear transformations, left-multiplication of $\boldsymbol{F}$ transforms the rows of the data (rows of column vectors say $\boldsymbol{f}$), such that

$$[\hat{\boldsymbol{f}}(x_1), \hat{\boldsymbol{f}}(x_2), ..., \hat{\boldsymbol{f}}(x_m)] = [\boldsymbol{F}\boldsymbol{f}(x_1), \boldsymbol{F}\boldsymbol{f}(x_2), ..., \boldsymbol{F}\boldsymbol{f}(x_m)] = \boldsymbol{F}[\boldsymbol{f}(x_1), \boldsymbol{f}(x_2), ..., \boldsymbol{f}(x_m)], \tag{C.159}$$

such as when the input has a spatial dependence on x. It is typically more convenient to write down the transpose of the transforms using a right multiplication (transforming the columns of $\boldsymbol{Q}$),

$$\hat{\boldsymbol{Q}} = [\hat{\boldsymbol{q}}_0, \hat{\boldsymbol{q}}_1, ..., \hat{\boldsymbol{q}}_{N-1}] = [\boldsymbol{q}_0, \boldsymbol{q}_1, ..., \boldsymbol{q}_{N-1}]\boldsymbol{F} = \boldsymbol{Q}\boldsymbol{F} \tag{C.160}$$

and

$$\boldsymbol{Q} = \hat{\boldsymbol{Q}}\overline{\boldsymbol{F}} \tag{C.161}$$

where $\boldsymbol{q} \in \mathbb{C}^m$, and the Fourier mode $\hat{\boldsymbol{q}}_k$ at the kth discrete frequency ($k = 0, 1, ...N-1$) is a linear combination of $\{\boldsymbol{q}_l\}_{l=0}^{N-1}$ weighted by

$$(1, \omega_N^k, \omega_N^{2k}, ..., \omega_N^{k(N-1)}),\ \omega_N^k = e^{-\mathrm{i}2\pi k/N} = e^{-i\tilde{\omega}_k^*}$$

where $\tilde{\omega}_k^* = 2\pi k/N = 2\pi \tilde{f}_k L_s$ is the non-dimensionalized kth (angular) frequency. and the right-multiplication with $\boldsymbol{F}$ transforms the columns of $\boldsymbol{Q}$ (each column is a row vector; note $\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}$).

C.4.2 Vandermonde matrix

A square $(n+1) \times (n+1)$ Vandermonde matrix is

$$\boldsymbol{V} = \boldsymbol{V}(t_0, x_1, ..., x_n) = \begin{bmatrix} 1 & t_0 & t_0^2 & ... & t_0^n \\ 1 & x_1 & x_1^2 & ... & x_1^n \\ 1 & x_2 & x_2^2 & ... & x_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & ... & x_n^n \end{bmatrix} \tag{C.162}$$

whose determinant (called Vandermonde determinant) is

$$\det(\boldsymbol{V}) = \prod_{0 \le i \le j \le n} (x_i - x_j) \tag{C.163}$$

is non-zero if and only if x_i are distinct.

The Vandermonde matrix can come from the Lagrange interpolation problem where an nth order polynomial

$$p(t) = a_0 + a_1 t + a_2 t^2 + ... + a_{n-1} t^{n-1} + a_n t^n \tag{C.164}$$

is used to represent the function that passes $n+1$ given points: $(t_0, y_0), (x_1, y_1), ..., (x_{n-1}, y_{n-1}), (x_n, y_n)$. The matrix form of the given problem is

$$\begin{bmatrix} 1 & t_0 & t_0^2 & ... & t_0^n \\ 1 & x_1 & x_1^2 & ... & x_1^n \\ 1 & x_2 & x_2^2 & ... & x_2^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & ... & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \tag{C.165}$$

When the number of parameters in the polynomial (order +1) matches the number of given points and the points are distinct, the above linear system can uniquely be solved for a_i ($\boldsymbol{a} = \boldsymbol{V}^{-1}\boldsymbol{y}$).

When the Fourier matrix (ignoring the normalization factor) is interpreted as a Vandermonde matrix, it is

$$\boldsymbol{F} = \boldsymbol{V}(\omega_N^0, \omega_N, ..., \omega_N^{(N-1)}) \tag{C.166}$$

C.4.3 An example

When $N = 4$, the forward Fourier transform is

$$\begin{bmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \hat{f}_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \end{bmatrix} \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \end{bmatrix} \tag{C.167}$$

where the DFT matrix can be factorized as

$$\boldsymbol{F}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \end{bmatrix} \tag{C.168}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -\mathrm{i} \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & \mathrm{i} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{C.169}$$

$$= \boldsymbol{B}_4 \boldsymbol{F}_{2*2} \boldsymbol{P}_4 \tag{C.170}$$

where $\boldsymbol{P}_4$ is an odd-even sorting, permutation matrix, $\boldsymbol{F}_{2*2}$ contains two $N = 2$ DFT matrices

$$\boldsymbol{F}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \tag{C.171}$$

and $\boldsymbol{B}_4$ is the butterfly matrix. We note that

$$\boldsymbol{F}_{2*2}\boldsymbol{P}_4 \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} f_0 + f_2 \\ f_0 - f_2 \\ f_1 + f_3 \\ f_1 - f_3 \end{bmatrix} \tag{C.172}$$

Here we can already see the reuse of a size $N/2$ DFT matrix and the sign of divide-and-conquer. These ideas will be shown more clearly in the following section.

Finally, we spell out the matrix multiplication

$$\begin{bmatrix} \hat{f}_0 \\ \hat{f}_1 \\ \hat{f}_2 \\ \hat{f}_3 \end{bmatrix} = \boldsymbol{B}_4 \begin{bmatrix} f_0 + f_2 \\ f_0 - f_2 \\ f_1 + f_3 \\ f_1 - f_3 \end{bmatrix} = \begin{bmatrix} (f_0 + f_2) + 1 * (f_1 + f_3) \\ (f_0 - f_2) - \mathrm{i} * (f_1 + f_3) \\ (f_0 + f_2) - 1 * (f_1 + f_3) \\ (f_0 - f_2) + \mathrm{i} * (f_1 + f_3) \end{bmatrix} = \begin{bmatrix} \hat{f}_0^{\text{even}} + \omega_4^0 \hat{f}_0^{\text{odd}} \\ \hat{f}_1^{\text{even}} + \omega_4^1 \hat{f}_1^{\text{odd}} \\ \hat{f}_0^{\text{even}} - \omega_4^0 \hat{f}_0^{\text{odd}} \\ \hat{f}_1^{\text{even}} - \omega_4^1 \hat{f}_1^{\text{odd}} \end{bmatrix} \tag{C.173}$$

where

$$\begin{bmatrix} \hat{f}_0^{\text{even}} \\ \hat{f}_1^{\text{even}} \end{bmatrix} = \boldsymbol{F}_2 \begin{bmatrix} f_0 \\ f_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_0 \\ f_2 \end{bmatrix} = \begin{bmatrix} f_0 + f_2 \\ f_0 - f_2 \end{bmatrix} \tag{C.174}$$

and

$$\begin{bmatrix} \hat{f}_0^{\text{odd}} \\ \hat{f}_1^{\text{odd}} \end{bmatrix} = \boldsymbol{F}_2 \begin{bmatrix} f_1 \\ f_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_3 \end{bmatrix} = \begin{bmatrix} f_1 + f_3 \\ f_1 - f_3 \end{bmatrix} \tag{C.175}$$

are the outputs of two size-$N/2$ DFTs of the even and odd elements of the input. This is a special case of (C.185)-(C.186). We can again see that $\hat{f}_k$ are nothing but linear combinations of f_j.

C.4.4 The idea of the FFT

First, we consider the time-complexity of/the operation counts for the DFT. Either the direct and the inverse DFT involves matrix multiplication of $\boldsymbol{F} \in \mathbb{C}^{N\times N}$ and a vector $\boldsymbol{f} \in \mathbb{R}^N$, requiring N^2 multiplications and $N(N-1)$ additions, totaling $O(N^2)$. The optimal operation count that can be achieved is $O(\frac{N}{2}\log_2 N)$, found by Cooley & Tukey (1965). We note that $\log_2 N$ grows almost like a constant when N is large, since it is beaten by any positive power of N. Hence, this scaling is also called the fast scaling. Below are the main ideas of the Cooley & Tukey (1965) FFT.

An observation can be made about the DFT:

$$\hat{f}_k = \sum_{j=0}^{N-1} \mathsf{F}_{kj} f_j = \sum_{j=0}^{N-1} \omega_N^{kj} f_j, \tag{C.176}$$

where

$$\omega_N = e^{\mathrm{i}2\pi/N}. \tag{C.177}$$

When N is an even number, the above sum can be split into odd and even terms as

$$\hat{f}_k = \sum_{j=0}^{N/2-1} \omega_N^{k(2j)} f_{2j} + \sum_{j=0}^{N/2-1} \omega_N^{k(2j+1)} f_{2j+1} \tag{C.178}$$

and providing that

$$\omega_N^{2kj} = e^{\mathrm{i}2\pi 2kj/N} = e^{\mathrm{i}2\pi kj/(N/2)} = \omega_{N/2}^{kj}, \tag{C.179}$$

we have

$$\hat{f}_k = \sum_{j=0}^{N/2-1} \omega_{N/2}^{kj} f_{2j} + \omega_N^k \sum_{j=0}^{N/2-1} \omega_{N/2}^{kj} f_{2j+1}, \tag{C.180}$$

which is exactly the sum of two DFTs of size $N/2$, of the even and odd elements of the input f_j, respectively. It is naturally to think of a recursion that divides the problem into smaller and smaller, easier and easier to solve ones.

We note that the same formula (C.180) remains for $k = 0, 1, ..., N/2-1$, but some considerations are needed for $k = N/2, N/2+1, ..., N-1$ since we want to constrain the weights $\omega_{N/2}^{kj}$ in $0 \leq k, j \leq N/2-1$. We can reuse (C.180) by a change of variable ($k = k' + N/2$, with $k' = 0, 1, ..., N/2-1$), such that, for $k = N/2, N/2+1, ..., N-1$,

$$\hat{f}_k = \sum_{j=0}^{N/2-1} \omega_{N/2}^{(k'+N/2)j} f_{2j} + \omega_N^{k'+N/2} \sum_{j=0}^{N/2-1} \omega_{N/2}^{(k'+N/2)j} f_{2j+1}, \tag{C.181}$$

with another observation that

$$\omega_{N/2}^{(k'+N/2)j} = e^{-\mathrm{i}2\pi(k'+N/2)j/(N/2)} = e^{-\mathrm{i}2\pi k'j/(N/2)} e^{-\mathrm{i}2\pi j} = e^{-\mathrm{i}2\pi k'j/(N/2)} = \omega_{N/2}^{k'j} \tag{C.182}$$

since $j = 0, 1, ..., N/2-1$. Similarly, we also have

$$\omega_N^{k'+N/2} = -\omega_N^{k'j}. \tag{C.183}$$

Thus, the above equation (C.181) can be simplified as

$$\hat{f}_k = \sum_{j=0}^{N/2-1} \omega_{N/2}^{k'j} f_{2j} - \omega_N^{k'} \sum_{j=0}^{N/2-1} \omega_{N/2}^{k'j} f_{2j+1}. \tag{C.184}$$

In summary, a formal representation is

$$\hat{f}_k^{\text{upper}} = \hat{f}_k^{\text{odd}} + \omega_N^k \hat{f}_k^{\text{even}} \; (0 \leq k \leq N/2-1) \tag{C.185}$$

$$\hat{f}_{k'+N/2}^{\text{lower}} = \hat{f}_{k'}^{\text{odd}} - \omega_N^{k'} \hat{f}_{k'}^{\text{even}} \; (N/2 \leq k \leq N-1, \, 0 \leq k' \leq N/2-1) \tag{C.186}$$

and such a structure, when written in matrix form, is call butterfly.

C.4.5 Miscellaneous

Note that matrix multiplication is not the only thing to do. If we just need to compute $\hat{\boldsymbol{f}}$ at several frequencies of interest, we can incrementally update (C.118) on-the-fly with the newest available snapshot multiplied with an appropriate weight. The time complexity is not favorable, partly compensated by fewer computed frequencies, but more importantly there is memory saving on the other hand, if there are large snapshots being FT'ed in one of the dimensions. This method is called streaming FFT.

Appendix D

Coordinate systems and their transformations

D.1 Cylindrical coordinates

Consider the cylindrical transformation

$$(x, y) \to (r, \theta) \tag{D.1}$$

where

$$x = r\cos\theta \tag{D.2}$$
$$y = r\sin\theta \tag{D.3}$$

or

$$r = \sqrt{x^2 + y^2} \tag{D.4}$$
$$\theta = \operatorname{actan}\left(\frac{y}{x}\right) \tag{D.5}$$

we have the corresponding relation between unit vectors

$$\begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\theta \end{bmatrix} \tag{D.6}$$

and

$$\begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \end{bmatrix}, \tag{D.7}$$

which can also be proven graphically. The transformation matrices in (D.6) and (D.7) are inverse to each other, and both are unitary. That being said, they are rotation matrices with $\det() = 1$.

The Jacobian of the forward transformation $(r, \theta) = F(x, y)$ is

$$J = \frac{\partial(r,\theta)}{\partial(x,y)} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\frac{1}{r}\sin\theta & \frac{1}{r}\cos\theta \end{bmatrix} \tag{D.8}$$

and of the backward transformation $(x, y) = G(r, \theta)$ is

$$J^{-1} = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \tag{D.9}$$

and it can be seen with a bit of algebra that these two are the inverse of each other.

We note that the directions of the unit vectors $\boldsymbol{e}_r, \boldsymbol{e}_\theta$ depend on space, i.e.,

$$\frac{\partial \boldsymbol{e}_r}{\partial r} = \frac{\partial \boldsymbol{e}_\theta}{\partial r} = 0 \tag{D.10}$$

$$\frac{\partial \boldsymbol{e}_r}{\partial \theta} = -\sin\theta \boldsymbol{e}_x + \cos\theta \boldsymbol{e}_y = \boldsymbol{e}_\theta \tag{D.11}$$

$$\frac{\partial \boldsymbol{e}_\theta}{\partial \theta} = -\cos\theta \boldsymbol{e}_x - \sin\theta \boldsymbol{e}_y = -\boldsymbol{e}_r \tag{D.12}$$

which can also be seen graphically. These relations are crucial to later derivations.

Consider the chain rule

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x} = \cos\theta \frac{\partial}{\partial r} - \sin\theta \frac{1}{r}\frac{\partial}{\partial \theta} \tag{D.13}$$

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial y} = \sin\theta \frac{\partial}{\partial r} + \cos\theta \frac{1}{r}\frac{\partial}{\partial \theta} \tag{D.14}$$

here we proof the cylindrical coordinate representation of $\nabla \cdot \boldsymbol{u} = 0$. Assume $\boldsymbol{u} = u\boldsymbol{e}_x + v\boldsymbol{e}_y + w\boldsymbol{e}_z = u_r\boldsymbol{e}_r + u_\theta \boldsymbol{e}_\theta + u_z \boldsymbol{e}_z$.

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases} \rightarrow \begin{cases} u = \dot{x} = \frac{\partial}{\partial t}(r\cos\theta) = u_r\cos\theta - u_\theta \sin\theta \\ v = \dot{y} = \frac{\partial}{\partial t}(r\sin\theta) = u_r \sin\theta + u_\theta \cos\theta \end{cases} \tag{D.15}$$

We have prepared

$$\begin{cases} \frac{\partial r}{\partial x} = \frac{x}{r} \\ \frac{\partial r}{\partial y} = \frac{y}{r} \\ \frac{\partial \theta}{\partial x} = -\frac{\sin\theta}{r} \\ \frac{\partial \theta}{\partial y} = \frac{\cos\theta}{r} \end{cases} . \tag{D.16}$$

Starting from

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \tag{D.17}$$

we use the chain rule to replace all coordinates and velocity components with their cylindrical counterparts, leading to (with a bit of algebra)

$$\frac{\partial u_r}{\partial r} + \frac{u_r}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0. \tag{D.18}$$

D.1.1 Operators in cylindrical coordinates

For a scalar function, say $f(x, y) = f(r, \theta)$, the gradient operator can be expressed as

$$\nabla = \boldsymbol{e}_x \frac{\partial}{\partial x} + \boldsymbol{e}_y \frac{\partial}{\partial y} + \boldsymbol{e}_z \frac{\partial}{\partial z} \tag{D.19}$$

$$= \left(\frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x}\right)(\cos\theta \boldsymbol{e}_r - \sin\theta \boldsymbol{e}_\theta) + \left(\frac{\partial}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial y}\right)(\sin\theta \boldsymbol{e}_r + \cos\theta \boldsymbol{e}_\theta) + \boldsymbol{e}_z\frac{\partial}{\partial z} \tag{D.20}$$

$$= \boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \boldsymbol{e}_z\frac{\partial}{\partial z} \tag{D.21}$$

The factor $r\partial\theta$ can be interpreted as infinitesimal length element in θ direction.

The Laplace operator

$$\nabla^2 = \nabla\cdot\nabla = \left(\boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \boldsymbol{e}_z\frac{\partial}{\partial z}\right)\cdot\left(\boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \boldsymbol{e}_z\frac{\partial}{\partial z}\right) \tag{D.22}$$

$$= \frac{\partial^2}{\partial r^2} + \frac{\partial^2}{\partial z^2} + \boldsymbol{e}_\theta\cdot\frac{1}{r}\left[\frac{\partial}{\partial \theta}\left(\boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta}\right)\right] \tag{D.23}$$

$$= \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \tag{D.24}$$

$$= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2} \tag{D.25}$$

Now consider a vector

$$\boldsymbol{u} = \boldsymbol{e}_r u + \boldsymbol{e}_\theta v + \boldsymbol{e}_z w \tag{D.26}$$

and its derivatives.

Its divergence is

$$\nabla\cdot\boldsymbol{u} = \left(\boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \boldsymbol{e}_z\frac{\partial}{\partial z}\right)\cdot(\boldsymbol{e}_r u + \boldsymbol{e}_\theta v + \boldsymbol{e}_z w) \tag{D.27}$$

$$= \frac{\partial u}{\partial r} + \frac{u}{r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} \tag{D.28}$$

$$= \frac{1}{r}\frac{\partial (ru)}{\partial r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} \tag{D.29}$$

The convection term

$$(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = \left(u\frac{\partial}{\partial r} + \frac{v}{r}\frac{\partial}{\partial \theta} + w\frac{\partial}{\partial z}\right)(\boldsymbol{e}_r u + \boldsymbol{e}_\theta v + \boldsymbol{e}_z w) \tag{D.30}$$

$$= \left(u\frac{\partial u}{\partial r} + \frac{v}{r}\frac{\partial u}{\partial \theta} + w\frac{\partial u}{\partial z} - \frac{v^2}{r}\right)\boldsymbol{e}_r \tag{D.31}$$

$$+ \left(u\frac{\partial v}{\partial r} + \frac{v}{r}\frac{\partial v}{\partial \theta} + w\frac{\partial v}{\partial z} + \frac{uv}{r}\right)\boldsymbol{e}_\theta \tag{D.32}$$

$$+ \left(u\frac{\partial w}{\partial r} + \frac{v}{r}\frac{\partial w}{\partial \theta} + w\frac{\partial w}{\partial z}\right)\boldsymbol{e}_z \tag{D.33}$$

Now we deal with $\nabla^2\boldsymbol{u}$.

$$\nabla^2\boldsymbol{u} = \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}\right)(\boldsymbol{e}_r u + \boldsymbol{e}_\theta v + \boldsymbol{e}_z w) \tag{D.34}$$

$$= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \boldsymbol{u}}{\partial r}\right) + \frac{\partial^2\boldsymbol{u}}{\partial z^2} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}(\boldsymbol{e}_r u + \boldsymbol{e}_\theta v) \tag{D.35}$$

$$= \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v}{\partial \theta} - \frac{u}{r^2} + \frac{\partial^2 u}{\partial z^2}\right)\boldsymbol{e}_r \tag{D.36}$$

$$+ \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial u}{\partial \theta} - \frac{v}{r^2} + \frac{\partial^2 v}{\partial z^2}\right)\boldsymbol{e}_\theta \tag{D.37}$$

$$+ \left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2} + \frac{\partial^2 w}{\partial z^2}\right)\boldsymbol{e}_z \tag{D.38}$$

with

$$\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}(\boldsymbol{e}_r u) = \frac{1}{r^2}\frac{\partial}{\partial \theta}\frac{\partial \boldsymbol{e}_r u}{\partial \theta} = \frac{1}{r^2}(2\frac{\partial u}{\partial \theta}\boldsymbol{e}_\theta - u\boldsymbol{e}_r + \frac{\partial^2 u}{\partial \theta^2}\boldsymbol{e}_r) \tag{D.39}$$

$$\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}(\boldsymbol{e}_\theta v) = \frac{1}{r^2}\frac{\partial}{\partial \theta}\frac{\partial \boldsymbol{e}_\theta v}{\partial \theta} = \frac{1}{r^2}(-2\frac{\partial v}{\partial \theta}\boldsymbol{e}_r - v\boldsymbol{e}_\theta + \frac{\partial^2 v}{\partial \theta^2}\boldsymbol{e}_\theta) \tag{D.40}$$

Moreover, the curl can be established (in a compact form) as

$$\nabla \times \boldsymbol{u} = \left(\boldsymbol{e}_r\frac{\partial}{\partial r} + \boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial \theta} + \boldsymbol{e}_z\frac{\partial}{\partial z}\right) \times (\boldsymbol{e}_r u + \boldsymbol{e}_\theta v + \boldsymbol{e}_z w) \tag{D.41}$$

$$= \begin{vmatrix} \boldsymbol{e}_r & \boldsymbol{e}_\theta & \boldsymbol{e}_z \\ \partial_r & \frac{1}{r}\partial_\theta & \partial_z \\ u & v & w \end{vmatrix} + \frac{1}{r}\boldsymbol{e}_\theta \times \frac{\partial(v\boldsymbol{e}_\theta)}{\partial \theta} \tag{D.42}$$

$$= \left(\frac{1}{r}\frac{\partial w}{\partial \theta} - \frac{\partial v}{\partial z}\right)\boldsymbol{e}_r + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial r}\right)\boldsymbol{e}_\theta + \left(\frac{\partial v}{\partial r} + \frac{v}{r} - \frac{1}{r}\frac{\partial u}{\partial \theta}\right)\boldsymbol{e}_z \tag{D.43}$$

$$= \left(\frac{1}{r}\frac{\partial w}{\partial \theta} - \frac{\partial v}{\partial z}\right)\boldsymbol{e}_r + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial r}\right)\boldsymbol{e}_\theta + \frac{1}{r}\left(\frac{\partial rv}{\partial r} - \frac{\partial u}{\partial \theta}\right)\boldsymbol{e}_z \tag{D.44}$$

$$= \frac{1}{r}\begin{vmatrix} \boldsymbol{e}_r & r\boldsymbol{e}_\theta & \boldsymbol{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ u & rv & w \end{vmatrix}. \tag{D.45}$$

Examples.

1. Axisymmetric flows ($u_\theta \neq 0$). We have the $\omega_z = 1/r\partial_r(ru_\theta) = u_\theta/r + \partial_r u_\theta$. The first term is called the curvature vorticity and the second called the shear vorticity.

2. Rigid body rotation with angular velocity Ω and $u_\theta = v = \Omega r$. Vorticity $\omega_z = 2\Omega$ (with equal contributions from curvature and shear) but there is no vortical motion.

3. Potential point vortex with $u_\theta = v = \Gamma/2\pi r$. Vorticity $\omega_z = 0$ according to (D.44) – there is no vorticity.

Now we turn our attention to velocity gradient and strain-rate tensors. In Cartesion coordinates,

$$\nabla \boldsymbol{u} = (\partial_i \boldsymbol{e}_i)(u_j \boldsymbol{e}_j) = \partial_i u_j \boldsymbol{e}_i \boldsymbol{e}_j. \tag{D.46}$$

In cylindrical coordinates,

$$\nabla \boldsymbol{u} = (\boldsymbol{e}_r\partial_r + \boldsymbol{e}_\theta 1/r\partial_\theta + \boldsymbol{e}_z\partial_z)(\boldsymbol{e}_r u_r + \boldsymbol{e}_\theta u_\theta + \boldsymbol{e}_z u_z) \tag{D.47}$$

$$= (\boldsymbol{e}_r\boldsymbol{e}_r\partial_r u + \boldsymbol{e}_r\boldsymbol{e}_\theta\partial_r v + \boldsymbol{e}_r\boldsymbol{e}_z\partial_r w) \tag{D.48}$$

$$+ [(\boldsymbol{e}_\theta\boldsymbol{e}_r 1/r\partial_\theta u + \boldsymbol{e}_\theta\boldsymbol{e}_\theta u/r) + (\boldsymbol{e}_\theta\boldsymbol{e}_\theta 1/r\partial_\theta v - \boldsymbol{e}_\theta\boldsymbol{e}_r v/r) + \boldsymbol{e}_\theta\boldsymbol{e}_z 1/r\partial_\theta w] \tag{D.49}$$

$$+ (\boldsymbol{e}_z\boldsymbol{e}_r\partial_z u + \boldsymbol{e}_z\boldsymbol{e}_\theta\partial_z v + \boldsymbol{e}_z\boldsymbol{e}_z\partial_z w) \tag{D.50}$$

$$= \begin{bmatrix} \frac{\partial u}{\partial r} & \frac{\partial v}{\partial r} & \frac{\partial w}{\partial r} \\ \frac{1}{r}\frac{\partial u}{\partial \theta} - \frac{v}{r} & \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{u}{r} & \frac{1}{r}\frac{\partial w}{\partial \theta} \\ \frac{\partial u}{\partial z} & \frac{\partial v}{\partial z} & \frac{\partial w}{\partial z} \end{bmatrix} \tag{D.51}$$

and the rate-of-strain tensor is

$$\boldsymbol{S} = \frac{1}{2}(\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T) = \begin{bmatrix} \frac{\partial u}{\partial r} & \frac{1}{2}\left(r\frac{\partial}{\partial r}\left(\frac{v}{r}\right) + \frac{1}{r}\frac{\partial u}{\partial \theta}\right) & \frac{1}{2}\left(\frac{\partial w}{\partial r} + \frac{\partial u}{\partial z}\right) \\ ... & \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{u}{r} & \frac{1}{2}\left(\frac{1}{r}\frac{\partial w}{\partial \theta} + \frac{\partial v}{\partial z}\right) \\ ... & ... & \frac{\partial w}{\partial z} \end{bmatrix} = \begin{bmatrix} S_{rr} & S_{r\theta} & S_{rz} \\ ... & S_{\theta\theta} & S_{\theta z} \\ ... & ... & S_{zz} \end{bmatrix} \tag{D.52}$$

We note the little tricks

$$\frac{\partial v}{\partial r} - \frac{v}{r} = r\frac{\partial}{\partial r}\left(\frac{v}{r}\right) \tag{D.53}$$

$$\frac{\partial v}{\partial r} + \frac{v}{r} = \frac{1}{r}\frac{\partial}{\partial r}(vr) \tag{D.54}$$

are used. If (u, v, w) are interpreted as stains instead of strain rates, the above results can also be found in books of solid mechanics, where both transformational (the one here) and geometrical derivations can be seen.

D.1.2 Navier–Stokes in cylindrical coordinates

With the preparation in the previous section, we are now able to write down the Navier–Stokes equations in cylindrical coordinates as

$$\frac{1}{r}\frac{\partial(ru)}{\partial r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} = 0 \tag{D.55}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r} + \frac{v}{r}\frac{\partial u}{\partial \theta} + w\frac{\partial u}{\partial z} - \frac{v^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} - \frac{2}{r^2}\frac{\partial v}{\partial \theta} - \frac{u}{r^2} + \frac{\partial^2 u}{\partial z^2}\right) \tag{D.56}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial r} + \frac{v}{r}\frac{\partial v}{\partial \theta} + w\frac{\partial v}{\partial z} + \frac{uv}{r} = -\frac{1}{\rho r}\frac{\partial p}{\partial \theta} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2} + \frac{2}{r^2}\frac{\partial u}{\partial \theta} - \frac{v}{r^2} + \frac{\partial^2 v}{\partial z^2}\right) \tag{D.57}$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial r} + \frac{v}{r}\frac{\partial w}{\partial \theta} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} + \nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 w}{\partial \theta^2} + \frac{\partial^2 w}{\partial z^2}\right) \tag{D.58}$$

Q.E.D.

The centrifugal force $(-v^2/r)$ and the advection of angular momentum (uv/r) both come from the nonlinear advection term and are the curvature/geometry effects (due to the derivative of unit vectors w.r.t. coordinates). These equations can also be written in conservative form as

$$\frac{\partial(ru)}{\partial r} + \frac{\partial v}{\partial \theta} + \frac{\partial(rw)}{\partial z} = 0 \tag{D.59}$$

$$\frac{\partial u}{\partial t}+\frac{1}{r}\frac{\partial (ru^2)}{\partial r}+\frac{1}{r}\frac{\partial (uv)}{\partial \theta}+\frac{\partial (uw)}{\partial z}-\frac{v^2}{r}=\text{RHSU} \tag{D.60}$$

$$\frac{\partial v}{\partial t}+\frac{1}{r}\frac{\partial (ruv)}{\partial r}+\frac{1}{r}\frac{\partial v^2}{\partial \theta}+\frac{\partial (vw)}{\partial z}+\frac{uv}{r}=\text{RHSV} \tag{D.61}$$

$$\frac{\partial w}{\partial t}+\frac{1}{r}\frac{\partial (ruw)}{\partial r}+\frac{1}{r}\frac{\partial (vw)}{\partial \theta}+\frac{\partial (w^2)}{\partial z}=\text{RHSW} \tag{D.62}$$

where we have used the cylindrical divergence formula

$$\nabla\cdot\boldsymbol{a}=\frac{1}{r}\frac{\partial (ra_r)}{\partial r}+\frac{1}{r}\frac{\partial a_\theta}{\partial \theta}+\frac{\partial a_z}{\partial z}. \tag{D.63}$$

D.1.3 Scalar equation in cylindrical coordinates

Additionally, the transport equation of a passive scalar (say ρ)

$$\frac{\partial \rho}{\partial t}+\boldsymbol{u}\cdot\nabla\rho=\nu\nabla^2\rho \tag{D.64}$$

can be cast in cylindrical coordinate as

$$\frac{\partial \rho}{\partial t}+u\frac{\partial \rho}{\partial r}+\frac{v}{r}\frac{\partial \rho}{\partial \theta}+w\frac{\partial \rho}{\partial z}=\nu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \rho}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\rho}{\partial\theta^2}+\frac{\partial^2\rho}{\partial z^2}\right] \tag{D.65}$$

D.1.4 Vorticity equation in cylindrical coordinates

The divergence-free condition for vorticity reads

$$\nabla\cdot\boldsymbol{\omega}=\frac{1}{r}\frac{\partial (r\omega_r)}{\partial r}+\frac{1}{r}\frac{\partial \omega_\theta}{\partial \theta}+\frac{\partial \omega_z}{\partial z}=0. \tag{D.66}$$

The convective term is

$$(\boldsymbol{u}\cdot\nabla)\boldsymbol{\omega}=\left(u\frac{\partial}{\partial r}+\frac{v}{r}\frac{\partial}{\partial\theta}+w\frac{\partial}{\partial z}\right)(\boldsymbol{e}_r\omega_r+\boldsymbol{e}_\theta\omega_\theta+\boldsymbol{e}_z\omega_z) \tag{D.67}$$

$$=\left(u\frac{\partial \omega_r}{\partial r}+\frac{v}{r}\frac{\partial \omega_r}{\partial\theta}+w\frac{\partial \omega_r}{\partial z}-\frac{v\omega_\theta}{r}\right)\boldsymbol{e}_r \tag{D.68}$$

$$+\left(u\frac{\partial \omega_\theta}{\partial r}+\frac{v}{r}\frac{\partial \omega_\theta}{\partial\theta}+w\frac{\partial \omega_\theta}{\partial z}+\frac{v\omega_r}{r}\right)\boldsymbol{e}_\theta \tag{D.69}$$

$$+\left(u\frac{\partial \omega_z}{\partial r}+\frac{v}{r}\frac{\partial \omega_z}{\partial\theta}+w\frac{\partial \omega_z}{\partial z}\right)\boldsymbol{e}_z. \tag{D.70}$$

The stretching term is

$$(\boldsymbol{\omega}\cdot\nabla)\boldsymbol{u}=\left(\omega_r\frac{\partial}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial}{\partial\theta}+\omega_z\frac{\partial}{\partial z}\right)(\boldsymbol{e}_r u+\boldsymbol{e}_\theta v+\boldsymbol{e}_z w) \tag{D.71}$$

$$=\left(\omega_r\frac{\partial u}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial u}{\partial\theta}+\omega_z\frac{\partial u}{\partial z}-\frac{v\omega_\theta}{r}\right)\boldsymbol{e}_r \tag{D.72}$$

$$+\left(\omega_r\frac{\partial v}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial v}{\partial\theta}+\omega_z\frac{\partial v}{\partial z}+\frac{u\omega_\theta}{r}\right)\boldsymbol{e}_\theta \tag{D.73}$$

$$+\left(\omega_r\frac{\partial w}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial w}{\partial\theta}+\omega_z\frac{\partial w}{\partial z}\right)\boldsymbol{e}_z. \tag{D.74}$$

The diffusion term is

$$\nabla^2\boldsymbol{\omega}=\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_r}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_r}{\partial\theta^2}-\frac{2}{r^2}\frac{\partial\omega_\theta}{\partial\theta}-\frac{\omega_r}{r^2}+\frac{\partial^2\omega_r}{\partial z^2}\right)\boldsymbol{e}_r \tag{D.75}$$

$$+\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_\theta}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_\theta}{\partial\theta^2}+\frac{2}{r^2}\frac{\partial\omega_r}{\partial\theta}-\frac{\omega_\theta}{r^2}+\frac{\partial^2\omega_\theta}{\partial z^2}\right)\boldsymbol{e}_\theta \tag{D.76}$$

$$+\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_z}{\partial\theta^2}+\frac{\partial^2\omega_z}{\partial z^2}\right)\boldsymbol{e}_z \tag{D.77}$$

Hence, the transport equation of vorticity, (2.22), in the absense of non-conservative external force and baroclinic torque, is cast in cylindrical coordinates as

$$\frac{\partial\omega_r}{\partial t}+u\frac{\partial\omega_r}{\partial r}+\frac{v}{r}\frac{\partial\omega_r}{\partial\theta}+w\frac{\partial\omega_r}{\partial z}=\omega_r\frac{\partial u}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial u}{\partial\theta}+\omega_z\frac{\partial u}{\partial z} \tag{D.78}$$

$$+\nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_r}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_r}{\partial\theta^2}-\frac{2}{r^2}\frac{\partial\omega_\theta}{\partial\theta}-\frac{\omega_r}{r^2}+\frac{\partial^2\omega_r}{\partial z^2}\right) \tag{D.79}$$

$$\frac{\partial\omega_\theta}{\partial t}+u\frac{\partial\omega_\theta}{\partial r}+\frac{v}{r}\frac{\partial\omega_\theta}{\partial\theta}+w\frac{\partial\omega_\theta}{\partial z}+\frac{v\omega_r}{r}=\omega_r\frac{\partial v}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial v}{\partial\theta}+\omega_z\frac{\partial v}{\partial z}+\frac{u\omega_\theta}{r} \tag{D.80}$$

$$+\nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_\theta}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_\theta}{\partial\theta^2}+\frac{2}{r^2}\frac{\partial\omega_r}{\partial\theta}-\frac{\omega_\theta}{r^2}+\frac{\partial^2\omega_\theta}{\partial z^2}\right) \tag{D.81}$$

$$\frac{\partial\omega_z}{\partial t}+u\frac{\partial\omega_z}{\partial r}+\frac{v}{r}\frac{\partial\omega_z}{\partial\theta}+w\frac{\partial\omega_z}{\partial z}=\omega_r\frac{\partial w}{\partial r}+\frac{\omega_\theta}{r}\frac{\partial w}{\partial\theta}+\omega_z\frac{\partial w}{\partial z} \tag{D.82}$$

$$+\nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\omega_z}{\partial\theta^2}+\frac{\partial^2\omega_z}{\partial z^2}\right) \tag{D.83}$$

D.1.5 Vorticity–streamfunction in 3D axisymmetric flow

With $\partial_\theta=0$, the continuity equation simplifies to

$$\frac{\partial(ru)}{\partial r}+\frac{\partial wr}{\partial z}=0 \tag{D.84}$$

such that an axisymmetric (Stokes) streamfunction can be defined as

$$u=-\frac{1}{r}\frac{\partial\psi}{\partial z},\ w=\frac{1}{r}\frac{\partial\psi}{\partial r}. \tag{D.85}$$

Further assuming swirl-free $v=0$, the vorticity–streamfunction equations are

$$\frac{\partial\omega_\theta}{\partial t}+u\frac{\partial\omega_\theta}{\partial r}+w\frac{\partial\omega_\theta}{\partial z}=\frac{u\omega_\theta}{r}+\nu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\omega_\theta}{\partial r}\right)-\frac{\omega_\theta}{r^2}+\frac{\partial^2\omega_\theta}{\partial z^2}\right) \tag{D.86}$$

$$-r\omega_\theta=\frac{\partial^2\psi}{\partial z^2}+\frac{\partial^2\psi}{\partial r^2}-\frac{1}{r}\frac{\partial\psi}{\partial r} \tag{D.87}$$

Also see Panton (2013), appendix 4 for a more complete list of streamfunctions in other coordinates.

D.2 Spherical coordinate

Consider the transformation

$$(x, y, z) \to (r, \phi, \theta) \tag{D.88}$$

where

$$x = r \sin\phi \cos\theta \tag{D.89}$$
$$y = r \sin\phi \sin\theta \tag{D.90}$$
$$z = r \cos\phi \tag{D.91}$$

or

$$r = \sqrt{x^2 + y^2 + z^2} \tag{D.92}$$
$$\phi = \arctan\left(\frac{\sqrt{x^2 + y^2}}{z}\right) \tag{D.93}$$
$$\theta = \arctan\left(\frac{y}{x}\right) \tag{D.94}$$

or thought of as from cylindrical with

$$z = r \cos\phi \tag{D.95}$$
$$r' = r \sin\phi \tag{D.96}$$
$$x = r' \sin\theta \tag{D.97}$$
$$y = r' \cos\theta \tag{D.98}$$

Here θ is the azimuthal angle with x-axis on the equatorial plane and ϕ is the polar angle with z-axis (North), for the convenience of going from cylindrical to polar and backwards.

We have the corresponding relation between unit vectors

$$\begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \\ \boldsymbol{e}_z \end{bmatrix} = \begin{bmatrix} \sin\phi\cos\theta & \cos\phi\cos\theta & -\sin\theta \\ \sin\phi\sin\theta & \cos\phi\sin\theta & \cos\theta \\ \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix} \tag{D.99}$$

and

$$\begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix} = \begin{bmatrix} \sin\phi\cos\theta & \sin\phi\sin\theta & \cos\phi \\ \cos\phi\cos\theta & \cos\phi\sin\theta & -\sin\phi \\ -\sin\theta & \cos\theta & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \\ \boldsymbol{e}_z \end{bmatrix}, \tag{D.100}$$

which can be proven graphically. We note that the grid transformation matrix is unitary and has $\det() = 1$ (rotation matrix).

D.2.1 From cylindrical to spherical

We have the transformation

$$\begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \\ \boldsymbol{e}_z \end{bmatrix} = \begin{bmatrix} \sin\phi\cos\theta & \cos\phi\cos\theta & -\sin\theta \\ \sin\phi\sin\theta & \cos\phi\sin\theta & \cos\theta \\ \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix} \tag{D.101}$$

that can be factorized as

$$\begin{bmatrix} \boldsymbol{e}_x \\ \boldsymbol{e}_y \\ \boldsymbol{e}_z \end{bmatrix} = \begin{bmatrix} \sin\phi\cos\theta & \cos\phi\cos\theta & -\sin\theta \\ \sin\phi\sin\theta & \cos\phi\sin\theta & \cos\theta \\ \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix} \tag{D.102}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \\ \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix} \tag{D.103}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_{r'} \\ \boldsymbol{e}_{\theta'} \\ \boldsymbol{e}_{z'} \end{bmatrix} \tag{D.104}$$

with

$$\begin{bmatrix} \boldsymbol{e}_{r'} \\ \boldsymbol{e}_{\theta'} \\ \boldsymbol{e}_{z'} \end{bmatrix} = \begin{bmatrix} \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \\ \cos\phi & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{e}_r \\ \boldsymbol{e}_\phi \\ \boldsymbol{e}_\theta \end{bmatrix}. \tag{D.105}$$

D.3 General curvilinear coordinates

Consider the coordinate transformations

$$q_i = q_i(x_1, x_2, x_3),\ x_i = x_i(q_1, q_2, q_3) \tag{D.106}$$

where (x_1, x_2, x_3) is the standard Cartesian coordinates and q_i are mutually independent.

D.3.1 Length, area, and volume

Consider the change of the vector

$$\boldsymbol{x} = x_1\boldsymbol{e}_{x_1} + x_2\boldsymbol{e}_{x_2} + x_3\boldsymbol{e}_{x_3} \tag{D.107}$$

$$= q_1\boldsymbol{h}_1 + q_2\boldsymbol{h}_2 + q_3\boldsymbol{h}_3 \tag{D.108}$$

where $\boldsymbol{x} = \boldsymbol{x}(x_i(q_j))$ as

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{e}_{x_1}\mathrm{d}x_1 + \boldsymbol{e}_{x_2}\mathrm{d}x_2 + \boldsymbol{e}_{x_3}\mathrm{d}x_3 \tag{D.109}$$

$$= \frac{\partial \boldsymbol{x}}{\partial q_1}\mathrm{d}q_1 + \frac{\partial \boldsymbol{x}}{\partial q_2}\mathrm{d}q_2 + \frac{\partial \boldsymbol{x}}{\partial q_3}\mathrm{d}q_3 \tag{D.110}$$

and

$$\boldsymbol{h}_i = \frac{\partial \boldsymbol{x}}{\partial q_i}. \tag{D.111}$$

We note that $\boldsymbol{h}_i$ is the change of $\boldsymbol{x}$ with only changing q_i, so it does define direction of coordinate lines of q_i. We denote with $\hat{(\cdot)}$ unit vectors and note that $\boldsymbol{h}_i$ are not necessary unit vectors.

Now consider the length of $\mathrm{d}\boldsymbol{x}$:

$$\mathrm{d}s^2 = \mathrm{d}\boldsymbol{x} \cdot \mathrm{d}\boldsymbol{x} \tag{D.112}$$

$$= \frac{\partial \boldsymbol{x}}{\partial q_j}\mathrm{d}q_j \cdot \frac{\partial \boldsymbol{x}}{\partial q_k}\mathrm{d}q_k \tag{D.113}$$

$$= \frac{\partial x_i}{\partial q_j}\mathrm{d}q_j \frac{\partial x_i}{\partial q_k}\mathrm{d}q_k \tag{D.114}$$

$$= g_{jk}\mathrm{d}q_j\mathrm{d}q_k \tag{D.115}$$

with

$$g_{ij} = \frac{\partial x_l}{\partial q_i}\frac{\partial x_l}{\partial q_j} \tag{D.116}$$

being the metric tensor. When q_i are orthogonal coordinates,

$$\frac{\partial \boldsymbol{x}}{\partial q_i} \cdot \frac{\partial \boldsymbol{x}}{\partial q_j} = \delta_{ij} \tag{D.117}$$

and g_{ij} only has diagonal elements and

$$\mathrm{d}s^2 = g_{11}(\mathrm{d}q_1)^2 + g_{22}(\mathrm{d}q_3)^2 + g_{33}(\mathrm{d}q_3)^2 \tag{D.118}$$

$$= h_1^2(\mathrm{d}q_1)^2 + h_2^2(\mathrm{d}q_3)^2 + h_3^2(\mathrm{d}q_3)^2 \tag{D.119}$$

Define the Lamé parameters as

$$h_1 = \sqrt{g_{11}} = \sqrt{\left(\frac{\partial x_1}{\partial q_1}\right)^2 + \left(\frac{\partial x_2}{\partial q_1}\right)^2 + \left(\frac{\partial x_3}{\partial q_1}\right)^2} = |\boldsymbol{h}_1| \tag{D.120}$$

$$h_2 = \sqrt{g_{22}} = \sqrt{\left(\frac{\partial x_1}{\partial q_2}\right)^2 + \left(\frac{\partial x_2}{\partial q_2}\right)^2 + \left(\frac{\partial x_3}{\partial q_2}\right)^2} = |\boldsymbol{h}_2| \tag{D.121}$$

$$h_3 = \sqrt{g_{33}} = \sqrt{\left(\frac{\partial x_1}{\partial q_3}\right)^2 + \left(\frac{\partial x_2}{\partial q_3}\right)^2 + \left(\frac{\partial x_3}{\partial q_3}\right)^2} = |\boldsymbol{h}_3| \tag{D.122}$$

and unit vectors in q_i directions as

$$\boldsymbol{h}_i = \frac{\boldsymbol{h}_i}{|\boldsymbol{h}_i|} = \frac{\boldsymbol{h}_i}{h_i}. \tag{D.123}$$

We note that the Lamé parameters can depend on the coordinates as

$$h_i = h_i(q_1, q_2, q_3). \tag{D.124}$$

The increment can be rewritten as

$$\begin{aligned} \mathrm{d}\boldsymbol{x} &= h_1 \mathrm{d}q_1 \boldsymbol{h}_1 + h_2 \mathrm{d}q_2 \boldsymbol{h}_2 + h_3 \mathrm{d}q_3 \boldsymbol{h}_3 && \text{(D.125)} \\ &= \mathrm{d}s_1 \boldsymbol{h}_1 + \mathrm{d}s_2 \boldsymbol{h}_2 + \mathrm{d}s_3 \boldsymbol{h}_3 && \text{(D.126)} \end{aligned}$$

with

$$\mathrm{d}s_i \tag{D.127}$$

being the projection of $\mathrm{d}\boldsymbol{x}$ on each coordinate.

Now consider the surface and volume of infinitesimal elements. The (directed) areas of surface elements are

$$\mathrm{d}\sigma_i = \boldsymbol{h}_i \cdot (h_j \mathrm{d}q_j \boldsymbol{h}_j \times h_k \mathrm{d}q_k \boldsymbol{h}_k) = h_j h_k \mathrm{d}q_j \mathrm{d}q_k \tag{D.128}$$

or

$$\begin{aligned} \mathrm{d}\sigma_1 &= h_1 h_2 \mathrm{d}q_1 \mathrm{d}q_2 && \text{(D.129)} \\ \mathrm{d}\sigma_2 &= h_1 h_3 \mathrm{d}q_1 \mathrm{d}q_3 && \text{(D.130)} \\ \mathrm{d}\sigma_3 &= h_1 h_2 \mathrm{d}q_1 \mathrm{d}q_2 && \text{(D.131)} \end{aligned}$$

The volume element (e.g. in volume integrals) spanned by the vector $\mathrm{d}\boldsymbol{x}$ is

$$\begin{aligned} \mathrm{d}V &= (h_1 \mathrm{d}q_1 \boldsymbol{h}_1) \cdot (h_2 \mathrm{d}q_2 \boldsymbol{h}_2 \times h_3 \mathrm{d}q_3 \boldsymbol{h}_3) && \text{(D.132)} \\ &= h_1 \mathrm{d}q_1 h_2 \mathrm{d}q_2 h_3 \mathrm{d}q_3 (\boldsymbol{h}_1) \cdot (\boldsymbol{h}_2 \times \boldsymbol{h}_3) && \text{(D.133)} \\ &= h_1 h_2 h_3 \mathrm{d}q_1 \mathrm{d}q_2 \mathrm{d}q_3 && \text{(D.134)} \end{aligned}$$

when $\boldsymbol{h}_i$ mutually orthogonal.

Example.

For cylindrical coordinate, by definition,

$$\begin{aligned} h_1 &= \sqrt{\left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2} = 1 && \text{(D.135)} \\ h_2 &= \sqrt{\left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2} = r && \text{(D.136)} \\ h_3 &= \sqrt{\left(\frac{\partial x}{\partial z}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2 + \left(\frac{\partial z}{\partial z}\right)^2} = 1 && \text{(D.137)} \end{aligned}$$

D.3.2 Jacobian

Now we consider the Jacobian of the backward transformation

$$(q_1, q_2, q_3) \to (x_1, x_2, x_3) \tag{D.138}$$

which reads

$$\boldsymbol{J} = \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix} \tag{D.139}$$

and the Jacobian determinant (with $\exists\, \boldsymbol{J}^{-1}$)

$$J = \det(\boldsymbol{J}) = \det(\boldsymbol{J}^{\mathrm{T}}) \tag{D.140}$$

$$= \begin{vmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_2}{\partial q_1} & \frac{\partial x_3}{\partial q_1} \\ \frac{\partial x_1}{\partial q_2} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_3}{\partial q_2} \\ \frac{\partial x_1}{\partial q_3} & \frac{\partial x_2}{\partial q_3} & \frac{\partial x_3}{\partial q_3} \end{vmatrix} \tag{D.141}$$

$$= \left(\frac{\partial x_1}{\partial q_1} \boldsymbol{x}_1 + \frac{\partial x_2}{\partial q_1} \boldsymbol{x}_2 + \frac{\partial x_3}{\partial q_1} \boldsymbol{x}_3 \right) \cdot \begin{vmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \boldsymbol{x}_3 \\ \frac{\partial x_1}{\partial q_2} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_3}{\partial q_2} \\ \frac{\partial x_1}{\partial q_3} & \frac{\partial x_2}{\partial q_3} & \frac{\partial x_3}{\partial q_3} \end{vmatrix} \tag{D.142}$$

$$= \frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial q_2} \times \frac{\partial \boldsymbol{x}}{\partial q_3} \right) \tag{D.143}$$

$$= \boldsymbol{h}_1 \cdot (\boldsymbol{h}_2 \times \boldsymbol{h}_3) \tag{D.144}$$

$$= h_1 h_2 h_3 \tag{D.145}$$

$$\neq 0 \tag{D.146}$$

Hence we have

$$\mathrm{d}V = \mathrm{d}x_1 \mathrm{d}x_2 \mathrm{d}x_3 = h_1 h_2 h_3 \mathrm{d}q_1 \mathrm{d}q_2 \mathrm{d}q_3 = J \mathrm{d}q_1 \mathrm{d}q_2 \mathrm{d}q_3. \tag{D.147}$$

D.3.3 Three major calculus theorems

Here we review the three major theorems of calculus and in the next section we will extend them in general curvilinear coordinates.

1. Gradient theorem:

$$\int_{\boldsymbol{l}:\boldsymbol{x}_1 \to \boldsymbol{x}_2} (\nabla f) \cdot \mathrm{d}\boldsymbol{l} = f(\boldsymbol{x}_2) - f(\boldsymbol{x}_1) \tag{D.148}$$

   The integral is independent of path since ∇f is potential (conservative, curl-free).

2. Divergence theorem:

$$\iint_{\Omega} (\nabla \times \boldsymbol{u}) \cdot \mathrm{d}\boldsymbol{A} = \oint_{l=\partial\Omega} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{l} \tag{D.149}$$

   Implication: vorticity is circulation per unit area.

3. Curl theorem:

$$\iiint_{V} (\nabla \cdot \boldsymbol{u})\, \mathrm{d}V = \iint_{\Omega=\partial V} \boldsymbol{u} \cdot \mathrm{d}\boldsymbol{A} \tag{D.150}$$

D.3.4 Differential operators in curvilinear coordinate systems

Next, let's consider differential operators in curvilinear coordinates. Consider a scalar $f = f(q_1, q_2, q_3)$ and its gradient ∇f. Starting from

$$\mathrm{d}f = \frac{\partial f}{\partial q_i}\mathrm{d}q_i, \tag{D.151}$$

due to the displacement $\boldsymbol{x}$. On the other hand,

$$\begin{aligned} \mathrm{d}f &= \nabla f \cdot \mathrm{d}\boldsymbol{x} && \text{(D.152)}\\ &= (\nabla f)_{q_i}\mathrm{d}s_i && \text{(D.153)}\\ &= (\nabla f)_{q_i} h_i \mathrm{d}q_i && \text{(D.154)} \end{aligned}$$

Compare (D.154) and (D.151) we have

$$(\nabla f)_{q_i} = \frac{1}{h_i}\frac{\partial f}{\partial q_i} \tag{D.155}$$

where

$$\begin{aligned} \nabla f &= (\nabla f)_{q_i}\boldsymbol{h}_i && \text{(D.156)}\\ &= \frac{1}{h_1}\frac{\partial f}{\partial q_1}\boldsymbol{h}_1 + \frac{1}{h_2}\frac{\partial f}{\partial q_2}\boldsymbol{h}_2 + \frac{1}{h_3}\frac{\partial f}{\partial q_3}\boldsymbol{h}_3 && \text{(D.157)}\\ &= \sum_{i=1}^{3}\frac{1}{h_i}\frac{\partial f}{\partial q_i}\boldsymbol{h}_i && \text{(D.158)} \end{aligned}$$

Consider the divergence of a vector $\boldsymbol{u}$ in a coordinate-free form:

$$\begin{aligned} \nabla \cdot \boldsymbol{u} &\triangleq \lim_{V\to 0}\frac{\oint_{\Omega=\partial V}\boldsymbol{u}\cdot\mathrm{d}\boldsymbol{\sigma}}{V} && \text{(D.159)}\\ &= \frac{1}{V}\left(\frac{\partial(u_1 h_2 h_3 \mathrm{d}q_2\mathrm{d}q_3)}{\partial q_1}\mathrm{d}q_1 + \frac{\partial(u_2 h_1 h_3 \mathrm{d}q_1\mathrm{d}q_3)}{\partial q_2}\mathrm{d}q_2 + \frac{\partial(u_3 h_1 h_2 \mathrm{d}q_1\mathrm{d}q_2)}{\partial q_3}\mathrm{d}q_3\right) && \text{(D.160)}\\ &= \frac{1}{h_1 h_2 h_3}\left(\frac{\partial(u_1 h_2 h_3)}{\partial q_1} + \frac{\partial(u_2 h_1 h_3)}{\partial q_2} + \frac{\partial(u_3 h_1 h_2)}{\partial q_3}\right) && \text{(D.161)}\\ &= \frac{1}{h_1 h_2 h_3}\sum_{i=1}^{3}\frac{\partial}{\partial q_i}\left(h_1 h_2 h_3 \frac{u_i}{h_i}\right) && \text{(D.162)} \end{aligned}$$

Consider the curl of a vector $\boldsymbol{u}$ in a coordinate-free form, its compoment along $\boldsymbol{n}$ (normal of the surface $\boldsymbol{S} = S\boldsymbol{n}$) is

$$(\nabla \times \boldsymbol{u})\cdot\boldsymbol{n} \triangleq \lim_{S\to 0}\frac{\oint_{l=\partial S}\boldsymbol{u}\cdot\mathrm{d}\boldsymbol{x}}{S} \tag{D.163}$$

and (consider the area spanned by $\mathrm{d}s_2 = h_2\mathrm{d}q_2$ and $\mathrm{d}s_3 = h_3\mathrm{d}q_3$)

$$(\nabla \times \boldsymbol{u})\cdot\boldsymbol{h}_1 = \frac{\oint_l \boldsymbol{u}\cdot\mathrm{d}\boldsymbol{x}}{\mathrm{d}\sigma_1} \tag{D.164}$$

$$= \frac{1}{h_2 h_3 \mathrm{d}q_2 \mathrm{d}q_3}[u_2 h_2 \mathrm{d}q_2 \tag{D.165}$$

$$- (u_2 h_2 + \frac{\partial u_2 h_2}{\partial q_3}\mathrm{d}q_3)\mathrm{d}q_2 \tag{D.166}$$

$$- u_3 h_3 \mathrm{d}q_3 \tag{D.167}$$

$$+ (u_3 h_3 + \frac{\partial u_3 h_3}{\partial q_2}\mathrm{d}q_2)\mathrm{d}q_3] \tag{D.168}$$

$$= \frac{1}{h_2 h_3}\left(\frac{\partial u_3 h_3}{\partial q_2} - \frac{\partial u_2 h_2}{\partial q_3}\right) \tag{D.169}$$

$$(\nabla \times \boldsymbol{u}) \cdot \boldsymbol{h}_2 = \frac{1}{h_1 h_3}\left(\frac{\partial u_1 h_1}{\partial q_3} - \frac{\partial u_3 h_3}{\partial q_1}\right) \tag{D.170}$$

$$(\nabla \times \boldsymbol{u}) \cdot \boldsymbol{h}_3 = \frac{1}{h_1 h_2}\left(\frac{\partial u_2 h_2}{\partial q_1} - \frac{\partial u_1 h_1}{\partial q_2}\right) \tag{D.171}$$

We note that the Lamé coefficient also changes as the coordinate changes.

In determinant form,

$$\nabla \times \boldsymbol{u} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\boldsymbol{h}_1 & h_2\boldsymbol{h}_2 & h_3\boldsymbol{h}_3 \\ \partial_{q_1} & \partial_{q_2} & \partial_{q_3} \\ h_1 u_1 & h_2 u_2 & h_3 u_3 \end{vmatrix} \tag{D.172}$$

The Laplacian can be obtained by taking the divergence of ∇f as combining (D.158) and (D.162)

$$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{1}{h_1 h_2 h_3}\left(\frac{\partial}{\partial q_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial q_2}\right) + \frac{\partial}{\partial q_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial q_3}\right)\right) \tag{D.173}$$

$$= \frac{1}{h_1 h_2 h_3}\sum_{i=1}^{3}\frac{\partial}{\partial q_i}\left(\frac{h_1 h_2 h_3}{h_i^2}\frac{\partial f}{\partial q_i}\right) \tag{D.174}$$

The convection operator can similarly be expanded as

$$(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \left(\sum_{j=1}^{3}\frac{u_j}{h_j}\frac{\partial}{\partial q_j}\right)(\sum_{i=1}^{3} u_i \boldsymbol{h}_i) \tag{D.175}$$

$$= \sum_{i,j=1}^{3}\frac{u_j}{h_j}\frac{\partial u_i}{\partial q_j}\boldsymbol{h}_i + \sum_{i,j=1}^{3}\frac{u_j}{h_j}\frac{\partial \boldsymbol{h}_i}{\partial q_j}u_i \tag{D.176}$$

$$= (\boldsymbol{u} \cdot \nabla u_i)\boldsymbol{h}_i + (\boldsymbol{u} \cdot \nabla \boldsymbol{h}_i)u_i \tag{D.177}$$

where the first and second terms are the component and geometrical derivatives, respectively. It's easy to see the geometrical derivative in cylindrical coordinates ($\boldsymbol{h}_1 = \boldsymbol{e}_r, \boldsymbol{h}_2 = \boldsymbol{e}_\theta, \boldsymbol{h}_3 = \boldsymbol{e}_z$, $q_1 = r, q_2 = \theta, q_3 = z$, $h_1 = h_3 = 1, h_2 = r$) is

$$\begin{bmatrix} u_r & u_\theta & u_z \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ \boldsymbol{e}_\theta/r & -\boldsymbol{e}_r/r & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix} = \frac{u_r u_\theta}{r}\boldsymbol{e}_\theta - \frac{u_\theta^2}{r}\boldsymbol{e}_r \tag{D.178}$$

D.3.5 Derivatives of unit vectors

In general curvilinear coordinates, the directions of unit vectors could change with coordinate as well. We are basically concerned about

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} \tag{D.179}$$

and we will establish that

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} // \boldsymbol{h}_j,\ i \neq j. \tag{D.180}$$

First we have

$$\boldsymbol{h}_i \cdot \frac{\partial \boldsymbol{h}_i}{\partial q_j} = \frac{\partial \boldsymbol{h}_i^2/2}{\partial q_j} = 0 \tag{D.181}$$

and hence

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} \perp \boldsymbol{h}_i,\ i \neq j. \tag{D.182}$$

According to the orthogonality we have

$$\boldsymbol{h}_1 \cdot \boldsymbol{h}_2 = \frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial \boldsymbol{x}}{\partial q_2} = 0 \tag{D.183}$$

and

$$0 = \frac{\partial}{\partial q_3}\left(\frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial \boldsymbol{x}}{\partial q_2}\right) = \frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_2 \partial q_3} + \frac{\partial^2 \boldsymbol{x}}{\partial q_1 \partial q_3} \cdot \frac{\partial \boldsymbol{x}}{\partial q_2} \tag{D.184}$$

$$= \frac{\partial \boldsymbol{x}}{\partial q_2} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_3 \partial q_1} + \frac{\partial^2 \boldsymbol{x}}{\partial q_2 \partial q_1} \cdot \frac{\partial \boldsymbol{x}}{\partial q_3} \tag{D.185}$$

$$= \frac{\partial \boldsymbol{x}}{\partial q_3} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_1 \partial q_2} + \frac{\partial^2 \boldsymbol{x}}{\partial q_3 \partial q_2} \cdot \frac{\partial \boldsymbol{x}}{\partial q_1} \tag{D.186}$$

then

$$\frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_2 \partial q_3} + \frac{\partial \boldsymbol{x}}{\partial q_2} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_3 \partial q_1} + \frac{\partial \boldsymbol{x}}{\partial q_3} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_1 \partial q_2} = 0 \tag{D.187}$$

and then

$$\frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_2 \partial q_3} = 0 \tag{D.188}$$

$$\frac{\partial \boldsymbol{x}}{\partial q_2} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_1 \partial q_3} = 0 \tag{D.189}$$

$$\frac{\partial \boldsymbol{x}}{\partial q_3} \cdot \frac{\partial^2 \boldsymbol{x}}{\partial q_1 \partial q_2} = 0 \tag{D.190}$$

From

$$0 = \frac{\partial \boldsymbol{x}}{\partial q_1} \cdot \frac{\partial}{\partial q_2}\left(\frac{\partial \boldsymbol{x}}{\partial q_3}\right) \tag{D.191}$$

$$= h_1 \boldsymbol{h}_1 \cdot \frac{\partial h_3 \boldsymbol{h}_3}{\partial q_2} \tag{D.192}$$

$$= h_1 \boldsymbol{h}_1 \cdot \left(h_3 \frac{\partial \boldsymbol{h}_3}{\partial q_2} + \boldsymbol{h}_3 \frac{\partial h_3}{\partial q_2} \right) \tag{D.193}$$

$$= h_1 h_3 \boldsymbol{h}_1 \cdot \frac{\partial \boldsymbol{h}_3}{\partial q_2} \tag{D.194}$$

we have (similarly)

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} \perp \boldsymbol{h}_k,\ i \neq j \neq k \neq i. \tag{D.195}$$

Combining (D.182) and (D.195) we have

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} // \boldsymbol{h}_j,\ i \neq j, \tag{D.196}$$

and using

$$\frac{\partial^2 \boldsymbol{x}}{\partial q_i \partial q_j} = \frac{\partial^2 \boldsymbol{x}}{\partial q_j \partial q_i} \tag{D.197}$$

we have

$$\frac{\partial}{\partial q_j} \left(\frac{\partial \boldsymbol{x}}{\partial q_i} \right) = \frac{\partial}{\partial q_i} \left(\frac{\partial \boldsymbol{x}}{\partial q_j} \right) \tag{D.198}$$

$$\boldsymbol{h}_i \frac{\partial h_i}{\partial q_j} + h_i \frac{\partial \boldsymbol{h}_i}{\partial q_j} = \boldsymbol{h}_j \frac{\partial h_j}{\partial q_i} + h_j \frac{\partial \boldsymbol{h}_j}{\partial q_i} \tag{D.199}$$

with repeated indices not implying summation. Since $i \neq j$, $\boldsymbol{h}_i$ and $\boldsymbol{h}_j$ are linearly independent, we have

$$\frac{\partial \boldsymbol{h}_i}{\partial q_j} = \frac{1}{h_i} \frac{\partial h_j}{\partial q_i} \boldsymbol{h}_j. \tag{D.200}$$

Now we turn back and consider $\partial \boldsymbol{h}_i / \partial q_j$.

$$\frac{\partial \boldsymbol{h}_i}{\partial q_i} = \frac{\partial (\boldsymbol{h}_j \times \boldsymbol{h}_k)}{\partial q_i} \tag{D.201}$$

$$= \frac{\partial \boldsymbol{h}_j}{\partial q_i} \times \boldsymbol{h}_k + \boldsymbol{h}_j \times \frac{\partial \boldsymbol{h}_k}{\partial q_i} \tag{D.202}$$

$$= \frac{1}{h_j} \frac{\partial h_i}{\partial q_j} \boldsymbol{h}_i \times \boldsymbol{h}_k + \boldsymbol{h}_j \times \boldsymbol{h}_i \frac{1}{h_k} \frac{\partial h_i}{\partial q_k} \tag{D.203}$$

$$= -\left(\frac{1}{h_j} \frac{\partial h_i}{\partial q_j} \boldsymbol{h}_j + \frac{1}{h_k} \frac{\partial h_i}{\partial q_k} \boldsymbol{h}_k \right) \tag{D.204}$$

without repeated indices being summed over.

Using the relations (D.200) and (D.204), gradient, curl, divergence, Laplacian, as well as operators like $\nabla \boldsymbol{u}$ and $\boldsymbol{u} \cdot \nabla \boldsymbol{u}$ can be expressed.

Example: $\nabla \cdot \boldsymbol{u}$.

We have before

$$\nabla = \frac{\boldsymbol{h}_1}{h_1} \frac{\partial}{\partial q_1} + \frac{\boldsymbol{h}_2}{h_2} \frac{\partial}{\partial q_2} + \frac{\boldsymbol{h}_3}{h_3} \frac{\partial}{\partial q_3} \tag{D.205}$$

and now consider $\nabla \boldsymbol{u}$ with

$$\boldsymbol{u} = u_1 \boldsymbol{h}_1 + u_2 \boldsymbol{h}_2 + u_3 \boldsymbol{h}_3 \tag{D.206}$$

and we have

$$\nabla \cdot \boldsymbol{u} = \left(\frac{\boldsymbol{h}_1}{h_1} \frac{\partial}{\partial q_1} + \frac{\boldsymbol{h}_2}{h_2} \frac{\partial}{\partial q_2} \boldsymbol{h}_2 + \frac{\boldsymbol{h}_3}{h_3} \frac{\partial}{\partial q_3} \boldsymbol{h}_3 \right) \cdot (u_1 \boldsymbol{h}_1 + u_2 \boldsymbol{h}_2 + u_3 \boldsymbol{h}_3) \tag{D.207}$$

$$= \frac{\boldsymbol{h}_1}{h_1} \frac{\partial}{\partial q_1} (u_1 \boldsymbol{h}_1 + u_2 \boldsymbol{h}_2 + u_3 \boldsymbol{h}_3) + ... \tag{D.208}$$

$$= \frac{1}{h_1} \left(\frac{\partial u_1}{\partial q_1} + \frac{u_2}{h_2} \frac{\partial h_1}{\partial q_2} + \frac{u_3}{h_3} \frac{\partial h_1}{\partial q_3} \right) \tag{D.209}$$

$$+ \frac{1}{h_2} \left(\frac{\partial u_2}{\partial q_2} + \frac{u_3}{h_3} \frac{\partial h_2}{\partial q_3} + \frac{u_1}{h_1} \frac{\partial h_2}{\partial q_1} \right) \tag{D.210}$$

$$+ \frac{1}{h_3} \left(\frac{\partial u_3}{\partial q_3} + \frac{u_1}{h_1} \frac{\partial h_3}{\partial q_1} + \frac{u_2}{h_2} \frac{\partial h_3}{\partial q_2} \right) \tag{D.211}$$

$$= \frac{1}{h_1 h_2 h_3} \left(\frac{\partial u_1 h_2 h_3}{\partial q_1} + \frac{\partial u_2 h_1 h_3}{\partial q_2} + \frac{\partial u_3 h_1 h_2}{\partial q_3} \right) \tag{D.212}$$

References: Appendices in Batchelor (1967); Griffiths (2013), and textbook of Wu (1982).

D.3.6 Examples

1. Cartesian. $(q_1, q_2, q_3) = (x_1, x_2, x_3)$
   Elements:

$$h_1 = h_2 = h_3 = 1 \tag{D.213}$$

$$\mathrm{d}s^2 = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 \tag{D.214}$$

$$\mathrm{d}V = \mathrm{d}x_1 \mathrm{d}x_2 \mathrm{d}x_3 \tag{D.215}$$

   Operators:

$$\nabla f = \frac{\partial f}{\partial x_1} \boldsymbol{e}_1 + \frac{\partial f}{\partial x_2} \boldsymbol{e}_2 + \frac{\partial f}{\partial x_3} \boldsymbol{e}_3 \tag{D.216}$$

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \tag{D.217}$$

$$\nabla \times \boldsymbol{u} = \begin{vmatrix} \boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z \\ \partial_x & \partial_y & \partial_z \\ u & v & w \end{vmatrix} \tag{D.218}$$

$$= (\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}) \boldsymbol{e}_x + (\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}) \boldsymbol{e}_y + (\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}) \boldsymbol{e}_z \tag{D.219}$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \tag{D.220}$$

2. Cylindrical. $(q_1, q_2, q_3) = (r, \theta, z)$
   Elements:

$$h_1 = h_3 = 1, h_2 = r \tag{D.221}$$

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + \mathrm{d}z^2 \tag{D.222}$$
$$\mathrm{d}V = r\mathrm{d}r\mathrm{d}\theta\mathrm{d}z \tag{D.223}$$

Operators:

$$\nabla f = \frac{\partial f}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\boldsymbol{e}_\theta + \frac{\partial f}{\partial z}\boldsymbol{e}_z \tag{D.224}$$
$$\nabla \cdot \boldsymbol{u} = \frac{1}{r}\left(\frac{\partial (ru)}{\partial r} + \frac{\partial v}{\partial \theta} + \frac{\partial (rw)}{\partial z}\right) \tag{D.225}$$
$$= \frac{1}{r}\frac{\partial (ru)}{\partial r} + \frac{1}{r}\frac{\partial v}{\partial \theta} + \frac{\partial w}{\partial z} \tag{D.226}$$
$$\nabla \times \boldsymbol{u} = \frac{1}{r}\begin{vmatrix} \boldsymbol{e}_r & r\boldsymbol{e}_\theta & \boldsymbol{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ u & rv & w \end{vmatrix} \tag{D.227}$$
$$= (\frac{1}{r}\frac{\partial w}{\partial \theta} - \frac{\partial v}{\partial z})\boldsymbol{e}_r + (\frac{\partial u}{\partial z} - \frac{\partial w}{\partial r})\boldsymbol{e}_\theta + \frac{1}{r}(\frac{\partial rv}{\partial r} - \frac{\partial u}{\partial \theta})\boldsymbol{e}_z \tag{D.228}$$
$$\nabla^2 f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2} \tag{D.229}$$

3. Spherical. $(q_1, q_2, q_3) = (r, \phi, \theta)$, ϕ is the polar angle and θ is the azimuthal.
   Elements:

$$h_1 = 1, h_2 = r, h_3 = r\sin\phi \tag{D.230}$$
$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\phi^2 + r^2\sin^2\phi\mathrm{d}\theta^2 \tag{D.231}$$
$$\mathrm{d}V = r^2\sin\phi\mathrm{d}r\mathrm{d}\phi\mathrm{d}\theta \tag{D.232}$$

Operators:

$$\nabla f = \frac{\partial f}{\partial r}\boldsymbol{e}_r + \frac{1}{r}\frac{\partial f}{\partial \phi}\boldsymbol{e}_\phi + \frac{1}{r\sin\phi}\frac{\partial f}{\partial \theta}\boldsymbol{e}_\theta \tag{D.233}$$
$$\nabla \cdot \boldsymbol{u} = \frac{1}{r^2\sin\phi}\left(\frac{\partial (r^2\sin\phi u)}{\partial r} + \frac{\partial (r\sin\phi v)}{\partial \phi} + \frac{\partial (rw)}{\partial \theta}\right) \tag{D.234}$$
$$= \frac{1}{r^2}\frac{\partial (r^2 u)}{\partial r} + \frac{1}{r\sin\phi}\frac{\partial (\sin\phi v)}{\partial \phi} + \frac{1}{r\sin\phi}\frac{\partial w}{\partial \theta} \tag{D.235}$$
$$\nabla \times \boldsymbol{u} = \frac{1}{r^2\sin\phi}\begin{vmatrix} \boldsymbol{e}_r & r\boldsymbol{e}_\phi & r\sin\phi\boldsymbol{e}_\theta \\ \partial_r & \partial_\phi & \partial_\theta \\ u & rv & r\sin\phi w \end{vmatrix} \tag{D.236}$$
$$= \frac{1}{r\sin\phi}(\frac{\partial \sin\phi w}{\partial \phi} - \frac{\partial v}{\partial \theta})\boldsymbol{e}_r + (\frac{1}{r\sin\phi}\frac{\partial u}{\partial \theta} - \frac{1}{r}\frac{\partial rw}{\partial r})\boldsymbol{e}_\phi + \frac{1}{r}(\frac{\partial rv}{\partial r} - \frac{\partial u}{\partial \phi})\boldsymbol{e}_\theta \tag{D.237}$$
$$\nabla^2 f = \frac{1}{r^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)\right) + \frac{1}{r^2\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial f}{\partial \phi}\right) + \frac{1}{r^2\sin\phi}\frac{\partial^2 f}{\partial \theta^2} \tag{D.238}$$

D.3.7 Galilean transformation and invariance

Appendix E

Special functions

E.1 Polynomials and Taylor's expansions

The Euler's identity

$$e^{i\pi} + 1 = 0, \tag{E.1}$$

is actualy a special case of the Euler's formula

$$e^{ix} = \cos x + i \sin x, \tag{E.2}$$

where $i = \sqrt{-1}$ is the imaginary unit.

It can be shown by Taylor expansions of $\cos x$ and $\sin x$ at $x = 0$ in Table E.1 and we have already observed some similarities to the Taylor expansion of an exponential

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + ... \tag{E.3}$$

Taking the Laurent expansion of e^{ix} at $x = 0$ we have proven that

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \tag{E.4}$$

$$= 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} - \frac{x^6}{6!} + ... \tag{E.5}$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + ...\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + ...\right) \tag{E.6}$$

$$= \cos x + i \sin x. \tag{E.7}$$

The Taylor expansion of an infinity continuously differentiable function $f \in \mathbb{C}$ around $x = x_0$ is

$$f(x) = \sum_{n=0}^{\infty} \frac{f(n)(x_0)}{n!}(x - x_0)^n, \tag{E.8}$$

and the expansions of commonly used functions are listed in the Table E.1.

Note that C_n^α is the combination number representing the total number of possibilities of selecting n from α and B_k are the Bernoulli numbers.

In this section, we will also connect the functions to their defining ODEs – the ODEs that they satisfy – to reveal their physical implications.

function (convergence radius R)	expansion sum	power series
Geometric functions ($\|x\| < 1$)		
$\frac{1}{1-x}$	$= \sum_{\infty}^{n=0} x^n$	$= 1 + x + x^2 + x^3 + ...$
$\frac{1}{(1-x)^2}$	$= \sum_{\infty}^{n=1} nx^{n-1}$	$= 1 + 2x + 3x^2 + 4x^3 + ...$
Binomial functions ($\|x\| < 1$)	$C_n^\alpha = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$	$\alpha \in \mathbb{C}$
$(1+x)^\alpha$	$= \sum_{\infty}^{n=0} C_n^\alpha x^n$	
$(1+x)^{1/2}$		$= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + ...$
$(1+x)^{-1/2}$		$= 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \frac{5}{16}x^3 + ...$
Trigonometric functions ($R = \infty$)		
$\sin x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1}$	$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + ...$
$\cos x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}x^{2n}$	$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + ...$
($\|x\| < \pi/2$), $\tan x$	$= \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!}x^{2n-1}$	$= x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + ...$
($\|x\| < \pi/2$), $\sec x$		$= 1 + \frac{x^2}{2} + \frac{5x^4}{24} + ...$
($\|x\| \leq 1$), $\arcsin x$		$= x + \frac{x^3}{6} + \frac{3x^5}{40} + ...$
($\|x\| \leq 1$), $\arccos x$		$= \frac{\pi}{2} - \arcsin x$
($\|x\| \leq 1, x \neq \pm i$), $\arctan x$	$= \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}x^{2n+1}$	$= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...$
Hyperbolic functions ($R = \infty$)		
$\sinh x$	$= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$	$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + ...$
$\cosh x$	$= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$	$= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + ...$
$\tanh x$	$= \sum_{n=1}^{\infty} \frac{B_{2n}4^n(4^n-1)}{(2n)!}x^{2n-1}$	$= x - \frac{x^3}{3} + \frac{2x^5}{15} - \frac{17x^7}{315} + ...$
$\operatorname{arctanh} x$	$= \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1}$	$= x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + ...$

Table E.1: Taylor (polynomial) expansion of some special functions. These expansions can be used to draw mental pictures of these functions. The similarity in the expansion of the hyperbolic functions to the corresponding trigonometric functions can be seen.

E.2 Trigonometric functions

E.2.1 Trigonometric relations

E.3 Hyperbolic functions

The hyperbolic functions are defined as

$$\sinh x = \frac{e^x - e^{-x}}{2} \tag{E.9}$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \tag{E.10}$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{E.11}$$

$$\tag{E.12}$$

and their first derivatives are listed in Table E.2.

$f(x)$	$f'(x)$
$\sinh x$	$\cosh x$
$\cosh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x \, (= 1/\cosh^2 x)$

Table E.2: Derivatives of hyperbolic functions.

It is easy to verify that both $y = \sinh kx$ and $y = \cosh kx$ are solutions to the second-order ODE

$$y'' - k^2 y = 0. \tag{E.13}$$

Similar to

$$\sin^2 x + \cos^2 x = 1, \tag{E.14}$$

for hyperbolic functions we have

$$\cosh^2 x - \sinh^2 x = 1. \tag{E.15}$$

Examples.
Mixing layer:

$$U(y) = U_0 \tanh(y/L). \tag{E.16}$$

Jet:

$$U(y) = U_0 \operatorname{sech}^2(y/L) = U_0 / \cosh^2(y/L). \tag{E.17}$$

We will see this in section xx as analytical solutions to lam/turb profiles? First derivatives, second derivatives, inflection points.

E.4 Chebyshev functions

The Chebyshev polynomials of the first kind are defined in $[-1, 1]$ as

$$T_n = \cos(n\theta),\ \theta = \arccos(x), \tag{E.18}$$

with the first several being

$$T_0(x) = 1 \tag{E.19}$$
$$T_1(x) = x \tag{E.20}$$
$$T_2(x) = 2x^2 - 1 \tag{E.21}$$
$$T_3(x) = 4x^3 - 3x \tag{E.22}$$
$$... \tag{E.23}$$
$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \tag{E.24}$$

The extrema of the N-th order Chebyshev polynomial are (Gauss–Lobato collocation points)

$$x_i = \cos\left(\frac{i\pi}{N}\right),\ i = 0, 1, ..., N \tag{E.25}$$

while the zeros are

$$x_i = \cos\left(\frac{\pi}{2}\frac{2i+1}{N}\right),\ i = 0, 1, ..., N+1. \tag{E.26}$$

E.4.1 Chebyshev spectral methods

E.5 Bessel functions

Bessel functions are eigenfunctions of the cylindrical Laplacian operator (2D for now):

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) - m^2 \tag{E.27}$$

such that

$$\nabla^2 f + \lambda^2 f = 0, \tag{E.28}$$

where $\lambda^2 > 0$ is the eigenvalue. Expand the derivatives we get

$$r^2 f'' + rf' + (\lambda^2 r^2 - m^2)f = 0. \tag{E.29}$$

By the change of variable $x = \lambda r$ we have the defining ODE (Bessel equation) of Bessel functions

$$x^2 f'' + xf' + (x^2 - \nu^2)f = 0, \tag{E.30}$$

where

$$m^2 = -\frac{G''}{G} \tag{E.31}$$

is the azimuthal wavenumber for separated variables $f(r, \theta) = F(r)G(\theta)$. The eigenfunctions of the Laplacian in the homogeneous direction are harmonic functions. ν is also called the order of the Bessel function J_ν.

An asymptotic behavior of the Bessel functions as $r = 0$ is approached is

$$J_\nu \to \begin{cases} 1 - \frac{r^2}{4} \to 1, \, \nu = 0 \\ \frac{1}{\nu!} \left(\frac{r}{2}\right)^\nu \sim r^\nu, \, \nu \neq 0 \end{cases} , \, r \to 0. \tag{E.32}$$

When the eigenvalue is negative $(-\lambda^2)$, or through a change of variable $\bar{x} = \mathrm{i}x$ we have the modified Bessel equation

$$x^2 f'' + x f' - (x^2 + \nu^2) f = 0, \tag{E.33}$$

leading to the modified Bessel functions I_ν and K_ν of the first and second kind, respectively.

Appendix F

Fundamentals of Numerical Mathematics

F.1 Numerical Approximation

F.2 Numerical Interpolation

F.3 Numerical Integration

Runge-Kutta, Adams-Bashforth;

F.4 Matrix Analysis

F.5 Filtering

Bibliography

Aris, R. 1989 *Vectors, tensors and the basic equations of fluid mechanics*. Dover Publications.

Arobone, E. & Sarkar, S. 2012 Evolution of a stratified rotating shear layer with horizontal shear. part i. linear stability. *Journal of fluid mechanics* **703**, 29–48.

Arratia, C. 2011 Non-modal instability mechanisms in stratified and homogeneous shear flow. PhD thesis, Ecole Polytechnique X.

Ashurst, W. T., Kerstein, A., Kerr, R. & Gibson, C. 1987 Alignment of vorticity and scalar gradient with strain rate in simulated navier–stokes turbulence. *The Physics of fluids* **30** (8), 2343–2353.

Batchelor, G. K. 1967 *An Introduction to Fluid Dynamics*. Cambridge university press.

Benzi, M., Golub, G. H. & Liesen, J. 2005 Numerical solution of saddle point problems. *Acta numerica* **14**, 1–137.

Burridge, D. & Drazin, P. 1969 Comments on "stability of pipe poiseuille flow". *The Physics of Fluids* **12** (1), 264–265.

Cardesa, J. & Lozano-Durán, A. 2019 Inter-scale energy transfer in turbulence from the viewpoint of subfilter scales. *arXiv preprint arXiv:1912.11143* .

Charru, F. 2011 *Hydrodynamic instabilities*, , vol. 37. Cambridge University Press.

Cho, M., Hwang, Y. & Choi, H. 2018 Scale interactions and spectral energy transfer in turbulent channel flow. *Journal of Fluid Mechanics* **854**, 474–504.

Choi, K.-S. & Lumley, J. L. 2001 The return to isotropy of homogeneous turbulence. *Journal of Fluid Mechanics* **436**, 59–84.

Chong, M. S., Perry, A. E. & Cantwell, B. J. 1990 A general classification of three-dimensional flow fields. *Physics of Fluids A: Fluid Dynamics* **2** (5), 765–777.

Chorin, A. J. 1968 Numerical solution of the navier-stokes equations. *Mathematics of computation* **22** (104), 745–762.

Chorin, A. J. & Marsden, J. E. 1990 *A Mathematical Introduction to Fluid Mechanics*, , vol. 3. Springer.

Chou, P.-Y. 1945 On velocity correlations and the solutions of the equations of turbulent fluctuation. *Quarterly of applied mathematics* **3** (1), 38–54.

Cooley, J. W. & Tukey, J. W. 1965 An algorithm for the machine calculation of complex fourier series. *Mathematics of computation* **19** (90), 297–301.

Drazin, P. G. 2002 *Introduction to Hydrodynamic Stability*, , vol. 32. Cambridge University Press.

Drazin, P. G. & Riley, N. 2006 *The Navier-Stokes equations: a classification of flows and exact solutions*. Cambridge University Press.

Frigo, M. & Johnson, S. G. 2020 Fftw user's manual. *Massachusetts Institute of Technology* .

Germano, M. 1986 A proposal for a redefinition of the turbulent stresses in the filtered navier-stokes equations. *Physics of Fluids* **29** (7), 2323–2324.

Germano, M., Piomelli, U., Moin, P. & Cabot, W. H. 1991 A dynamic subgrid-scale eddy viscosity model. *Physics of Fluids A: Fluid Dynamics* **3** (7), 1760–1765.

Gresho, P. M. & Sani, R. L. 1987 On pressure boundary conditions for the incompressible navier-stokes equations. *International journal for numerical methods in fluids* **7** (10), 1111–1145.

Gresho, P. M. & Sani, R. L. 1998 Incompressible flow and the finite element method. volume 1: Advection-diffusion and isothermal laminar flow .

Griffiths, D. J. 2013 *Introduction to Electrodynamics*, 4th edn. Cambridge University Press.

Heaton, C. & Peake, N. 2007 Transient growth in vortices with axial flow. *Journal of Fluid Mechanics* **587**, 271–301.

Holton, J. R. 1972 *An Introduction to Dynamic Meteorology*. Academic Press.

Howard, L. N. 1961 Note on a paper of john w. miles. *Journal of Fluid Mechanics* **10** (4), 509–512.

Hunt, J. C., Wray, A. A. & Moin, P. 1988 Eddies, streams, and convergence zones in turbulent flows. *Studying turbulence using numerical simulation databases, 2. Proceedings of the 1988 summer program* .

Hussain, A. K. M. F. & Reynolds, W. C. 1970 The mechanics of an organized wave in turbulent shear flow. *Journal of Fluid mechanics* **41** (2), 241–258.

Jeong, J. & Hussain, F. 1995 On the identification of a vortex. *Journal of fluid mechanics* **285**, 69–94.

Johnson, P. L. 2021 On the role of vorticity stretching and strain self-amplification in the turbulence energy cascade. *Journal of Fluid Mechanics* **922**, A3.

Kaminski, A., Caulfield, C. & Taylor, J. 2014 Transient growth in strongly stratified shear layers. *Journal of Fluid Mechanics* **758**, R4.

Kim, J., Moin, P. & Moser, R. 1987 Turbulence statistics in fully developed channel flow at low reynolds number. *Journal of fluid mechanics* **177**, 133–166.

Kloosterziel, R. & Van Heijst, G. 1991 An experimental study of unstable barotropic vortices in a rotating fluid. *Journal of Fluid Mechanics* **223**, 1–24.

Kundu, P. K., Cohen, I. M. & Dowling, D. R. 2015 *Fluid mechanics*. Academic Press.

Leonard, A. 1975 Energy cascade in large-eddy simulations of turbulent fluid flows. In *Advances in geophysics*, , vol. 18, pp. 237–248. Elsevier.

Luchini, P. & Bottaro, A. 2014 Adjoint equations in stability analysis. *Annual Review of fluid mechanics* **46**, 493–517.

Lumley, J. L. & Newman, G. R. 1977 The return to isotropy of homogeneous turbulence. *Journal of Fluid Mechanics* **82** (1), 161–178.

Manzoni, A., Quarteroni, A. & Salsa, S. 2021 *Optimal control of partial differential equations*. Springer.

Meneveau, C. 2011 Lagrangian dynamics and models of the velocity gradient tensor in turbulent flows. *Annual Review of Fluid Mechanics* **43**, 219–245.

Mizuno, Y. 2016 Spectra of energy transport in turbulent channel flows for moderate reynolds numbers. *Journal of Fluid Mechanics* **805**, 171–187.

Orszag, S. A. 1971 Accurate solution of the orr–sommerfeld stability equation. *Journal of Fluid Mechanics* **50** (4), 689–703.

Panton, R. L. 2013 *Incompressible flow*, 4th edn. John Wiley & Sons.

Pedlosky, J. 2013 *Geophysical Fluid Dynamics*. Springer Science & Business Media.

Pope, S. B. 2001 *Turbulent Flows*. Cambridge University Press.

Schlichting, H. & Gersten, K. 2016 *Boundary-layer Theory*. Springer.

Schmid, P. J. 2007 Nonmodal stability theory. *Annu. Rev. Fluid Mech.* **39**, 129–162.

Schmid, P. J. & Henningson, D. S. 2001 *Stability and transition in shear flows*, , vol. 142. Springer-Verlag New Work.

Shi, X. 1994 *Turbulent flows*. Tianjin University Press.

Welch, P. 1967 The use of fast fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Transactions on audio and electroacoustics* **15** (2), 70–73.

White, F. M. 2008 *Fluid Mechanics*, 6th edn. McGraw-Hill.

Wu, W.-Y. 1982 *Fluid Mechanics*. Peking University Press.

Zhou, J., Adrian, R. J., Balachandar, S. & Kendall, T. 1999 Mechanisms for generating coherent packets of hairpin vortices in channel flow. *Journal of fluid mechanics* **387**, 353–396.